

Lecture Notes on Mathematical Analysis

Single and Multivariable Calculus

Tommy Chen

Version 3.1.0 (revision draft)

Revision date: September 12, 2026

© 2026 Tommy Chen

This work is licensed under the [Creative Commons Attribution 4.0 International License](https://creativecommons.org/licenses/by/4.0/) (CC BY 4.0).

Contents

Preface	v
How to Read These Notes	vi
I Single-Variable Calculus and Analysis	1
1 Motivation and the Transition from Calculus to Analysis	2
1.1 Why Calculus Needs Foundations	2
1.2 A First Look at Limits	2
1.3 The Structural Ideas of Analysis	3
1.4 Warnings That Motivate Hypotheses	3
1.5 Road Map	4
Exercises	4
2 Basic Set Theory, Logic, Functions, and Relations	5
2.1 Sets	5
2.2 Logic and Proofs	6
2.2.1 Writing Proofs	7
2.3 Relations and Functions	8
2.4 The Natural Numbers, Integers, and Rational Numbers	11
2.5 Countability	14
2.6 Choice	16
Exercises	16
3 The Real Number System	18
3.1 Ordered Fields and the Problem to Be Solved	18
3.2 Rational Numbers, Completeness, and Uniqueness	20
3.3 Cauchy Sequences of Rational Numbers	24
3.4 Order and Completeness	26
3.5 A Small Toolkit of Inequalities	30
Exercises	31
4 Sequences of Real Numbers	32
4.1 Convergence	32
4.2 Algebra and Order of Limits	34
4.3 Subsequences and Elementary Divergence	35
4.4 Completeness Consequences	36
4.5 Equivalent Completeness Principles	37
Exercises	40

5	Limits and Continuity of Functions	41
5.1	Limits at a Point	41
5.2	Asymptotic Notation	44
5.3	Continuity	45
5.4	Uniform Continuity	47
5.5	Intermediate Values	48
5.6	Compact Intervals	49
	Exercises	50
6	Sequences and Families of Functions	51
6.1	Pointwise and Uniform Convergence	51
6.2	Continuous Limits	52
6.3	Polynomial Approximation	54
6.4	Equicontinuity	55
	Exercises	57
7	Infinite Series	59
7.1	Series, Partial Sums, and the Cauchy Criterion	59
7.2	Nonnegative Series and Comparison	60
7.3	Absolute Convergence and Products	62
7.4	Ratio, Root, and Alternating Tests	65
7.5	Power Series	69
	Exercises	70
8	Differentiation	72
8.1	Derivative and Differentiability	72
8.2	Chain Rule and Mean Value Theorems	73
8.3	Definitions of Elementary Functions	77
8.4	Differentiating Limits and Power Series	77
8.5	The Standard Elementary Functions	80
8.6	A Smooth Cutoff and the Limits of Taylor Series	86
8.7	Convexity and a Continuous Nondifferentiable Example	86
	Exercises	88
9	Riemann Integration and the Fundamental Theorem	89
9.1	Partitions and Riemann Sums	89
9.2	Continuous Functions Are Integrable	91
9.2.1	Tagged sums: the calculus meaning of the integral	95
9.3	Uniform Limits and Integration	95
9.3.1	Pointwise bounded convergence requires a different argument	96
9.4	The Fundamental Theorem of Calculus	99
9.5	Improper Integrals	102
9.6	The Lebesgue Criterion for Riemann Integrability	105
9.7	Limits of the Riemann Theory	110
	Exercises	111
II	Multivariable Calculus and Analysis	113
10	Necessary Topology and Linear Algebra	114
10.1	Distances, Balls, and Neighborhoods	114
10.2	Open and Closed Sets in an Ambient Space	116
10.2.1	Subspaces: change the universe before testing openness	116
10.2.2	Interior, closure, and boundary	116
10.3	Convergence, Completeness, and Metric Continuity	117
10.4	Topological Spaces and Finite Products	119
10.4.1	Why rectangles suffice	120

10.5	Compactness: From Local Pieces to Global Control	120
10.6	Sequential Compactness, Finite Nets, and a Common Scale	122
10.7	Connectedness and Paths	123
10.8	Vector Spaces, Bases, and Dimension	124
10.8.1	What can be added and rescaled?	124
10.8.2	Subspaces	125
10.8.3	Linear combinations and span	125
10.8.4	Independence means no redundancy	126
10.8.5	Coordinates: existence and uniqueness	127
10.8.6	Why dimension is a theorem before it is a number	127
10.9	Finite Direct Sums	129
10.9.1	External sums: keep the components in separate positions	129
10.9.2	Internal sums: uniqueness is an extra requirement	129
10.10	Linear Transformations, Kernels, and Images	131
10.11	Matrices as Coordinate Representations	133
10.12	Matrix Multiplication Comes from Composition	134
10.13	Changing Coordinates while Keeping the Map	135
10.14	Inner Products and Norms	137
10.15	Operator Norms and Finite-Dimensional Estimates	140
10.16	Dual Spaces and Multilinear Maps	141
10.17	Determinants and Orientation	143
10.18	Perspective and Transition	148
	Exercises	149
11	Differentiation in Finite-Dimensional Spaces	152
11.1	Differentiability and the Derivative	152
11.2	Higher Derivatives and Local Estimates	157
11.3	The Contraction Principle	161
11.4	Inverse and Implicit Functions	162
	Exercises	171
11.5	Perspective and Transition	172
12	Multivariable Riemann Integration	173
12.1	Rectangles, Partitions, and Darboux Sums	173
12.2	Iterated Integration	175
12.3	Jordan Sets and Integration over Bounded Regions	177
12.3.1	Simple planar regions and changing the order	181
12.4	Linear Changes of Variables	182
12.5	Nonlinear Changes of Variables	187
12.5.1	Alternative Viewpoint: Pullbacks and Manifold Integration	195
12.6	Uniform Limits and Multiple Integration	197
	Exercises	198
12.7	Limits of the Riemann Theory	199
13	Differential Forms, Stokes' Theorem, and Calculus on Manifolds	200
13.1	Parametrized Curves and Line Integrals	200
13.1.1	A curve is a traversal, not just a set	200
13.2	Green's Theorem in the Plane	202
13.3	Parametrized Surfaces and Surface Integrals	203
13.3.1	A small parameter rectangle becomes a tangent parallelogram	204
13.4	Classical Stokes for a Parametrized Surface Patch	205
13.5	Why Differential Forms?	206
13.6	Alternating Multilinear Algebra	206
13.7	Differential Forms, Pullbacks, and Exterior Derivatives	209
13.7.1	One form through all its types	212
13.7.2	Pullback as a workflow, including a surface calculation	213

13.8	Integration and the Local Stokes Theorem	213
13.9	Smooth Manifolds, Tangent Spaces, and Orientation	216
13.9.1	Charts undo parametrizations	216
13.9.2	Deriving the tangent-coordinate rule from a moving point	218
13.9.3	The differential: intrinsic map, changing coordinate matrices	219
13.9.4	Pullbacks, wedge products, and exterior derivatives on manifolds	221
13.10	Partitions of Unity and Integration on Manifolds	222
13.11	The General Stokes Theorem	227
13.12	Classical Theorems Revisited	228
	Exercises	230
13.13	Perspective and Further Directions	232
A	Optional: Permutations and Determinants over a Field	234
A.1	What Survives over an Arbitrary Field?	234
A.2	Cycles, Transpositions, and Parity	235
A.2.1	Reading and composing bijections	235
A.2.2	Following a permutation until it closes	235
A.2.3	Adjacent swaps and the parity question	236
A.2.4	Counting inversions instead of choosing swaps	237
A.2.5	Permutation matrices connect the notation to linear maps	237
A.3	Constructing the Determinant Form	238
B	Optional: Tensor Products and Exterior Powers	240
B.1	Quotient Vector Spaces and Universal Factorization	240
B.1.1	What a universal property does	242
B.2	Bilinear Maps and the Tensor-Product Problem	242
B.3	Constructing the Tensor Product	244
B.3.1	Formal symbols before relations	244
B.3.2	Spanning is not surjectivity of the input map	245
B.4	Tensor Bases and a First Canonical Isomorphism	245
B.5	Symmetry, Induced Maps, and Direct Sums	246
B.5.1	Interchanging factors	246
B.5.2	Applying maps to the two factors	247
B.5.3	Distributing over a finite direct sum	247
B.6	Finite Multivariable Tensor Products	249
B.6.1	Induced maps in all the factors	251
B.7	Associativity: Constructing Both Nested Maps	251
B.8	Tensor–Hom Correspondences	252
B.8.1	A map whose values are maps	252
B.8.2	A covector and a vector make a linear map	253
B.9	Exterior Powers: Imposing Alternation	254
B.9.1	Begin with two inputs	254
B.9.2	Expand, remove repeats, sort, and collect	255
B.9.3	Representations need not be unique	256
B.9.4	Induced exterior maps and quotient compatibility	256
B.10	Constructing Exterior Multiplication	256
B.11	The Exterior Algebra and Its Grading	258
B.12	Duality: Alternating Rules and Linear Functionals	258
B.12.1	First identification: factor an alternating map	258
B.12.2	Second identification: pair covector wedges with vector wedges	259
B.13	The Top Exterior Power and the Determinant	260
	Exercises	261
	References and Further Reading	262

Preface

These notes are for students who know the main ideas of single-variable calculus and want to understand why its central statements are true. They develop rigorous real analysis before rebuilding calculus from limits, completeness, continuity, differentiation, and Riemann integration. The intended audience is an advanced undergraduate beginning proof-based mathematics. Familiar algebra and ordinary calculus are assumed, but the set-theoretic, logical, topological, and finite-dimensional linear-algebraic tools needed later are developed in the text.

Part I (Chapters 1–9) constructs the real number system and develops one-variable analysis through Riemann integration and the Fundamental Theorem of Calculus. Part II (Chapters 10–13) develops the finite-dimensional topology and linear algebra needed for multivariable differentiation and Riemann integration, and ends with differential forms and the general Stokes theorem. Measure theory, Lebesgue integration, complex and functional analysis, partial differential equations, and deeper differential geometry are deliberately outside the scope; the final chapter points toward them as natural continuations.

AI-assisted preparation. OpenAI Codex was used extensively in drafting, revising, proofreading, and preparing the $\text{\LaTeX}$ source for these notes. The author determined the scope and mathematical organization, made the final decisions about retained content, and is responsible for approving and verifying the correctness of the published version.

How to Read These Notes

These notes start from familiar calculus questions but use a different standard of explanation: every formal assertion has hypotheses, and every main claim is justified from material already established or from an explicitly identified external theorem. A definition fixes meaning; a theorem requires proof unless it is marked as external. Examples provide intuition, but not proof of a general assertion. Occasionally, an explicitly identified informal preview uses familiar notation to motivate a later formal development; such previews are not used as formal justification.

The chapters are designed to be read in order. Exercises are part of the course: they supply routine practice, short proofs, counterexamples, and connections among the main results. A reader should attempt them after the relevant section, consulting the stated hypotheses and earlier results rather than relying on calculus folklore. The bibliography distinguishes the one substantial imported topological result from further-reading references.

Read a substantial proof twice with different aims. On the first pass, identify the problem, the main construction, and the purpose of each estimate. On the second, reconstruct the quantifiers and check where every hypothesis enters. Understanding a proof's architecture and reproducing all its details are related but distinct skills.

In Chapter 3, the construction of $\mathbb{R}$ is foundationally important: it proves that the system we use exists. Difficulty reproducing every equivalence-class verification should not block further study. Retain its working consequences: ordered-field structure, the Archimedean property, rational density, completeness, Cauchy convergence, and the least-upper-bound property. Return to the implementation as these ideas become familiar. In Part II, watch these same tools reappear as norm estimates, compactness arguments, linear approximation, and local-to-global integration.

Chapter 10 assumes no previous topology or linear algebra course. Its distance examples lead to open sets and metric continuity before abstract topology; its coordinate calculations lead to bases, direct sums, linear maps, and matrices. Appendix B is a separate first introduction to quotient spaces, universal properties, tensors, and exterior powers over a field. At the first serious use of a proof method, look for the actual map, sequence, or estimate being constructed. Later uses refer back to that mechanism. In particular, existence of a map is a different task from proving equality of two maps, and choosing a finite index is a different task from subsequently taking a limit.

Part I

Single-Variable Calculus and Analysis

Chapter 1

Motivation and the Transition from Calculus to Analysis

1.1 Why Calculus Needs Foundations

Calculus describes change and accumulation. Its central calculations often begin with an informal phrase: “make the increment arbitrarily small,” “add infinitely many pieces,” or “approach a point.” These phrases are useful, but they conceal questions that computation alone cannot answer. What does it mean to approach? Why is a limiting value unique? Why may a limiting process be interchanged with addition, differentiation, or integration?

Mathematical analysis supplies precise meanings and proofs. The precision is not a bureaucratic replacement for intuition. It identifies the hypotheses that make a calculation reliable and exposes plausible-looking arguments that fail.

Example 1.1 (A number suggested by geometry). The diagonal of a square of side length 1 has a length customarily denoted by $\sqrt{2}$. If this length were a rational number p/q in lowest terms, then $p^2 = 2q^2$. Thus p^2 is even. An odd integer has the form $2k + 1$, whose square is $4k(k + 1) + 1$ and is odd; hence p is even. Write $p = 2r$. Then $q^2 = 2r^2$, so the same argument makes q even. This contradicts that p/q was in lowest terms. Hence no rational number has square 2.

The example does not yet construct $\sqrt{2}$; rather, it points out why rational numbers are not sufficient for familiar geometry. Chapter 3 will construct a number system containing such limits and isolate the property that makes it adequate: completeness.

1.2 A First Look at Limits

Consider the values $1, 1/2, 1/3, \dots$. Their terms become as small as we wish. A rigorous definition should express exactly that statement without appealing to a last term, because there is no last positive integer.

Informal preview of convergence. A sequence $(a_n)_{n \in \mathbb{N}}$ of real numbers **converges to** a real number L when, for every positive tolerance ε , all sufficiently late terms a_n lie within distance ε of L . In symbols, the intended statement is

for every $\varepsilon > 0$ there is $N \in \mathbb{N}$ such that $n \geq N \implies |a_n - L| < \varepsilon$.

This is not yet a formal definition: the quantifiers, real numbers, and natural numbers will be developed formally in Chapters 2–4.

For instance, $1/n$ converges to 0: given $\varepsilon > 0$, choose a natural number $N > 1/\varepsilon$. Then $n \geq N$ implies $0 < 1/n \leq 1/N < \varepsilon$. The existence of such an N is an Archimedean property, which will be proved from the construction of the real numbers rather than silently assumed.

1.3 The Structural Ideas of Analysis

Limits

A limit records stable behavior at arbitrarily fine scales. The order of the quantifiers matters: a proposed tolerance is given first, and the required stage may depend on it.

Completeness

A number system is complete when a process that should determine a number does not leave a gap. The least-upper-bound principle and the assertion that Cauchy sequences converge are two important formulations.

Continuity

A function is continuous at a point when small changes in input force small changes in output. The exact definition uses the same quantifier pattern as a limit.

Differentiation

The derivative is the best linear approximation to a function at a point. In one variable it is a number; in several variables it is a linear map.

Integration

Integration formalizes the limiting process of approximating accumulated quantities by finite sums. The existence of an integral, and its relationship to derivatives, are the substance of the Fundamental Theorem of Calculus.

1.4 Warnings That Motivate Hypotheses

Example 1.2 (Pointwise convergence need not preserve continuity). For $n \in \mathbb{N}$, define $f_n : [0, 1] \rightarrow \mathbb{R}$ by $f_n(x) = x^n$. Every f_n is continuous. For $0 \leq x < 1$, the values x^n tend to 0, whereas $f_n(1) = 1$ for every n . Thus the pointwise limiting function is

$$f(x) = \begin{cases} 0, & 0 \leq x < 1, \\ 1, & x = 1, \end{cases}$$

which is not continuous at 1. Chapter 6 will prove that **uniform** convergence supplies the missing hypothesis.

Example 1.3 (A bounded set may lack a rational supremum). Let $S = \{q \in \mathbb{Q} : q > 0 \text{ and } q^2 < 2\}$. The set is nonempty (for example $1 \in S$) and bounded above in $\mathbb{Q}$ (for example 2 is an upper bound). It has no least rational upper bound. Informally, any such bound would have to be $\sqrt{2}$, which is not rational. The least-upper-bound argument in Chapter 3 constructs positive square roots; its perturbation estimates also show why no rational candidate can be the least upper bound of this rational set.

1.5 Road Map

Chapter 2 establishes the language of proof. Chapters 3–4 build $\mathbb{R}$ and its sequence theory. Chapters 5–9 reconstruct one-variable calculus from limits. The second part then enlarges the setting to finite-dimensional spaces, develops multivariable differentiation and integration, and first develops work along curves and flux through surfaces. Green and parametrized-patch Stokes then motivate differential forms; the general Stokes theorem unifies these formulas on manifolds.

Exercises

Exercise 1.1. Let (a_n) be a sequence of real numbers that converges to both L and M . Using the informal definition above, formulate a strategy for proving $L = M$. The complete proof appears after the necessary order facts have been proved.

Exercise 1.2. For each claim, identify an existence question and a separate computational question: finding a square root, summing an infinite series, and solving an equation for one variable. Explain why a plausible formula alone need not answer the existence question.

Chapter 2

Basic Set Theory, Logic, Functions, and Relations

2.1 Sets

Mathematical analysis constantly describes collections: the possible inputs of a function, the points of an interval, or the terms of a sequence. The language of sets lets us describe such collections precisely and compare them without relying on a picture or an informal list.

We take

$$\mathbb{N} = \{1, 2, 3, \dots\}$$

as the primitive discrete number system. Its usual arithmetic means that addition and multiplication have their familiar laws and keep natural numbers natural. Its induction property says that a statement true for (1), and inherited from each number to its successor, is true for every natural number. Its well-ordering property says that every nonempty collection of natural numbers has a smallest member. These principles will be used repeatedly, and induction is stated formally in [Theorem 2.4](#). We write

$$\mathbb{N}_0 = \mathbb{N} \cup \{0\}.$$

The integers and rational numbers will be constructed in this chapter, and the real numbers in [Chapter 3](#). A **set** is a collection of objects, called its elements. We write $x \in A$ when x is an element of A .

Definition 2.1 (Subsets and set equality). For sets A, B , write $A \subseteq B$ if every element of A is an element of B . Define $A = B$ to mean both $A \subseteq B$ and $B \subseteq A$.

The empty set $\emptyset$ has no elements, and is a subset of every set. Here “and” requires both stated conditions, “or” allows either condition (including both), and $x \notin B$ means that x is not an element of B . For any sets A, B , define

$$\begin{aligned} A \cup B &= \{x : x \in A \text{ or } x \in B\}, & A \cap B &= \{x : x \in A \text{ and } x \in B\}, \\ A \setminus B &= \{x : x \in A \text{ and } x \notin B\}. \end{aligned}$$

If $A \subseteq X$, its complement is $A^c = X \setminus A$. The power set is $\mathcal{P}(A) = \{E : E \subseteq A\}$.

Proposition 2.2 (De Morgan’s laws). If $A, B \subseteq X$, then $(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$.

Proof. For $x \in X$, membership in $(A \cup B)^c$ means that it is false that $x \in A$ or $x \in B$. This is equivalent to $x \notin A$ and $x \notin B$, namely $x \in A^c \cap B^c$. The second identity follows identically with “and” and “or” exchanged. $\square$

Definition 2.3 (Cartesian product). For sets A, B , define $A \times B = \{(a, b) : a \in A, b \in B\}$, where $(a, b) = (a', b')$ means $a = a'$ and $b = b'$.

2.2 Logic and Proofs

Definitions specify the objects of mathematics, while proofs explain why their asserted properties follow. This section introduces the small amount of logical language needed to state definitions and organize proofs clearly.

A **proposition** is a statement that is either true or false. A **predicate** is a statement with a variable whose truth depends on the value of that variable. For example, let $P(n)$ mean “ n is even.” Then $P(4)$ is a true proposition, whereas $P(n)$ is not a complete proposition until n is assigned a value or is quantified.

If P and Q are propositions, then P and Q means that both are true, and P or Q means that at least one is true. This is the inclusive mathematical meaning of “or.” For instance,

$$n > 0 \text{ and } n < 5$$

requires both inequalities, while “ n is even or divisible by 3” permits either property or both. The **negation** of P , written

$$\neg P,$$

means that P is false. Thus the negation of $x > 3$ is $x \leq 3$. Negating a mathematical statement means negating its full meaning, not merely placing the word “not” at an arbitrary point.

The implication

$$P \Rightarrow Q$$

means “if P , then Q ”; P is its **hypothesis** (or antecedent), and Q is its **conclusion** (or consequent). For example,

$$n \text{ is divisible by } 4 \implies n \text{ is even.}$$

Its **contrapositive** is

$$\neg Q \Rightarrow \neg P,$$

and an implication is logically equivalent to its contrapositive. The **converse** is $Q \Rightarrow P$; it is a different assertion and does not follow automatically. The notation

$$P \iff Q$$

means “ P if and only if Q ,” often abbreviated “iff”: both $P \Rightarrow Q$ and $Q \Rightarrow P$ hold. A proof of an iff statement normally proves the two directions separately.

Quantifiers turn predicates into propositions. The statement

$$\forall x \in A \, P(x)$$

means that $P(x)$ holds for every $x \in A$, while

$$\exists x \in A \, P(x)$$

means that there is at least one $x \in A$ for which $P(x)$ holds. For example,

$$\forall n \in \mathbb{N} \quad n + 1 > n,$$

whereas

$$\exists n \in \mathbb{N} \quad n^2 = 4.$$

The first says something about every natural number; the second asserts the existence of a witness, here $n = 2$. After predicates, quantifiers, and negation have been introduced, their basic interaction can be written as

$$\neg(\forall x \in A \ P(x)) \iff \exists x \in A \ \neg P(x),$$

$$\neg(\exists x \in A \ P(x)) \iff \forall x \in A \ \neg P(x).$$

In words, “every student passed” is false exactly when there is a student who did not pass; “there is a student who passed” is false exactly when every student failed.

2.2.1 Writing Proofs

Logic becomes useful here as a guide to proof writing. To prove a universal implication

$$\forall x \in A \quad P(x) \Rightarrow Q(x),$$

begin by letting $x \in A$ be arbitrary, suppose $P(x)$, and show $Q(x)$. The word “arbitrary” matters: the argument must work for every allowed x , not only for a convenient example. To prove an existential statement, give a **witness** and verify that it has the required property. To prove that an object exists uniquely, first prove existence and then show that any two objects with the property are equal. This pattern will recur for limits, least upper bounds, inverse functions, and implicit functions.

Set equality is proved by showing both inclusions

$$A \subseteq B \quad \text{and} \quad B \subseteq A.$$

Similarly, two functions with the same domain and codomain are equal when their values agree at every point of the domain. A direct proof derives the conclusion from the hypotheses. A contrapositive proof establishes $\neg Q \Rightarrow \neg P$ in order to prove $P \Rightarrow Q$. A proof by contradiction assumes the hypotheses and the negation of the desired conclusion, then derives an impossibility.

A proof should indicate why each nontrivial step follows: from a hypothesis, a definition, an earlier theorem, or an algebraic calculation. Calculations are part of a proof only when their purpose is clear, so short explanatory sentences around a nontrivial chain of formulas are often helpful. The aim is clarity and visible logical dependence, not rigid prose templates. The next theorem records the induction method: prove a base case, assume the statement at an arbitrary stage as the induction hypothesis, and then prove the induction step.

Theorem 2.4 (Induction). Let $P(n)$ be a proposition for each $n \in \mathbb{N}$. If $P(1)$ is true and $P(n) \Rightarrow P(n+1)$ for every $n \in \mathbb{N}$, then $P(n)$ is true for all $n \in \mathbb{N}$.

Proof. This is the induction axiom for $\mathbb{N}$. Equivalently, it follows from the well-ordering principle that each nonempty subset of $\mathbb{N}$ has a least element: a least n for which $P(n)$ failed could be neither 1 nor a successor of a number for which P holds. $\square$

2.3 Relations and Functions

Many mathematical statements compare two objects: two numbers may be equal, one number may divide another, or one set may be contained in another. A relation records exactly which ordered pairs of objects are being regarded as related.

Definition 2.5 (Binary relation). Let A and B be sets. A **binary relation** R from A to B is a subset

$$R \subseteq A \times B.$$

If $A = B$, then R is a **binary relation on** A . When

$$(a, b) \in R,$$

we commonly write

$$a R b.$$

Example 2.6 (Familiar relations). Equality, the usual numerical order, divisibility on $\mathbb{N}$, and inclusion between subsets of a fixed set are all binary relations. For example, on

$$A = \{1, 2, 3, 4\},$$

the divisibility relation is the subset

$$R = \{(a, b) \in A \times A : a \text{ divides } b\}.$$

It contains, among others,

$$(1, 2), \quad (1, 4), \quad (2, 4), \quad (3, 3),$$

but

$$(3, 4) \notin R.$$

Thus the statement that a relation is a set of ordered pairs has concrete content. Not every relation is an equivalence relation or an order: for example, on $\mathbb{N}$, the relation

$$m R n \iff n = m + 1$$

is neither, since it is not reflexive.

Definition 2.7 (Equivalence relation). An **equivalence relation** is a binary relation $\sim$ on A such that, for all $a, b, c \in A$, it is **reflexive**, **symmetric**, and **transitive**:

$$a \sim a, \quad a \sim b \implies b \sim a, \quad a \sim b \text{ and } b \sim c \implies a \sim c.$$

Its equivalence class at a is

$$[a] = \{b \in A : b \sim a\}.$$

Equivalence relations classify elements by grouping those regarded as the same for a specified purpose. Equality is the simplest example. Congruence modulo a natural number, used after the construction of $\mathbb{Z}$ below, is a further familiar example.

Proposition 2.8. If $\sim$ is an equivalence relation, then A is the union of its equivalence classes, and two classes are either equal or disjoint.

Proof. Reflexivity gives $a \in [a]$ for every a . If $x \in [a] \cap [b]$, symmetry and transitivity imply $a \sim b$. Then $y \sim a$ implies $y \sim b$, and conversely, so $[a] = [b]$. $\square$

Definition 2.9 (Partial order). An **order relation** or **partial order** on a set X is a binary relation $\leq$ on X that is reflexive, antisymmetric, and transitive: for all $x, y, z \in X$,

$$x \leq x,$$

$$x \leq y \text{ and } y \leq x \implies x = y,$$

and

$$x \leq y \text{ and } y \leq z \implies x \leq z.$$

A set equipped with a partial order is a **partially ordered set**, or **poset**. Antisymmetric does not mean “not symmetric”; it is the specific implication displayed above.

Example 2.10 (Partial orders and incomparability). The power set $\mathcal{P}(X)$, ordered by inclusion, is a poset. If X contains 1 and 2, then

$$\{1\} \not\subseteq \{2\}, \quad \{2\} \not\subseteq \{1\},$$

so these two sets are not comparable. Divisibility is another partial order: on positive integers, define

$$m \leq_d n \iff m \text{ divides } n.$$

Definition 2.11 (Total order and strict order). A **total order**, or **linear order**, is a partial order $\leq$ on X such that every two elements are comparable:

$$x \leq y \text{ or } y \leq x \quad (x, y \in X).$$

For a total order, define its associated strict order by

$$x < y \iff x \leq y \text{ and } x \neq y.$$

We take $\leq$ as the primitive relation. The usual order on $\mathbb{N}$ is a total order, whereas inclusion on $\mathcal{P}(X)$ is generally only a partial order.

Thus binary relations have two importantly different specializations: equivalence relations classify or group elements, whereas partial orders compare elements; total orders are the partial orders with no incomparability.

Definition 2.12 (Function and inverse images). A function $f : A \rightarrow B$ assigns exactly one $f(a) \in B$ to every $a \in A$. Its graph

$$\{(a, f(a)) : a \in A\} \subseteq A \times B$$

is therefore a relation for which every $a \in A$ occurs in exactly one ordered pair. For $E \subseteq A$ and $F \subseteq B$, set $f(E) = \{f(x) : x \in E\}$ and $f^{-1}(F) = \{x \in A : f(x) \in F\}$. It is **injective** if $f(a) = f(a')$ implies $a = a'$, **surjective** if $f(A) = B$, and **bijective** if both.

Definition 2.13 (Composition and identity functions). Let $f : A \rightarrow B$ and $g : B \rightarrow C$. Their **composition** is the function $g \circ f : A \rightarrow C$ defined by

$$(g \circ f)(a) = g(f(a)) \quad (a \in A).$$

For every set A , the **identity function** $\text{id}_A : A \rightarrow A$ is defined by $\text{id}_A(a) = a$.

Theorem 2.14 (Inverse function). If $f : A \rightarrow B$ is bijective, there is a unique function $f^{-1} : B \rightarrow A$ satisfying $f^{-1}(f(a)) = a$ and $f(f^{-1}(b)) = b$.

Proof. For each $b \in B$, surjectivity supplies at least one $a \in A$ with $f(a) = b$. If a' is another, injectivity gives $a' = a$. Thus there is exactly one such preimage; define $g(b)$ to be it. This defines a function $g : B \rightarrow A$, not a choice depending on an arbitrary preimage. By its definition $f(g(b)) = b$ for every b . For $a \in A$, the point a is a preimage of $f(a)$, and its uniqueness gives $g(f(a)) = a$. Hence $fg = \text{id}_B$ and $gf = \text{id}_A$. If $h : B \rightarrow A$ also satisfies the two identities, then $f(h(b)) = b = f(g(b))$, so injectivity gives $h(b) = g(b)$ for every b . The inverse is unique, and we denote it f^{-1} . This inverse function is distinct from the inverse-image notation $f^{-1}(E)$, which is defined even when f is not bijective. $\square$

Proposition 2.15 (Basic laws of composition). Let $f : A \rightarrow B$, $g : B \rightarrow C$, and $h : C \rightarrow D$ be functions.

- (a) $h \circ (g \circ f) = (h \circ g) \circ f$.
- (b) $f \circ \text{id}_A = f$ and $\text{id}_B \circ f = f$.
- (c) If f and g are injective, then $g \circ f$ is injective; if f and g are surjective, then $g \circ f$ is surjective. In particular, a composition of bijections is a bijection.
- (d) If f and g are bijections, then $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Proof. For $a \in A$, both functions in the first assertion have value $h(g(f(a)))$, proving equality. The identity assertions follow from $\text{id}_A(a) = a$ and $\text{id}_B(f(a)) = f(a)$. If $(g \circ f)(a) = (g \circ f)(a')$, injectivity of g gives $f(a) = f(a')$, and injectivity of f gives $a = a'$. If $c \in C$, surjectivity of g gives $b \in B$ with $g(b) = c$, and surjectivity of f gives $a \in A$ with $f(a) = b$; then $(g \circ f)(a) = c$. Finally,

$$(f^{-1} \circ g^{-1}) \circ (g \circ f) = f^{-1} \circ (g^{-1} \circ g) \circ f = f^{-1} \circ \text{id}_B \circ f = \text{id}_A,$$

In the other order,

$$(g \circ f) \circ (f^{-1} \circ g^{-1}) = g \circ (f \circ f^{-1}) \circ g^{-1} = g \circ \text{id}_B \circ g^{-1} = \text{id}_C.$$

A two-sided inverse is unique: if u and v are both two-sided inverses of a function w , then

$$u = u \circ (w \circ v) = (u \circ w) \circ v = v.$$

This proves the last assertion without using a later result. $\square$

Proposition 2.16 (Inverse-image rules). For $f : A \rightarrow B$ and $F, G \subseteq B$, one has $f^{-1}(F \cup G) = f^{-1}(F) \cup f^{-1}(G)$, $f^{-1}(F \cap G) = f^{-1}(F) \cap f^{-1}(G)$, and $f^{-1}(B \setminus F) = A \setminus f^{-1}(F)$.

Proof. For example, $x \in f^{-1}(F \cap G)$ if and only if $f(x) \in F$ and $f(x) \in G$, precisely when $x \in f^{-1}(F) \cap f^{-1}(G)$. The other two identities use the same membership calculation. $\square$

Proposition 2.17 (Image and inverse-image rules). Let $f : A \rightarrow B$, let $E, H \subseteq A$, and let $F, G \subseteq B$. Then

$$f(E \cup H) = f(E) \cup f(H), \quad f(E \cap H) \subseteq f(E) \cap f(H),$$

and equality holds in the second formula when f is injective. Moreover,

$$E \subseteq f^{-1}(f(E)), \quad f(f^{-1}(F)) \subseteq F,$$

and equality holds in the first inclusion when f is injective and in the second when f is surjective.

Proof. Each assertion follows by unpacking membership. For example, if

$$y \in f(E \cap H),$$

then $y = f(x)$ for some $x \in E \cap H$, so y belongs to both $f(E)$ and $f(H)$. Conversely, if f is injective and

$$y = f(e) = f(h)$$

with $e \in E$ and $h \in H$, then $e = h \in E \cap H$. The remaining statements follow similarly; the equality cases use respectively injectivity to recover a unique preimage and surjectivity to supply one. $\square$

2.4 The Natural Numbers, Integers, and Rational Numbers

The natural numbers are the starting point for the arithmetic used throughout the notes. A fully set-theoretic construction of $\mathbb{N}$, for example by finite ordinals, belongs to foundations rather than analysis. We therefore take $\mathbb{N}$ and $\mathbb{N}_0$ with their usual arithmetic, induction, and well-ordering structure as primitive. The next two quotient constructions explain the algebraic chain

$$\mathbb{N} \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q}$$

before Chapter 3 treats the analytic extension

$$\mathbb{Q} \longrightarrow \mathbb{R}.$$

Passing from $\mathbb{N}_0$ to $\mathbb{Z}$ permits subtraction, and passing from $\mathbb{Z}$ to $\mathbb{Q}$ permits division by nonzero elements.

Definition 2.18 (The integers). On $\mathbb{N}_0 \times \mathbb{N}_0$, define

$$(a, b) \sim_{\mathbb{Z}} (c, d)$$

if and only if

$$a + d = b + c.$$

The set of equivalence classes is denoted by

$$\mathbb{Z} = (\mathbb{N}_0 \times \mathbb{N}_0) / \sim_{\mathbb{Z}}.$$

Write $[a, b]_{\mathbb{Z}}$ for the class of (a, b) , interpreted as the formal difference $a - b$. Define

$$[a, b]_{\mathbb{Z}} + [c, d]_{\mathbb{Z}} = [a + c, b + d]_{\mathbb{Z}},$$

$$[a, b]_{\mathbb{Z}}[c, d]_{\mathbb{Z}} = [ac + bd, ad + bc]_{\mathbb{Z}},$$

and

$$-[a, b]_{\mathbb{Z}} = [b, a]_{\mathbb{Z}}.$$

Set

$$0_{\mathbb{Z}} = [0, 0]_{\mathbb{Z}}, \quad 1_{\mathbb{Z}} = [1, 0]_{\mathbb{Z}},$$

and define

$$\iota_{\mathbb{Z}} : \mathbb{N}_0 \longrightarrow \mathbb{Z}, \quad \iota_{\mathbb{Z}}(n) = [n, 0]_{\mathbb{Z}}.$$

Finally, define the order by

$$[a, b]_{\mathbb{Z}} < [c, d]_{\mathbb{Z}} \iff a + d < b + c.$$

Proposition 2.19 (Integer arithmetic). The relation $\sim_{\mathbb{Z}}$ is an equivalence relation, and the operations and order in the preceding definition are independent of representatives. The map $\iota_{\mathbb{Z}}$ is injective and preserves addition, multiplication, and order. With these operations, $\mathbb{Z}$ has commutative, associative, distributive addition and multiplication, identities $0_{\mathbb{Z}}, 1_{\mathbb{Z}}$, additive inverses, and no zero divisors. Its displayed order is total and is compatible with addition and multiplication by positive elements.

Proof. Reflexivity and symmetry of $\sim_{\mathbb{Z}}$ are immediate. If

$$a + d = b + c, \quad c + f = d + e,$$

then

$$a + d + f = b + c + f = b + d + e.$$

Cancellation in $\mathbb{N}_0$ gives $a + f = b + e$, proving transitivity.

An operation written with representatives gives an operation on equivalence classes only when changing representatives does not change the resulting class. This is what **well defined** means here. For the displayed sum, suppose

$$(a, b) \sim_{\mathbb{Z}} (a', b'), \quad (c, d) \sim_{\mathbb{Z}} (c', d').$$

Then

$$\begin{aligned} (a + c) + (b' + d') &= (a + b') + (c + d') \\ &= (b + a') + (d + c') \\ &= (b + d) + (a' + c'). \end{aligned}$$

Thus

$$(a + c, b + d) \sim_{\mathbb{Z}} (a' + c', b' + d'),$$

so addition is well defined. For multiplication we must avoid assuming integer subtraction before integers have been constructed. First change only (a, b) , using $a + b' = b + a'$. Multiply that equality by c and by d :

$$ac + b'c = bc + a'c, \quad ad + b'd = bd + a'd.$$

Adding these equalities in the required order gives

$$ac + bd + a'd + b'c = ad + bc + a'c + b'd.$$

This is precisely $(ac + bd, ad + bc) \sim_{\mathbb{Z}} (a'c + b'd, a'd + b'c)$. Next change only (c, d) , using $c + d' = d + c'$. Multiplication by a' and b' gives $a'c + a'd' = a'd + a'c'$ and $b'c + b'd' = b'd + b'c'$. Thus

$$a'c + b'd + a'd' + b'c' = a'd + b'c + a'c' + b'd',$$

the equivalence between the new product and $(a'c' + b'd', a'd' + b'c')$. Transitivity proves independence when both representatives change. A formula on representatives was only a proposed operation; these checks turn it into an operation on classes. The comparison defining the order is likewise unchanged after adding the equalities associated with equivalent representatives. These checks show that all formulas descend to classes.

The associative, commutative, and distributive laws now follow from the corresponding arithmetic identities in $\mathbb{N}_0$. The formulas for $0_{\mathbb{Z}}, 1_{\mathbb{Z}}$, and $-[a, b]_{\mathbb{Z}}$ give the required identities and additive inverses. The order says exactly that a formal difference is positive when its first entry exceeds

its second; the order laws follow by adding inequalities and multiplying positive differences in $\mathbb{N}_0$. Consequently, a product can vanish only when one factor does. Finally,

$$[m, 0]_{\mathbb{Z}} + [n, 0]_{\mathbb{Z}} = [m + n, 0]_{\mathbb{Z}},$$

$$[m, 0]_{\mathbb{Z}}[n, 0]_{\mathbb{Z}} = [mn, 0]_{\mathbb{Z}},$$

and

$$[m, 0]_{\mathbb{Z}} < [n, 0]_{\mathbb{Z}} \iff m < n,$$

which proves the assertions about $\iota_{\mathbb{Z}}$. □

Definition 2.20 (The rational numbers). On

$$\mathbb{Z} \times (\mathbb{Z} \setminus \{0_{\mathbb{Z}}\}),$$

define

$$(a, b) \sim_{\mathbb{Q}} (c, d)$$

if and only if

$$ad = bc.$$

The set of equivalence classes is denoted by

$$\mathbb{Q} = (\mathbb{Z} \times (\mathbb{Z} \setminus \{0_{\mathbb{Z}}\})) / \sim_{\mathbb{Q}}.$$

Write $[a, b]_{\mathbb{Q}}$, or simply a/b , for the class of (a, b) . Define

$$[a, b]_{\mathbb{Q}} + [c, d]_{\mathbb{Q}} = [ad + bc, bd]_{\mathbb{Q}},$$

$$[a, b]_{\mathbb{Q}}[c, d]_{\mathbb{Q}} = [ac, bd]_{\mathbb{Q}},$$

and

$$\iota_{\mathbb{Q}} : \mathbb{Z} \longrightarrow \mathbb{Q}, \quad \iota_{\mathbb{Q}}(a) = [a, 1]_{\mathbb{Q}}.$$

The additive inverse of $[a, b]_{\mathbb{Q}}$ is $[-a, b]_{\mathbb{Q}}$; if $a \neq 0_{\mathbb{Z}}$, its multiplicative inverse is $[b, a]_{\mathbb{Q}}$. Define positivity by

$$[a, b]_{\mathbb{Q}} > 0 \iff ab > 0_{\mathbb{Z}}.$$

As in every ordered arithmetic system, define

$$r < s \iff s - r > 0.$$

Proposition 2.21 (Rational arithmetic). The relation $\sim_{\mathbb{Q}}$, the operations, and the positivity relation in the preceding definition are well defined. The map $\iota_{\mathbb{Q}}$ is an injective order-preserving arithmetic embedding. The resulting system $\mathbb{Q}$ satisfies the ordered-field axioms stated in Chapter 3.

Proof. The only new point needed for the equivalence relation is cancellation of a nonzero integer. If

$$ad = bc, \quad cf = de,$$

then

$$adf = bcf = bde.$$

Cancelling the nonzero factor d gives $af = be$, proving transitivity. Again, well definedness asks whether a formula has the same output class after representatives are changed. For addition, suppose

$$ab' = a'b, \quad cd' = c'd.$$

Then

$$\begin{aligned} (ad + bc)b'd' &= ab'dd' + bb'cd' \\ &= a'bdd' + bb'c'd \\ &= (a'd' + b'c')bd. \end{aligned}$$

Hence

$$(ad + bc, bd) \sim_{\mathbb{Q}} (a'd' + b'c', b'd'),$$

so rational addition is well defined. For multiplication the proposed pairs are (ac, bd) and $(a'c', b'd')$. Their denominators are nonzero, and the hypotheses give

$$(ac)(b'd') = (ab')(cd') = (a'b)(c'd) = (a'c')(bd).$$

This is the defining equality of their rational classes. Thus multiplication too is independent of both representatives.

The displayed formulas for additive and multiplicative inverses verify the inverse laws directly. If

$$[a, b]_{\mathbb{Q}} = [c, d]_{\mathbb{Q}},$$

then multiplying $ad = bc$ by bd shows that ab and cd have the same sign, because nonzero squares are positive in $\mathbb{Z}$. Thus positivity is well defined. It is closed under addition and multiplication, and every nonzero rational is either positive or has positive negative; these facts follow by choosing the signs of numerator and denominator. Hence the order is compatible with the operations. The remaining field identities descend from the integer identities through the displayed fraction formulas. Finally, the definition of $\iota_{\mathbb{Q}}$ immediately gives injectivity and preservation of arithmetic and order. $\square$

2.5 Countability

Definition 2.22 (Countability). Sets A, B have the same cardinality if a bijection $A \rightarrow B$ exists. A set is **finite** if it is empty or bijective with $\{1, \dots, n\}$ for some $n \in \mathbb{N}$; it is **countably infinite** if it is bijective with $\mathbb{N}$; and **countable** if finite or countably infinite.

Proposition 2.23 (Elementary finite-set facts). Every subset of a finite set is finite. If A and B are finite, then

$$A \cup B, \quad A \times B$$

are finite. If A has m elements and B has n elements, then

$$A \times B$$

has mn elements.

Proof. Transport the question along bijections with initial segments of $\mathbb{N}$. A subset of

$$\{1, \dots, m\}$$

can be listed in increasing order and therefore has at most m elements. The union is listed by first listing A and then retaining the members of B not already listed. Finally, the ordered pairs

$$(i, j), \quad 1 \leq i \leq m, \quad 1 \leq j \leq n,$$

are enumerated by

$$(i, j) \mapsto n(i-1) + j,$$

which is a bijection onto $\{1, \dots, mn\}$. $\square$

Theorem 2.24 (Diagonal enumeration). The set $\mathbb{N} \times \mathbb{N}$ is countable. If, for every $n \in \mathbb{N}$, a listing of a set A_n is given, then

$$\bigcup_{n=1}^{\infty} A_n$$

is countable.

Proof. For $s \geq 2$, the s th diagonal is the finite list of pairs (n, k) with $n + k = s$, ordered by increasing n . Every $(n, k) \in \mathbb{N} \times \mathbb{N}$ belongs to the unique diagonal $n + k$, so concatenating these finite lists gives a listing of $\mathbb{N} \times \mathbb{N}$ without omissions or repetitions.

Now let $a_{n,1}, a_{n,2}, \dots$ be the given listing of A_n . Replace the pair (n, k) in the preceding diagonal list by $a_{n,k}$. If x belongs to the displayed union, then $x \in A_n$ for some n and hence $x = a_{n,k}$ for some k ; therefore this new sequence contains x . It can contain repetitions, either because a listing repeats an element or because two sets overlap. Read the sequence from left to right and retain an entry exactly when it has not occurred earlier. If infinitely many entries remain, their positions give a bijection from $\mathbb{N}$ onto the union. If only finitely many remain, they give a finite listing. Thus the union is countable. $\square$

Corollary 2.25. The rational numbers are countable.

Proof. The explicitly constructed integer classes are listed exactly once by

$$0, 1, -1, 2, -2, \dots,$$

so $\mathbb{Z}$ is countable. Hence $\mathbb{Z} \times \mathbb{N}$ is countable: compose its first-coordinate listing with the diagonal enumeration from [Theorem 2.24](#). The map

$$\mathbb{Z} \times \mathbb{N} \longrightarrow \mathbb{Q}, \quad (p, q) \mapsto p/q,$$

is surjective because every rational class has a numerator in $\mathbb{Z}$ and, after changing both signs if necessary, a positive denominator in $\mathbb{N}$. Applying this map to a listing of $\mathbb{Z} \times \mathbb{N}$ gives a listing of $\mathbb{Q}$; retaining the first occurrence of each value makes it a finite listing or a bijection with $\mathbb{N}$. $\square$

Theorem 2.26 (Cantor's diagonal argument—informal preview). The interval $(0, 1)$ is uncountable.

Proof. This familiar decimal argument is an informal preview: Chapter 3 constructs the real numbers, and Chapter 4 later interprets decimal notation through limits of finite truncations. It is included here to display the diagonal idea before that analytic machinery is available. The formal result is [Theorem 4.4](#); later proofs use that result.

Suppose, for contradiction, that $x_1, x_2, \dots$ listed $(0, 1)$. Give each x_n the decimal expansion that does not end in infinitely many 9s, and let d_n be its n th digit. Define $y = 0.e_1e_2\dots$, where $e_n = 1$ if $d_n \neq 1$, and $e_n = 2$ if $d_n = 1$. This decimal uses only 1 and 2, so it represents a number strictly between 0 and 1 and does not have an alternative expansion ending in infinitely many 9s. Its n th digit differs from d_n , so $y \neq x_n$ for every n . This contradicts the assumed listing. $\square$

2.6 Choice

Definition 2.27 (Axiom of choice). The **axiom of choice** asserts that for every indexed family $(A_i)_{i \in I}$ of nonempty sets, there is a function

$$c : I \rightarrow \bigcup_{i \in I} A_i$$

such that $c(i) \in A_i$ for every $i \in I$. Such a function is a **choice function**.

The assertion is invisible for a finite family when one can explicitly name elements. Its content is the right to make infinitely many selections even when no rule distinguishes a preferred member of each set. A useful mental image is an unlabelled collection of nonempty boxes: a choice function selects one item from every box at once, without requiring an algorithm for the selections.

Why should analysis care? Choice packages many local existence statements into a global object. For example, it yields the familiar unrestricted form of the countable-union theorem: for a sequence of nonempty countable sets, it permits one to select a listing for each set, after which the proof of [Theorem 2.24](#) applies. It also supports the standard existence of vector-space bases and, in more advanced analysis, the extension principles underlying Hahn–Banach and the existence of ultrafilters. Those later results are not prerequisites for this text.

There is a trade-off. The advantage is powerful general existence theorems and concise structural arguments. The disadvantage is that a choice argument often supplies no explicit object and no procedure for finding one. In some foundational settings, very strong forms of choice also yield counterintuitive decompositions of sets. Accordingly, we distinguish an explicit construction from an existence proof using choice, and state the latter use when it occurs. The standard equivalents of the full axiom are Zorn’s lemma and the well-ordering theorem; their proofs are outside the scope of these notes.

For the core development through [Chapters 3–9](#), the proofs will be arranged to use explicit sequences and finite choices whenever possible. The construction of $\mathbb{R}$ uses a specified rational representative at a time, and the Bolzano–Weierstrass proof in [Chapter 4](#) selects indices recursively from explicitly nonempty sets. If a later chapter invokes a general basis, a nonconstructive global selection, or a result equivalent to choice, the dependence will be identified at that point.

Exercises

Exercise 2.1. Prove that the set of all finite strings on a fixed finite alphabet is countable.

Exercise 2.2. Let $f : A \rightarrow B$ and let $E \subseteq A$.

(a) Prove that

$$f(f^{-1}(f(E))) = f(E).$$

(b) Prove that

$$E \subseteq f^{-1}(f(E)).$$

(c) Prove that

$$f^{-1}(f(E)) = E$$

whenever f is injective.

- (d) Give an explicit example showing that the equality in the preceding part can fail when f is not injective.

Exercise 2.3. Suppose that

$$(a, b) \sim_{\mathbb{Z}} (a', b') \quad \text{and} \quad (c, d) \sim_{\mathbb{Z}} (c', d').$$

Verify directly that

$$(a + c, b + d) \sim_{\mathbb{Z}} (a' + c', b' + d').$$

Thus the addition of integer classes is well defined.

Exercise 2.4. Prove directly from the equivalence relation defining $\mathbb{Q}$ that

$$\iota_{\mathbb{Q}} : \mathbb{Z} \longrightarrow \mathbb{Q}, \quad a \longmapsto [a, 1_{\mathbb{Z}}]_{\mathbb{Q}},$$

is injective and preserves order.

Exercise 2.5. Use induction to prove

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2} \quad (n \in \mathbb{N}).$$

State clearly the base case and the induction step.

Exercise 2.6. Give an explicit bijection between

$$\mathbb{N}$$

and the set of even positive integers.

Chapter 3

The Real Number System

3.1 Ordered Fields and the Problem to Be Solved

Chapter 2 constructs the arithmetic chain

$$\mathbb{N} \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q}.$$

The first extension permits subtraction, and the second permits division by a nonzero element. The rational numbers nevertheless have gaps: as Chapter 1 observed, no rational number has square 2. This chapter completes the chain

$$\mathbb{N} \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{R}$$

by constructing a system without those analytic and order-theoretic gaps. We begin by isolating the algebraic and order structure that the final system must possess.

Definition 3.1 (Field). A **field** is a set F with elements $0, 1$ ($0 \neq 1$) and operations $+$ and $\cdot$ such that, for all $a, b, c \in F$, addition and multiplication are associative and commutative, multiplication distributes over addition, $0 + a = a$, $1a = a$, every a has an additive inverse $-a$, and every nonzero a has a multiplicative inverse a^{-1} .

Definition 3.2 (Ordered field). An **ordered field** is a field F equipped with a total order $\leq$, in the sense of Chapter 2, that is compatible with its field operations: for all $a, b, c \in F$,

$$a \leq b \implies a + c \leq b + c,$$

and

$$0 \leq a \text{ and } 0 \leq b \implies 0 \leq ab.$$

Proposition 3.3 (Positive-cone characterization). Let F be a field. An ordered-field structure on F is equivalently specified by a subset

$$P = \{x \in F : x > 0\}$$

such that

$$P + P \subseteq P, \quad PP \subseteq P,$$

where

$$P + P = \{p + q : p, q \in P\}, \quad PP = \{pq : p, q \in P\},$$

and, for every $x \in F$, exactly one of

$$x = 0, \quad x \in P, \quad -x \in P$$

holds. Conversely, such a subset defines an ordered-field structure by

$$x \leq y \iff x = y \text{ or } y - x \in P.$$

Proof. Suppose first that F is an ordered field. If $a, b > 0$, compatibility with addition gives $a + b \geq a > 0$; equality $a + b = a$ would force $b = 0$. Compatibility with multiplication gives $ab \geq 0$. If $ab = 0$, multiplication by a^{-1} gives $b = 0$, contradicting $b > 0$; hence $ab > 0$. Totality gives the displayed trichotomy.

Conversely, let P satisfy the stated conditions. The trichotomy for zero shows that $0 \notin P$. The displayed relation is reflexive. If both $x \leq y$ and $y \leq x$ hold and $x \neq y$, then both $y - x$ and $x - y$ belong to P , so closure under addition would put 0 in P , a contradiction; hence the relation is antisymmetric. Closure under addition also gives transitivity, and the trichotomy for $y - x$ gives totality. Its compatibility with addition is immediate from

$$(y + c) - (x + c) = y - x.$$

Finally, if $0 \leq a$ and $0 \leq b$, then either factor is zero or both belong to P , in which case $ab \in P$. Thus multiplication preserves nonnegativity, as required. $\square$

Definition 3.4 (Absolute value). In an ordered field, define $|x| = x$ if $x \geq 0$ and $|x| = -x$ if $x < 0$.

Proposition 3.5 (Basic order rules). In an ordered field, $<$ is the strict order associated with a transitive total order. If $a < b$, then $a + c < b + c$ for every c ; if $a < b$ and $0 < c$, then $ac < bc$. Also $a^2 \geq 0$ for every a , and

$$|a + b| \leq |a| + |b|, \quad |ab| = |a||b|$$

for all a, b .

Proof. The total-order axioms give totality and transitivity. Addition compatibility gives $a + c \leq b + c$, and equality would imply $a = b$ after cancellation; hence $a + c < b + c$. If $a < b$ and $0 < c$, then

$$0 \leq (b - a)c.$$

If this product were zero, multiplication by $(b - a)^{-1}$ would give $c = 0$, a contradiction. Thus $bc - ac > 0$ and $ac < bc$. Totality gives either $a \geq 0$ or $-a \geq 0$, and multiplication compatibility then gives $a^2 \geq 0$. The four possible sign combinations for a and b prove the product formula for absolute values. Applying the order rules to

$$-|a| \leq a \leq |a|, \quad -|b| \leq b \leq |b|$$

gives

$$-|a| - |b| \leq a + b \leq |a| + |b|,$$

which is the triangle inequality. $\square$

An upper bound is a ceiling, not necessarily a member of the set. For $S = (0, 1)$ in $\mathbb{R}$, every number at least 1 is an upper bound. The smallest ceiling is 1, yet S has no largest member: between $x < 1$ and 1 lies $(x + 1)/2$. For $[0, 1]$ the same ceiling is also a member, so it is a maximum. A supremum is the least upper bound; a maximum is a member larger than or equal to all other members. Existence of a supremum does not assert that it is attained.

Definition 3.6 (Upper bounds and completeness). Let $S \subseteq F$. An element $u \in F$ is an **upper bound** of S if $s \leq u$ for every $s \in S$. A **least upper bound** of S is an upper bound u with $u \leq v$ for every upper bound v of S . An ordered field has the **least-upper-bound property** if every nonempty set having an upper bound has a least upper bound.

Proposition 3.7 (Approaching the least upper bound from below). If $s = \sup S$ for a nonempty set S , then for every $\varepsilon > 0$ there is $x \in S$ with $s - \varepsilon < x \leq s$.

Proof. Every $x \in S$ satisfies $x \leq s$. If no member exceeded $s - \varepsilon$, that smaller number would be an upper bound, contradicting leastness. $\square$

This is how a supremum supplies actual elements for estimates. For $S = \{1 - 1/n : n \geq 1\}$, choose $n > 1/\varepsilon$ to obtain such an element. The supremum is 1, even though no term is 1.

Definition 3.8 (Complete ordered field). An ordered field having the least-upper-bound property is called a **complete ordered field**. This is a characterization, not a construction: it says what the real numbers are like, but does not yet exhibit them.

3.2 Rational Numbers, Completeness, and Uniqueness

The preceding axioms describe the target. Before constructing a model, we show that they already determine its arithmetic and order uniquely. The key is that every ordered field contains a forced copy of the rational numbers.

Definition 3.9 (Field homomorphisms and isomorphisms). Let F and G be fields. A **field homomorphism** is a map

$$\phi : F \longrightarrow G$$

such that

$$\phi(1_F) = 1_G, \quad \phi(x + y) = \phi(x) + \phi(y), \quad \phi(xy) = \phi(x)\phi(y).$$

A bijective field homomorphism is a **field isomorphism**. Between ordered fields, a homomorphism is **order preserving** if

$$x < y \implies \phi(x) < \phi(y).$$

An order-preserving field isomorphism is called an **ordered-field isomorphism**.

Theorem 3.10 (Canonical rational copy). Every ordered field F has a unique order-preserving field homomorphism

$$\iota_F : \mathbb{Q} \longrightarrow F.$$

It is injective, and its image is an ordered subfield of F . Thus we identify $\mathbb{Q}$ with its image in F .

Proof. For $n \in \mathbb{N}$, let n_F be the sum of n copies of 1_F . For negative integers, use additive inverses to define m_F . Induction gives $n_F > 0$, hence $n_F \neq 0$. Define

$$\iota_F(m/n) = m_F(n_F)^{-1} \quad (m \in \mathbb{Z}, n \in \mathbb{N}).$$

To check well definedness, suppose

$$\frac{m}{n} = \frac{p}{q}.$$

Then $mq = np$, so the corresponding integer elements of F satisfy

$$m_F q_F = n_F p_F.$$

Multiplying by $(n_F)^{-1}(q_F)^{-1}$ gives

$$m_F(n_F)^{-1} = p_F(q_F)^{-1}.$$

Thus the definition does not depend on the chosen fraction. The field laws then directly show that ι_F preserves addition and multiplication.

If $u > 0$ in an ordered field, then $u^{-1} > 0$: it is nonzero, and if $u^{-1} < 0$, multiplication of this inequality by $u > 0$ would give

$$1 = u^{-1}u < 0,$$

contrary to $1 = 1^2 > 0$. A positive rational has a representation m/n with $m, n \in \mathbb{N}$, so $m_F > 0$ and $(n_F)^{-1} > 0$; hence its image is positive. Therefore ι_F is order preserving and thus injective. Any field homomorphism fixes 1, then every integer, and then every quotient of integers; uniqueness follows. $\square$

Theorem 3.11 (Archimedean properties). Let F be a complete ordered field. Then:

(a) for every $x \in F$, there is $N \in \mathbb{N}$ such that

$$x < N;$$

(b) for every $\varepsilon > 0$, there is $N \in \mathbb{N}$ such that

$$\frac{1}{N} < \varepsilon.$$

Proof. Identify $\mathbb{Q}$ with its canonical copy. If $\mathbb{N}$ had an upper bound, let $s = \sup \mathbb{N}$. Since $s - 1$ is not an upper bound, some $n \in \mathbb{N}$ satisfies

$$s - 1 < n.$$

Then $s < n + 1$, contradicting that s is an upper bound. Thus $\mathbb{N}$ is unbounded above, which proves (a). Apply (a) to $1/\varepsilon$ to obtain $N > 1/\varepsilon$, proving (b). $\square$

Theorem 3.12 (Rational density). Let F be a complete ordered field and let $x < y$ in F . Then there is $q \in \mathbb{Q}$ such that

$$x < q < y.$$

Proof. Choose $N \in \mathbb{N}$ with

$$N(y - x) > 1.$$

Choose $M \in \mathbb{N}$ with $M > -Nx$. The set

$$E = \{n \in \mathbb{N} : n > Nx + M\}$$

is nonempty and has a least member k . Minimality gives

$$k - 1 \leq Nx + M < k,$$

and hence

$$Nx < k - M \leq Nx + 1 < Ny.$$

Divide by the positive integer N and take

$$q = \frac{k - M}{N}.$$

$\square$

Lemma 3.13 (Rational cuts determine points). If F is a complete ordered field and $x, y \in F$ satisfy

$$\{q \in \mathbb{Q} : q < x\} = \{q \in \mathbb{Q} : q < y\},$$

then $x = y$.

Proof. If $x < y$, rational density gives a rational strictly between them, which belongs to exactly one of the two sets. The case $y < x$ is symmetric. $\square$

Theorem 3.14 (Uniqueness of complete ordered fields). If F and G are complete ordered fields, there is a unique order-preserving field isomorphism

$$\Phi : F \longrightarrow G$$

which identifies their canonical rational copies.

Proof. Strategy. Rational density means that rational numbers below a point determine that point. Match these lower cuts in the two fields, then use rational approximations to show that this matching preserves arithmetic. Any successful construction of a complete ordered field must therefore give this same system up to the unique canonical isomorphism.

Identify both rational copies with $\mathbb{Q}$. For $x \in F$, let

$$L_F(x) = \{q \in \mathbb{Q} : q < x\}.$$

The Archimedean property and rational density make this set nonempty and bounded above in G . Define

$$\Phi(x) = \sup_G L_F(x).$$

For every rational q ,

$$q < \Phi(x) \iff q < x. \quad (*)$$

Indeed, if $q < x$, choose $r \in \mathbb{Q}$ with $q < r < x$, so $q < r \leq \Phi(x)$. Conversely, if $x \leq q$, then q is an upper bound of $L_F(x)$.

The equivalence $(*)$ shows that Φ is strictly order preserving and fixes $\mathbb{Q}$. It also proves additivity: for a rational q ,

$$q < x + y$$

holds exactly when there are rationals $p < x$ and $r < y$ with

$$q = p + r.$$

For the nontrivial implication, choose $p \in \mathbb{Q}$ between $q - y$ and x and put $r = q - p$. Apply $(*)$ in both fields and then use the preceding lemma to conclude

$$\Phi(x + y) = \Phi(x) + \Phi(y).$$

For positive x, y , we next compare the lower cuts of the products. If

$$q < 0,$$

then automatically $q < xy$, and the same is true of

$$\Phi(x)\Phi(y).$$

Now let $q \in \mathbb{Q}$ satisfy

$$q \geq 0.$$

We claim that

$$q < xy$$

if and only if there are positive rationals p, r such that

$$0 < p < x, \quad 0 < r < y, \quad q < pr. \quad (**)$$

The reverse implication follows from the order rules. Conversely, if

$$q < xy,$$

then

$$\frac{q}{y} < x.$$

Rational density gives a positive rational p with

$$\frac{q}{y} < p < x.$$

Because $p > q/y$, one has $q/p < y$; rational density again gives a positive rational r with

$$\frac{q}{p} < r < y.$$

Thus $q < pr$, proving (**).

The equivalence (*) transfers the inequalities

$$p < x, \quad r < y$$

exactly to

$$p < \Phi(x), \quad r < \Phi(y).$$

Applying (**) in F and in G therefore shows that xy and $\Phi(x)\Phi(y)$ have the same rational lower cut. By [Lemma 3.13](#),

$$\Phi(xy) = \Phi(x)\Phi(y)$$

when $x, y > 0$. Additivity gives

$$\Phi(-x) = -\Phi(x).$$

If one factor is zero, multiplicativity is immediate. If exactly one factor is negative, apply the positive case to its negative and use this identity; if both are negative, apply it twice. Hence multiplicativity holds for all

$$x, y \in F.$$

Thus Φ is an order-preserving field homomorphism.

Define the reverse map explicitly by

$$\Psi(y) = \sup_F \{q \in \mathbb{Q} : q < y \text{ in } G\}, \quad y \in G.$$

The rational set is nonempty and bounded by the Archimedean property, so completeness of F supplies its supremum. The proof of (*) applied to this definition gives $q < \Psi(y)$ if and only if $q < y$. Consequently for every rational q and $x \in F$,

$$q < \Psi(\Phi(x)) \iff q < \Phi(x) \iff q < x.$$

The rational-cut lemma implies $\Psi\Phi(x) = x$. For $y \in G$,

$$q < \Phi(\Psi(y)) \iff q < \Psi(y) \iff q < y,$$

so the same lemma gives $\Phi\Psi(y) = y$. Both composites are identities, and hence the already proved field homomorphism Φ is an isomorphism. Finally, any order-preserving field isomorphism fixing $\mathbb{Q}$ has property (*), because it preserves rational lower cuts; therefore it equals Φ . $\square$

Definition 3.15 (The real number system). The **real number system**, denoted by $\mathbb{R}$, is a complete ordered field. The preceding theorem makes this definition unambiguous up to the unique order-preserving field isomorphism fixing $\mathbb{Q}$. The construction below proves that such a field exists.

3.3 Cauchy Sequences of Rational Numbers

The preceding discussion gives the intended abstract meaning of $\mathbb{R}$ and proves that such a field, if it exists, is unique. We now supply existence by constructing a concrete complete ordered field from rational Cauchy sequences. The uniqueness theorem will then identify this model with the real-number system just characterized.

We use the usual ordered-field properties of $\mathbb{Q}$, which Chapter 2 establishes from its construction as equivalence classes of integer pairs (p, q) with $q \neq 0$. No completeness property of $\mathbb{Q}$ will be used.

The construction in one picture

A real number will be an equivalence class of rational Cauchy approximation processes. Different processes represent the same number when their difference tends to zero. We can test whether approximations settle down without already knowing a limit: this is why we use the Cauchy condition in $\mathbb{Q}$. For example, $(1 + 1/n)$ and $(1 - 1/n)$ should represent the same number, $(1/n)$ should represent zero, and the constant sequences (1) and (2) must remain different. Eventual equality would be much too restrictive: the first two sequences never agree, although their difference $2/n$ becomes smaller than every positive rational tolerance.

The work has three layers. First identify equivalent processes and do arithmetic on their classes. Next build order, keeping nonzero classes separated from zero. Finally show that rational approximations converge to their class and that this supplies limits for every real Cauchy sequence.

Definition 3.16 (Cauchy sequence in $\mathbb{Q}$). A sequence $(q_n)_{n \in \mathbb{N}}$ in $\mathbb{Q}$ is **Cauchy** if, for every rational $\varepsilon > 0$, there is $N \in \mathbb{N}$ such that $m, n \geq N$ implies $|q_m - q_n| < \varepsilon$.

Definition 3.17 (Null equivalence). For rational Cauchy sequences (q_n) and (r_n) , write $(q_n) \sim (r_n)$ if, for every $\varepsilon \in \mathbb{Q}$ with $\varepsilon > 0$, there is N such that $n \geq N$ implies $|q_n - r_n| < \varepsilon$.

Lemma 3.18. The relation $\sim$ is an equivalence relation on the set of rational Cauchy sequences.

Proof. Reflexivity and symmetry are immediate. If $q \sim r$ and $r \sim s$, then for a given $\varepsilon > 0$, both $|q_n - r_n| < \varepsilon/2$ and $|r_n - s_n| < \varepsilon/2$ eventually hold. The triangle inequality gives $|q_n - s_n| < \varepsilon$, proving transitivity. $\square$

Lemma 3.19 (Elementary Cauchy facts). Let (q_n) and (r_n) be Cauchy sequences in $\mathbb{Q}$.

- (a) Each sequence is bounded: there is $M \in \mathbb{Q}$ with $|q_n| \leq M$ for all n .

(b) The sequences $(q_n + r_n)$ and $(q_n r_n)$ are Cauchy.

(c) If $(q_n) \sim (q'_n)$ and $(r_n) \sim (r'_n)$, then

$$(q_n + r_n) \sim (q'_n + r'_n) \quad \text{and} \quad (q_n r_n) \sim (q'_n r'_n).$$

Proof. Choose N such that $m, n \geq N$ gives $|q_m - q_n| < 1$. Then $|q_n| \leq |q_N| + 1$ for $n \geq N$; enlarge this rational bound to cover the finitely many earlier terms. For sums, make each difference less than $\varepsilon/2$ and apply the triangle inequality. For products, first bound the potentially large factors, then make the differences small. Take a common rational bound $M \geq 1$ for both sequences and use

$$|q_m r_m - q_n r_n| \leq |q_m| |r_m - r_n| + |r_n| |q_m - q_n| \leq M(|r_m - r_n| + |q_m - q_n|).$$

Given $\varepsilon > 0$, choose a stage after which both differences are less than $\varepsilon/(2M)$. The displayed bound is then less than ε , proving Cauchyness. For the final assertion, the triangle inequality gives the sum statement. Choose a common rational bound $M \geq 1$ for the four sequences. The identity

$$q_n r_n - q'_n r'_n = q_n(r_n - r'_n) + r'_n(q_n - q'_n)$$

then gives

$$|q_n r_n - q'_n r'_n| \leq M|r_n - r'_n| + M|q_n - q'_n|,$$

which is less than ε once both null differences are less than $\varepsilon/(2M)$. Thus the product class does not depend on the representatives. $\square$

Lemma 3.20 (Finite changes). Changing finitely many entries preserves Cauchyness and null-equivalence, and therefore does not change a represented class.

Proof. For each tolerance choose the original stage, then increase it beyond every changed index. All inequalities in the definitions are then unchanged. $\square$

The next lemma solves the denominator problem. If a Cauchy process does not represent zero, one sufficiently late term stays a definite distance from zero; all later terms must stay near it, on the same side of zero.

Lemma 3.21 (Eventual separation). Let (q_n) be a rational Cauchy sequence that is not null-equivalent to the zero sequence. Then there are $\delta > 0$ in $\mathbb{Q}$ and $N \in \mathbb{N}$ such that either $q_n \geq \delta$ for every $n \geq N$, or $q_n \leq -\delta$ for every $n \geq N$.

Proof. Null-equivalence to zero means

$$\forall \eta \in \mathbb{Q}_{>0} \exists N \forall n \geq N : |q_n| < \eta.$$

Its negation is $\exists \eta \in \mathbb{Q}_{>0} \forall N \exists n \geq N$ with $|q_n| \geq \eta$. Thus there is $\eta > 0$ for which $|q_n| \geq \eta$ for infinitely many n . Choose N so that $|q_m - q_n| < \eta/2$ whenever $m, n \geq N$, and choose $k \geq N$ with $|q_k| \geq \eta$. If $q_k \geq \eta$, then $q_n > \eta/2$ for every $n \geq N$. If $q_k \leq -\eta$, then $q_n < -\eta/2$ for every $n \geq N$. Take $\delta = \eta/2$. $\square$

Construction 3.22 (The set of real numbers). Define $\mathbb{R}$ to be the set of equivalence classes $[(q_n)]$ under $\sim$. Define

$$[(q_n)] + [(r_n)] = [(q_n + r_n)], \quad [(q_n)] [(r_n)] = [(q_n r_n)], \quad -[(q_n)] = [(-q_n)].$$

For $a \in \mathbb{Q}$, define $\iota(a) = [(a, a, a, \dots)]$.

Proposition 3.23 (Algebra of the construction). The operations in [construction 3.22](#) are well defined. With $0 = \iota(0)$ and $1 = \iota(1)$, they make $\mathbb{R}$ a field, and $\iota : \mathbb{Q} \rightarrow \mathbb{R}$ is an injective field homomorphism.

Proof. The last part of [Lemma 3.19](#) proves well-definedness of addition and multiplication; it also proves that the displayed sequences are Cauchy. All associative, commutative, distributive, and identity laws hold term by term for rational sequences and therefore for their classes. Additive inverses are termwise. If $x = [(q_n)] \neq 0$, [Lemma 3.21](#) gives $\delta > 0$ and N such that $|q_n| \geq \delta$ for every $n \geq N$. Define

$$p_n = \begin{cases} 1, & n < N, \\ q_n^{-1}, & n \geq N. \end{cases}$$

The estimate

$$|p_m - p_n| = \frac{|q_m - q_n|}{|q_m q_n|} \leq \frac{1}{\delta^2} |q_m - q_n|$$

for $m, n \geq N$ proves Cauchyness: given $\varepsilon > 0$, use the Cauchy tolerance $\varepsilon\delta^2$ for (q_n) and a stage at least N . The finitely many initial choices are harmless by the finite-changes lemma. Since $q_n p_n = 1$ for every $n \geq N$, its product class is 1; hence $[(p_n)]$ is the multiplicative inverse of x . Finally, constant-sequence operations show that ι is a homomorphism; if $\iota(a) = \iota(b)$, then the constant difference $a - b$ is smaller in absolute value than every positive rational, which is possible only when $a = b$. $\square$

3.4 Order and Completeness

Eventual positivity of the entries alone cannot define a positive class: $1/n > 0$ at every stage, but $(1/n)$ represents zero. We need a positive margin that survives arbitrarily accurate changes of representative.

Definition 3.24 (Order on $\mathbb{R}$). For $x = [(q_n)] \in \mathbb{R}$, write $x > 0$ if there are a rational $\delta > 0$ and $N \in \mathbb{N}$ such that $q_n > \delta$ for every $n \geq N$. Define $x < y$ to mean $y - x > 0$.

Lemma 3.25. The preceding definition is independent of the chosen representative. It makes $\mathbb{R}$ an ordered field, and ι preserves order.

Proof. If $q_n > \delta$ eventually and $q \sim r$, then $|q_n - r_n| < \delta/2$ eventually, so $r_n > \delta/2$ eventually. Thus positivity is well defined. Positive eventual lower bounds are closed under addition and multiplication. For a nonzero class, [Lemma 3.21](#) gives exactly one eventual sign. Therefore [Proposition 3.3](#) makes the resulting relation a compatible total order, and [Proposition 3.5](#) supplies its basic rules. Constant sequences visibly preserve rational order. $\square$

Theorem 3.26 (Archimedean property). For every $x \in \mathbb{R}$, there is $N \in \mathbb{N}$ with $x < N$. Consequently, for every $\varepsilon > 0$, there is $N \in \mathbb{N}$ with $1/N < \varepsilon$.

Proof. Let $x = [(q_n)]$. By [Lemma 3.19](#), the rational sequence (q_n) has a rational upper bound M . Choose an integer $N > M$, using the Archimedean property of $\mathbb{Q}$. Then $N - q_n \geq N - M > 0$ for all n , so the definition of positivity, with margin $(N - M)/2$, gives $x < N$. For the consequence choose $N > 1/\varepsilon$. $\square$

Lemma 3.27 (Integer-part lemma). For every $x \in \mathbb{R}$, there is an integer $k \in \mathbb{Z}$ such that

$$k \leq x < k + 1.$$

Proof. By [Theorem 3.26](#), choose $M \in \mathbb{N}$ with $M > -x$. The nonempty set

$$E = \{n \in \mathbb{N} : n > x + M\}$$

has a least element n by well ordering. Since $n - 1 \notin E$ when $n > 1$, one has $n - 1 \leq x + M < n$. The case $n = 1$ gives the same left inequality because $x + M > 0$. Thus $k = n - M - 1$ is an integer satisfying $k \leq x < k + 1$. $\square$

Theorem 3.28 (Density of the rational numbers). If $x, y \in \mathbb{R}$ and $x < y$, then there is $q \in \mathbb{Q}$ such that

$$x < \iota(q) < y.$$

Proof. By [Theorem 3.26](#), choose $N \in \mathbb{N}$ with $N(y - x) > 1$. Apply [Lemma 3.27](#) to Nx to obtain $k \in \mathbb{Z}$ with $k \leq Nx < k + 1$. Then

$$Nx < k + 1 \leq Nx + 1 < Ny.$$

Dividing by the positive integer N gives the result for $q = (k + 1)/N$. $\square$

Definition 3.29 (Convergence of a real sequence). Let (x_n) be a sequence in $\mathbb{R}$, and let $x \in \mathbb{R}$. We say that (x_n) **converges to** x , and write

$$\lim_{n \rightarrow \infty} x_n = x,$$

if for every $\varepsilon > 0$ there is $N \in \mathbb{N}$ such that

$$n \geq N \implies |x_n - x| < \varepsilon.$$

Definition 3.30 (Cauchy sequence in $\mathbb{R}$). A real sequence (x_n) is **Cauchy** if, for every $\varepsilon > 0$, there is $N \in \mathbb{N}$ such that $m, n \geq N$ implies $|x_m - x_n| < \varepsilon$.

Lemma 3.31 (Bounds for classes). If $a \leq u_n \leq b$ eventually, with rational a, b , then $\iota(a) \leq [(u_n)] \leq \iota(b)$.

Proof. If $[(u_n)] > \iota(b)$, positivity would require $u_n - b > \delta > 0$ eventually, contradicting the upper bound. The lower bound is identical. $\square$

Strict bounds need not remain strict: $u_n = 1 - 1/(n + 1) < 1$ represents 1. To prove a strict conclusion, reserve a margin in the approximation.

Lemma 3.32 (Rational approximation from a representative). Let $x = [(q_n)] \in \mathbb{R}$. Then

$$\lim_{n \rightarrow \infty} \iota(q_n) = x.$$

Consequently, for every $\varepsilon > 0$ there is a rational q such that

$$|\iota(q) - x| < \varepsilon.$$

Proof. Let $\varepsilon > 0$. By [Theorem 3.28](#), choose $e \in \mathbb{Q}$ with

$$0 < \iota(e) < \varepsilon.$$

Cauchyness of (q_n) supplies N such that

$$m, n \geq N \implies |q_m - q_n| < e/2.$$

For a fixed $n \geq N$, the real number $\iota(q_n) - x$ is represented by the sequence $(q_n - q_k)_{k \in \mathbb{N}}$, whose tail has absolute value less than $e/2$. By [Lemma 3.31](#),

$$|\iota(q_n) - x| \leq \iota(e/2) < \iota(e) < \varepsilon.$$

This proves the convergence. Taking one sufficiently large n proves the final assertion. $\square$

Four levels of notation in the completeness argument. The construction begins with rational sequences (q_k) , not sequences of already available real numbers. A constructed real is a class $[(q_k)]$ of equivalent rational Cauchy sequences. A rational r is embedded as the class of the constant sequence $(r, r, \dots)$. Finally, the sequence whose completeness is now being proved is a sequence of these classes.

$$\begin{array}{ll} q_k \in \mathbb{Q} & \longrightarrow (q_k)_{k \geq 1} \text{ rational Cauchy sequence} \\ & (q_k) \longmapsto [(q_k)] \text{ one constructed real} \\ X_1, X_2, \dots & \text{a sequence of constructed reals.} \end{array}$$

To approximate its n th term X_n , choose one embedded rational r_n close to that class. The diagonal choice (r_n) is a new rational sequence, not a choice of one fixed representative sequence for all X_n . The proof checks it is Cauchy, takes its class as the candidate limit, and compares that class back to X_n . This avoids assuming the very completeness being established.

Theorem 3.33 (Cauchy completeness of the construction). Every Cauchy sequence in $\mathbb{R}$ converges to an element of $\mathbb{R}$.

Proof. Idea. Replace the j th real approximation by one rational number within $1/j$. The resulting rational Cauchy process is itself a real number:

$$x_j \approx \iota(r_j) \longrightarrow [(r_j)].$$

No summable error is needed, only an error tending to zero.

By [Lemma 3.32](#), choose $r_j \in \mathbb{Q}$ with $|x_j - \iota(r_j)| < 1/j$. This can be explicit: fix a listing of $\mathbb{Q}$ and take its first entry satisfying the inequality for each j . Given rational $e > 0$, choose J so that $1/J < e/3$ and $|x_j - x_k| < \iota(e/3)$ for $j, k \geq J$. Then

$$|\iota(r_j) - \iota(r_k)| \leq |\iota(r_j) - x_j| + |x_j - x_k| + |x_k - \iota(r_k)| < \iota(e).$$

Order preservation gives $|r_j - r_k| < e$, so (r_j) is rational Cauchy. Set $x = [(r_j)]$. The representative-approximation lemma gives $\iota(r_j) \rightarrow x$. Given real $\varepsilon > 0$, choose J so that, for $j \geq J$, both $1/j < \varepsilon/2$ and $|\iota(r_j) - x| < \varepsilon/2$. Hence

$$|x_j - x| \leq |x_j - \iota(r_j)| + |\iota(r_j) - x| < \varepsilon.$$

Thus $x_j \rightarrow x$, which is exactly what completing $\mathbb{Q}$ was meant to achieve. $\square$

Theorem 3.34 (Completeness of the constructed field). The ordered field $\mathbb{R}$ of [construction 3.22](#) has the least-upper-bound property.

Proof. Roadmap. Keep one endpoint that is not an upper bound and one that is. Bisect to shrink their distance to zero, use Cauchy completeness to find their common limiting boundary, and check that it is the least upper bound.

Let $S \subseteq \mathbb{R}$ be nonempty and bounded above. Choose $s_0 \in S$ and an upper bound u_0 . If s_0 is an upper bound, then it is a maximum of S and hence its least upper bound. Otherwise s_0 is not an upper bound. Since every element of S is at most u_0 , one has $s_0 < u_0$. Bisect $[s_0, u_0]$: at stage n , let $m_n = (s_n + u_n)/2$; if m_n is an upper bound of S , put $(s_{n+1}, u_{n+1}) = (s_n, m_n)$, and otherwise put $(s_{n+1}, u_{n+1}) = (m_n, u_n)$. Then s_n is not an upper bound, u_n is an upper bound, and $u_n - s_n = 2^{-n}(u_0 - s_0)$.

The sequence (s_n) is Cauchy: for $m \geq n$, $0 \leq s_m - s_n \leq u_n - s_n = 2^{-n}(u_0 - s_0)$, and the right side tends to 0 by [Theorem 3.26](#). By [Theorem 3.33](#), let $s_n \rightarrow a$. The triangle inequality shows directly that $u_n \rightarrow a$, because $|u_n - a| \leq |u_n - s_n| + |s_n - a|$. Finally, for fixed n , the inequalities

$s_n \leq s_m \leq u_m \leq u_n$ for $m \geq n$ pass to the limit: if, for example, $s_n > a$, then s_m cannot lie within $(s_n - a)/2$ of a for any $m \geq n$. Hence $s_n \leq a \leq u_n$.

If $t \in S$ and $t > a$, choose n with $u_n - s_n < t - a$; then $t > u_n$, contradicting that u_n is an upper bound. Thus a is an upper bound. If $v < a$, choose n with $u_n - s_n < a - v$; then $v < s_n$. Since s_n is not an upper bound, neither is v . Hence a is the least upper bound. $\square$

The construction has now supplied the existence promised in [Definition 3.15](#). Therefore the ordered field constructed in [construction 3.22](#) is, up to the unique order-preserving field isomorphism fixing $\mathbb{Q}$, the real number system. The remaining results record useful equivalent forms and consequences of its completeness.

Other standard descriptions—Dedekind cuts, suitable decimal expansions, and nested-interval models—are not constructed in full here. The uniqueness theorem does not by itself prove that such descriptions are equivalent to this model: for each one, it remains necessary to prove the field, order, and completeness properties. Once that has been done, however, the uniqueness theorem supplies its canonical order-preserving field isomorphism with the Cauchy-sequence model and hence with $\mathbb{R}$.

Proposition 3.35 (Positive square roots). For every $a > 0$ there is a unique $s > 0$ with $s^2 = a$, denoted $\sqrt{a}$.

Proof. Let $S = \{x \geq 0 : x^2 < a\}$. It contains a positive element, for example $\min\{1, a\}/2$, and is bounded above by $a + 1$. Let $s = \sup S > 0$. If $s^2 < a$, we seek a slightly larger member of S . The error is $(s + h)^2 - s^2 = 2sh + h^2$. First require $h < 1$, so this error is less than $(2s + 1)h$; then require the latter to be smaller than the gap $a - s^2$. Accordingly choose $0 < h < \min\{1, (a - s^2)/(2s + 1)\}$. Then $(s + h)^2 < s^2 + (2s + 1)h < a$, contradicting that s is an upper bound. If $s^2 > a$, we instead seek a smaller upper bound. Since $s^2 - (s - h)^2 = 2sh - h^2 \leq 2sh$, it suffices to keep $2sh$ smaller than $s^2 - a$, while $h < s$ keeps $s - h$ positive. Choose $0 < h < \min\{s, (s^2 - a)/(2s)\}$. Then $(s - h)^2 > s^2 - 2sh > a$. Any $x \in S$ must satisfy $x < s - h$, since for nonnegative $x \geq s - h$ we would have $x^2 \geq (s - h)^2 > a$. Thus $s - h$ is a smaller upper bound, again a contradiction. Thus $s^2 = a$. For $0 < u < v$, $v^2 - u^2 = (v - u)(v + u) > 0$, which proves uniqueness. $\square$

In particular $\sqrt{2}$ exists and is irrational by the initial parity argument. The same perturbation estimates, with rational s, h , show directly that $\{q \in \mathbb{Q} : q > 0, q^2 < 2\}$ has no rational least upper bound.

Theorem 3.36 (Greatest-lower-bound property). Every nonempty subset of $\mathbb{R}$ that is bounded below has a greatest lower bound.

Proof. Let $S \subseteq \mathbb{R}$ be nonempty and bounded below. Define

$$-S = \{-s : s \in S\}.$$

Then $-S$ is nonempty and bounded above. By [Theorem 3.34](#), let $u = \sup(-S)$, and put $\ell = -u$. For every $s \in S$, one has $-s \leq u$, hence $\ell \leq s$; thus ℓ is a lower bound. If v is any lower bound of S , then $-s \leq -v$ for every $s \in S$, so $-v$ is an upper bound of $-S$. Therefore $u \leq -v$, or $v \leq \ell$. Hence ℓ is the greatest lower bound. $\square$

Theorem 3.37 (Nested-interval property). Let

$$[a_1, b_1] \supseteq [a_2, b_2] \supseteq \cdots$$

be nested closed intervals in $\mathbb{R}$, and suppose

$$\lim_{n \rightarrow \infty} (b_n - a_n) = 0.$$

Then there is exactly one real number belonging to every interval.

Proof. The sequence (a_n) is increasing and bounded above by b_1 , so its range has a least upper bound c . For every fixed n , nesting gives $a_m \leq a_n \leq b_n$ when $m < n$, and $a_m \leq b_m \leq b_n$ when $m \geq n$. Thus b_n is an upper bound of the whole range, so $c \leq b_n$. Also $a_n \leq c$ by definition of upper bound. Hence $c \in [a_n, b_n]$ for every n .

If d belongs to every interval, then $|c - d| \leq b_n - a_n$ for every n . Taking the limit gives $|c - d| = 0$, so $c = d$. Thus the common point is unique. $\square$

Remark 3.38 (What to retain). Uniqueness identifies $\mathbb{R}$ up to canonical isomorphism; the Cauchy construction proves existence. Subsequent analysis uses the ordered-field laws, Archimedean property, rational density, and completeness far more often than it uses equivalence classes. Understanding their construction matters; reproducing every verification on a first reading is a separate achievement.

3.5 A Small Toolkit of Inequalities

Exact computation is often unnecessary in analysis. The decisive step is usually an estimate that replaces a complicated quantity by one already known to be small or bounded.

Proposition 3.39 (Basic absolute-value estimates). For $x, y \in \mathbb{R}$,

$$|x + y| \leq |x| + |y|, \quad ||x| - |y|| \leq |x - y|.$$

Consequently, $|x_1 + \cdots + x_n| \leq \sum_{j=1}^n |x_j|$.

Proof. The triangle inequality was established with the ordered-field absolute value. Applying it to $x = (x - y) + y$ gives $|x| - |y| \leq |x - y|$; exchange x and y for the reverse bound. The finite form follows by induction. $\square$

Proposition 3.40 (Quadratic and Bernoulli inequalities). For $u, v \in \mathbb{R}$,

$$2uv \leq u^2 + v^2.$$

If $t \geq -1$ and $n \in \mathbb{N}$, then

$$(1 + t)^n \geq 1 + nt.$$

Proof. The first inequality is $(u - v)^2 \geq 0$. For the second, induction begins at $n = 1$. If the assertion holds for n , multiplication by $1 + t \geq 0$ gives

$$(1 + t)^{n+1} \geq 1 + (n + 1)t + nt^2 \geq 1 + (n + 1)t.$$

$\square$

Remark 3.41 (How estimates are used). The triangle inequality splits an error into controllable pieces; the reverse triangle inequality keeps denominators away from zero; the quadratic estimate turns products into sums; and Bernoulli compares powers with linear growth. For nonnegative

inputs, the quadratic estimate later yields the usual arithmetic–geometric mean inequality after roots are available. Chapter 10 adds Cauchy–Schwarz and the norm comparisons

$$|x_i| \leq \|x\|_2 \leq \sqrt{n} \|x\|_\infty$$

for multivariable estimates. Later examples will repeatedly seek a sufficiently strong bound rather than an exact formula.

Exercises

Exercise 3.1. Give two unequal rational Cauchy sequences representing 1, and a positive rational sequence representing zero. Explain why eventual equality and eventual positivity would give the wrong equivalence relation and order.

Exercise 3.2. Adapt the square-root construction to prove existence of a positive cube root of 2. First show that $\{x \geq 0 : x^3 < 2\}$ is nonempty and bounded; then use $(s+h)^3 - s^3 = 3s^2h + 3sh^2 + h^3$ to rule out either strict inequality for the cube of its supremum.

Exercise 3.3. Let F be a complete ordered field. Prove that every nonempty open interval in F contains a rational element.

Exercise 3.4. Suppose rational approximations satisfy $|x_j - r_j| < 1/j$. Explain why Cauchyness of (x_j) forces Cauchyness of (r_j) , and why replacing $1/j$ by any positive error tending to zero leaves the argument valid.

Chapter 4

Sequences of Real Numbers

4.1 Convergence

In this chapter the real numbers are the ordered field constructed in Chapter 3. A sequence in $\mathbb{R}$ is a function $a : \mathbb{N} \rightarrow \mathbb{R}$; its value at n is written a_n .

Calculus repeatedly asks what happens after indefinitely many successive approximations. Convergence turns that informal question into a test with a clear responsibility: after a requested accuracy is named, all sufficiently late terms must meet it.

Definition 4.1 (Limit of a sequence). Let (a_n) be a real sequence and let $L \in \mathbb{R}$. We say that (a_n) **converges to** L , and write

$$\lim_{n \rightarrow \infty} a_n = L,$$

or equivalently $a_n \rightarrow L$, if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that, for every $n \geq N$, one has $|a_n - L| < \varepsilon$. A sequence that converges to some real number is **convergent**; otherwise it is **divergent**.

The number N may depend on ε , but it must work for every $n \geq N$. This order of quantifiers is essential.

Decimals are limits, not new numbers

Decimal notation involves three separate tasks. First, a prescribed digit string must have convergent truncations. Second, every real must admit such digits. Third, different digit strings may name the same real. The geometric tail bound addresses the first task, successive decimal intervals address the second, and the equality of a terminating expansion with a 9-tail addresses the third.

Example 4.2 (Constructing digits by successive intervals). For $x = 1/7$, integer division successively gives $10 = 1 \cdot 7 + 3$, $30 = 4 \cdot 7 + 2$, $20 = 2 \cdot 7 + 6$, $60 = 8 \cdot 7 + 4$, $40 = 5 \cdot 7 + 5$, and $50 = 7 \cdot 7 + 1$. Thus the digits repeat 142857. The first three truncations are 0.1, 0.14, 0.142, and the n th truncation d_n satisfies $d_n \leq 1/7 < d_n + 10^{-n}$. The shrinking error, not merely the repeating pattern, proves that the digit string represents $1/7$.

Proposition 4.3 (Decimal representations). For digits $d_j \in \{0, \dots, 9\}$, the truncations $s_n = \sum_{j=1}^n d_j 10^{-j}$ converge to a real number in $[0, 1]$, denoted $0.d_1 d_2 \dots$. Every $x \in [0, 1)$ has such an expansion which is not eventually 9. Two different digit strings represent the same number only when, at their first differing place, their digits differ by one and the smaller digit is followed entirely by 9's, the larger entirely by 0's.

Proof. For $m > n$, a finite geometric sum gives

$$0 \leq s_m - s_n \leq \sum_{j=n+1}^m 9 \cdot 10^{-j} = 10^{-n} - 10^{-m} < 10^{-n}.$$

Given $\varepsilon > 0$, choose N with $10^{-N} < \varepsilon$ (use $10^N \geq N + 1$). The truncations are Cauchy and converge by [Theorem 3.33](#). Their bounds $0 \leq s_n \leq 1$ also bound their limit: a limit outside that interval would have a positive distance from every truncation.

To construct digits for $x < 1$, set $r_0 = x$ and recursively choose the integer d_n with $d_n \leq 10r_{n-1} < d_n + 1$, putting $r_n = 10r_{n-1} - d_n$. Then $0 \leq r_n < 1$, $d_n \in \{0, \dots, 9\}$, and $x = s_n + 10^{-n}r_n$, so $s_n \rightarrow x$. This expansion cannot have only 9's after position n : that tail would force $r_n = 1$.

For uniqueness, suppose two strings first differ at place k , with $a_k < b_k$. Their later tails can compensate by at most $\sum_{j>k} 9 \cdot 10^{-j} = 10^{-k}$. Equality of the represented numbers therefore forces $b_k - a_k = 1$ and the maximal compensating tails. If even one later digit fails to be maximal, the compensation loses a fixed positive amount. Thus the tails must be respectively all 9's and all 0's. $\square$

The first useful example is

$$0.\underbrace{99 \dots 9}_{n \text{ digits}} = 1 - 10^{-n}, \quad 0.999 \dots = 1.$$

Likewise $0.24999 \dots = 0.25$. A decimal expression represents a number; it is not the number itself. For a canonical representation use the one that is not eventually 9 (and the usual integer part for numbers outside $[0, 1)$).

Theorem 4.4 (Cantor's diagonal argument, now formal). The interval $(0, 1)$, and hence $\mathbb{R}$, is uncountable.

Proof. If (x_n) listed $(0, 1)$, write each entry in its canonical decimal expansion. Choose the n th digit c_n to be 2 if the n th digit of x_n is 1, and 1 otherwise. The preceding proposition supplies a real number $c = 0.c_1c_2 \dots$ in $(0, 1)$. Its digits are only 1 and 2, so no terminating/repeating-9 ambiguity can identify it with an x_n whose n th digit differs. This contradicts the claimed listing. $\square$

This discharges the informal decimal preview in [Chapter 2](#): existence and representation ambiguity have now both been justified.

Lemma 4.5. If $x \in \mathbb{R}$ satisfies $|x| < \varepsilon$ for every $\varepsilon > 0$, then $x = 0$.

Proof. If $x \neq 0$, then $|x| > 0$. Taking $\varepsilon = |x|/2$ contradicts the hypothesis. $\square$

Theorem 4.6 (Uniqueness of limits). A convergent real sequence has exactly one limit.

Proof. Suppose $a_n \rightarrow L$ and $a_n \rightarrow M$. Given $\varepsilon > 0$, choose N_1, N_2 such that $|a_n - L| < \varepsilon/2$ for $n \geq N_1$ and $|a_n - M| < \varepsilon/2$ for $n \geq N_2$. For $n \geq \max\{N_1, N_2\}$, the triangle inequality gives

$$|L - M| \leq |L - a_n| + |a_n - M| < \varepsilon.$$

By [Lemma 4.5](#), $L - M = 0$. $\square$

Example 4.7. The sequence $a_n = 1/n$ converges to 0. For a given $\varepsilon > 0$, [Theorem 3.26](#) provides $N > 1/\varepsilon$, and then $n \geq N$ implies

$$0 < \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

Definition 4.8 (Divergence to infinity). We write

$$a_n \longrightarrow +\infty$$

if, for every $M \in \mathbb{R}$, there is $N \in \mathbb{N}$ such that

$$n \geq N \implies a_n > M.$$

We write $a_n \rightarrow -\infty$ if, for every $M \in \mathbb{R}$, there is N such that

$$n \geq N \implies a_n < M.$$

These statements describe two particular forms of divergence. Neither $+\infty$ nor $-\infty$ is a real number, and neither statement says that (a_n) converges in $\mathbb{R}$.

Example 4.9 (Examples and nonexamples). The sequence n^2 tends to $+\infty$: given $M \in \mathbb{R}$, choose $N > \max\{M, 1\}$. The sequence $-n$ tends to $-\infty$. By contrast, $(-1)^n n$ is unbounded in both directions but tends to neither infinity; for every proposed stage, later terms occur with each sign. Thus “unbounded” is much weaker than “eventually larger than every real bound.”

Remark 4.10 (A useful proof pattern). To prove $a_n \rightarrow +\infty$, start with an arbitrary target height M and solve the inequality $a_n > M$ for a sufficient condition on n . This is the infinite-limit analogue of solving $|a_n - L| < \varepsilon$ in an ordinary ε - N proof.

4.2 Algebra and Order of Limits

Once limits have been defined, the next question is whether familiar algebra survives passage to the limit. The following rules justify the calculations that will later occur inside functions, series, derivatives, and integrals.

Theorem 4.11 (Algebra of limits). Let (a_n) and (b_n) be real sequences with $a_n \rightarrow a$ and $b_n \rightarrow b$, and let $c \in \mathbb{R}$. Then

$$\lim_{n \rightarrow \infty} (a_n + b_n) = a + b, \quad \lim_{n \rightarrow \infty} ca_n = ca, \quad \lim_{n \rightarrow \infty} a_n b_n = ab.$$

If $b \neq 0$, then

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{a}{b}$$

whenever $b_n \neq 0$ for all sufficiently large n .

Proof. For the sum, choose stages at which $|a_n - a| < \varepsilon/2$ and $|b_n - b| < \varepsilon/2$, then apply the triangle inequality. Scalar multiplication follows from $|ca_n - ca| = |c||a_n - a|$; for $c = 0$ it is immediate, and otherwise use tolerance $\varepsilon/|c|$.

Convergent sequences are eventually bounded: choosing N with $|a_n - a| < 1$ gives $|a_n| \leq |a| + 1$ for $n \geq N$, and the finitely many earlier terms can be included in a larger bound. Let M bound both $|a_n|$ and $|b|$. Then

$$|a_n b_n - ab| \leq |a_n| |b_n - b| + |b| |a_n - a| \leq M(|b_n - b| + |a_n - a|),$$

which proves the product assertion.

For division, choose N_0 so that $|b_n - b| < |b|/2$ for $n \geq N_0$. Then $|b_n| \geq |b|/2$, so such b_n are nonzero. Thus

$$\left| \frac{a_n}{b_n} - \frac{a}{b} \right| \leq \frac{2}{|b|} |a_n - a| + \frac{2|a|}{|b|^2} |b_n - b|.$$

Choose stages making the two terms on the right sum to less than ε . □

Theorem 4.12 (Order preservation). Let $a_n \rightarrow a$ and $b_n \rightarrow b$. If $a_n \leq b_n$ for every sufficiently large n , then $a \leq b$.

Proof. If $a > b$, let $\varepsilon = (a - b)/3 > 0$. For all sufficiently large n , $|a_n - a| < \varepsilon$ and $|b_n - b| < \varepsilon$. Hence $a_n > a - \varepsilon = b + 2\varepsilon > b_n$, a contradiction. $\square$

Corollary 4.13 (Squeeze theorem). If $a_n \leq b_n \leq c_n$ eventually and $a_n \rightarrow L$, $c_n \rightarrow L$, then $b_n \rightarrow L$.

Proof. Given $\varepsilon > 0$, take a stage after which the ordering holds and both $L - \varepsilon < a_n$ and $c_n < L + \varepsilon$. Then $L - \varepsilon < b_n < L + \varepsilon$, proving the limit. $\square$

Example 4.14 (Algebraic simplification and rationalization). The candidate limit in

$$a_n = \frac{3n^2 - n}{2n^2 + 5}$$

becomes visible after division by n^2 ; the limit laws then give $a_n \rightarrow 3/2$. For

$$b_n = \sqrt{n^2 + n} - n,$$

direct substitution hides the answer. Multiplying by the conjugate gives

$$b_n = \frac{n}{\sqrt{n^2 + n} + n} = \frac{1}{\sqrt{1 + 1/n} + 1} \rightarrow \frac{1}{2}.$$

Here square roots exist by [Proposition 3.35](#); their needed limit follows directly from

$$0 \leq \sqrt{1 + 1/n} - 1 = \frac{1/n}{\sqrt{1 + 1/n} + 1} \leq \frac{1}{n}.$$

The algebra does not itself prove a limit; it rewrites the expression into terms to which already proved limit laws apply.

Remark 4.15 (Strategy for sequence limits). Separate discovery from proof. First simplify, compute a few terms, or compare dominant terms to guess a candidate. Then prove it using the definition, the limit laws, a squeeze estimate, or monotonicity and boundedness. To prove that no finite limit exists, look for incompatible subsequences, failure of the Cauchy condition, or divergence to $\pm\infty$. A failed attempt to find a limit is not a proof of nonexistence.

4.3 Subsequences and Elementary Divergence

A subsequence samples an infinite process along a thinner infinite set of stages. It is especially useful for detecting divergence: incompatible long-term behaviors cannot belong to one convergent sequence.

Definition 4.16 (Subsequence). Let (a_n) be a sequence. A **subsequence** is a sequence $(a_{n_k})_{k \in \mathbb{N}}$, where $n_1 < n_2 < \dots$ are natural numbers.

Proposition 4.17 (Subsequences preserve limits). If $a_n \rightarrow L$, then every subsequence of (a_n) converges to L .

Proof. For $n_1 < n_2 < \dots$, induction gives $n_k \geq k$. Given $\varepsilon > 0$, choose N such that $n \geq N$ implies $|a_n - L| < \varepsilon$. Then $k \geq N$ implies $n_k \geq N$, hence $|a_{n_k} - L| < \varepsilon$. $\square$

Corollary 4.18. If a sequence has two subsequences converging to distinct limits, then it is divergent.

Proof. Otherwise [Proposition 4.17](#) would make both subsequences converge to the sequence's limit, contradicting [Theorem 4.6](#). $\square$

Example 4.19. The sequence $(-1)^n$ diverges. Its even-indexed subsequence is constantly 1, while its odd-indexed subsequence is constantly -1 .

4.4 Completeness Consequences

The order completeness of the real numbers now becomes a practical tool. It converts bounded monotone approximation into a limit, prevents Cauchy approximations from leaving a gap, and forces bounded sequences to have convergent subsequences.

Definition 4.20 (Monotone sequence). A sequence (a_n) is **increasing** if $a_n \leq a_{n+1}$ for all n , **decreasing** if $a_{n+1} \leq a_n$ for all n , and **monotone** if it is increasing or decreasing.

Theorem 4.21 (Monotone convergence). An increasing sequence converges if and only if it is bounded above; in that event, its limit is $\sup \{a_n : n \in \mathbb{N}\}$. A decreasing sequence converges if and only if it is bounded below, and its limit is the infimum of its range.

Proof. If $a_n \rightarrow L$, then $a_n \leq L + 1$ eventually, and the finitely many earlier terms have a common upper bound. Conversely, let $s = \sup \{a_n : n \in \mathbb{N}\}$. Given $\varepsilon > 0$, $s - \varepsilon$ is not an upper bound, so $a_N > s - \varepsilon$ for some N . For $n \geq N$, $s - \varepsilon < a_N \leq a_n \leq s$, and hence $|a_n - s| < \varepsilon$. The decreasing statement follows on applying the increasing statement to $(-a_n)$. $\square$

Theorem 4.22 (Cauchy criterion). A real sequence converges if and only if it is Cauchy.

Proof. If $a_n \rightarrow L$, choose N with $|a_n - L| < \varepsilon/2$ for $n \geq N$ and use the triangle inequality. Conversely, the rational-Cauchy construction of $\mathbb{R}$ proves exactly this assertion in [Theorem 3.33](#). $\square$

Theorem 4.23 (Bolzano–Weierstrass). Every bounded real sequence has a convergent subsequence.

Proof. Let the sequence (a_j) lie in $[a, b]$. Bisect and retain a closed half containing infinitely many terms; repeat to obtain nested intervals $I_n = [\ell_n, u_n]$ of length $(b-a)2^{-n}$. Choose strictly increasing indices k_n with $a_{k_n} \in I_n$. Here “infinitely many terms” means infinitely many sequence indices, not necessarily distinct values. At least one closed half retains infinitely many indices, since the union of two finite index sets would be finite. After choosing k_{n-1} , only finitely many indices are at most k_{n-1} , so I_n still contains a term with an index $k_n > k_{n-1}$. Start with $k_0 = 0$. The nested-interval property gives a common point x . Given $\varepsilon > 0$, choose N with $(b-a)2^{-N} < \varepsilon$. For $n \geq N$, both a_{k_n} and x lie in I_N , so $|a_{k_n} - x| \leq u_N - \ell_N < \varepsilon$. This proves convergence. $\square$

Definition 4.24 (Limsup and liminf). Let (a_n) be bounded. For $n \in \mathbb{N}$, put

$$s_n = \sup \{a_k : k \geq n\}, \quad i_n = \inf \{a_k : k \geq n\}.$$

Define

$$\limsup_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} s_n, \quad \liminf_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} i_n.$$

Theorem 4.25 (Basic limsup and liminf facts). For every bounded sequence, limsup and liminf exist and

$$\liminf_{n \rightarrow \infty} a_n \leq \limsup_{n \rightarrow \infty} a_n.$$

Moreover, $a_n \rightarrow L$ if and only if the two displayed limits both equal L .

Proof. The sequence (s_n) is decreasing and bounded below, while (i_n) is increasing and bounded above. Their limits therefore exist by [Theorem 4.21](#); since $i_n \leq s_n$, their limits have the stated order by [Theorem 4.12](#). If $a_n \rightarrow L$, then for every $\varepsilon > 0$, all sufficiently late terms lie in

$$\left(L - \frac{\varepsilon}{2}, L + \frac{\varepsilon}{2} \right).$$

For every tail beginning after that stage, its infimum is at least $L - \varepsilon/2$ and its supremum is at most $L + \varepsilon/2$. Their limits therefore equal L . Conversely, if both limits equal L , then for a given $\varepsilon > 0$, eventually $i_n > L - \varepsilon$ and $s_n < L + \varepsilon$. Since $i_n \leq a_n \leq s_n$, the squeeze theorem gives $a_n \rightarrow L$. $\square$

Why boundedness remains in this definition.

Example 4.26 (Tail ceilings and floors). For $a_n = (-1)^n + 1/n$, the even subsequence tends to 1 and the odd subsequence to -1 . Removing finitely many terms cannot remove either recurring behavior. The tail suprema decrease to 1, while the tail infima increase to -1 . Thus $\limsup a_n = 1$ and $\liminf a_n = -1$. For $b_n = 2 + (-1)^n/n$, both tail barriers tend to 2, so the sequence converges despite oscillating. The limsup records the eventual ceiling, not the largest early spike in the whole sequence.

For a bounded sequence, every subsequential limit lies between liminf and limsup: its late terms lie between every fixed pair of tail bounds. Each barrier itself is a subsequential limit. To reach the upper one s , make two choices at stage j . First choose a tail index $N_j > k_{j-1}$ so late that $|s_{N_j} - s| < 1/j$, where $k_0 = 0$. Then the definition of supremum supplies $k_j \geq N_j$ with $s_{N_j} - 1/j < a_{k_j} \leq s_{N_j}$. Thus $k_j > k_{j-1}$ and $|a_{k_j} - s| < 2/j$, proving convergence. For the lower barrier i , first choose $N_j > k_{j-1}$ with $|i_{N_j} - i| < 1/j$, then choose $k_j \geq N_j$ with $i_{N_j} \leq a_{k_j} < i_{N_j} + 1/j$. Again the error is less than $2/j$. One can develop conventions for limsup and liminf of unbounded sequences, but they are unnecessary here. We retain real-valued suprema and infima and use [Definition 4.8](#) when a sequence eventually escapes every real bound.

4.5 Equivalent Completeness Principles

The least-upper-bound property is not the only possible formulation of completeness. The equivalences below explain why several apparently different principles may all serve as the decisive feature distinguishing the real numbers from the rationals.

Theorem 4.27 (Equivalence of standard completeness principles). Let F be an Archimedean ordered field. The following assertions are equivalent.

- (a) Every nonempty subset of F having an upper bound has a least upper bound.
- (b) Every Cauchy sequence in F converges.
- (c) Every bounded sequence in F has a convergent subsequence.
- (d) Every increasing sequence in F that is bounded above converges.

(e) Whenever

$$[a_1, b_1] \supseteq [a_2, b_2] \supseteq \cdots$$

and

$$\lim_{n \rightarrow \infty} (b_n - a_n) = 0,$$

the intervals have a common point.

Consequently, each assertion may be used as a completeness axiom for an Archimedean ordered field.

Proof. All definitions of convergence, Cauchy-ness, boundedness, and intervals, as well as the elementary order-limit lemmas used below, are obtained verbatim by replacing $\mathbb{R}$ with F ; their proofs use only the ordered-field laws.

We prove a cycle of implications. Suppose first that (a) holds. The proof of [Theorem 4.21](#) uses only (a), so (d) follows. A Cauchy sequence is bounded: choose N so that

$$m, n \geq N \implies |x_m - x_n| < 1,$$

and bound the tail by $|x_N| + 1$ and the finitely many earlier terms separately. For the tail sets

$$T_n = \{x_k : k \geq n\},$$

put

$$s_n = \sup T_n, \quad i_n = \inf T_n.$$

The sequences (s_n) and (i_n) are respectively decreasing and increasing, so (d) gives limits s and i . Given $\varepsilon > 0$, Cauchy-ness gives N such that every two terms of T_N differ in absolute value by less than ε . Hence

$$s_N - i_N \leq 2\varepsilon.$$

Passing to the limit in this inequality gives

$$0 \leq s - i \leq 2\varepsilon.$$

Since this holds for every $\varepsilon > 0$, the small-positive-number argument shows $s = i$. Since $i_n \leq x_n \leq s_n$, the squeeze theorem gives $x_n \rightarrow s$. Thus (a) implies (b).

Assume (b), and let S be nonempty and bounded above. Choose $s_0 \in S$ and an upper bound u_0 . If s_0 is already an upper bound, it is the maximum and hence the least upper bound. Otherwise, bisect the interval $[s_0, u_0]$ repeatedly: retain its lower half when its midpoint is an upper bound of S , and retain its upper half otherwise. This produces numbers s_n, u_n such that

$$s_n \text{ is not an upper bound,} \quad u_n \text{ is an upper bound,}$$

and

$$u_n - s_n = 2^{-n}(u_0 - s_0).$$

The Archimedean property implies that $2^{-n} \rightarrow 0$: choose an integer larger than any prescribed positive reciprocal, and use $2^n \geq n + 1$. Thus (s_n) is Cauchy, since

$$0 \leq s_m - s_n \leq u_n - s_n \quad (m \geq n).$$

Let $s_n \rightarrow c$ by (b). The same estimate gives $u_n \rightarrow c$. If $t \in S$ were greater than c , then for sufficiently large n one would have

$$u_n - s_n < t - c,$$

and the order-limit argument gives

$$s_n \leq c \leq u_n.$$

These inequalities force $t > u_n$, a contradiction. Hence c is an upper bound. If $v < c$, then the same inequality with $c - v$ in place of $t - c$, together with $s_n \leq c \leq u_n$, gives $v < s_n$ for a sufficiently large n ; since s_n is not an upper bound, neither is v . Therefore $c = \sup S$, proving (a).

Still under (a), the nested-interval argument in [Theorem 3.37](#) proves (e): the supremum of the left endpoints belongs to every interval. The bisection argument used for [Theorem 4.23](#), together with this nested-interval conclusion, proves (c). More explicitly, repeatedly retain a half of a bounded interval containing infinitely many terms, choose a strictly increasing sequence of indices from the retained intervals, and use the common point of the nested intervals as the limit of that subsequence.

Assume now (c), and let (a_n) be increasing and bounded above. A subsequence (a_{n_k}) converges, say to L . For a fixed n , all sufficiently late members of this subsequence are at least a_n ; [Theorem 4.12](#) gives $a_n \leq L$. Given $\varepsilon > 0$, choose k with

$$|a_{n_k} - L| < \varepsilon.$$

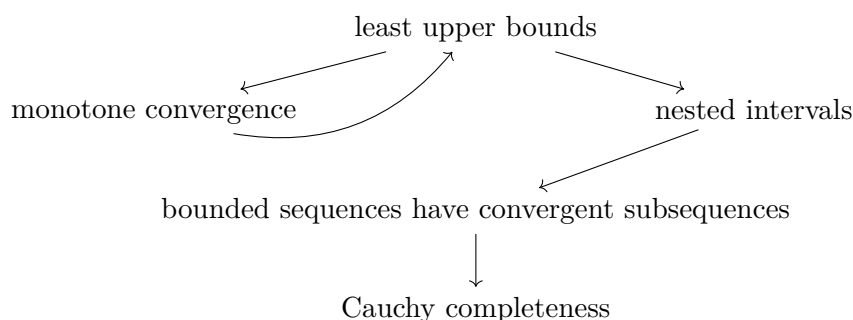
Then, for every $n \geq n_k$,

$$L - \varepsilon < a_{n_k} \leq a_n \leq L,$$

so $a_n \rightarrow L$. This proves (d).

Assume (d). The bisection construction in the second paragraph gives an increasing, bounded-above sequence (s_n) , hence a limit c . The same shrinking-width argument proves that c is the least upper bound of S . Thus (d) implies (a).

Finally, assume (e). Apply the same bisection construction to a nonempty bounded-above set S . Its intervals are nested and their lengths tend to zero, so they have a common point c . The two final paragraphs of the bisection argument show that c is the least upper bound of S . Thus (e) implies (a), completing the proof. $\square$



This diagram highlights proof routes established above in an Archimedean ordered field. Monotone convergence handles increasing approximations; nested intervals locate an unknown point by bisection; subsequences extract limits from bounded data; Cauchy completeness establishes a limit without knowing its value in advance. The equivalence theorem supplies the remaining reverse implications; the diagram is a guide to use, not a substitute proof.

Exercises

Exercise 4.1. Let (a_n) be increasing and bounded above, and let s be the supremum of its range. Adapt the proof of [Theorem 4.21](#) to show directly that

$$\lim_{n \rightarrow \infty} a_n = s.$$

Identify the exact use of the least-upper-bound property.

Exercise 4.2. Give an example of an unbounded sequence for which

$$\lim_{n \rightarrow \infty} \sup a_n$$

cannot be interpreted as a real number. Explain the purpose of the postponed extended convention.

Exercise 4.3. Assuming the least-upper-bound property, prove that every decreasing sequence that is bounded below converges, without appealing to the increasing case as a black box.

Exercise 4.4. Explain why changing the diagonal digits to only 0 and 9 would require an extra ambiguity check. Prove $0.12999 \dots = 0.13$ from truncations.

Chapter 5

Limits and Continuity of Functions

5.1 Limits at a Point

Sequence limits describe processes indexed by natural numbers. A function limit asks the same question when the input can approach a point through all nearby domain values, so the value assigned at the target itself must be kept logically separate.

Definition 5.1 (Limit point). Let $D \subseteq \mathbb{R}$ and $a \in \mathbb{R}$. The point a is a **limit point** of D if, for every $\delta > 0$, there is an $x \in D$ such that $0 < |x - a| < \delta$.

There are points of D other than a arbitrarily close to a . Membership and approachability are different questions. For $D = (0, 1)$, the point 0 is a limit point although $0 \notin D$: for any $\delta > 0$, $x = \min\{\delta/2, 1/2\}$ belongs to D and satisfies $0 < x < \delta$. Thus a right-hand limit at this missing endpoint is meaningful. For $D = \{0\} \cup [1, 2]$, however, $0 \in D$ is not a limit point: no point of D satisfies $0 < |x| < 1/2$.

Definition 5.2 (Isolated point). A point $a \in D$ is **isolated in D** if some $\delta > 0$ satisfies $D \cap (a - \delta, a + \delta) = \{a\}$. For $a \in D$, negating the limit-point definition shows that being isolated is equivalent to not being a limit point of D .

Proposition 5.3 (Sequential test for a limit point). The point a is a limit point of D if and only if there is a sequence $x_n \in D \setminus \{a\}$ with $x_n \rightarrow a$.

Proof. Choose $0 < |x_n - a| < 1/n$ in one direction. Conversely, convergence puts some $x_n \neq a$ in every prescribed punctured neighborhood of a . $\square$

This makes sequences a natural way to test function limits below; the limit-point hypothesis guarantees that there really are approaching inputs.

Definition 5.4 (Function limit). Let $f : D \rightarrow \mathbb{R}$, let a be a limit point of D , and let $L \in \mathbb{R}$. We write

$$\lim_{x \rightarrow a} f(x) = L$$

if, for every $\varepsilon > 0$, there is $\delta > 0$ such that every $x \in D$ satisfying $0 < |x - a| < \delta$ also satisfies $|f(x) - L| < \varepsilon$.

Why require a limit point? If a is not a limit point, some punctured neighborhood contains no domain points. The implication in the limit definition would then be true for *every* proposed L , because there is no admissible x to test. It could not determine a unique value. In the uniqueness proof below, the limit-point assumption is exactly what supplies an actual $x \neq a$ close enough to compare both proposed limits. Notice that $f(a)$ is not needed and, when defined, has no effect on the punctured limit.

Example 5.5 (Localize before choosing the tolerance). To prove $x^2 \rightarrow a^2$, discovery begins with $|x^2 - a^2| = |x - a||x + a|$. The second factor is not fixed. First impose $|x - a| < 1$, which gives $|x + a| < 1 + 2|a|$. Now choose

$$\delta = \min\{1, \varepsilon/(1 + 2|a|)\}.$$

For $0 < |x - a| < \delta$, the product is less than ε . The reusable idea is to bound the uncontrolled factor locally and only then choose the accuracy.

Theorem 5.6 (Uniqueness of function limits). Under the hypotheses of [Definition 5.4](#), there is at most one value L for which the displayed limit holds.

Proof. Suppose L and M both satisfy the definition. Given $\varepsilon > 0$, obtain positive numbers δ_L, δ_M for tolerance $\varepsilon/2$. Since a is a limit point, choose $x \in D$ with

$$0 < |x - a| < \min\{\delta_L, \delta_M\}.$$

Then

$$|L - M| \leq |L - f(x)| + |f(x) - M| < \varepsilon.$$

By [Lemma 4.5](#), $L = M$. □

Theorem 5.7 (Sequential criterion). Let $f : D \rightarrow \mathbb{R}$ and let a be a limit point of D . Then

$$\lim_{x \rightarrow a} f(x) = L$$

if and only if, for every sequence (x_n) in $D \setminus \{a\}$ with $x_n \rightarrow a$, one has $f(x_n) \rightarrow L$.

Proof. Suppose first that the function limit exists. Given $\varepsilon > 0$, take δ from [Definition 5.4](#). Since $x_n \rightarrow a$, eventually $|x_n - a| < \delta$, so eventually $|f(x_n) - L| < \varepsilon$.

Conversely, suppose the sequential statement holds but the function limit fails. Negating the definition gives

$$\exists \varepsilon_0 > 0 \forall \delta > 0 \exists x \in D : 0 < |x - a| < \delta \quad \text{and} \quad |f(x) - L| \geq \varepsilon_0.$$

The order of quantifiers supplies one fixed ε_0 for all choices of δ . For each n , set $\delta = 1/n$ and choose the corresponding point $x_n \in D$, so

$$0 < |x_n - a| < 1/n \quad \text{and} \quad |f(x_n) - L| \geq \varepsilon_0.$$

The first inequality and the Archimedean property imply $x_n \rightarrow a$, while the second prevents $f(x_n)$ from converging to L , a contradiction. □

Definition 5.8 (One-sided and infinite limits). Let $f : D \rightarrow \mathbb{R}$. For a right-hand limit assume that a is a limit point of $D \cap (a, \infty)$; use $D \cap (-\infty, a)$ for a left-hand limit. The assertion

$$\lim_{x \rightarrow a^+} f(x) = L$$

means that for every $\varepsilon > 0$ there is $\delta > 0$ such that

$$x \in D, \quad 0 < x - a < \delta \implies |f(x) - L| < \varepsilon.$$

The left-hand limit is defined with $0 < a - x < \delta$. We write $f(x) \rightarrow +\infty$ as $x \rightarrow a^+$ if for every $M \in \mathbb{R}$ there is $\delta > 0$ such that the same hypotheses imply $f(x) > M$; replace the last inequality by $f(x) < M$ for a limit of $-\infty$. The analogous two-sided definitions use $0 < |x - a| < \delta$.

At an unbounded end of the domain (assume D is unbounded in the stated direction),

$$\lim_{x \rightarrow +\infty} f(x) = L$$

means that for every $\varepsilon > 0$ there is $A \in \mathbb{R}$ such that $x \in D$ and $x > A$ imply $|f(x) - L| < \varepsilon$. Replacing this conclusion by $f(x) > M$ defines $f(x) \rightarrow +\infty$ as $x \rightarrow +\infty$; changing the direction of the input or output inequalities gives the other cases.

Remark 5.9 (Infinity is behavior, not a value). The notation $f(x) \rightarrow +\infty$ does not assert convergence to a member of $\mathbb{R}$. It abbreviates an eventual-inequality statement. For example, $1/x \rightarrow +\infty$ as $x \rightarrow 0^+$, but the two-sided limit at 0 does not tend to either sign of infinity because $1/x \rightarrow -\infty$ from the left.

Theorem 5.10 (Algebraic function-limit laws). Let a be a limit point of a common domain D , and suppose $f(x) \rightarrow L$ and $g(x) \rightarrow M$ as $x \rightarrow a$ through D . For constants c, d , $cf + dg \rightarrow cL + dM$, $fg \rightarrow LM$, and if $M \neq 0$, $f/g \rightarrow L/M$ on a sufficiently small punctured domain.

Proof. Strategy. Transfer the sequence laws through the sequential criterion. For every $x_n \in D \setminus \{a\}$ tending to a , the hypotheses give $f(x_n) \rightarrow L$, $g(x_n) \rightarrow M$. The sequence sum and product laws give the asserted limits on every such sequence, hence the function limits. If $M \neq 0$, impose $|g(x) - M| < |M|/2$ first; then $|g(x)| > |M|/2$ and the quotient is defined locally. Apply the sequence quotient law and the same criterion on that punctured domain. $\square$

Example 5.11 (Direct proof and rationalization). To prove

$$\lim_{x \rightarrow 2} (3x - 1) = 5,$$

observe that $|(3x - 1) - 5| = 3|x - 2|$ and choose $\delta = \varepsilon/3$. For a less direct example,

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x} = \lim_{x \rightarrow 0} \frac{1}{\sqrt{1+x} + 1} = \frac{1}{2},$$

where the equality for $x \neq 0$ follows by rationalization and the last step follows directly from an estimate, without invoking continuity. For $|x| < 1/2$, the positive root is defined by the real-number root construction, and

$$\left| \frac{1}{\sqrt{1+x} + 1} - \frac{1}{2} \right| = \frac{|x|}{2(\sqrt{1+x} + 1)^2} \leq \frac{|x|}{2}.$$

Choosing $\delta = \min\{1/2, 2\varepsilon\}$ proves the limit.

Example 5.12 (One-sided limits detect nonexistence). For $f(x) = x/|x|$ on $\mathbb{R} \setminus \{0\}$,

$$\lim_{x \rightarrow 0^-} f(x) = -1, \quad \lim_{x \rightarrow 0^+} f(x) = 1.$$

Hence the two-sided limit does not exist. Indeed, the sequences $x_n = -1/n$ and $y_n = 1/n$ approach the same point while their images have different limits.

Remark 5.13 (A working strategy for function limits). First discover a plausible answer by substitution, simplification, a graph, or comparison. Next prove it: use limit laws only after their hypotheses are met; cancel only on the punctured neighborhood; rationalize radicals; or bound the error by a simpler expression and squeeze. For nonexistence, compare one-sided limits or produce two approaching sequences with different image limits. The exact value $f(a)$ is irrelevant to $\lim_{x \rightarrow a} f(x)$.

5.2 Asymptotic Notation

Limit arguments often need only a comparison of sizes rather than an exact formula. The following compact notation records two common kinds of comparison without replacing the underlying estimates.

Definition 5.14 (Big-O and little-o). Let f, g be real-valued functions defined sufficiently near a . We write

$$f(x) = O(g(x)) \quad (x \rightarrow a)$$

if there are $C > 0$ and a punctured neighborhood of a on which

$$|f(x)| \leq C|g(x)|.$$

We write

$$f(x) = o(g(x)) \quad (x \rightarrow a)$$

if $g(x) \neq 0$ sufficiently near a and

$$\frac{f(x)}{g(x)} \rightarrow 0.$$

For sequences, $a_n = O(b_n)$ as $n \rightarrow \infty$ and $a_n = o(b_n)$ as $n \rightarrow \infty$ have the identical eventual-bound and quotient-limit meanings.

The limiting regime is part of the assertion. For example, $x^2 = o(x)$ as $x \rightarrow 0$, but not as $x \rightarrow \infty$; and $n^{-2} = o(n^{-1})$ as $n \rightarrow \infty$. All rules below refer to one fixed regime, with comparison functions eventually nonzero when quotients occur.

Translating the notation. The assertion $r = o(g)$ means $r = \epsilon(x)g(x)$ with $\epsilon(x) \rightarrow 0$. Equivalently, for every $\varepsilon > 0$, $|r(x)| \leq \varepsilon|g(x)|$ sufficiently near the limit. The assertion $r = O(g)$ uses one fixed constant C , rather than every positive tolerance. When g is eventually nonzero it is equivalent to $r = C(x)g(x)$ with $C(x)$ bounded. These are functions obeying estimates, not mysterious individual objects called $O(g)$ or $o(g)$. Thus $f = g + o(g)$ means that there exists a remainder $r = o(g)$ for which $f = g + r$. Different occurrences of $o(g)$ may denote different remainders.

Proposition 5.15 (When does a remainder vanish?). One has $r = o(1)$ exactly when $r \rightarrow 0$. The assertion $r = O(1)$ says only that r is bounded. If $g \rightarrow 0$, either $r = O(g)$ or $r = o(g)$ implies $r \rightarrow 0$.

Proof. The first two statements translate the definitions with comparison scale 1. For big-O use $|r| \leq C|g|$ and squeeze. Little-o implies big-O by choosing tolerance 1, so the same argument applies. $\square$

For example $\sin x = O(1)$ as $x \rightarrow \infty$ but has no limit (use $x = 2\pi n + \pi/2$ and $x = 2\pi n + 3\pi/2$). As $x \rightarrow \infty$, $x = o(x^2)$ and $\sqrt{x} = o(x)$, although both remainders diverge. Little-o means small *relative to a scale*, not necessarily small in absolute size.

Near $h = 0$, the three conditions are strictly ordered:

$$r(h) = o(|h|) \implies r(h) = O(|h|) \implies r(h) \rightarrow 0.$$

Neither converse holds: $|h|$ is $O(|h|)$ but not $o(|h|)$, while $\sqrt{|h|} \rightarrow 0$ but is not $O(|h|)$. In contrast, $|h|^2 = o(|h|)$. Chapter 10 replaces absolute values by norms with the same examples.

Proposition 5.16 (A calculus of remainder estimates). In a fixed regime,

$$\begin{aligned} O(g) + O(g) &= O(g), & o(g) + o(g) &= o(g), \\ O(f)O(g) &= O(fg), & o(f)O(g) &= o(fg), & o(f)o(g) &= o(fg). \end{aligned}$$

Also $r = O(f)$, $f = O(g)$ imply $r = O(g)$; replacing either of these two hypotheses by little-o gives $r = o(g)$.

Proof. Idea. Translate each remainder into its comparison scale times a bounded or vanishing coefficient. For sums the coefficients add; for products they multiply. Sums of bounded coefficients are bounded, sums of vanishing coefficients vanish, and a vanishing coefficient times a bounded one vanishes. A vanishing coefficient is itself eventually bounded. For transitivity, multiply r/f by f/g ; these same facts apply. For big-O sums even at zeros of g , the triangle inequality gives $|r_1 + r_2| \leq (C_1 + C_2)|g|$; product bounds work in the same way. $\square$

These displayed “equalities” assert estimates for arbitrary remainders; they do not license cancellation as if $O(g)$ were a fixed number. In particular, $O(g) - O(g)$ need not be zero, and the reciprocal of an $O(1)$ function need not be $O(1)$ without a bound away from zero.

For eventually nonzero g , write $f \sim g$ when $f/g \rightarrow 1$. Equivalently $f = g + o(g)$, since $(f - g)/g = f/g - 1$. Once the elementary sine limit is established, it says $\sin x = x + o(x)$ and $\sin x \sim x$ as $x \rightarrow 0$.

Preview: why differentiation uses little-o. Chapter 8’s derivative condition is equivalent to

$$f(a + h) = f(a) + f'(a)h + o(|h|) \quad (h \rightarrow 0).$$

Indeed, for $r(h) = f(a + h) - f(a) - f'(a)h$, the absolute error quotient is $|r(h)|/|h| = |(f(a + h) - f(a))/h - f'(a)|$. Thus the nonlinear error is negligible compared with the first-order scale, not merely a function tending to zero. The multivariable condition $f(a + h) = f(a) + Df(a)h + o(\|h\|)$ has exactly the same meaning.

5.3 Continuity

Continuity compares nearby function values with the value at a domain point. At a limit point it removes the separation between the limiting value and the assigned value; isolated points require a separate explanation. Open and closed sets provide a second language for this stability and prepare for compactness.

Definition 5.17 (Neighborhoods, open sets, and closed sets). For $a \in \mathbb{R}$ and $r > 0$, the open interval

$$B(a, r) = \{x \in \mathbb{R} : |x - a| < r\}$$

is the **open ball** of radius r about a . A set $U \subseteq \mathbb{R}$ is **open** if every $a \in U$ has some $r > 0$ for which

$$B(a, r) \subseteq U.$$

A set $C \subseteq \mathbb{R}$ is **closed** if its complement is open.

Theorem 5.18 (Sequential characterization of closed sets). A set $C \subseteq \mathbb{R}$ is closed if and only if every convergent sequence in C has its limit in C .

Proof. Suppose first that C is closed and that $x_n \in C$ converges to x . If $x \notin C$, then the open set C^c contains a ball

$$B(x, r).$$

Eventually $x_n \in B(x, r)$, contradicting $x_n \in C$. Conversely, suppose the sequential condition holds and let $x \in C^c$. If no ball about x were contained in C^c , then for every n one could choose

$$x_n \in C, \quad |x_n - x| < 1/n.$$

Then $x_n \rightarrow x$, contradicting the sequential condition. Hence C^c is open. $\square$

Definition 5.19 (Continuity). Let $f : D \rightarrow \mathbb{R}$ and let $a \in D$. The function f is **continuous at a** if, for every $\varepsilon > 0$, there is $\delta > 0$ such that every $x \in D$ satisfying $|x - a| < \delta$ satisfies $|f(x) - f(a)| < \varepsilon$. It is **continuous on D** if it is continuous at every point of D .

Limits and continuity have different domain requirements. A function limit requires a to be a limit point but need not have $a \in D$; its test $0 < |x - a| < \delta$ excludes $x = a$. Continuity requires $a \in D$ because $f(a)$ must exist, but need not require a limit point; its test $|x - a| < \delta$ includes a .

For $D = \{0\} \cup [1, 2]$, choose $\delta = 1/2$. Any $x \in D$ with $|x| < \delta$ must equal 0. The only comparison required for continuity is therefore $|f(0) - f(0)| = 0 < \varepsilon$, for every $\varepsilon > 0$, regardless of the other values of f . Within that neighborhood the function is defined only at the isolated point itself. This differs from a punctured limit, where there would be no admissible points at all.

Consequently, if $a \in D$ is a limit point, f is continuous at a if and only if $\lim_{x \rightarrow a} f(x) = f(a)$: the excluded case $x = a$ adds only the comparison $0 < \varepsilon$. If $a \in D$ is isolated, every function on D is continuous there, while its punctured function limit is not defined under the convention of these notes.

Theorem 5.20 (Sequential criterion for continuity). Let $f : D \rightarrow \mathbb{R}$ and $a \in D$. Then f is continuous at a if and only if every sequence (x_n) in D satisfying $x_n \rightarrow a$ also satisfies $f(x_n) \rightarrow f(a)$.

Proof. The proof is the same as [Theorem 5.7](#), except that terms equal to a require no exclusion: for such a term, $|f(x_n) - f(a)| = 0$. Conversely, failure of continuity gives, for each n , an $x_n \in D$ with $|x_n - a| < 1/n$ and $|f(x_n) - f(a)| \geq \varepsilon_0$. $\square$

Theorem 5.21 (Algebra of continuous functions). Let $f, g : D \rightarrow \mathbb{R}$ be continuous at $a \in D$, and let $c \in \mathbb{R}$. Then $f + g$, cf , and fg are continuous at a . If $g(a) \neq 0$, then f/g is continuous at a on the set where it is defined.

Proof. For every sequence $x_n \rightarrow a$ in D , continuity gives $f(x_n) \rightarrow f(a)$ and $g(x_n) \rightarrow g(a)$. Apply the sequence limit laws of [Theorem 4.11](#) and then [Theorem 5.20](#). $\square$

Example 5.22. Every polynomial is continuous on $\mathbb{R}$: constant and identity functions are continuous directly from [Definition 5.19](#), and repeated application of [Theorem 5.21](#) proves the claim. Every rational function is continuous wherever its denominator is nonzero.

5.4 Uniform Continuity

Ordinary continuity permits the required input tolerance to vary from point to point. Uniform continuity asks for one tolerance that works everywhere on the domain, precisely the stronger control needed for approximation and integration.

Definition 5.23 (Uniform continuity). Let $f : D \rightarrow \mathbb{R}$. The function f is **uniformly continuous on D** if, for every $\varepsilon > 0$, there is $\delta > 0$ such that, for all $x, y \in D$,

$$|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon.$$

The difference from ordinary continuity is that δ may depend on ε , but not on the point of D . Schematically, the quantifiers are

$$\text{continuity: } \forall a \in D \forall \varepsilon > 0 \exists \delta = \delta(a, \varepsilon) > 0,$$

whereas

$$\text{uniform continuity: } \forall \varepsilon > 0 \exists \delta = \delta(\varepsilon) > 0 \forall x, y \in D.$$

Moving the point quantifiers past $\exists \delta$ is the whole strengthening.

Proposition 5.24 (Lipschitz functions). Let $f : D \rightarrow \mathbb{R}$. If there is $K \geq 0$ such that

$$|f(x) - f(y)| \leq K|x - y| \quad \text{for all } x, y \in D,$$

then f is uniformly continuous on D .

Proof. If $K = 0$, the function is constant. If $K > 0$, choose $\delta = \varepsilon/K$. Then the displayed inequality gives the conclusion of [Definition 5.23](#). $\square$

The displayed hypothesis defines a **Lipschitz function**, and K is a Lipschitz constant. Thus

$$\text{Lipschitz} \implies \text{uniformly continuous} \implies \text{continuous}.$$

Neither converse holds in general.

Example 5.25 (The three levels of control). The function $x \mapsto x^2$ is not uniformly continuous on $\mathbb{R}$: with $x_n = n$ and $y_n = n + 1/n$, the inputs approach one another while

$$y_n^2 - x_n^2 = 2 + 1/n^2.$$

On $[-A, A]$, however,

$$|x^2 - y^2| \leq 2A|x - y|,$$

so it is Lipschitz there. The function $\sqrt{x}$ on $[0, 1]$ is uniformly continuous because

$$|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|},$$

but it is not Lipschitz: if a constant K worked, then $\sqrt{x} \leq Kx$ for every $x > 0$, equivalently $1/\sqrt{x} \leq K$, which fails near 0.

Example 5.26 (Continuity need not be uniform). The function $f : (0, 1) \rightarrow \mathbb{R}$ given by $f(x) = 1/x$ is continuous by [Theorem 5.21](#), but not uniformly continuous. Indeed, put $x_n = 1/n$ and $y_n = 1/(n+1)$. Then $|x_n - y_n| \rightarrow 0$, whereas $|f(x_n) - f(y_n)| = 1$ for every n .

On $[a, \infty)$ with $a > 0$, the same function $1/x$ is Lipschitz, since

$$\left| \frac{1}{x} - \frac{1}{y} \right| = \frac{|x - y|}{xy} \leq \frac{1}{a^2} |x - y|.$$

This contrast shows that uniform continuity depends on the domain as well as on the formula.

5.5 Intermediate Values

Continuity has a qualitative consequence beyond limiting estimates: it prevents a function from jumping over a value. Completeness supplies the point at which this intuitive crossing must occur.

Theorem 5.27 (Intermediate value theorem). Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous, with $a < b$. If u lies between $f(a)$ and $f(b)$, then there is $c \in [a, b]$ such that $f(c) = u$.

Proof. It is enough to treat $f(a) \leq u \leq f(b)$. Let

$$S = \{x \in [a, b] : f(x) \leq u\}.$$

The set is nonempty and bounded above, so let $c = \sup S$. If $f(c) < u$, then $c \neq b$ because

$$f(b) \geq u.$$

Continuity gives a positive δ such that $f(x) < u$ whenever

$$|x - c| < \delta.$$

Choose

$$0 < r < \min(\delta, b - c).$$

Then $c + r \in S$, contradicting that c is an upper bound of S . If $f(c) > u$, then $c \neq a$ because

$$f(a) \leq u.$$

Continuity gives a positive δ such that $f(x) > u$ whenever

$$|x - c| < \delta.$$

By the defining property of the supremum, there is

$$x \in S$$

with

$$c - \delta < x \leq c,$$

which contradicts $f(x) \leq u$. Therefore $f(c) = u$. □

Theorem 5.28 (Existence and uniqueness of roots). For every $c \geq 0$ and $n \in \mathbb{N}$, there is a unique $r \geq 0$ such that $r^n = c$.

Proof. The polynomial $p(t) = t^n - c$ is continuous. On $[0, \max\{1, c\}]$ its endpoint values have opposite weak signs, so the Intermediate Value Theorem gives a zero. If $0 \leq r < s$, then

$$s^n - r^n = (s - r)(s^{n-1} + s^{n-2}r + \cdots + r^{n-1}) > 0,$$

so $t \mapsto t^n$ is strictly increasing on $[0, \infty)$ and the root is unique. $\square$

Remark 5.29 (Looking back at familiar radicals). The notation $\sqrt[n]{c}$ now names the unique root supplied by the theorem; it was not justified merely by writing the symbol. For even n , a real n th root requires $c \geq 0$. For odd n , a negative input has the unique negative root $-\sqrt[n]{-c}$. Chapter 7 uses these roots to define rational powers consistently, and Chapter 8 later defines arbitrary real powers of positive bases through $\exp$ and $\log$.

5.6 Compact Intervals

Closed bounded intervals are the domains on which local information can be made global. Compactness will turn subsequential control into maxima, minima, and uniform continuity.

Definition 5.30 (Sequential compactness). A set $K \subseteq \mathbb{R}$ is **sequentially compact** if every sequence in K has a subsequence converging to an element of K .

Theorem 5.31 (Compactness of closed bounded intervals). If $a \leq b$, then $[a, b]$ is sequentially compact.

Proof. Let (x_n) be a sequence in $[a, b]$. It is bounded, so Theorem 4.23 gives a subsequence converging to some $x \in \mathbb{R}$. Since $a \leq x_{n_k} \leq b$ for every k , Theorem 4.12 gives $a \leq x \leq b$. Thus $x \in [a, b]$. $\square$

Theorem 5.32 (Extreme value theorem). Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous, where $a \leq b$. Then there are points $x_{\min}, x_{\max} \in [a, b]$ such that

$$f(x_{\min}) \leq f(x) \leq f(x_{\max}) \quad \text{for every } x \in [a, b].$$

Proof. Let $s = \sup f([a, b])$. To justify that this supremum exists, suppose first that $f([a, b])$ were unbounded above. For each n , choose $x_n \in [a, b]$ with $f(x_n) > n$. By Theorem 5.31, a subsequence x_{n_k} converges to $x \in [a, b]$. Continuity gives $f(x_{n_k}) \rightarrow f(x)$, but $f(x_{n_k}) \geq n_k \rightarrow \infty$, an impossibility. Hence the image is bounded above, and it is nonempty. For each n , choose $x_n \in [a, b]$ with $f(x_n) > s - 1/n$. A convergent subsequence has limit $x_{\max} \in [a, b]$. Continuity and the squeeze theorem give $f(x_{\max}) = s$. Applying the same argument to $-f$ gives $x_{\min}$. $\square$

Theorem 5.33 (Heine–Cantor theorem). Every continuous function $f : [a, b] \rightarrow \mathbb{R}$, where $a \leq b$, is uniformly continuous.

Proof. Suppose not. Then some $\varepsilon_0 > 0$ has the property that, for every n , there are $x_n, y_n \in [a, b]$ such that

$$|x_n - y_n| < 1/n \quad \text{and} \quad |f(x_n) - f(y_n)| \geq \varepsilon_0.$$

By Theorem 5.31, a subsequence x_{n_k} converges to $x \in [a, b]$. Retain those same indices in the paired sequence. Then

$$|y_{n_k} - x| \leq |y_{n_k} - x_{n_k}| + |x_{n_k} - x| < 1/n_k + |x_{n_k} - x| \rightarrow 0.$$

Thus $y_{n_k} \rightarrow x$ as well. Choosing a different subsequence of the y_n would lose the paired estimate, which concerns x_n and y_n at one index. Continuity gives both $f(x_{n_k}) \rightarrow f(x)$ and $f(y_{n_k}) \rightarrow f(x)$, so their difference tends to 0, contradicting the second inequality. $\square$

Theorem 5.34 (Continuity of a monotone inverse). If f is continuous and strictly monotone on an interval I , then $f : I \rightarrow f(I)$ is bijective and its inverse is continuous.

Proof. Strict monotonicity gives injectivity, and the intermediate value theorem makes $f(I)$ an interval. Suppose f is increasing, fix $a \in I$, and let $\varepsilon > 0$. If I contains points to the left of a , choose such a point ℓ with $a - \varepsilon < \ell < a$; similarly choose $r \in I$ with $a < r < a + \varepsilon$ when the right side exists. Each available difference $f(a) - f(\ell)$ or $f(r) - f(a)$ is positive. Choose δ smaller than these differences. For $y \in f(I)$ with $|y - f(a)| < \delta$, monotonicity forces $\ell < f^{-1}(y) < r$ on the available sides; a missing side is automatically bounded by the endpoint a . Thus $|f^{-1}(y) - a| < \varepsilon$. For a singleton interval the assertion is immediate. For decreasing f , apply the argument to $-f$. $\square$

Remark 5.35 (Relative size is not absolute smallness). The statement $r = o(1)$ means $r \rightarrow 0$, while $r = O(1)$ means boundedness near the limiting point. If $g \rightarrow 0$, either $r = O(g)$ or $r = o(g)$ forces $r \rightarrow 0$, by the respective absolute-value estimates. Without $g \rightarrow 0$ there is no such conclusion: as $x \rightarrow +\infty$, $x = o(x^2)$ although x diverges. Little- o compares scales. For vector-valued remainders the same definitions use the norm of r ; Chapter 10 proves the norm estimates needed for composition.

Exercises

Exercise 5.1. For $D = \{0\} \cup \{1/n : n \in \mathbb{N}\} \cup [2, 3]$, determine which of 0, 1, 2, 3 belong to D , are limit points of D , or are isolated in D . At which of these points is every function on D automatically continuous?

Exercise 5.2. As $h \rightarrow 0$, classify $|h|^{1/2}$, $|h|$, $|h|^2$ according to whether they are $o(1)$, $O(|h|)$, or $o(|h|)$. Translate each answer into a bound or quotient limit. Repeat the relative-size comparison of x with x^2 as $x \rightarrow \infty$ and explain why the remainder does not vanish.

Exercise 5.3. Let $r = o(g)$ and $s = O(g)$ in one fixed regime. Prove $r + s = O(g)$, and give an example showing that $r + s = o(g)$ need not hold.

Exercise 5.4. Use the sequential criterion to prove that

$$f(x) = x^2$$

is continuous at every real number.

Exercise 5.5. Let

$$f(x) = \begin{cases} 0, & x \leq 0, \\ 1, & x > 0. \end{cases}$$

Determine all points at which f is continuous, and explain your answer using sequences.

Exercise 5.6. Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous and suppose

$$f(a)f(b) < 0.$$

Use the Intermediate Value Theorem to prove that f has a zero in (a, b) .

Chapter 6

Sequences and Families of Functions

6.1 Pointwise and Uniform Convergence

For functions, convergence has two distinct quantifier patterns. Pointwise convergence lets the stage of approximation depend on the input; uniform convergence does not. This distinction explains exactly when a limit may inherit analytic properties from its approximants.

Definition 6.1 (Pointwise convergence). Let $D \subseteq \mathbb{R}$, and let $f_n : D \rightarrow \mathbb{R}$ and $f : D \rightarrow \mathbb{R}$. We say that (f_n) **converges pointwise** to f on D if, for every $x \in D$,

$$\lim_{n \rightarrow \infty} f_n(x) = f(x).$$

Definition 6.2 (Uniform convergence). Under the same hypotheses, (f_n) **converges uniformly** to f on D if, for every $\varepsilon > 0$, there is $N \in \mathbb{N}$ such that, for every $n \geq N$ and every $x \in D$,

$$|f_n(x) - f(x)| < \varepsilon.$$

Definition 6.3 (Sup norm). If $h : D \rightarrow \mathbb{R}$ is bounded, define

$$\|h\|_\infty = \sup_{x \in D} |h(x)|.$$

Whenever $f_n - f$ is bounded, uniform convergence is equivalently

$$\|f_n - f\|_\infty \rightarrow 0.$$

Indeed, the inequality $\|f_n - f\|_\infty < \varepsilon$ gives the same error bound at every x , while a uniform error bound of $\varepsilon/2$ forces the supremum to be at most $\varepsilon/2 < \varepsilon$.

The dependence of the index is worth displaying:

$$\begin{aligned} \text{pointwise: } & \forall x \in D \forall \varepsilon > 0 \exists N = N(x, \varepsilon), \\ \text{uniform: } & \forall \varepsilon > 0 \exists N = N(\varepsilon) \forall x \in D. \end{aligned}$$

Uniform convergence implies pointwise convergence, because one may fix x after choosing the common stage N . The converse fails: $f_n(x) = x^n$ on $[0, 1]$ converges pointwise to the discontinuous function described in Chapter 1, and hence cannot converge uniformly by [Theorem 6.5](#) below.

More directly, the pointwise limit of x^n on $[0, 1]$ equals 0 on $[0, 1)$ and 1 at 1. For every n , choosing $x_n = 2^{-1/n}$ gives $x_n < 1$ and hence $f(x_n) = 0$, but

$$|f_n(x_n) - f(x_n)| = \frac{1}{2}.$$

Thus the sup-norm error never tends to zero. In contrast, $g_n(x) = x/n$ on $[0, 1]$ converges uniformly to zero because

$$\|g_n\|_\infty = \frac{1}{n}.$$

Proposition 6.4 (Uniform Cauchy criterion). Let $f_n : D \rightarrow \mathbb{R}$. There is a function $f : D \rightarrow \mathbb{R}$ to which f_n converges uniformly if and only if, for every $\varepsilon > 0$, there is N such that, for all $m, n \geq N$ and all $x \in D$,

$$|f_n(x) - f_m(x)| < \varepsilon.$$

Proof. If $f_n \rightarrow f$ uniformly, apply the triangle inequality with tolerance $\varepsilon/2$. Conversely, the displayed condition makes $(f_n(x))$ a Cauchy real sequence for each fixed x . By [Theorem 4.22](#), define $f(x)$ to be its limit. Fix $\varepsilon > 0$ and choose N from the displayed condition. For $n \geq N$, take the limit as $m \rightarrow \infty$ in the inequality with fixed n, x ; the order preservation of limits gives $|f_n(x) - f(x)| \leq \varepsilon$. Repeating with $\varepsilon/2$ proves the strict inequality required by [Definition 6.2](#). $\square$

6.2 Continuous Limits

The principal benefit of uniform convergence is that one approximation index works simultaneously near every point. The three-term estimate below makes this control visible and explains why pointwise convergence is insufficient.

Theorem 6.5 (Uniform limit theorem). Let $D \subseteq \mathbb{R}$. If every $f_n : D \rightarrow \mathbb{R}$ is continuous at $a \in D$ and $f_n \rightarrow f$ uniformly on D , then f is continuous at a .

Proof. Given $\varepsilon > 0$, choose N such that $|f_N(x) - f(x)| < \varepsilon/3$ for every $x \in D$. Continuity of f_N at a supplies $\delta > 0$ such that $|x - a| < \delta$ implies $|f_N(x) - f_N(a)| < \varepsilon/3$. For such x ,

$$|f(x) - f(a)| \leq |f(x) - f_N(x)| + |f_N(x) - f_N(a)| + |f_N(a) - f(a)| < \varepsilon.$$

$\square$

Remark 6.6 (What uniform convergence permits). Uniform limits of continuous functions are continuous by the theorem above. Chapter 9 proves that a uniform limit may also be passed through a Riemann integral. Differentiation is more delicate: uniform convergence of f_n alone does not control the difference quotients, so Chapter 8 imposes uniform convergence on derivatives and fixes one anchor value. The general lesson is to identify the estimate needed by the operation before interchanging two limiting processes.

Proposition 6.7 (Uniform convergence respects algebra). Let $f_n \rightarrow f$ and $g_n \rightarrow g$ uniformly on D . Then $f_n + g_n \rightarrow f + g$ uniformly. If both (f_n) and (g_n) are uniformly bounded, then

$$f_n g_n \rightarrow f g$$

uniformly on D .

Proof. The sum follows from the triangle inequality. Let

$$|f_n(x)| \leq M, \quad |g_n(x)| \leq K \quad (n \in \mathbb{N}, x \in D).$$

Taking pointwise limits gives

$$|g(x)| \leq K \quad (x \in D).$$

Then

$$|f_n(x)g_n(x) - f(x)g(x)| \leq M|g_n(x) - g(x)| + K|f_n(x) - f(x)|.$$

Choose a common stage making both terms uniformly small. $\square$

Proposition 6.8 (Uniform limits preserve boundedness). Let $f_n \rightarrow f$ uniformly on D . If every function f_n is bounded on D , then f is bounded on D .

Proof. Choose N so that

$$|f(x) - f_N(x)| < 1 \quad (x \in D).$$

If $|f_N(x)| \leq M$ throughout D , then

$$|f(x)| \leq M + 1 \quad (x \in D).$$

$\square$

No common bound on the approximating functions is required: only the chosen late function needs to supply a bound. An arbitrary early bounded term cannot do this. On $\mathbb{R}$, take $f_1 = 0$ and $f_n(x) = x$ for $n \geq 2$. Convergence to x is uniform, but the limit is unbounded.

Example 6.9 (A continuous limit without uniform convergence). On $[0, 1]$, put $f_n(x) = \max\{1 - n|x - 1/n|, 0\}$. Each graph is a continuous triangular spike. At 0 its value is zero; for fixed $x > 0$, it is zero once $n > 2/x$. Thus $f_n \rightarrow 0$ pointwise, with a continuous limit. Nevertheless $f_n(1/n) = 1$, so $\|f_n\|_\infty = 1$ for every n . Continuity of the limit does not supply the missing uniform control.

Example 6.10 (Pointwise limits need not preserve boundedness). On

$$D = (0, 1],$$

define

$$f_n(x) = \begin{cases} n, & 0 < x \leq 1/n, \\ 1/x, & 1/n < x \leq 1. \end{cases}$$

Each f_n is bounded by n , while for every $x \in D$,

$$\lim_{n \rightarrow \infty} f_n(x) = \frac{1}{x}.$$

The pointwise limit is unbounded. Thus the uniform hypothesis in [Proposition 6.8](#) cannot be weakened to pointwise convergence.

Example 6.11 (Two limits need not be interchanged). For

$$f_n(x) = x^n \quad (0 \leq x < 1),$$

one has

$$\lim_{n \rightarrow \infty} \left(\lim_{x \rightarrow 1^-} f_n(x) \right) = 1,$$

whereas

$$\lim_{x \rightarrow 1^-} \left(\lim_{n \rightarrow \infty} f_n(x) \right) = 0.$$

The convergence in n is pointwise but not uniform near $x = 1$. This is the basic obstruction behind unjustified interchanges of limits.

6.3 Polynomial Approximation

Polynomials are among the simplest continuous functions: they can be added, multiplied, and differentiated exactly. A natural question is how much of $C([a, b])$ they occupy. The answer is strikingly strong: every continuous function can be approximated everywhere on a compact interval by one polynomial with one common error bound.

Reading the Bernstein weights

For degree two the approximation is

$$B_2(f)(x) = f(0)(1-x)^2 + 2f(1/2)x(1-x) + f(1)x^2.$$

At $x = 1/2$ its weights are $1/4, 1/2, 1/4$. For $f(t) = t^2$ this gives $B_2(f)(x) = (x + x^2)/2$, not x^2 itself. The approximating polynomial samples nearby values heavily; exact reproduction of every quadratic is not the aim. The proof below establishes concentration using finite algebraic sums, with no probability prerequisite.

Theorem 6.12 (Weierstrass approximation theorem). If $f \in C([a, b])$, then for every $\varepsilon > 0$ there is a polynomial p such that

$$\|f - p\|_\infty < \varepsilon.$$

Equivalently, polynomials are uniformly dense in $C([a, b])$.

Proof. We first work on $[0, 1]$. For $f \in C([0, 1])$, define its n th Bernstein polynomial by

$$B_n(f)(x) = \sum_{k=0}^n f(k/n) \binom{n}{k} x^k (1-x)^{n-k}.$$

It is a polynomial. Its nonnegative weights have sum 1 by the binomial theorem, so $B_n(f)(x)$ is a weighted average of sampled values $f(k/n)$. Before estimating a general function, check the samples $1, t, t^2$:

$$B_n(1) = 1, \quad B_n(t) = x, \quad B_n(t^2) = x^2 + \frac{x(1-x)}{n}.$$

For weights $w_{n,k} = \binom{n}{k} x^k (1-x)^{n-k}$, the first moment is

$$\sum_{k=0}^n \frac{k}{n} w_{n,k} = x \sum_{k=1}^n \binom{n-1}{k-1} x^{k-1} (1-x)^{n-k} = x.$$

The factorial second moment is similarly $\sum k(k-1)w_{n,k} = n(n-1)x^2$. Since $k^2 = k(k-1) + k$, division by n^2 gives the displayed quadratic identity.

These follow without differentiation: use $k \binom{n}{k} = n \binom{n-1}{k-1}$ and $k(k-1) \binom{n}{k} = n(n-1) \binom{n-2}{k-2}$, sum with the binomial theorem, and write $k^2 = k(k-1) + k$ (the case $n = 1$ is direct). Expanding the square now shows that the weights concentrate near $k/n = x$:

$$\sum_{k=0}^n \left(\frac{k}{n} - x \right)^2 \binom{n}{k} x^k (1-x)^{n-k} = \frac{x(1-x)}{n} \leq \frac{1}{4n}.$$

Let $M = \|f\|_\infty$. Given $\varepsilon > 0$, uniform continuity of f supplies $\delta > 0$ such that $|s - t| < \delta$ implies $|f(s) - f(t)| < \varepsilon/2$. Split the defining weighted average for $B_n(f)(x) - f(x)$ according as

$|k/n - x| < \delta$ or not. The first part has absolute value at most $\varepsilon/2$. On the second part, the error is at most $2M$, while

$$\sum_{|k/n - x| \geq \delta} \binom{n}{k} x^k (1-x)^{n-k} \leq \frac{1}{\delta^2} \sum_{k=0}^n \left(\frac{k}{n} - x\right)^2 \binom{n}{k} x^k (1-x)^{n-k} \leq \frac{1}{4n\delta^2}.$$

For sufficiently large n , this second contribution is below $\varepsilon/2$, uniformly in x . Hence $B_n(f) \rightarrow f$ uniformly on $[0, 1]$.

For a general interval $[a, b]$, apply this result to $F(t) = f(a + (b-a)t)$ and substitute $t = (x-a)/(b-a)$. The resulting expression is a polynomial in x and has the required uniform error. $\square$

The error estimate separates two reasons a contribution can be small:

samples	function error	total weight
$ k/n - x < \delta$	$< \varepsilon/2$	≤ 1
$ k/n - x \geq \delta$	$\leq 2M$	$\leq 1/(4n\delta^2)$

The near samples are accurate because of uniform continuity. The far samples need not be accurate, but together carry little weight because each contributes at least δ^2 times its weight to the squared distance sum. First choose δ from f , then choose n large. Neither choice depends on the evaluation point x .

Example 6.13 (Approximating a corner). The function $f(x) = |x|$ on $[-1, 1]$ is not a polynomial and is not differentiable at 0. Nevertheless, the theorem provides polynomials p_n with $\|p_n - |x|\|_\infty \rightarrow 0$. This does not contradict the fact that uniform limits of differentiable functions need not be differentiable: uniform convergence preserves continuity, but derivatives require the stronger hypotheses discussed in Chapter 8.

Remark 6.14 (Optional perspective: Stone–Weierstrass). For a compact Hausdorff space X , the real Stone–Weierstrass theorem says that a subalgebra of $C(X, \mathbb{R})$ which contains the constants and separates points is dense in the sup norm. On $[a, b]$, polynomials have these properties, so the classical theorem is its first example. The general result belongs naturally to functional analysis and is not used here.

6.4 Equicontinuity

Uniform convergence controls variation in the sequence index. Equicontinuity instead controls variation in the input uniformly across a family; together with compactness it is a central compactness principle for function spaces.

Definition 6.15 (Equicontinuity). Let $\mathcal{F}$ be a family of functions $D \rightarrow \mathbb{R}$. It is **equicontinuous at** $a \in D$ if, for every $\varepsilon > 0$, there is $\delta > 0$ such that every $f \in \mathcal{F}$ satisfies

$$|x - a| < \delta \implies |f(x) - f(a)| < \varepsilon \quad (x \in D).$$

It is equicontinuous on D if it is equicontinuous at every point of D .

Proposition 6.16. If $f_n \rightarrow f$ pointwise on D and the family $\{f_n : n \in \mathbb{N}\}$ is equicontinuous at a , then f is continuous at a .

Proof. Strategy. Choose the common input neighborhood first, then take pointwise limits at the two fixed points. Equicontinuity supplies δ such that $|f_n(x) - f_n(a)| < \varepsilon/2$ for every n whenever $x \in D$ and $|x - a| < \delta$. Fix such x and let $n \rightarrow \infty$. The inequality becomes $|f(x) - f(a)| \leq \varepsilon/2 < \varepsilon$. This is continuity at a ; no uniform convergence was needed. $\square$

Example 6.17 (A shared Lipschitz bound and moving spikes). If $|f(x) - f(y)| \leq L|x - y|$ for every member of a family, then $\delta = \varepsilon/(L+1)$ proves equicontinuity at every point. The functions $x \mapsto c + sx$ with $|s| \leq 3$ give an example even when their constants c are unbounded. Equicontinuity controls variation, not absolute heights. The triangular spikes introduced earlier are not equicontinuous at zero: $f_n(0) = 0$, $f_n(1/n) = 1$, and $1/n \rightarrow 0$. Every proposed common δ fails for a late spike.

Lemma 6.18 (Uniform equicontinuity on a compact interval). Let $\mathcal{F}$ be an equicontinuous family of functions

$$[a, b] \rightarrow \mathbb{R}.$$

Then, for every $\varepsilon > 0$, there is $\delta > 0$ such that

$$|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon$$

for every $f \in \mathcal{F}$ and every $x, y \in [a, b]$.

Proof. If not, there would be $\varepsilon > 0$, functions $h_n \in \mathcal{F}$, and points $x_n, y_n \in [a, b]$ such that

$$|x_n - y_n| < \frac{1}{n}, \quad |h_n(x_n) - h_n(y_n)| \geq \varepsilon.$$

By compactness of $[a, b]$, pass to a subsequence with $x_n \rightarrow x$. Then also $y_n \rightarrow x$. Relabel this subsequence as (x_n, y_n, h_n) . Equicontinuity at x , with tolerance $\varepsilon/3$, makes both $|h_n(x_n) - h_n(x)|$ and $|h_n(y_n) - h_n(x)|$ less than $\varepsilon/3$ for all sufficiently large n . The triangle inequality gives a contradiction. $\square$

Theorem 6.19 (Arzelà–Ascoli on an interval). Let (f_n) be an equicontinuous sequence in $C([a, b])$ for which there is $M > 0$ satisfying

$$|f_n(x)| \leq M \quad (n \in \mathbb{N}, x \in [a, b]).$$

Then it has a subsequence that converges uniformly on $[a, b]$ to a continuous function.

Proof. The idea is to first obtain convergence on a countable dense set and then use equicontinuity to make that convergence uniform. Enumerate a countable dense subset of $[a, b]$ as (q_j) . Uniform boundedness and Bolzano–Weierstrass allow us to choose successively a subsequence converging at q_1 , then a subsequence of it converging at q_2 , and so on. The diagonal subsequence can be made explicit. Write the nested increasing index lists as $(n_k^{(j)})_{k \geq 1}$ and take $g_k = f_{n_k^{(k)}}$. Because list $k+1$ is a subsequence of list k , $n_{k+1}^{(k+1)} \geq n_{k+1}^{(k)} > n_k^{(k)}$. For fixed j , all entries with $k \geq j$ belong to list j , in increasing order. Thus $(g_k(q_j))$ converges. The remaining step upgrades these separate convergences to one uniform estimate by using only finitely many nearby test points.

The diagonal selection can be pictured as follows; each row is a subsequence of the preceding row, and the marked diagonal gives g_k :

	1	2	3	...	convergence guaranteed at
1	$n_1^{(1)}$	$n_2^{(1)}$	$n_3^{(1)}$	...	q_1
2	$n_1^{(2)}$	$n_2^{(2)}$	$n_3^{(2)}$	...	q_1, q_2
3	$n_1^{(3)}$	$n_2^{(3)}$	$n_3^{(3)}$	...	q_1, q_2, q_3

Given $\varepsilon > 0$, [Lemma 6.18](#) supplies $\delta > 0$ such that

$$|x - y| < \delta \implies |g_k(x) - g_k(y)| < \varepsilon/3 \quad (k \in \mathbb{N}).$$

Using density and a finite partition of $[a, b]$, choose finitely many of the q_j so that every $x \in [a, b]$ lies within δ of one of them. Since (g_k) is Cauchy at each of these finitely many points, there is N such that for $r, s \geq N$ its values there differ by less than $\varepsilon/3$. For a nearby selected point q , the three-term estimate is

$$|g_r(x) - g_s(x)| \leq \underbrace{|g_r(x) - g_r(q)|}_{\text{equicontinuity}} + \underbrace{|g_r(q) - g_s(q)|}_{\text{finite test points}} + \underbrace{|g_s(q) - g_s(x)|}_{\text{equicontinuity}}.$$

There are only finitely many test points, so taking the maximum of their Cauchy stages gives one N . The triangle inequality then gives

$$|g_r(x) - g_s(x)| < \varepsilon \quad (x \in [a, b]).$$

Thus (g_k) is uniformly Cauchy, so [Proposition 6.4](#) gives a uniform limit. It is continuous by [Theorem 6.5](#). $\square$

Exercises

Exercise 6.1. On

$$[0, 1],$$

let

$$f_n(x) = x^n.$$

Determine its pointwise limit and prove directly from [Definition 6.2](#) that the convergence is not uniform.

Exercise 6.2. Let

$$f_n(x) = \frac{x}{n} \quad (x \in \mathbb{R}).$$

Prove that f_n converges pointwise to zero but not uniformly on $\mathbb{R}$.

Exercise 6.3. Suppose that

$$f_n \rightarrow f$$

uniformly on D and that every f_n is bounded by the same number M . Prove that

$$|f(x)| \leq M \quad (x \in D).$$

Exercise 6.4. For $f_n(x) = nx/(1 + n^2x^2)$ on $[0, 1]$, find the pointwise limit and a moving point preventing uniform convergence. Prove uniform convergence on $[\delta, 1]$ for each $\delta > 0$.

Exercise 6.5. Compute $B_2(f)$ for $f(t) = |2t - 1|$. Compare its error at $0, 1/2, 1$. Which part of the Bernstein argument depends on continuity, and which part only on the binomial identities?

Exercise 6.6. Show that pointwise convergence to f and equicontinuity on $[a, b]$ imply uniform convergence there. Use the continuity of f proved above and a finite set of nearby test points. Contrast the moving-spike example.

Exercise 6.7. Give a uniformly bounded family that is not equicontinuous, and an equicontinuous family that is not uniformly bounded. Explain why neither alone supplies the hypotheses of Arzelà–Ascoli.

Exercise 6.8. Suppose $f_n(0) = 0$ and $|f_n(x) - f_n(y)| \leq 2|x - y|$ on $[0, 1]$. Use Arzelà–Ascoli and prove that every uniform subsequential limit also has this Lipschitz bound. Must the original sequence converge?

Chapter 7

Infinite Series

7.1 Series, Partial Sums, and the Cauchy Criterion

An infinite sum is never formed by adding infinitely many terms at once. Its mathematical meaning is the limit of finite partial sums, so all questions about series begin with the sequence theory already developed.

Definition 7.1 (Series). Let (a_n) be a real sequence. Its N th **partial sum** is

$$s_N = \sum_{n=1}^N a_n.$$

The **series** with terms (a_n) is denoted by

$$\sum_{n=1}^{\infty} a_n.$$

It **converges** when (s_N) converges; its sum is then

$$\sum_{n=1}^{\infty} a_n = \lim_{N \rightarrow \infty} s_N.$$

Theorem 7.2 (Cauchy criterion for series). The series

$$\sum_{n=1}^{\infty} a_n$$

converges if and only if, for every $\varepsilon > 0$, there is $N \in \mathbb{N}$ such that

$$m > n \geq N \implies \left| \sum_{k=n+1}^m a_k \right| < \varepsilon.$$

Proof. For $m > n$,

$$s_m - s_n = \sum_{k=n+1}^m a_k.$$

Thus the stated condition says exactly that the partial sums are Cauchy. Apply [Theorem 4.22](#). $\square$

Proposition 7.3 (Finite changes and telescoping). Changing finitely many terms of a series does not affect convergence. If

$$a_n = b_n - b_{n+1},$$

then

$$\sum_{n=1}^N a_n = b_1 - b_{N+1}.$$

Consequently, this telescoping series converges if and only if (b_n) converges, and its sum is

$$b_1 - \lim_{n \rightarrow \infty} b_n.$$

Proof. Partial sums after a finite change differ by a constant from some point onward. The displayed finite identity follows by cancellation; its limit proves the last assertion. $\square$

Proposition 7.4 (Necessary condition). If a series converges, then its terms tend to zero.

Proof. If $s_N \rightarrow s$, then

$$a_N = s_N - s_{N-1} \quad (N \geq 2).$$

The algebra of limits gives $a_N \rightarrow 0$. $\square$

Remark 7.5 (Preliminary strategy for choosing a test). Before trying a sophisticated theorem, check the necessary condition $a_n \rightarrow 0$; failure settles divergence, although success proves nothing by itself. Next look for exact geometric or telescoping structure. For nonnegative terms, compare with an already understood benchmark. Factorials and exponentials often favor the Ratio Test, while expressions raised to the n th power often favor the Root Test. Alternating signs suggest checking absolute convergence first and then the Alternating-Series Test. An inconclusive test is information about the test, not a verdict about the series.

7.2 Nonnegative Series and Comparison

When terms are nonnegative, partial sums can only increase. This lets order and completeness convert estimates by simpler benchmark series into decisive convergence tests.

Theorem 7.6 (Comparison test). Let

$$0 \leq a_n \leq b_n \quad (n \in \mathbb{N}).$$

If

$$\sum_{n=1}^{\infty} b_n$$

converges, then

$$\sum_{n=1}^{\infty} a_n$$

converges. If the latter series diverges, then the former diverges.

Proof. The partial sums of (a_n) are increasing and bounded above by the sum of the series with terms b_n . Apply [Theorem 4.21](#). The second statement is the contrapositive. $\square$

Theorem 7.7 (Limit comparison test). Let $a_n, b_n > 0$ eventually, and suppose

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L$$

with

$$0 < L < \infty.$$

Then the series with terms a_n converges if and only if the series with terms b_n converges.

Proof. Eventually,

$$\frac{L}{2}b_n < a_n < \frac{3L}{2}b_n.$$

The conclusion follows from two applications of [Theorem 7.6](#) and [Proposition 7.3](#). $\square$

Theorem 7.8 (Geometric series). For $r \in \mathbb{R}$, the series with terms r^n , beginning at $n = 0$, converges if and only if

$$|r| < 1.$$

In that case,

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}.$$

Proof. For $r \neq 1$,

$$\sum_{n=0}^N r^n = \frac{1 - r^{N+1}}{1 - r}.$$

If $r = 0$, the conclusion is immediate. If $0 < r < 1$, write $r = 1/(1+h)$ with $h > 0$. Bernoulli's inequality gives

$$(1+h)^N \geq 1 + Nh,$$

so

$$0 \leq r^N \leq \frac{1}{1 + Nh} \rightarrow 0.$$

The same conclusion holds whenever $|r| < 1$, because

$$|r^N| = |r|^N.$$

Taking limits proves the formula. If $|r| \geq 1$, the terms do not tend to zero, so [Proposition 7.4](#) proves divergence. $\square$

The condensation test groups terms into blocks of doubling length:

$$\underbrace{b_1}_{1 \text{ term}}, \quad \underbrace{b_2 + b_3}_{2 \text{ terms}}, \quad \underbrace{b_4 + \cdots + b_7}_{4 \text{ terms}}, \quad \underbrace{b_8 + \cdots + b_{15}}_{8 \text{ terms}}.$$

On the k th block all 2^k terms lie between the values at its ends. Comparing one block at a time gives the inequalities in the proof; it does not replace every term by an asymptotic guess.

Theorem 7.9 (Cauchy condensation test). Let (b_n) be a nonincreasing sequence of nonnegative real numbers. Then

$$\sum_{n=1}^{\infty} b_n$$

converges if and only if

$$\sum_{k=0}^{\infty} 2^k b_{2^k}$$

converges.

Proof. For every $k \geq 0$,

$$2^k b_{2^{k+1}} \leq \sum_{n=2^k}^{2^{k+1}-1} b_n \leq 2^k b_{2^k}.$$

Summing the right inequalities proves that convergence of the condensed series implies convergence of the original series. Summing the left inequalities gives

$$\frac{1}{2} \sum_{k=1}^{m+1} 2^k b_{2^k} \leq \sum_{n=1}^{2^{m+1}-1} b_n.$$

Hence boundedness of the original partial sums implies boundedness of the condensed partial sums. Both series have nonnegative terms, so monotone convergence completes the proof. $\square$

7.3 Absolute Convergence and Products

Signs permit cancellation, which is useful but dangerous: a series can look small only because large positive and negative tails offset each other. Absolute convergence rules out that instability and will justify changing orders and multiplying series.

Definition 7.10 (Absolute and conditional convergence). A series with real terms a_n **converges absolutely** if

$$\sum_{n=1}^{\infty} |a_n|$$

converges. It **converges conditionally** if it converges but does not converge absolutely.

Absolute convergence controls the total size of the terms; this is why it will justify rearrangement. Conditional convergence is weaker, so changing the order of its terms requires separate care.

Theorem 7.11 (Absolute convergence test). Absolute convergence implies convergence.

Proof. For $m > n$,

$$\left| \sum_{k=n+1}^m a_k \right| \leq \sum_{k=n+1}^m |a_k|.$$

The Cauchy criterion for the absolute-value series and [Theorem 7.2](#) show that the original partial sums are Cauchy. $\square$

Theorem 7.12 (Weierstrass M-test). Let $f_n : D \rightarrow \mathbb{R}$ be functions, and suppose that there are numbers $M_n \geq 0$ such that

$$|f_n(x)| \leq M_n \quad (x \in D, n \in \mathbb{N}).$$

If

$$\sum_{n=1}^{\infty} M_n$$

converges, then

$$\sum_{n=1}^{\infty} f_n$$

converges uniformly and absolutely on D .

Proof. For $q \geq p$,

$$\left| \sum_{n=p}^q f_n(x) \right| \leq \sum_{n=p}^q |f_n(x)| \leq \sum_{n=p}^q M_n.$$

The final bound is independent of x and tends uniformly to zero as $p, q \rightarrow \infty$ by the numerical Cauchy criterion. The uniform Cauchy criterion of Chapter 6 proves uniform convergence. The same estimate applied to $|f_n|$ proves absolute convergence at every point. $\square$

Example 7.13 (A uniform majorant). On $[-1, 1]$,

$$\left| \frac{x^n}{2^n} \right| \leq \frac{1}{2^n}.$$

Since $\sum_{n=1}^{\infty} 2^{-n}$ converges, the M-test shows that

$$\sum_{n=1}^{\infty} \frac{x^n}{2^n}$$

converges uniformly and absolutely on $[-1, 1]$.

Theorem 7.14 (Rearrangement of an absolutely convergent series). Let

$$\sum_{n=1}^{\infty} a_n$$

converge absolutely, and let $(a_{\sigma(n)})$ be a rearrangement, where $\sigma : \mathbb{N} \rightarrow \mathbb{N}$ is a bijection. Then the rearranged series converges absolutely and has the same sum.

Proof. Let

$$A = \sum_{n=1}^{\infty} |a_n|.$$

Every finite rearranged absolute partial sum is at most A , so the rearranged absolute-value series converges. Given $\varepsilon > 0$, choose N such that

$$\sum_{n=N+1}^{\infty} |a_n| < \frac{\varepsilon}{2}.$$

Choose M so that

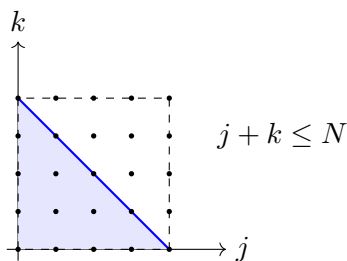
$$\{1, \dots, N\} \subseteq \{\sigma(1), \dots, \sigma(M)\}.$$

For $m \geq M$, put $I_m = \{\sigma(1), \dots, \sigma(m)\}$ and choose $K \geq \max I_m$. Compare the finite sums first:

$$\sum_{j=1}^K a_j - \sum_{j=1}^m a_{\sigma(j)} = \sum_{j \in \{1, \dots, K\} \setminus I_m} a_j.$$

The missing indices all exceed N , so the absolute value is at most $\sum_{j>N} |a_j| < \varepsilon/2$. Keep m fixed and let $K \rightarrow \infty$; the rearranged partial sum stays within $\varepsilon/2$ of the original sum. This holds for every $m \geq M$, proving equality of the sums without any unjustified manipulation of an infinite reordering. $\square$

A product of finite partial sums sums over a rectangle of index pairs (j, k) . A Cauchy product instead collects pairs along diagonals $j+k = n$; its partial sums fill a triangle. Absolute convergence controls the terms in the difference of those two shapes.



Theorem 7.15 (Cauchy product). Suppose that

$$\sum_{n=0}^{\infty} a_n \quad \text{and} \quad \sum_{n=0}^{\infty} b_n$$

converge absolutely. Define

$$c_n = \sum_{k=0}^n a_k b_{n-k}.$$

Then

$$\sum_{n=0}^{\infty} c_n$$

converges absolutely and

$$\left(\sum_{n=0}^{\infty} a_n \right) \left(\sum_{n=0}^{\infty} b_n \right) = \sum_{n=0}^{\infty} c_n.$$

Proof. The partial sums of

$$\sum_{n=0}^{\infty} |c_n|$$

are bounded by the product of the two absolute sums, because

$$\sum_{n=0}^N |c_n| \leq \sum_{\substack{j, k \geq 0 \\ j+k \leq N}} |a_j| |b_k|.$$

Thus the Cauchy product converges absolutely. Let $\varepsilon > 0$. Choose K so large that

$$\left(\sum_{j=K+1}^{\infty} |a_j| \right) \left(\sum_{k=0}^{\infty} |b_k| \right) + \left(\sum_{j=0}^{\infty} |a_j| \right) \left(\sum_{k=K+1}^{\infty} |b_k| \right) < \varepsilon.$$

For $N \geq 2K$, every pair (j, k) with

$$0 \leq j, k \leq N, \quad j + k > N$$

has either $j > K$ or $k > K$. Distributivity of finite sums gives the exact identity

$$\left(\sum_{j=0}^N a_j \right) \left(\sum_{k=0}^N b_k \right) - \sum_{n=0}^N c_n = \sum_{\substack{0 \leq j, k \leq N \\ j+k > N}} a_j b_k.$$

The absolute value of the right side is bounded by

$$\left(\sum_{j>K} |a_j| \right) \left(\sum_{k \geq 0} |b_k| \right) + \left(\sum_{j \geq 0} |a_j| \right) \left(\sum_{k>K} |b_k| \right) < \varepsilon.$$

The finite product tends to the product of the two original sums. Its difference from the Cauchy partial sum tends to zero, proving the formula. No infinite double series has been rearranged in this argument. $\square$

7.4 Ratio, Root, and Alternating Tests

Many series are best compared not term by term with a fixed model, but by examining how consecutive terms or their roots behave. The following tests make that asymptotic comparison precise.

Definition 7.16 (Positive roots). For $x \geq 0$ and $n \in \mathbb{N}$, $n \geq 1$, write

$$x^{1/n}$$

for the unique nonnegative number whose n th power is x . Existence follows from the Intermediate Value Theorem applied to

$$t \longmapsto t^n$$

on a sufficiently large compact interval; uniqueness follows from strict increase on $[0, \infty)$.

Definition 7.17 (Rational powers). Let

$$x > 0, \quad p \in \mathbb{Q}.$$

Choose integers a and b with

$$b > 0, \quad p = \frac{a}{b},$$

and define

$$x^p = (x^a)^{1/b}.$$

The next lemma proves that this does not depend on the chosen representation of p .

Lemma 7.18 (Elementary laws of rational powers). For $x > 0$ and $p, q \in \mathbb{Q}$, rational powers are well defined and satisfy

$$x^{p+q} = x^p x^q, \quad (x^m)^p = x^{mp} \quad (m \in \mathbb{Z}).$$

If

$$0 < x < y \quad \text{and} \quad 0 < p,$$

then

$$x^p < y^p.$$

Proof. Suppose

$$\frac{a}{b} = \frac{c}{d} \quad (b, d > 0).$$

If

$$u = (x^a)^{1/b}, \quad v = (x^c)^{1/d},$$

then $u, v > 0$ and

$$u^{bd} = x^{ad} = x^{bc} = v^{bd}.$$

Uniqueness of positive roots gives $u = v$, so the definition is independent of the representation. Writing rational exponents over a common positive denominator and raising both sides to that denominator proves the two displayed laws, again by uniqueness of positive roots.

For the order statement, put

$$z = \frac{y}{x} > 1.$$

The power laws give

$$\frac{y^p}{x^p} = z^p.$$

Writing

$$p = \frac{a}{b} \quad (a, b \in \mathbb{N}),$$

one has

$$(z^p)^b = z^a > 1.$$

Thus $z^p > 1$, for otherwise its positive b th power would be at most 1. Hence $y^p > x^p$. □

Theorem 7.19 (Rational p -series test). Let

$$p \in \mathbb{Q}.$$

Then

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

converges if and only if

$$p > 1.$$

Proof. Suppose first that

$$p > 0.$$

By [Lemma 7.18](#), the sequence

$$b_n = n^{-p}$$

is nonincreasing. The Cauchy Condensation Test gives convergence precisely when

$$\sum_{k=0}^{\infty} 2^k (2^k)^{-p} = \sum_{k=0}^{\infty} 2^{k(1-p)}$$

converges. This is a geometric series with positive ratio

$$2^{1-p};$$

by the order statement in [Lemma 7.18](#), that ratio is less than 1 exactly when

$$p > 1.$$

If

$$p \leq 0,$$

then

$$n^{-p} \geq 1 \quad (n \in \mathbb{N}),$$

so the terms do not tend to zero. The necessary condition in [Proposition 7.4](#) proves divergence. $\square$

Remark 7.20 (From calculus to analysis: an area preview). The familiar calculus picture compares a positive decreasing sequence $a_n = f(n)$ with rectangles under and over the graph of f . It suggests that

$$\sum_{n=1}^{\infty} f(n) \quad \text{and} \quad \int_1^{\infty} f(x) \, dx$$

should have the same convergence behavior. This is only a preview here: improper integrals have not yet been defined, so the picture is not used as a theorem. [Chapter 9](#) defines the integral as a limit of proper integrals, proves the Integral Test, and recovers the full real p -series rule

$$\sum \frac{1}{n^p} \text{ converges exactly when } p > 1.$$

The preceding theorem has already established the rational-exponent case without integration.

Theorem 7.21 (Ratio test). Let $a_n \neq 0$ eventually. If there are $r < 1$ and N such that

$$|a_{n+1}/a_n| \leq r \quad (n \geq N),$$

then the series with terms a_n converges absolutely. If there are $r > 1$ and N such that

$$|a_{n+1}/a_n| \geq r \quad (n \geq N),$$

then the series diverges.

Proof. The first inequality gives

$$|a_{N+k}| \leq |a_N| r^k,$$

so geometric comparison applies. The second makes the absolute values eventually increasing and nonzero, so the terms cannot tend to zero. $\square$

Theorem 7.22 (Root test). Let (a_n) be a real sequence. If there are r and N such that

$$0 \leq r < 1$$

and

$$|a_n|^{1/n} \leq r \quad (n \geq N),$$

then the series with terms a_n converges absolutely. If there are $r > 1$ and N such that

$$|a_n|^{1/n} \geq r \quad (n \geq N),$$

then the series diverges.

Proof. The first inequality gives

$$|a_n| \leq r^n \quad (n \geq N),$$

so geometric comparison applies. The second gives

$$|a_n| \geq r^n \geq 1$$

eventually, so the terms do not tend to zero. □

Example 7.23 (When a test says nothing). For $a_n = 1/n^2$,

$$\frac{a_{n+1}}{a_n} = \left(\frac{n}{n+1} \right)^2 \rightarrow 1,$$

so the common limit form of the Ratio Test is inconclusive. The series nevertheless converges by the rational p -series theorem. The same ratio limit is 1 for the divergent harmonic series. A value of 1 therefore cannot distinguish the two cases; comparison or the p -series theorem is the appropriate tool.

Theorem 7.24 (Alternating-series test). Let $b_n \geq 0$ decrease to 0. Then

$$\sum_{n=1}^{\infty} (-1)^{n+1} b_n$$

converges.

Proof. The even partial sums are increasing and the odd partial sums are decreasing. Each even sum is at most each later odd sum, so both are bounded. Their limits differ by b_{2n+1} , which tends to zero. Hence all partial sums have one common limit. □

Proposition 7.25 (Alternating remainder estimate). Let $b_n \geq 0$ decrease to zero and set

$$S = \sum_{n=1}^{\infty} (-1)^{n-1} b_n.$$

Then $|S - S_N| \leq b_{N+1}$.

Proof. Grouping adjacent terms in the tail shows that its even partial sums, after factoring out $(-1)^N$, are nonnegative; grouping after the first term shows all tail partial sums are at most b_{N+1} . Passing to the limit gives $0 \leq (-1)^N(S - S_N) \leq b_{N+1}$. □

Forward reference: the Integral Test and real exponents. The Integral Test compares a decreasing sequence of positive terms with an improper integral, so it belongs in Chapter 9. The rational p -series criterion above uses only Chapter 7 material. Arbitrary real powers have not yet been defined; after Chapter 8 constructs them from $\exp$ and $\log$, Chapter 9 gives an analytic rederivation of the p -series criterion for every real exponent.

Remark 7.26 (Expanded test-selection guide). Once the full toolkit is available, rational functions of n usually invite comparison with the power determined by their leading degrees. A positive decreasing term given by a manageable function invites the Integral Test. Factorials and exponentials invite ratios; n th powers invite roots; signs invite absolute convergence and then alternation. Limit comparison is often cleaner than forcing a termwise inequality. Always record the hypotheses (positivity, monotonicity, or eventual nonvanishing) before computing, and switch tests when the conclusion is inconclusive.

7.5 Power Series

Power series are infinite polynomials. Their special feature is a sharp radius inside which convergence is uniform on smaller intervals, allowing algebra and later differentiation to be recovered from finite polynomials.

Definition 7.27 (Power series). For real coefficients (c_n) and center $a \in \mathbb{R}$, a **power series** is

$$\sum_{n=0}^{\infty} c_n(x-a)^n.$$

Theorem 7.28 (Radius of convergence). Every power series has a number

$$R \in [0, \infty]$$

such that it converges absolutely for

$$|x-a| < R$$

and diverges for

$$|x-a| > R.$$

Proof. Let E be the set of $t \geq 0$ for which

$$\sum_{n=0}^{\infty} |c_n| t^n$$

converges, with $0^0 = 1$. Then $0 \in E$. If $t \in E$ and $0 \leq u < t$, comparison shows that $u \in E$. Let

$$R = \sup E,$$

with $R = \infty$ if E is unbounded. If $|x-a| < R$, choose $t \in E$ with

$$|x-a| < t;$$

comparison gives absolute convergence.

Conversely, suppose the series converges at a point with

$$z = |x-a| > 0.$$

Its terms tend to zero and are eventually bounded by some M . Thus, for $0 \leq t < z$ and all sufficiently large n ,

$$|c_n|t^n \leq M \left(\frac{t}{z}\right)^n.$$

Geometric comparison gives $t \in E$. If $z > R$, choosing t strictly between R and z contradicts the definition of R . Hence the series diverges outside its radius. $\square$

The theorem intentionally makes no assertion at points satisfying

$$|x - a| = R.$$

Endpoint behavior depends on the particular coefficients and must be examined separately.

Theorem 7.29 (Uniform convergence inside the radius). Let a power series have radius R , and let

$$0 \leq \rho < R.$$

Then it converges uniformly on

$$[a - \rho, a + \rho].$$

Proof. Choose t with

$$\rho < t < R.$$

The series

$$\sum_{n=0}^{\infty} |c_n|t^n$$

converges. For $|x - a| \leq \rho$,

$$|c_n(x - a)^n| \leq |c_n|t^n.$$

The Weierstrass M-test, [Theorem 7.12](#), proves the assertion. $\square$

Example 7.30 (The endpoints are separate problems). Consider $\sum_{n \geq 1} x^n/n$. For $|x| < 1$, comparison with the geometric series proves absolute convergence, and on $[-r, r]$, $r < 1$, the M-test gives uniform convergence. For $|x| > 1$ the terms do not tend to zero, so the radius is 1. At $x = 1$ the harmonic series diverges; at $x = -1$ the alternating-series test gives convergence, which is not absolute. Thus the convergence interval is $[-1, 1)$. The radius theorem settles neither endpoint. Uniform convergence on smaller closed intervals also does not assert uniform convergence on the entire interval $(-1, 1)$.

Power-series operations deferred. On every closed interval strictly inside its radius, a power series is uniformly convergent. Termwise differentiation and integration require additional theorems and are proved in [Chapters 8 and 9](#).

Exercises

Exercise 7.1. Use [Theorem 7.19](#) to prove that

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

converges and that the harmonic series diverges. Identify the dyadic blocks responsible for the latter conclusion.

Exercise 7.2. Use [Theorem 7.24](#) and [Theorem 7.19](#) to show that

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$$

converges conditionally.

Exercise 7.3. Find the radius of convergence of

$$\sum_{n=0}^{\infty} 3^n (x-2)^n.$$

What does [Theorem 7.28](#) leave undecided at the two endpoints?

Exercise 7.4. Let

$$\sum_{n=0}^{\infty} c_n (x-a)^n$$

have radius of convergence $R > 0$. Prove that it converges absolutely at every point satisfying

$$|x-a| < R.$$

Which step in the proof of [Theorem 7.28](#) gives this conclusion?

Exercise 7.5. How many terms of $\sum_{n \geq 1} (-1)^{n-1}/n$ guarantee error at most 10^{-3} ? Explain why a bound on the next term controls the entire tail.

Exercise 7.6. Compare the radius and the behavior at both endpoints for $\sum_{n \geq 1} x^n$, $\sum_{n \geq 1} x^n/n$, and $\sum_{n \geq 1} x^n/n^2$. For the last numerical benchmark use condensation. Which series converge uniformly on the entire closed interval $[-1, 1]$? Justify each answer.

Chapter 8

Differentiation

8.1 Derivative and Differentiability

The derivative formalizes instantaneous rate of change and the slope of the best local linear approximation. Its definition is a function limit, so the earlier theory of limits and continuity supplies its logical foundation.

Definition 8.1 (Derivative). Let $f : D \rightarrow \mathbb{R}$ and let a be a limit point of $D \cap \{x : x \neq a\}$. Assume also $a \in D$. The function is **differentiable at a** if the following limit exists:

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}.$$

The number $f'(a)$ is the **derivative** of f at a .

Theorem 8.2 (Differentiability implies continuity). If f is differentiable at a , then it is continuous at a .

Proof. For $x \neq a$,

$$f(x) - f(a) = (x - a) \frac{f(x) - f(a)}{x - a}.$$

The first factor tends to zero and the second tends to $f'(a)$, so the product tends to zero by the algebra of limits. Thus $f(x) \rightarrow f(a)$. $\square$

Theorem 8.3 (Derivative rules). Let $f, g : D \rightarrow \mathbb{R}$ be differentiable at a , and let $c \in \mathbb{R}$. Then $f + g$, cf , and fg are differentiable at a , with

$$(f + g)'(a) = f'(a) + g'(a), \quad (cf)'(a) = cf'(a),$$

$$(fg)'(a) = f'(a)g(a) + f(a)g'(a).$$

If $g(a) \neq 0$, then f/g is differentiable at a and

$$\left(\frac{f}{g}\right)'(a) = \frac{f'(a)g(a) - f(a)g'(a)}{g(a)^2}.$$

Proof. Expand each difference quotient. For products, use

$$f(x)g(x) - f(a)g(a) = f(x)(g(x) - g(a)) + g(a)(f(x) - f(a)).$$

Divide by $x - a$, then use continuity of differentiable functions and the algebra of limits. Continuity and $g(a) \neq 0$ keep $g(x)$ away from zero near a . Directly,

$$\frac{1/g(x) - 1/g(a)}{x - a} = -\frac{1}{g(x)g(a)} \frac{g(x) - g(a)}{x - a} \longrightarrow -\frac{g'(a)}{g(a)^2}.$$

This establishes differentiability of $1/g$ before any rule is applied to it. The product rule applied to $f(1/g)$ now gives the quotient formula. $\square$

Example 8.4 (Exceptional points require the definition). Away from 0, the absolute-value function is given by a polynomial and has derivative 1 or -1 . At 0, however,

$$\frac{|h| - |0|}{h} = \begin{cases} 1, & h > 0, \\ -1, & h < 0, \end{cases}$$

so the one-sided derivatives disagree and $|x|$ is not differentiable there. It is nevertheless continuous. This shows both that continuity is necessary but insufficient and that a formula valid on separate pieces does not settle a junction point.

Example 8.5 (A piecewise differentiability check). Let

$$f(x) = \begin{cases} x^2 \sin(1/x), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Temporarily using the familiar prior-calculus bound $|\sin t| \leq 1$ (proved later from the power-series construction),

$$\left| \frac{f(h) - f(0)}{h} \right| = |h \sin(1/h)| \leq |h| \longrightarrow 0.$$

Thus $f'(0) = 0$. The correct workflow is to use differentiation rules where their hypotheses apply and return to the limit definition at exceptional points.

8.2 Chain Rule and Mean Value Theorems

Derivative rules explain how local linear approximations combine. The mean value theorems add a global principle: a net change across an interval is recorded by a derivative at some interior point.

Theorem 8.6 (Chain rule). Let $f : D \rightarrow \mathbb{R}$ be differentiable at a , let $g : E \rightarrow \mathbb{R}$ be differentiable at $f(a)$, and suppose $f(x) \in E$ for all $x \in D$ sufficiently near a . Then $g \circ f$ is differentiable at a , with

$$(g \circ f)'(a) = g'(f(a))f'(a).$$

Proof. Put $b = f(a)$ and define an auxiliary function on E by

$$H(y) = \begin{cases} (g(y) - g(b))/(y - b), & y \neq b, \\ g'(b), & y = b. \end{cases}$$

Differentiability of g at b says exactly that the first expression tends to the assigned value $H(b)$; hence H is continuous at b . For every x sufficiently near a , including those with $f(x) = b$,

$$g(f(x)) - g(b) = H(f(x))(f(x) - b).$$

The identity is valid in the equal-value case because both sides are zero. For $x \neq a$ divide only by $x - a$. Continuity of f at a gives $H(f(x)) \rightarrow H(b) = g'(b)$, while $(f(x) - b)/(x - a) \rightarrow f'(a)$. The product limit gives $(g \circ f)'(a) = g'(b)f'(a)$. The auxiliary function removes the division-by-zero problem before the limit is taken. $\square$

Theorem 8.7 (Rolle's theorem). Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) , where $a < b$. If $f(a) = f(b)$, then some $c \in (a, b)$ satisfies $f'(c) = 0$.

Proof. By [Theorem 5.32](#), f attains a maximum and a minimum. If both values equal $f(a)$, then f is constant and every interior point works. Otherwise an extremum occurs at an interior point c . At an interior maximum, the difference quotient is nonnegative for $x < c$ and nonpositive for $x > c$; since its two-sided limit exists, it is zero. The minimum case is identical with signs reversed. $\square$

Theorem 8.8 (Mean value theorem). Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) , where $a < b$. Then some $c \in (a, b)$ satisfies

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof. Define

$$h(x) = f(x) - f(a) - \frac{f(b) - f(a)}{b - a}(x - a).$$

Then $h(a) = h(b) = 0$. Apply [Theorem 8.7](#) to h and compute $h'(c) = 0$. $\square$

Definition 8.9 (Monotonicity). Let $I \subseteq \mathbb{R}$ be an interval and $f : I \rightarrow \mathbb{R}$. The function f is **increasing** (or **nondecreasing**) if

$$x < y \implies f(x) \leq f(y),$$

and **strictly increasing** if the final inequality is strict. It is **decreasing** (or **nonincreasing**) if

$$x < y \implies f(x) \geq f(y),$$

and **strictly decreasing** if the final inequality is strict. A function is **monotone** if it is increasing or decreasing, and **strictly monotone** if it is strictly increasing or strictly decreasing. Throughout these notes, “increasing” and “decreasing” are used in the non-strict sense; strict inequalities are indicated by “strictly increasing” and “strictly decreasing.”

Corollary 8.10 (Bounded derivative criterion for Lipschitz continuity). If f is continuous on an interval I , differentiable in its interior, and $|f'| \leq M$ there, then

$$|f(y) - f(x)| \leq M|y - x| \quad (x, y \in I).$$

Hence f is Lipschitz, and therefore uniformly continuous, on I .

Proof. Apply the Mean Value Theorem on the closed interval with endpoints x and y ; endpoint cases follow by continuity. $\square$

The ordinary Mean Value Theorem compares the net change of one function with one derivative. Cauchy's version compares the changes of two functions at once, which is exactly the structure needed later when L'Hôpital's Rule relates a quotient of function increments to a quotient of derivatives.

Theorem 8.11 (Cauchy's mean value theorem). Let $f, g : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) , where $a < b$. Then some $c \in (a, b)$ satisfies

$$(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c).$$

Proof. Apply Theorem 8.7 to

$$h(x) = (f(b) - f(a))g(x) - (g(b) - g(a))f(x).$$

The endpoint values of h agree, and $h'(c) = 0$ is exactly the displayed identity. $\square$

Infinite limits. We use the eventual-inequality definitions from Chapter 5. In particular, expressions such as ∞/∞ name an **indeterminate limiting form**; they are not arithmetic quotients of numbers called infinity. The absolute-divergence formulation permits either sign for each function.

Theorem 8.12 (L'Hôpital's rule: the $0/0$ and ∞/∞ forms). Let $f, g : (a, b) \rightarrow \mathbb{R}$ be differentiable, where $a < b$. Assume that

$$g'(x) \neq 0 \quad \text{for every } x \in (a, b),$$

and that one of the following alternatives holds:

(a) f and g extend continuously to a with $f(a) = g(a) = 0$;

(b)

$$\lim_{x \rightarrow a^+} |f(x)| = \infty \quad \text{and} \quad \lim_{x \rightarrow a^+} |g(x)| = \infty.$$

If $L \in \mathbb{R}$ and

$$\lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = L.$$

then the quotient is defined sufficiently close to a and

$$\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L.$$

Proof. First suppose that $f(a) = g(a) = 0$. If $g(x) = 0$ for some $x > a$, Rolle's theorem applied to g on $[a, x]$ would give a point where g' vanishes. Thus the quotient is defined. Fix $x \in (a, b)$. Cauchy's theorem on $[a, x]$ gives $c_x \in (a, x)$ such that

$$\frac{f(x)}{g(x)} = \frac{f'(c_x)}{g'(c_x)}.$$

As $x \rightarrow a^+$, one has $a < c_x < x$, hence $c_x \rightarrow a^+$. The assumed derivative-quotient limit therefore gives the conclusion.

Now suppose that both absolute values diverge. Let $\varepsilon > 0$. Choose $\delta > 0$ so that

$$\left| \frac{f'(t)}{g'(t)} - L \right| < \frac{\varepsilon}{4} \quad \text{whenever} \quad a < t < a + \delta,$$

and choose $c \in (a, a + \delta)$. For $x \in (a, c)$, Rolle's theorem shows that $g(x) \neq g(c)$, and Cauchy's Mean Value Theorem on $[x, c]$ gives a point $\xi_x \in (x, c)$ for which

$$\frac{f(x) - f(c)}{g(x) - g(c)} = \frac{f'(\xi_x)}{g'(\xi_x)}.$$

Thus this quotient differs from L by less than $\varepsilon/4$. Since $|g(x)| \rightarrow \infty$, for all x sufficiently close to a we also have

$$\left| \frac{g(x) - g(c)}{g(x)} \right| \leq \frac{3}{2} \quad \text{and} \quad \left| \frac{f(c) - Lg(c)}{g(x)} \right| < \frac{\varepsilon}{2}.$$

Writing the desired difference as

$$\frac{f(x)}{g(x)} - L = \frac{g(x) - g(c)}{g(x)} \left(\frac{f(x) - f(c)}{g(x) - g(c)} - L \right) + \frac{f(c) - Lg(c)}{g(x)},$$

the triangle inequality proves that its absolute value is less than ε . $\square$

Infinite derivative-quotient variants. Under the same hypotheses, if the derivative quotient tends to $+\infty$ (respectively, $-\infty$), then so does $f(x)/g(x)$. For the $0/0$ form this follows from the displayed Cauchy-Mean-Value identity in the proof. For the ∞/∞ form, use the second displayed decomposition there: the first factor tends to 1, while the second term tends to 0. Taking the derivative quotient arbitrarily large positive (respectively, negative) therefore gives the stated conclusion.

Other domains and forms. The same proof works for limits as $x \rightarrow a^-$ and as $x \rightarrow \pm\infty$, after the evident changes of domain. For example,

$$\lim_{x \rightarrow \infty} \frac{3x^2 + x}{2x^2 - 1} = \lim_{x \rightarrow \infty} \frac{6x + 1}{4x} = \frac{3}{2}.$$

Here both the numerator and denominator have absolute value tending to infinity; their signs can affect the resulting quotient, but they require no separate theorem. L'Hôpital's Rule applies directly only to quotient forms of type $0/0$ and ∞/∞ . Forms such as $0 \cdot \infty$, $\infty - \infty$, 1^∞ , 0^0 , and ∞^0 must first be transformed into a suitable quotient or logarithmic form.

Example 8.13 (Two legitimate applications). Once the elementary-function derivatives below have been established, L'Hôpital's Rule gives

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = \lim_{x \rightarrow 0} \frac{e^x}{1} = 1$$

from the $0/0$ form, and

$$\lim_{x \rightarrow +\infty} \frac{\log x}{x} = \lim_{x \rightarrow +\infty} \frac{1/x}{1} = 0$$

from the ∞/∞ form. In each calculation the equal sign relates two limits under the theorem's hypotheses; it is not the pointwise identity $f/g = f'/g'$.

Remark 8.14 (When not to apply the rule). The quotient $(x^2 + 1)/x$ tends to $+\infty$ as $x \rightarrow 0^+$, but it is not a $0/0$ or ∞/∞ form, so differentiating numerator and denominator would incorrectly suggest the limit 0. For $x^2 \sin(1/x)/x$ at 0, the derivative quotient is more complicated than the original expression, while the estimate $|x \sin(1/x)| \leq |x|$ settles the limit immediately. Finally, if f'/g' has no limit, the theorem gives no conclusion; the original quotient may still have a limit by another method. Always identify the form and verify the hypotheses before differentiating.

8.3 Definitions of Elementary Functions

Remark 8.15 (From familiar formulas to rigorous objects). Polynomials

$$a_0 + a_1x + \cdots + a_nx^n$$

are available as soon as the field operations on $\mathbb{R}$ are available, and rational functions are defined wherever their denominators are nonzero. Roots require the existence and uniqueness theorem proved from continuity and completeness in Chapter 5; rational powers then use those roots. The expressions a^x , $\log x$, $\sin x$, and $\cos x$ for arbitrary real x require more analysis. Earlier calculus treated them as familiar computational objects; here we now give definitions and prove the identities and derivative formulas previously used informally.

Power series provide one coherent construction that does not appeal to geometry, areas, or already-known calculus formulas. Their continuity, identities, periodicity, and derivatives will be theorems rather than parts of the definitions.

Proposition 8.16 (Convergence of the elementary power series). For every $x \in \mathbb{R}$, the three series

$$\sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \quad \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

converge absolutely.

Proof. For the exponential series, the ratio of consecutive absolute terms is

$$\frac{|x|}{n+1} \rightarrow 0.$$

For the sine and cosine series, the corresponding ratios are

$$\frac{|x|^2}{(2n+2)(2n+3)} \rightarrow 0, \quad \frac{|x|^2}{(2n+1)(2n+2)} \rightarrow 0.$$

The Ratio Test proves the claim in all three cases. □

Definition 8.17 (Exponential and trigonometric functions). For $x \in \mathbb{R}$, define

$$\begin{aligned} \exp(x) &= \sum_{n=0}^{\infty} \frac{x^n}{n!}, & \sin(x) &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \\ \cos(x) &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}. \end{aligned}$$

The base- e exponential function is defined by $e^x = \exp(x)$.

8.4 Differentiating Limits and Power Series

Differentiating a limit term by term is not automatic. Uniform control of the derivative series, together with one anchored value, supplies exactly the hypothesis that prevents differentiation from being an unjustified interchange of limits.

Theorem 8.18 (Differentiation of a sequence of functions). Let F_n be differentiable on $I = [\alpha, \beta]$, and suppose that $F_n(x_0)$ converges at one $x_0 \in I$ and $F'_n \rightarrow g$ uniformly on I . Then F_n converges uniformly to a function f , differentiable in the interior, with $f' = g$.

Proof. Idea. Derivatives control variation but cannot detect additive constants. Convergence at one anchor fixes those constants; uniform Cauchy control of the derivatives then controls variation away from the anchor.

Uniform Cauchyness of (F'_N) and the Mean Value Theorem imply that, for m, n sufficiently large,

$$|(F_m(x) - F_n(x)) - (F_m(x_0) - F_n(x_0))| \leq \frac{\varepsilon}{1 + \beta - \alpha} |x - x_0| < \varepsilon$$

throughout I . Since $(F_N(x_0))$ converges, this proves that (F_N) is uniformly Cauchy, hence uniformly convergent, to f .

Fix an interior point x and define, for $h \neq 0$ with $x + h \in I$,

$$Q_f(h) = \frac{f(x+h) - f(x)}{h}, \quad Q_N(h) = \frac{F_N(x+h) - F_N(x)}{h}.$$

Given $\varepsilon > 0$, first choose N so large that $|F'_m(u) - F'_N(u)| < \varepsilon/3$ for all $m \geq N, u \in I$, and also $|F'_N(x) - g(x)| < \varepsilon/3$. The Mean Value Theorem for $F_m - F_N$ gives

$$\left| \frac{F_m(x+h) - F_m(x)}{h} - Q_N(h) \right| < \varepsilon/3.$$

For each admissible h , let $m \rightarrow \infty$, using the already established convergence of the functions. We obtain $|Q_f(h) - Q_N(h)| \leq \varepsilon/3$ for every such h , with the same N .

Now keep that N fixed. Differentiability of this one function F_N at x supplies $\delta > 0$ such that $0 < |h| < \delta$ implies $|Q_N(h) - F'_N(x)| < \varepsilon/3$, and shrink δ so $x + h \in I$. Then

$$|Q_f(h) - g(x)| \leq |Q_f(h) - Q_N(h)| + |Q_N(h) - F'_N(x)| + |F'_N(x) - g(x)| < \varepsilon.$$

This directly proves $Q_f(h) \rightarrow g(x)$, so $f'(x)$ exists and equals $g(x)$. The order is essential: choose N first, freeze it, then choose h small using differentiability of that fixed F_N . $\square$

For instance $F_n(x) = n$ has derivative zero for every n , yet the functions do not converge. Derivatives cannot detect additive constants; the anchor hypothesis is indispensable. The series version is now a direct application to its partial sums.

Corollary 8.19 (Termwise differentiation). If each f_n is differentiable on I , $\sum f_n(x_0)$ converges at one point, and $\sum f'_n$ converges uniformly to g , then $\sum f_n$ converges uniformly to a differentiable sum f on the interior and $f' = g$ there.

Proof. Set $F_N = \sum_{n=0}^N f_n$. The three hypotheses and the conclusions are exactly those of [Theorem 8.18](#) for (F_N) . $\square$

Theorem 8.20 (Power-series differentiation). Let

$$f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n$$

have radius of convergence R . Then, for

$$|x-a| < R,$$

the function f is differentiable and

$$f'(x) = \sum_{n=1}^{\infty} n c_n (x - a)^{n-1}.$$

The derivative series has the same radius of convergence. Repeated termwise differentiation is valid at every point strictly inside the radius.

Proof. Fix numbers ρ, t with

$$0 < \rho < t < R.$$

The numerical series

$$\sum_{n=0}^{\infty} |c_n| t^n$$

converges. Since the sequence

$$n \left(\frac{\rho}{t} \right)^{n-1}$$

is bounded, comparison gives uniform convergence of the derivative series on

$$[a - \rho, a + \rho].$$

The original series converges at a , so [Corollary 8.19](#) gives the formula on that interval. Since every point with distance less than R from a lies in such an interval, the formula holds throughout the open convergence interval. This proves that the derivative radius R' is at least R . For the reverse inequality, if $0 < t < R'$, then

$$\sum_{n \geq 1} |c_n| t^n \leq t \sum_{n \geq 1} n |c_n| t^{n-1} < \infty.$$

Thus every such t is within the original absolute-convergence range, so $R \geq R'$. The same implication handles $R = 0$ or infinite radii. The radii are equal. Repeating the argument for the differentiated coefficients proves successive differentiation inside this common radius. $\square$

This theorem is the analytic justification for formal power-series differentiation: the formal derivative represents the derivative of the summed function only after uniform convergence of the derivative series has been verified on smaller closed intervals.

Definition 8.21 (Higher derivatives). If f' is differentiable, write

$$f'' = (f')'.$$

Inductively, $f^{(n)}$ denotes the n th derivative when it exists, with

$$f^{(0)} = f.$$

Successive derivatives provide increasingly accurate local polynomial models:

$$f(x) \approx f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \cdots.$$

Taylor's theorem makes this familiar calculus approximation rigorous. Its new content is the remainder term, which quantifies the error and foreshadows the local linearization estimates used in multivariable analysis.

Theorem 8.22 (Taylor's theorem with Lagrange remainder). Let $f : [a, b] \rightarrow \mathbb{R}$ have $n + 1$ continuous derivatives on $[a, b]$, where $n \geq 0$. For every $x \in [a, b]$ there is a number ξ between a and x such that

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k + \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-a)^{n+1}.$$

Proof. The case $x = a$ is immediate. Fix $x \neq a$ and write $P_n(t) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (t-a)^k$. Choose

$$C = \frac{f(x) - P_n(x)}{(x-a)^{n+1}}.$$

Here x is fixed; t will be the variable. The auxiliary function

$$H(t) = f(t) - \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (t-a)^k - C(t-a)^{n+1}$$

satisfies

$$H(a) = H(x) = 0.$$

By the definition of the Taylor polynomial,

$$H(a) = H'(a) = \cdots = H^{(n)}(a) = 0.$$

Rolle's theorem first gives a zero of H' between a and x . Together with $H'(a) = 0$, that gives two zeros of H' ; applying Rolle again gives a zero of H'' . Continuing in this way is legitimate because each derivative through order n is continuous on the closed interval between a and x and differentiable in its interior. After $n + 1$ applications, there is a point ξ between a and x for which

$$H^{(n+1)}(\xi) = 0.$$

Solving this identity for C gives

$$C = \frac{f^{(n+1)}(\xi)}{(n+1)!}.$$

The equation $H(x) = 0$ is the required formula. □

8.5 The Standard Elementary Functions

The preceding machinery now turns the power-series definitions into the familiar exponential, logarithmic, trigonometric, and inverse trigonometric functions. Their standard properties are consequences, not assumptions.

Theorem 8.23 (Derivatives of the power-series definitions). The functions $\exp$, $\sin$, and $\cos$ defined above satisfy

$$\exp'(x) = \exp(x), \quad \sin'(x) = \cos(x), \quad \cos'(x) = -\sin(x).$$

Proof. [Proposition 8.16](#) shows that all three series have infinite radius of convergence. [Theorem 8.20](#) now gives the three derivative formulas. □

Theorem 8.24 (Addition formulas and fundamental identities). For all $x, y \in \mathbb{R}$,

$$\exp(x + y) = \exp(x) \exp(y),$$

$$\sin(x + y) = \sin x \cos y + \cos x \sin y,$$

$$\cos(x + y) = \cos x \cos y - \sin x \sin y,$$

and

$$\sin^2 x + \cos^2 x = 1.$$

Proof. The Cauchy Product Theorem and the binomial theorem give the exponential identity. Fix y and let the difference between the two sides of the proposed sine addition formula be $h(x)$. By [Theorem 8.23](#),

$$h'(x) = \cos(x + y) + \sin x \sin y - \cos x \cos y$$

is the difference between the two sides of the proposed cosine addition formula. Differentiating the latter difference gives the negative of h . Thus the sum of the squares of these two differences is constant. At $x = 0$, both differences vanish, because

$$\sin 0 = 0, \quad \cos 0 = 1,$$

as follows from their series. Hence both differences vanish identically. Finally, the derivative of

$$\sin^2 x + \cos^2 x$$

is zero, and its value at 0 is 1; the Mean Value Theorem proves the last identity. $\square$

Theorem 8.25 (Exponential bijection). The function $\exp$ is a strictly increasing bijection from $\mathbb{R}$ onto

$$(0, \infty).$$

Proof. The exponential identity gives

$$\exp(x/2) \exp(-x/2) = \exp(0) = 1.$$

Thus $\exp(x/2) \neq 0$, and consequently

$$\exp(x) = \exp(x/2)^2 > 0.$$

Its derivative is positive, so the Mean Value Theorem makes it strictly increasing. Its series gives

$$\exp(n) \geq 1 + n \quad (n \in \mathbb{N}),$$

so monotonicity implies that $\exp(x)$ tends to infinity as $x \rightarrow \infty$. For $t > 0$, the addition law gives

$$\exp(-t) = \frac{1}{\exp(t)},$$

and hence $\exp(x)$ tends to 0 as $x \rightarrow -\infty$. If $y > 0$, choose real numbers $A < B$ such that

$$\exp(A) < y < \exp(B).$$

Continuity and the Intermediate Value Theorem then give an $x \in (A, B)$ with $\exp(x) = y$. Positivity shows that no other values occur, proving the asserted range. $\square$

Definition 8.26 (Natural logarithm). Define

$$\log : (0, \infty) \rightarrow \mathbb{R}, \quad \log = (\exp)^{-1}.$$

Theorem 8.27 (Inverse derivative theorem). Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \rightarrow \mathbb{R}$ be continuously differentiable and strictly monotone. If

$$f'(x) \neq 0 \quad (x \in I),$$

then its inverse on $f(I)$ is differentiable, with

$$(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}.$$

In particular,

$$\log'(x) = \frac{1}{x} \quad (x > 0).$$

Proof. Continuity and strict monotonicity make f^{-1} continuous on its image. For $y = f(x)$ and $z = f(u)$, the Mean Value Theorem gives a point c between x and u such that

$$\frac{f^{-1}(z) - f^{-1}(y)}{z - y} = \frac{u - x}{f(u) - f(x)} = \frac{1}{f'(c)}.$$

As $z \rightarrow y$, [Theorem 5.34](#) gives $u \rightarrow x$, hence $c \rightarrow x$; continuity of f' proves the formula. Apply it to $f = \exp$ to obtain

$$\log'(x) = \frac{1}{\exp(\log x)} = \frac{1}{x}.$$

□

Definition 8.28 (Real powers of positive numbers). For $x > 0$ and $p \in \mathbb{R}$, define

$$x^p = \exp(p \log x).$$

The definition agrees with rational powers constructed earlier. For $r = m/n$, $n > 0$, set $y = \exp((m/n) \log x) > 0$. The exponential addition law gives $y^n = \exp(m \log x) = x^m$, also for negative m by the reciprocal identity. Uniqueness of a positive n th root therefore identifies y with the earlier rational power $x^{m/n}$.

Theorem 8.29 (Derivative of a real power). For every $p \in \mathbb{R}$, the function

$$x \mapsto x^p$$

is differentiable on $(0, \infty)$ and

$$\frac{d}{dx} x^p = p x^{p-1}.$$

Proof. The chain rule and [Theorem 8.27](#) give

$$\frac{d}{dx} \exp(p \log x) = \exp(p \log x) \frac{p}{x} = p x^{p-1}.$$

□

Theorem 8.30 (Construction of π and the trigonometric ranges). There is a least positive zero c of $\cos$. Put

$$\pi = 2c.$$

Then

$$\cos c = 0, \quad \sin c = 1, \quad \cos \pi = -1, \quad \sin \pi = 0,$$

and

$$\cos(x + 2\pi) = \cos x, \quad \sin(x + 2\pi) = \sin x.$$

Moreover, $\cos x > 0$ for $-c < x < c$, and $\sin$ maps $[-c, c]$ continuously and strictly increasingly onto $[-1, 1]$.

Proof. Taylor's theorem applied to $\cos$ at 0 with degree 3 gives, for some ξ between 0 and 3,

$$\cos 3 = 1 - \frac{3^2}{2} + \frac{\cos \xi}{4!} 3^4.$$

Since the identity in [Theorem 8.24](#) gives $|\cos \xi| \leq 1$, the right side is at most

$$1 - \frac{9}{2} + \frac{81}{24} < 0.$$

Thus $\cos$ has a zero in $(0, 3)$ by the Intermediate Value Theorem. Since $\cos 0 = 1$, continuity gives a number

$$\delta > 0$$

such that

$$\cos x > 0 \quad (0 \leq x \leq \delta).$$

Consequently the set

$$Z = \{x \in [\delta, 3] : \cos x = 0\}$$

is nonempty: the zero just obtained cannot lie in $[0, \delta]$. Let

$$c = \inf Z.$$

For every $n \in \mathbb{N}$, the defining property of the infimum gives a point

$$z_n \in Z$$

with

$$c \leq z_n < c + \frac{1}{n}.$$

Thus $z_n \rightarrow c$, and continuity gives

$$\cos c = \lim_{n \rightarrow \infty} \cos z_n = 0.$$

Also $c \geq \delta > 0$. Every positive zero of cosine is at least c : a zero larger than 3 is certainly so, and a zero in $(0, 3]$ belongs to Z . Thus c is the least positive zero of cosine.

If $0 \leq x < c$ and $\cos x \leq 0$, the Intermediate Value Theorem applied on $[0, x]$ would give a zero smaller than c . Thus $\cos x > 0$ on $[0, c)$. The power-series definitions give that cosine is even and sine is odd. Hence cosine is positive on $(-c, c)$. By the Mean Value Theorem, sine is strictly increasing on $[-c, c]$. The identity

$$\sin^2 c + \cos^2 c = 1$$

and monotonicity give $\sin c = 1$. The addition formulas now give

$$\sin(2c) = 0, \quad \cos(2c) = -1,$$

and, after one further doubling,

$$\sin(4c) = 0, \quad \cos(4c) = 1.$$

The addition formulas therefore give the stated $2\pi = 4c$ periodicity. Finally, oddness and the endpoint values show that the continuous strictly increasing function sine maps $[-c, c]$ onto $[-1, 1]$. $\square$

Definition 8.31 (Tangent). On

$$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right),$$

define

$$\tan x = \frac{\sin x}{\cos x}.$$

Theorem 8.32 (Trigonometric restrictions). The function

$$\sin : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow [-1, 1]$$

is continuous, strictly increasing, and onto. The function

$$\cos : [0, \pi] \rightarrow [-1, 1]$$

is continuous, strictly decreasing, and onto. Tangent is continuous, strictly increasing, and maps

$$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

onto $\mathbb{R}$.

Proof. The sine assertion is part of [Theorem 8.30](#). For cosine on $(0, \pi)$, sine is positive: on

$$\left(0, \frac{\pi}{2}\right)$$

this follows from strict increase, and on

$$\left(\frac{\pi}{2}, \pi\right)$$

it follows from

$$\sin x = \sin(\pi - x).$$

Hence

$$\cos' = -\sin < 0,$$

so the endpoint values and the Intermediate Value Theorem give the cosine assertion.

[Theorem 8.30](#) gives positivity of cosine on the tangent domain, and hence the quotient rule gives

$$\tan'(x) = \frac{1}{\cos^2 x} > 0.$$

Thus tangent is strictly increasing. As $x \rightarrow (\pi/2)^-$, continuity gives $\sin x \rightarrow 1$ and $\cos x \rightarrow 0^+$. Given $M > 0$, choose x sufficiently close to $\pi/2$ that

$$\sin x > \frac{1}{2}, \quad 0 < \cos x < \frac{1}{2M};$$

then $\tan x > M$. Oddness gives the corresponding limit $-\infty$ at $-\pi/2$. The Intermediate Value Theorem proves that tangent is onto $\mathbb{R}$. $\square$

The identities $\cos^2 t + \sin^2 t = 1$ and $(\cos t, \sin t)' = (-\sin t, \cos t)$ put the series-defined functions back on the unit circle. The curve starts at $(1, 0)$, travels with speed one, and initially moves counterclockwise. The sign and periodicity results above describe the successive quadrants. Thus t is the usual radian parameter, providing the bridge needed for polar coordinates later.

Definition 8.33 (Inverse trigonometric functions). Define

$$\arcsin : [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], \quad \arcsin = (\sin|_{[-\pi/2, \pi/2]})^{-1},$$

$$\arccos : [-1, 1] \rightarrow [0, \pi], \quad \arccos = (\cos|_{[0, \pi]})^{-1},$$

and

$$\arctan : \mathbb{R} \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \quad \arctan = (\tan|_{(-\pi/2, \pi/2)})^{-1}.$$

Theorem 8.34 (Derivatives of the inverse trigonometric functions). The derivative formulas are

$$(\arcsin)'(x) = \frac{1}{\sqrt{1-x^2}}, \quad (\arccos)'(x) = -\frac{1}{\sqrt{1-x^2}} \quad (-1 < x < 1),$$

and

$$(\arctan)'(x) = \frac{1}{1+x^2} \quad (x \in \mathbb{R}).$$

Proof. The inverse-derivative theorem and

$$\cos(\arcsin x) > 0, \quad \sin(\arccos x) > 0$$

give the first two formulas using $\sin^2 + \cos^2 = 1$. The tangent formula follows from the inverse-derivative theorem and

$$\tan'(x) = 1 + \tan^2 x.$$

$\square$

Example 8.35 (A numerical Taylor approximation with certified error). The cubic approximation to $e^{0.1}$ is $1 + 0.1 + 0.1^2/2 + 0.1^3/6 = 1.105166666\dots$. On $[0, 0.1]$, the exponential series gives $e^x \leq \sum_{k \geq 0} 0.1^k = 10/9$. Taylor's remainder is therefore at most $(10/9)0.1^4/24 < 0.00000463$. This supplies a numerical error bound, not just a formal series expression.

There are two different limiting questions. With degree n fixed and $h \rightarrow 0$, a remainder of order h^{n+1} improves a local approximation. With the evaluation point fixed and $n \rightarrow \infty$, equality to the infinite Taylor series requires a new bound tending to zero in n .

Definition 8.36 (Analytic at a point). A function is **real analytic** at a if on some open interval about a it equals a convergent power series in $x - a$. Termwise differentiation shows that its coefficients must be $f^{(n)}(a)/n!$. Smooth means that all derivatives exist and are continuous; it does not by itself assert equality to that series.

8.6 A Smooth Cutoff and the Limits of Taylor Series

Finite Taylor approximation is a local assertion with a remainder. Equality with an infinite Taylor series additionally requires that the remainders tend to zero as the degree grows. Smoothness alone does not give this.

Theorem 8.37 (The smooth cutoff building block). The function

$$\theta(t) = \begin{cases} \exp(-1/t), & t > 0, \\ 0, & t \leq 0 \end{cases}$$

is C^∞ on $\mathbb{R}$, with $\theta^{(k)}(0) = 0$ for every $k \geq 0$.

Proof. Idea. Each differentiation creates powers of $1/t$, but exponential decay dominates every fixed power, even after the extra division by t needed for a derivative at zero. For an integer $m \geq 0$ and $u > 0$, the exponential series gives

$$e^u \geq \frac{u^{m+1}}{(m+1)!}, \quad 0 \leq u^m e^{-u} \leq \frac{(m+1)!}{u} \longrightarrow 0.$$

The first derivatives on $t > 0$ are

$$\theta'(t) = t^{-2}e^{-1/t}, \quad \theta''(t) = (t^{-4} - 2t^{-3})e^{-1/t}.$$

Each is a polynomial in $1/t$ times the same decaying exponential. On $t > 0$, induction gives $\theta^{(k)}(t) = P_k(1/t)e^{-1/t}$, where $P_0 = 1$ and $P_{k+1}(u) = u^2(P_k(u) - P'_k(u))$. Define g_k by this formula on $t > 0$ and by zero on $t \leq 0$. The estimate makes each g_k continuous at zero. Moreover $g_k(t)/t \rightarrow 0$ as $t \rightarrow 0^+$ by the same estimate with one extra power, while the quotient is zero on the negative side. Thus $g'_k(0) = 0$; away from zero $g'_k = g_{k+1}$. Inductively $\theta^{(k)} = g_k$ everywhere. $\square$

Every Taylor polynomial of θ at zero is zero, yet $\theta(t) > 0$ for $t > 0$. This smooth function is not analytic at zero. It is also useful:

$$b(x) = \theta(1 - \|x\|_2^2)$$

will be a smooth function supported in the closed unit ball once the multivariable chain rule is available. Chapter 13 uses translations and rescalings of this same building block to localize forms.

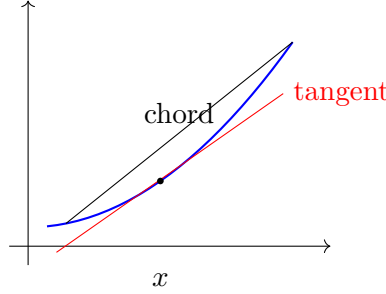
Exercise 8.1. Show that $\theta(t)/(\theta(t) + \theta(1-t))$ is smooth, equals zero for $t \leq 0$, and equals one for $t \geq 1$. Explain why its denominator never vanishes. This constructs a smooth transition between two constant values.

8.7 Convexity and a Continuous Nondifferentiable Example

Definition 8.38 (Convex function). Let I be an interval. A function $f : I \rightarrow \mathbb{R}$ is **convex** if

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y)$$

for all $x, y \in I$ and $0 \leq t \leq 1$. Geometrically, the graph lies below each of its chords.



For $x < z < y$, the chord slope across $[x, y]$ is the weighted average of the two smaller chord slopes, with weights $(z-x)/(y-x)$ and $(y-z)/(y-x)$. Convexity orders the smaller slopes; their weighted average lies between them. Passing to a tangent means taking a limit of these ordered secant slopes.

Theorem 8.39 (Derivative criteria for convexity). Let f be differentiable on an open interval I . Then f is convex if and only if f' is nondecreasing. In that case every tangent line supports the graph:

$$f(y) \geq f(x) + f'(x)(y - x) \quad (x, y \in I).$$

If f'' exists throughout I and $f'' \geq 0$, then f is convex.

Proof. For $x < z < y$, convexity at z rearranges to

$$\frac{f(z) - f(x)}{z - x} \leq \frac{f(y) - f(x)}{y - x} \leq \frac{f(y) - f(z)}{y - z}.$$

Keep x, y fixed. Let $z \downarrow x$ in the left inequality and $z \uparrow y$ in the right one to get

$$f'(x) \leq \frac{f(y) - f(x)}{y - x} \leq f'(y).$$

In particular f' is nondecreasing. The first bound gives $f(y) \geq f(x) + f'(x)(y - x)$ for $y > x$. For $y < x$, apply the second bound to the ordered pair $y < x$ and rearrange to the same tangent inequality.

Conversely suppose f' is nondecreasing. For $x < y$ and $0 < t < 1$, put $z = tx + (1 - t)y$, so $x < z < y$. The Mean Value Theorem on $[x, z]$ and $[z, y]$ supplies $\xi_1 \in (x, z)$ and $\xi_2 \in (z, y)$ with

$$\frac{f(z) - f(x)}{z - x} = f'(\xi_1) \leq f'(\xi_2) = \frac{f(y) - f(z)}{y - z}.$$

Multiply by the positive denominators and collect the $f(z)$ terms: $(y - x)f(z) \leq (y - z)f(x) + (z - x)f(y)$. Since $(y - z)/(y - x) = t$ and $(z - x)/(y - x) = 1 - t$, this is the chord inequality. The cases $t = 0, 1$ are equalities; reversing x, y covers the other order. Finally, if $f'' \geq 0$, applying the Mean Value Theorem to f' shows that f' is nondecreasing, so the result applies. $\square$

Thus x^2 , e^x , and $-\log x$ are convex on their natural domains, by their nonnegative second derivatives. The function $-x^2$ is not convex; at the midpoint of -1 and 1 its value 0 lies above the chord value -1 . Applying the tangent inequality to e^x at 0 gives the useful estimate

$$e^y \geq 1 + y.$$

Example 8.40 (A continuous trigonometric series). For $0 < a < 1$ and any positive integer b , the series

$$W(x) = \sum_{n=0}^{\infty} a^n \cos(\pi b^n x)$$

converges uniformly on $\mathbb{R}$ by the bound $|a^n \cos(\pi b^n x)| \leq a^n$. Its sum is therefore continuous. This argument establishes only continuity; it proves no assertion about differentiability of this family. The formal derivative terms have bounds $\pi(ab)^n$, which supply no convergent geometric majorant when $ab \geq 1$. Failure of this bound is not itself a proof of nondifferentiability. To separate continuity from differentiability within results proved here, recall $|x|$: its one-sided slopes at zero are -1 and 1 , so it is continuous there but not differentiable.

Exercises

Exercise 8.2. Use the Mean Value Theorem to prove that if

$$|f'(x)| \leq M \quad (x \in (a, b)),$$

where f is continuous on $[a, b]$ and differentiable on (a, b) , then

$$|f(y) - f(x)| \leq M|y - x| \quad (x, y \in [a, b]).$$

Exercise 8.3. Use [Theorem 8.20](#) to differentiate

$$\sum_{n=0}^{\infty} (x-1)^n$$

on its interval of convergence. Explain why this is an analytic theorem rather than merely a formal manipulation of symbols.

Exercise 8.4. Prove from the definitions that

$$\arcsin(-x) = -\arcsin(x) \quad (-1 \leq x \leq 1).$$

Exercise 8.5. For $F_n(x) = x + x^2/n$ on $[-1, 1]$, verify the sequence differentiation theorem including the anchor. Then explain why $F_n(x) = n + x$ fails it despite uniformly convergent derivatives.

Exercise 8.6. Use the degree-two Taylor polynomial of $\log(1+x)$ at zero to approximate $\log(1.1)$ with an explicit third-derivative error bound.

Exercise 8.7. If all derivatives of a smooth function vanish at zero, must it vanish near zero? Give a counterexample. What changes if it is analytic there?

Exercise 8.8. Use a supporting tangent line to prove $\log x \leq x - 1$ for $x > 0$. State precisely which function is convex or concave in your argument.

Chapter 9

Riemann Integration and the Fundamental Theorem of Calculus

9.1 Partitions and Riemann Sums

Integration formalizes approximation of accumulated quantities by finite sums. Darboux upper and lower sums avoid choosing sample points: they record the best guaranteed under- and over-approximations on every subinterval.

Definition 9.1 (Partition). Let $a < b$. A **partition** of $[a, b]$ is a finite ordered set

$$P = \{x_0, x_1, \dots, x_n\}$$

with $a = x_0 < x_1 < \dots < x_n = b$. Its norm is

$$\|P\| = \max_{1 \leq i \leq n} (x_i - x_{i-1}).$$

Definition 9.2 (Upper and lower sums). Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded and let P be a partition. Put

$$M_i = \sup \{f(x) : x \in [x_{i-1}, x_i]\}, \quad m_i = \inf \{f(x) : x \in [x_{i-1}, x_i]\}.$$

The **upper sum** and **lower sum** are

$$U(f, P) = \sum_{i=1}^n M_i(x_i - x_{i-1}), \quad L(f, P) = \sum_{i=1}^n m_i(x_i - x_{i-1}).$$

Definition 9.3 (Riemann integrability). Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded. It is **Riemann integrable** if

$$\inf_P U(f, P) = \sup_P L(f, P),$$

where the infimum and supremum range over all partitions of $[a, b]$. Their common value is the **Riemann integral** of f and is denoted by

$$\int_a^b f(x) \, dx.$$

Example 9.4 (All the Darboux sums for a simple function). For $f(x) = x$ on $[0, 1]$, use $x_j = j/n$. On cell j , the minimum is $(j-1)/n$, the maximum j/n , and the width $1/n$. Therefore

$$L(f, P_n) = \frac{1}{n^2} \sum_{j=1}^n (j-1) = \frac{n-1}{2n}, \quad U(f, P_n) = \frac{1}{n^2} \sum_{j=1}^n j = \frac{n+1}{2n}.$$

The gap is $1/n$ and both sums tend to $1/2$. Every tagged sum for this partition lies between them, regardless of the tags. The next criterion abstracts the shrinking-gap mechanism.

Proposition 9.5 (Darboux criterion). A bounded function $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable if and only if, for every $\varepsilon > 0$, there is a partition P with

$$U(f, P) - L(f, P) < \varepsilon.$$

Proof. We first record what refinement does to the sums. If R refines a partition P , then each interval of R lies in an interval of P . On a smaller interval the infimum of f cannot be smaller than the infimum on its parent interval, so splitting intervals can only increase the lower sum. Similarly, the supremum on a smaller interval cannot be larger than the supremum on its parent interval, so splitting intervals can only decrease the upper sum. Thus

$$L(f, P) \leq L(f, R) \leq U(f, R) \leq U(f, P),$$

where the middle inequality follows from $\inf_I f \leq \sup_I f$ on each subinterval I , after multiplying by its positive length and summing.

Now let P and Q be arbitrary partitions, and let $R = P \cup Q$ be their common refinement. The preceding observation gives

$$L(f, P) \leq L(f, R) \leq U(f, R) \leq U(f, Q).$$

Hence $L(f, P) \leq U(f, Q)$ for every pair of partitions P, Q . Therefore

$$\sup_P L(f, P) \leq U(f, Q)$$

for every Q , and taking the infimum over Q yields

$$\sup_P L(f, P) \leq \inf_Q U(f, Q).$$

Put $L^* = \sup_P L(f, P)$ and $U^* = \inf_Q U(f, Q)$. If f is integrable, then $L^* = U^* = I$. Given $\varepsilon > 0$, choose partitions P, Q such that

$$U(f, P) < I + \frac{\varepsilon}{2}, \quad L(f, Q) > I - \frac{\varepsilon}{2}.$$

For their common refinement R ,

$$L(f, R) \geq L(f, Q) > I - \frac{\varepsilon}{2}, \quad U(f, R) \leq U(f, P) < I + \frac{\varepsilon}{2}.$$

Consequently $U(f, R) - L(f, R) < \varepsilon$.

Conversely, suppose that for every $\varepsilon > 0$ there is a partition P with $U(f, P) - L(f, P) < \varepsilon$. Then

$$L(f, P) \leq L^* \leq U^* \leq U(f, P),$$

and hence

$$0 \leq U^* - L^* \leq U(f, P) - L(f, P) < \varepsilon.$$

Since this holds for every $\varepsilon > 0$, Lemma 4.5 gives $U^* - L^* = 0$. Thus $L^* = U^*$, so f is Riemann integrable. $\square$

9.2 Continuous Functions Are Integrable

Continuity prevents large oscillation on sufficiently short intervals. Uniform continuity on a compact interval makes this local control uniform, which is exactly what the Darboux criterion requires.

Theorem 9.6 (Integrability of continuous functions). Every continuous function $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable.

Proof. By Theorem 5.33, f is uniformly continuous. Given $\varepsilon > 0$, choose $\delta > 0$ such that

$$|x - y| < \delta \implies |f(x) - f(y)| < \frac{\varepsilon}{b - a}.$$

Choose a partition with norm less than δ . On every subinterval, $M_i - m_i \leq \varepsilon/(b - a)$; hence

$$U(f, P) - L(f, P) = \sum_{i=1}^n (M_i - m_i)(x_i - x_{i-1}) \leq \varepsilon.$$

The Darboux criterion proves integrability. □

Theorem 9.7 (Basic properties of the integral). If $f, g : [a, b] \rightarrow \mathbb{R}$ are Riemann integrable and $c \in \mathbb{R}$, then $f + g$ and cf are Riemann integrable, and

$$\int_a^b (f + g) \, dx = \int_a^b f \, dx + \int_a^b g \, dx, \quad \int_a^b cf \, dx = c \int_a^b f \, dx.$$

Moreover, $f \leq g$ implies that their integrals have the same order.

Proof. Fix $\varepsilon > 0$. By the Darboux criterion, choose partitions P_f and P_g such that

$$U(f, P_f) - L(f, P_f) < \frac{\varepsilon}{2}, \quad U(g, P_g) - L(g, P_g) < \frac{\varepsilon}{2}.$$

Let P be their common refinement. Refinement can only decrease upper sums and increase lower sums. On each interval I_i of P ,

$$\sup_{I_i} (f + g) \leq \sup_{I_i} f + \sup_{I_i} g, \quad \inf_{I_i} (f + g) \geq \inf_{I_i} f + \inf_{I_i} g.$$

Multiplying by the length of I_i and summing gives

$$U(f + g, P) \leq U(f, P) + U(g, P), \quad L(f + g, P) \geq L(f, P) + L(g, P)$$

and therefore

$$U(f + g, P) - L(f + g, P) \leq [U(f, P) - L(f, P)] + [U(g, P) - L(g, P)] < \varepsilon.$$

Thus $f + g$ is integrable. Write

$$I_f = \int_a^b f \, dx, \quad I_g = \int_a^b g \, dx.$$

Because P refines P_f ,

$$L(f, P) \geq L(f, P_f).$$

Moreover, $I_f = \sup_Q L(f, Q) \leq U(f, P_f)$, so

$$0 \leq I_f - L(f, P_f) \leq U(f, P_f) - L(f, P_f) < \frac{\varepsilon}{2}.$$

Also $L(f, P_f) \leq I_f$, and since P refines P_f ,

$$0 \leq U(f, P) - I_f \leq U(f, P_f) - I_f \leq U(f, P_f) - L(f, P_f) < \frac{\varepsilon}{2}.$$

Thus

$$L(f, P) > I_f - \frac{\varepsilon}{2}, \quad U(f, P) < I_f + \frac{\varepsilon}{2},$$

and the same argument, with P_g in place of P_f , gives

$$L(g, P) > I_g - \frac{\varepsilon}{2}, \quad U(g, P) < I_g + \frac{\varepsilon}{2}.$$

The lower and upper sum inequalities above now imply

$$I_f + I_g - \varepsilon < L(f + g, P) \leq \int_a^b (f + g) \, dx \leq U(f + g, P) < I_f + I_g + \varepsilon.$$

Since this holds for every $\varepsilon > 0$, [Lemma 4.5](#) gives the asserted equality.

For a scalar $c > 0$ and every interval I ,

$$\sup_I(cf) = c \sup_I f, \quad \inf_I(cf) = c \inf_I f.$$

Thus $U(cf, P) = cU(f, P)$ and $L(cf, P) = cL(f, P)$, which prove both integrability and the scalar formula by the Darboux criterion. If $c < 0$, multiplication reverses order, so

$$\sup_I(cf) = c \inf_I f, \quad \inf_I(cf) = c \sup_I f.$$

Hence $U(cf, P) = cL(f, P)$ and $L(cf, P) = cU(f, P)$, and the same conclusion follows; the case $c = 0$ is immediate. Finally, if $f \leq g$, then on every partition interval I ,

$$\sup_I f \leq \sup_I g.$$

Therefore $U(f, P) \leq U(g, P)$ for every partition P . Taking infima and using integrability gives

$$\int_a^b f \, dx \leq \int_a^b g \, dx.$$

□

Theorem 9.8 (Additivity and absolute-value estimates). Let $a < c < b$.

- (a) If $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable, then its restrictions to $[a, c]$ and $[c, b]$ are Riemann integrable and

$$\int_a^b f(x) \, dx = \int_a^c f(x) \, dx + \int_c^b f(x) \, dx.$$

- (b) If $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable, then $|f|$ is Riemann integrable and

$$\left| \int_a^b f(x) \, dx \right| \leq \int_a^b |f(x)| \, dx.$$

(c) If f is Riemann integrable and

$$|f(x)| \leq M \quad (x \in [a, b]),$$

then

$$\left| \int_a^b f(x) \, dx \right| \leq M(b-a).$$

Proof. For (a), let $\varepsilon > 0$ and choose a partition P of $[a, b]$ with $U(f, P) - L(f, P) < \varepsilon$. Refine P by inserting c ; the Darboux gap cannot increase under refinement. Let P_{left} and P_{right} be the induced partitions of $[a, c]$ and $[c, b]$. The sums split as

$$U_{[a,b]}(f, P) = U_{[a,c]}(f, P_{\text{left}}) + U_{[c,b]}(f, P_{\text{right}})$$

and

$$L_{[a,b]}(f, P) = L_{[a,c]}(f, P_{\text{left}}) + L_{[c,b]}(f, P_{\text{right}}).$$

Consequently,

$$U_{[a,b]} - L_{[a,b]} = (U_{[a,c]} - L_{[a,c]}) + (U_{[c,b]} - L_{[c,b]}).$$

Both terms on the right are nonnegative, so each is less than ε . The Darboux criterion proves that both restrictions are integrable.

Write $I = \int_a^b f$, $I_{\text{left}} = \int_a^c f$, and $I_{\text{right}} = \int_c^b f$. For every partition containing c , the split lower and upper sums give

$$L_{[a,b]}(f, P) \leq I_{\text{left}} + I_{\text{right}} \leq U_{[a,b]}(f, P).$$

Choose such a partition with $U_{[a,b]}(f, P) - L_{[a,b]}(f, P) < \varepsilon$. Since $L_{[a,b]}(f, P) \leq I \leq U_{[a,b]}(f, P)$, it follows that

$$I - \varepsilon < I_{\text{left}} + I_{\text{right}} < I + \varepsilon.$$

As this holds for every $\varepsilon > 0$, the desired additivity identity follows.

For (b), the inequality

$$||u| - |v|| \leq |u - v|$$

implies that, for x, y in a partition interval I_i ,

$$||f(x)| - |f(y)|| \leq |f(x) - f(y)|.$$

Thus the oscillation of $|f|$ on I_i satisfies

$$\sup_{I_i} |f| - \inf_{I_i} |f| \leq \sup_{I_i} f - \inf_{I_i} f,$$

because the right-hand side is the supremum of the possible differences $|f(x) - f(y)|$ for $x, y \in I_i$. Multiplying by the interval lengths and summing gives

$$U(|f|, P) - L(|f|, P) \leq U(f, P) - L(f, P).$$

The Darboux criterion proves integrability of $|f|$. Since

$$-|f| \leq f \leq |f|,$$

[Theorem 9.7](#) gives

$$-\int_a^b |f| \, dx \leq \int_a^b f \, dx \leq \int_a^b |f| \, dx,$$

which yields the stated inequality. For (c), order preservation applied to

$$0 \leq |f| \leq M.$$

gives

$$\int_a^b |f| \, dx \leq \int_a^b M \, dx = M(b-a).$$

Combine this with (b). □

Theorem 9.9 (Integrability of monotone functions). Every monotone function $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable.

Proof. It is enough to consider an increasing function; the decreasing case follows by applying the result to $-f$. For a partition P , monotonicity gives

$$M_i = f(x_i), \quad m_i = f(x_{i-1}).$$

Thus

$$U(f, P) - L(f, P) = \sum_{i=1}^n (f(x_i) - f(x_{i-1}))(x_i - x_{i-1}) \leq \|P\| \sum_{i=1}^n (f(x_i) - f(x_{i-1})).$$

Here $x_i - x_{i-1} \leq \|P\|$ and each difference $f(x_i) - f(x_{i-1})$ is nonnegative. The sum telescopes:

$$\sum_{i=1}^n (f(x_i) - f(x_{i-1})) = f(b) - f(a).$$

Therefore

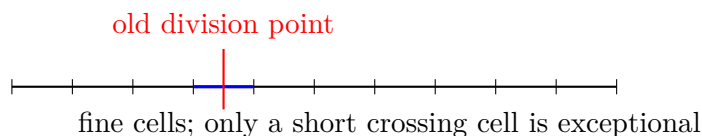
$$U(f, P) - L(f, P) \leq \|P\| (f(b) - f(a)).$$

Choose a partition with norm less than

$$\frac{\varepsilon}{f(b) - f(a)}$$

when $f(b) > f(a)$; if $f(b) = f(a)$, the function is constant and the upper and lower sums are equal. The Darboux criterion proves integrability. □

Fix a coarse partition first. A new, fine cell wholly inside one old cell inherits its oscillation bound. Only fine cells crossing old division points need the crude global bound. Each old point belongs to at most two fine cells, whose total length is at most twice the mesh.



9.2.1 Tagged sums: the calculus meaning of the integral

For a partition $P : a = x_0 < \cdots < x_m = b$, choose tags $\xi_i \in [x_{i-1}, x_i]$. Its tagged sum is $S(f, P, \xi) = \sum_i f(\xi_i)(x_i - x_{i-1})$; its mesh is $|P| = \max_i (x_i - x_{i-1})$.

Theorem 9.10 (Fine-mesh characterization). A bounded function $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable with integral I if and only if for every $\varepsilon > 0$ there is $\delta > 0$ such that every tagged partition of mesh less than δ satisfies $|S(f, P, \xi) - I| < \varepsilon$.

Proof. Idea. Start with one good Darboux partition. A different fine partition can cross its finitely many dividing points only on intervals of small total length. Suppose f is integrable and $|f| \leq M$. Choose a partition P_0 with $U(f, P_0) - L(f, P_0) < \varepsilon/2$, and let r be its number of interior dividing points. For another partition P , intervals whose interiors contain such points have total length at most $r|P|$. The remaining intervals lie in single cells of P_0 . Their oscillation contributions are at most the gap for P_0 ; the crossing intervals contribute at most $2Mr|P|$. Thus $U(f, P) - L(f, P) < \varepsilon/2 + 2Mr|P|$. Choose $\delta = \varepsilon/(4(M+1)(r+1))$. Both I and every tagged sum lie between $L(f, P)$ and $U(f, P)$, proving the assertion.

Conversely, choose a partition of mesh less than the stated δ for tolerance $\varepsilon/4$. Tags in each of its finitely many cells can approximate the supremum and infimum as closely as desired. Taking limits of such sums gives $U(f, P) \leq I + \varepsilon/4$ and $L(f, P) \geq I - \varepsilon/4$. The Darboux criterion proves integrability. Its integral differs from I by at most $\varepsilon/4$ for every ε , so it equals I . $\square$

This is why sampled values times interval widths approximate accumulated quantities: the limit is independent of all sample choices, once the mesh is sufficiently fine.

9.3 Uniform Limits and Integration

An integral is itself a limit, so a second limiting process must be handled with care. In the Riemann theory, uniform convergence is the clean condition that permits a limit to pass through an integral: one error bound works on the whole interval and hence controls the accumulated error.

Theorem 9.11 (Uniform interchange of limits and Riemann integrals). Let

$$f_n : [a, b] \longrightarrow \mathbb{R}$$

be Riemann integrable for every n , and suppose that f_n converges uniformly on $[a, b]$ to a function f . Then f is Riemann integrable and

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x) \, dx = \int_a^b f(x) \, dx.$$

In fact,

$$\lim_{n \rightarrow \infty} \int_a^b |f_n(x) - f(x)| \, dx = 0.$$

Proof. Fix $\varepsilon > 0$. Choose n such that

$$\sup_{x \in [a, b]} |f_n(x) - f(x)| < \frac{\varepsilon}{4(b-a)}.$$

Choose a partition P for which the Darboux gap of f_n is less than $\varepsilon/2$. On each subinterval of P , the oscillation of f is at most the oscillation of f_n plus twice the displayed uniform error. Consequently,

$$U(f, P) - L(f, P) < \varepsilon,$$

and f is integrable by [Proposition 9.5](#).

The basic integral estimates give

$$\left| \int_a^b f_n \, dx - \int_a^b f \, dx \right| \leq \int_a^b |f_n - f| \, dx \leq (b-a) \sup_{x \in [a,b]} |f_n(x) - f(x)|.$$

The right-hand side tends to zero. The same final estimate proves the last assertion. $\square$

9.3.1 Pointwise bounded convergence requires a different argument

Uniform convergence provides a single bound on all pointwise errors. Pointwise convergence does not: a bad location may move with n . Nevertheless bounded heights and an integrable limit suffice for integral convergence. The missing mechanism is an elementary fact about intervals. We prove it explicitly, without introducing Lebesgue integration.

Before using finite subcovers for interval unions, we establish the needed open-cover mechanism from the sequential compactness already proved.

Lemma 9.12 (Lebesgue number for a closed interval). Let $\mathcal{U}$ be an open cover of $[a, b]$. There is a number $\lambda > 0$ such that every interval contained in $[a, b]$ and having length less than λ is contained in a member of $\mathcal{U}$.

Proof. Suppose no such λ exists. For every $n \in \mathbb{N}$, choose an interval

$$J_n \subseteq [a, b]$$

of length less than $1/n$ that is contained in no member of $\mathcal{U}$, and choose $x_n \in J_n$. By [Theorem 4.23](#), some subsequence of (x_n) converges to a point $x \in [a, b]$. Choose $U \in \mathcal{U}$ containing x . Since U is open, there is $r_0 > 0$ such that

$$(x - r_0, x + r_0) \cap [a, b] \subseteq U.$$

For sufficiently large terms of the convergent subsequence, both the distance from x_n to x and the length of J_n are less than $r_0/2$. Then

$$J_n \subseteq U,$$

a contradiction. $\square$

Corollary 9.13 (Finite subcovers of closed subsets). Let $K \subseteq [a, b]$ be closed in $[a, b]$. Every cover of K by open intervals has a finite subcover.

Proof. Let $\mathcal{V}$ be an open cover of K . For every point of

$$[a, b] \setminus K,$$

there is an open interval in $\mathbb{R}$ whose intersection with K is empty and which contains that point. Let W be the union of all open intervals with this property. Then

$$\mathcal{V} \cup \{W\}$$

is an open cover of $[a, b]$. By [Lemma 9.12](#), a finite partition of $[a, b]$ having sufficiently small mesh has each of its intervals contained in one member of this cover. For every partition interval meeting K , that member cannot be W , so it belongs to $\mathcal{V}$. The finitely many members thereby selected cover K . $\square$

For an open subset G of (a, b) , define its length $\ell(G)$ as the sum of the lengths of its disjoint component intervals. These intervals are countable: each contains a different rational number. To obtain a component containing x , take the union of all open intervals containing x that lie in G ; overlapping such intervals merge into an interval. Two resulting maximal intervals are either identical or disjoint. Every finite sum of their lengths is at most $b - a$, so the full sum exists. This definition is only a tool for open interval covers.

Lemma 9.14 (Two elementary facts about open-set lengths). If an open G is contained in $\bigcup_j G_j$ with the G_j open, then $\ell(G) \leq \sum_j \ell(G_j)$. Also, for every $\eta > 0$ an open $G \subset (a, b)$ contains a finite union K of compact intervals such that $\ell(G \setminus K) < \eta$.

Proof. Strategy. Reduce comparisons to finite interval covers by compactness. The length of G is the supremum of lengths of finite unions of compact intervals contained in it: retain finitely many component intervals and shorten their endpoints slightly. Any such compact union has a finite subcover from the component intervals of all the G_j . A finite family of intervals covering disjoint compact intervals has total length at least their combined length. To verify this elementary inequality, sort all endpoints and subdivide: every covered subinterval is counted at least once in the sum of covering lengths. Thus the compact union has length at most $\sum_j \ell(G_j)$, and taking the supremum proves the first assertion. For the second, retain finitely many components whose omitted tail has length less than $\eta/2$; shorten the retained finitely many intervals by a combined length less than $\eta/2$. Their compact union is K , and the components of $G \setminus K$ consist of the discarded tail and endpoint strips. Their total length is less than η . $\square$

Lemma 9.15 (Nested open sets of uniformly positive length). If $G_n \subset (a, b)$ are decreasing open sets and $\ell(G_n) \geq c > 0$ for every n , then $\bigcap_n G_n \neq \emptyset$.

Proof. Open sets are not compact, so the ordinary nested-compact-set theorem cannot be applied directly. Choose compact finite interval unions $K_n \subset G_n$ with $\ell(G_n \setminus K_n) < c/2^{n+1}$. The approximations K_n need not be nested: shortening and retaining different intervals at each stage makes independent choices. Set $H_m = K_1 \cap \cdots \cap K_m$ instead. It is a finite union of closed bounded intervals (possibly degenerate), hence has the finite-subcover property just proved, and $H_{m+1} \subseteq H_m$. We must still show nonemptiness; the summable errors were chosen for exactly this purpose. If H_m were empty, every point of G_m would fail to belong to some K_i , $i \leq m$. Since $G_m \subset G_i$, this would give

$$G_m \subset \bigcup_{i=1}^m (G_i \setminus K_i), \quad \ell(G_m) < \sum_{i=1}^m c/2^{i+1} < c/2,$$

contradicting its lower bound. The nonempty compact sets H_m decrease inside $[a, b]$, so their intersection is nonempty. Explicitly, otherwise the open complements of the H_m would finitely cover H_1 , forcing one late H_m to be empty. A point in every H_m belongs to every G_n . $\square$

Theorem 9.16 (Arzelà bounded convergence). Let $f_n : [a, b] \rightarrow \mathbb{R}$ be Riemann integrable, $f_n \rightarrow f$ pointwise, and $|f_n(x)| \leq M$ for all n, x . If f is Riemann integrable, then

$$\int_a^b |f_n - f| \longrightarrow 0, \quad \int_a^b f_n \longrightarrow \int_a^b f.$$

Proof. Strategy. Large integrals with bounded heights force open intervals of substantial combined length on which the errors stay large. The two lemmas force one point to lie in infinitely many such sets, contradicting pointwise convergence. Let $g_n = |f_n - f|$. These are integrable by the integral algebra already proved, satisfy $0 \leq g_n \leq 2M$, and tend pointwise to zero. The cases $a = b$

or $M = 0$ are immediate. Otherwise rescale domain and height to reduce to $0 \leq g_n \leq 1$ on $[0, 1]$. If the integrals do not tend to zero, pass to a subsequence with $\int g_n > 2\varepsilon$ for a fixed $\varepsilon > 0$. For each n , integrability supplies a lower Darboux sum greater than 2ε . Cells whose lower height is at most ε contribute at most ε in total; the remaining cells therefore contribute more than ε . Since heights are at most 1, their combined lengths exceed ε . Let U_n be the union of the interiors of these remaining cells. Then $\ell(U_n) > \varepsilon$ and $g_n(x) > \varepsilon$ throughout U_n . The open sets $G_N = \bigcup_{n \geq N} U_n$ decrease and each has length greater than ε . By [Lemma 9.15](#), some x belongs to every G_N . Unpacking the union and intersection,

$$x \in \bigcap_N G_N \iff \text{for every } N \text{ there exists } n \geq N \text{ with } x \in U_n.$$

It is one fixed point x lying in arbitrarily late bad sets, so $g_n(x) > \varepsilon$ at arbitrarily large indices. Pointwise convergence at this fixed x says all sufficiently late errors are less than ε , a contradiction. The point cannot vary with N in this last step. This proves convergence of the absolute error integrals. Finally $|\int f_n - \int f| \leq \int |f_n - f|$ proves the second assertion. $\square$

Example 9.17 (The limit's integrability cannot be dropped). Enumerate the rationals in $[0, 1]$ as $q_1, q_2, \dots$ and set $f_n = \mathbf{1}_{\{q_1, \dots, q_n\}}$. A finite set is covered by finitely many intervals of arbitrarily small total length, so each f_n has integral zero. Also $0 \leq f_n \leq 1$ and f_n converges pointwise to $\mathbf{1}_{\mathbb{Q}}$. Every nondegenerate interval contains both rational and irrational points, so the limit has lower sum 0 and upper sum 1 for every partition. It is not Riemann integrable.

Uniform Riemann interchange automatically supplies an integrable limit. Arzelà's theorem permits pointwise convergence but separately assumes that integrability. Neither theorem permits arbitrary unbounded moving spikes. Bartle [\[1\]](#) is a classical companion for comparing these convergence hypotheses; the interval-cover proof above is internal.

Remark 9.18 (The stronger Lebesgue theorem). The Lebesgue dominated convergence theorem is substantially stronger. In its usual form, pointwise almost-everywhere convergence of measurable functions, together with domination by one integrable function, implies convergence of the integrals; uniform convergence is not required. This result depends on measure and Lebesgue integration and is therefore not proved or used in these notes.

Pointwise convergence is generally too weak to preserve continuity or to justify passing a limit through an integral. Dini's theorem is a striking special case in which compactness, continuity, and monotonicity cooperate to upgrade pointwise convergence to uniform convergence.

Theorem 9.19 (Dini's theorem). Let

$$f_n : [a, b] \longrightarrow \mathbb{R}$$

be continuous functions such that

$$f_n(x) \leq f_{n+1}(x) \quad (x \in [a, b], n \in \mathbb{N}),$$

and suppose that $f_n(x)$ converges pointwise to a continuous function f on $[a, b]$. Then f_n converges uniformly to f , and hence

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x) \, dx = \int_a^b f(x) \, dx.$$

The same conclusion holds for a pointwise decreasing sequence.

Proof. Consider the increasing case and put

$$h_n = f - f_n.$$

Each h_n is continuous, nonnegative, and the sequence (h_n) is pointwise decreasing to zero. If convergence were not uniform, some $\varepsilon > 0$ and points

$$x_n \in [a, b]$$

would satisfy

$$h_n(x_n) \geq \varepsilon \quad (n \in \mathbb{N}).$$

By [Theorem 5.31](#), a subsequence x_{n_j} converges to some $x \in [a, b]$. For every fixed m , eventually $n_j \geq m$, and monotonicity gives

$$h_m(x_{n_j}) \geq h_{n_j}(x_{n_j}) \geq \varepsilon.$$

Monotonicity has frozen the function index at m , while the point x_{n_j} varies. Continuity applies to this one fixed function h_m ; it cannot be applied directly to the varying functions h_{n_j} . Continuity of h_m implies $h_m(x) \geq \varepsilon$. This holds for every m , contradicting $h_m(x) \rightarrow 0$. Thus convergence is uniform, and [Theorem 9.11](#) proves the integral formula. Apply the increasing case to $-f_n$ for a decreasing sequence. $\square$

Remark 9.20 (Dini versus Lebesgue monotone convergence). Dini's theorem uses a compact interval, continuity of every f_n and of the limit, and monotone pointwise convergence to produce **uniform** convergence. The preceding integral formula follows only after applying [Theorem 9.11](#). By contrast, the Lebesgue Monotone Convergence Theorem concerns nonnegative measurable functions and gives convergence of integrals without continuity or uniform convergence. It is a measure-theoretic result and is not used in the Riemann development.

9.4 The Fundamental Theorem of Calculus

Differentiation measures local change, while integration accumulates local change. The Fundamental Theorem identifies these two operations under the continuity hypotheses that make both constructions stable.

Proposition 9.21 (Accumulation at continuity points). For Riemann-integrable f on $[a, b]$, $F(x) = \int_a^x f$ is continuous. At every interior point c where f is continuous, $F'(c) = f(c)$.

Proof. If $|f| \leq M$, additivity and the integral bound give $|F(y) - F(x)| \leq M|y - x|$, so F is continuous. For $h > 0$ with $c + h \in [a, b]$, additivity first gives the increment identity

$$\frac{F(c+h) - F(c)}{h} - f(c) = \frac{1}{h} \int_c^{c+h} (f(t) - f(c)) \, dt.$$

For $h < 0$ use only forward-endpoint integrals at this stage:

$$\frac{F(c+h) - F(c)}{h} - f(c) = \frac{1}{-h} \int_{c+h}^c (f(t) - f(c)) \, dt.$$

In either case the absolute error is at most $\sup\{|f(t) - f(c)| : |t - c| \leq |h|, t \in [a, b]\}$. Given $\varepsilon > 0$, continuity at c supplies a neighborhood on which $|f(t) - f(c)| < \varepsilon/2$. Choose $|h|$ small enough that the whole interval of integration lies there. The supremum is at most $\varepsilon/2 < \varepsilon$, so the difference quotient tends to $f(c)$. $\square$

Corollary 9.22 (Fundamental theorem of calculus, first part). Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous and define

$$F(x) = \int_a^x f(t) \, dt.$$

Then F is differentiable on (a, b) and $F'(x) = f(x)$.

Proof. Continuity makes f Riemann integrable and makes every interior point a continuity point. Apply [Proposition 9.21](#) at each such point. $\square$

Theorem 9.23 (Fundamental theorem of calculus, second part). Let $F : [a, b] \rightarrow \mathbb{R}$ be differentiable on (a, b) and continuous on $[a, b]$. Suppose that its derivative extends to a continuous function $f : [a, b] \rightarrow \mathbb{R}$, so that

$$F'(x) = f(x) \quad \text{for every } x \in (a, b).$$

Then

$$\int_a^b f(x) \, dx = F(b) - F(a).$$

Proof. Define

$$G(x) = \int_a^x f(t) \, dt.$$

By [Corollary 9.22](#), $G'(x) = f(x)$ for $x \in (a, b)$. Therefore the function $H = F - G$ is continuous on $[a, b]$, differentiable on (a, b) , and satisfies $H'(x) = 0$ throughout (a, b) . For $x = a$, the identity

$$H(x) = H(a)$$

is immediate. If $x \in (a, b]$, the Mean Value Theorem applied on $[a, x]$ gives a point $c \in (a, x)$ for which

$$H(x) - H(a) = H'(c)(x - a) = 0.$$

Thus H is constant. Since $G(a) = 0$, evaluating at b gives

$$F(b) - F(a) = G(b) = \int_a^b f(x) \, dx.$$

$\square$

Definition 9.24 (Oriented interval integrals). If f is Riemann integrable on the closed interval with endpoints r and s , retain the preceding Riemann integral when

$$r \leq s,$$

and, when

$$s < r,$$

define

$$\int_r^s f(x) \, dx = - \int_s^r f(x) \, dx.$$

In particular,

$$\int_r^r f(x) \, dx = 0.$$

Thus reversing endpoints reverses the orientation of an integral.

Theorem 9.25 (Substitution). Let

$$\phi : [\alpha, \beta] \longrightarrow \mathbb{R}$$

be of class C^1 , and let f be continuous on an open interval containing

$$\phi([\alpha, \beta]).$$

Then

$$\int_{\alpha}^{\beta} f(\phi(t))\phi'(t) \, dt = \int_{\phi(\alpha)}^{\phi(\beta)} f(u) \, du.$$

Proof. Choose c in the open interval on which f is defined and set

$$F(x) = \int_c^x f(u) \, du,$$

using [Definition 9.24](#). The first part of the Fundamental Theorem of Calculus, applied locally on closed subintervals, gives

$$F'(x) = f(x).$$

The chain rule therefore gives

$$(F \circ \phi)'(t) = f(\phi(t))\phi'(t).$$

Applying [Theorem 9.23](#) to $F \circ \phi$ yields

$$\int_{\alpha}^{\beta} f(\phi(t))\phi'(t) \, dt = F(\phi(\beta)) - F(\phi(\alpha)) = \int_{\phi(\alpha)}^{\phi(\beta)} f(u) \, du.$$

□

Remark 9.26 (Substitution, Jacobians, and pullbacks). In one variable, substitution is a short consequence of the chain rule and the Fundamental Theorem of Calculus; no injectivity of ϕ is required. In several variables, the analogous transformation factor is

$$\phi'(t) \longrightarrow \det D\Phi(x),$$

and its rigorous Riemann/Jordan treatment is the genuinely nontrivial change-of-variables theorem of Chapter 12. Chapter 13 then interprets both formulas as pullback: one-variable substitution leads to multivariable change of variables and then to the pullback of differential forms.

Theorem 9.27 (Integration by parts). Let

$$u, v : [a, b] \longrightarrow \mathbb{R}$$

be of class C^1 . Then

$$\int_a^b u(x)v'(x) \, dx = [u(x)v(x)]_{x=a}^{x=b} - \int_a^b u'(x)v(x) \, dx.$$

Proof. The product rule gives

$$(uv)' = u'v + uv'.$$

Integrating and applying [Theorem 9.23](#) to uv gives

$$u(b)v(b) - u(a)v(a) = \int_a^b u'(x)v(x) \, dx + \int_a^b u(x)v'(x) \, dx.$$

Rearrangement proves the formula. The mnemonic notation

$$\int u \, dv = uv - \int v \, du$$

records this identity but is not its formal statement. □

Example 9.28 (A short integration-by-parts computation). Taking

$$u(x) = x, \quad v(x) = \exp x$$

on $[0, 1]$ gives

$$\int_0^1 x \exp x \, dx = [x \exp x]_{x=0}^{x=1} - \int_0^1 \exp x \, dx = 1.$$

Theorem 9.29 (Mean value theorem for integrals). Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous, where $a < b$. There is a point

$$c \in [a, b]$$

such that

$$\int_a^b f(x) \, dx = f(c)(b - a).$$

Proof. By the Extreme Value Theorem, f has a minimum m and maximum M . Order preservation of the integral gives

$$m(b - a) \leq \int_a^b f(x) \, dx \leq M(b - a).$$

Dividing by $b - a > 0$ gives a number between m and M . The Intermediate Value Theorem supplies $c \in [a, b]$ at which f takes that value. $\square$

Proposition 9.30 (Newton–Leibniz with an integrable derivative). If F is continuous on $[a, b]$, differentiable on (a, b) , and F' extends to a Riemann-integrable function g on $[a, b]$, then $F(b) - F(a) = \int_a^b g$.

Proof. On each partition interval the mean value theorem supplies an interior tag ξ_i with $F(x_i) - F(x_{i-1}) = g(\xi_i)(x_i - x_{i-1})$. Summing telescopes to $F(b) - F(a)$. Let the mesh tend to zero and apply [Theorem 9.10](#). $\square$

9.5 Improper Integrals

Finite-interval Riemann integration does not directly cover infinite domains or unbounded behavior. Improper integrals retain the finite theory and add an explicit limiting process whose convergence must be established.

Definition 9.31 (Improper Riemann integral). Let $f : [a, \infty) \rightarrow \mathbb{R}$ be Riemann integrable on every finite interval $[a, b]$. The **improper integral** of f converges if

$$\int_a^\infty f(x) \, dx = \lim_{b \rightarrow \infty} \int_a^b f(x) \, dx$$

exists as a finite real number. Define an improper integral at a finite endpoint analogously by a one-sided limit.

Thus an improper integral is not a new kind of finite-interval integral: it is a limit of ordinary Riemann integrals whose existence must be proved.

At a singular left endpoint, the definition is

$$\int_a^b f(x) \, dx = \lim_{c \rightarrow a^+} \int_c^b f(x) \, dx,$$

when this limit exists finitely; a right-endpoint singularity is analogous. An interior singularity at c requires two separately convergent integrals,

$$\int_a^c f + \int_c^b f,$$

and cancellation between divergent sides is not allowed. Likewise,

$$\int_{-\infty}^{\infty} f = \int_{-\infty}^c f + \int_c^{\infty} f$$

means that both one-sided improper integrals converge, independently of the chosen finite splitting point c . Infinity is never an endpoint in $\mathbb{R}$.

Example 9.32 (p -integrals at infinity and at zero). For real p , direct evaluation of the truncated proper integrals gives

$$\int_1^{\infty} x^{-p} dx \text{ converges exactly when } p > 1,$$

whereas

$$\int_0^1 x^{-p} dx \text{ converges exactly when } p < 1.$$

In particular,

$$\int_0^1 \frac{1}{\sqrt{x}} dx = \lim_{c \rightarrow 0^+} 2(1 - \sqrt{c}) = 2,$$

while $\int_0^1 x^{-1} dx$ and $\int_1^{\infty} x^{-1} dx$ diverge. The opposite thresholds are a useful reminder to identify which endpoint causes the improper behavior.

Definition 9.33 (Absolute convergence of an improper integral). An improper integral of f is **absolutely convergent** if the corresponding improper integral of $|f|$ converges.

Proposition 9.34 (Absolute convergence implies convergence). If f is integrable on every finite truncation and its improper integral converges absolutely, then its improper integral converges.

Proof. For $a \leq c < d$, the proper-integral estimate gives

$$\left| \int_a^d f - \int_a^c f \right| = \left| \int_c^d f \right| \leq \int_c^d |f|.$$

If $\int_a^{\infty} |f|$ converges, its truncated integrals are Cauchy, so the right side is less than any prescribed $\varepsilon > 0$ whenever $d > c$ are sufficiently large. Hence the truncations of f satisfy the same Cauchy condition. To see that this gives a limit, take integer endpoints $a + n$: their integrals form a real Cauchy sequence, hence tend to L . For any sufficiently large real endpoint b , compare its integral with one sufficiently late integer endpoint. The Cauchy estimate and closeness of that integer truncation to L put it within ε of L as well. At a finite improper endpoint use a sequence of truncation points tending to that endpoint and the same tail estimate. For two improper endpoints apply the argument to each independently. This is the integral analogue of bounding a series tail by the tail of its absolute-value series. $\square$

Theorem 9.35 (Comparison for improper integrals). Let $0 \leq f \leq g$ on $[a, \infty)$, with both functions Riemann integrable on finite intervals. If the improper integral of g converges, then that of f converges.

Proof. The truncated integrals of f are increasing and bounded above by those of g . Monotone convergence supplies their finite limit. $\square$

Corollary 9.36 (Limit comparison for improper integrals). Suppose $f, g > 0$ eventually on $[a, \infty)$ and

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = L, \quad 0 < L < \infty.$$

Then their improper integrals have the same convergence behavior.

Proof. Eventually $(L/2)g \leq f \leq (3L/2)g$. Apply comparison in both directions; changing a finite initial interval does not affect convergence. $\square$

Theorem 9.37 (Integral test). Let $f : [1, \infty) \rightarrow \mathbb{R}$ be continuous, nonnegative, and decreasing, and put $a_n = f(n)$. Then

$$\sum_{n=1}^{\infty} a_n$$

converges if and only if the improper integral

$$\int_1^{\infty} f(x) \, dx$$

converges. More precisely, for every $N \in \mathbb{N}$,

$$\int_1^{N+1} f(x) \, dx \leq \sum_{n=1}^N f(n) \leq f(1) + \int_1^N f(x) \, dx.$$

Proof. For $x \in [n, n+1]$, monotonicity gives

$$f(n+1) \leq f(x) \leq f(n).$$

Integrating these inequalities over the unit intervals and summing gives the displayed bounds. The partial sums and truncated integrals are increasing, so each is bounded above if and only if the other is bounded above. Monotone convergence then proves the equivalence. $\square$

Theorem 9.38 (p -series test: analytic rederivation). For $p \in \mathbb{R}$, the series

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

converges if and only if $p > 1$.

Proof. For rational p , this conclusion was already proved without integration in [Theorem 7.19](#). We now give an analytic proof that extends the criterion to every real exponent, using the real powers constructed in [Chapter 8](#).

For $p > 0$, the function

$$f(x) = x^{-p}$$

is continuous, nonnegative, and decreasing on $[1, \infty)$ by [Theorem 8.29](#). If $p \neq 1$, that theorem gives

$$\frac{d}{dx} \left(\frac{x^{1-p}}{1-p} \right) = x^{-p};$$

if $p = 1$, [Theorem 8.27](#) gives

$$\frac{d}{dx} \log x = \frac{1}{x}.$$

Thus [Theorem 9.23](#) gives

$$\int_1^b x^{-p} dx = \begin{cases} \frac{b^{1-p} - 1}{1-p}, & p \neq 1, \\ \log b, & p = 1. \end{cases}$$

Thus the improper integral converges exactly when $p > 1$, and [Theorem 9.37](#) gives the conclusion. If $p \leq 0$, the terms do not tend to zero, so the series diverges by [Proposition 7.4](#). $\square$

Example 9.39 (Cancellation at a singularity). The two one-sided integrals of $1/x$ across zero diverge: the negative side gives $\log \delta \rightarrow -\infty$, the positive side gives $-\log \eta \rightarrow +\infty$. Choosing $\delta = \eta$ cancels them at each stage but does not define an improper integral. That linked-cutoff limit is a Cauchy principal value. Choosing $\eta = \delta^2$ instead makes the sum $-\log \delta$ diverge. Independent one-sided convergence is essential.

9.6 The Lebesgue Criterion for Riemann Integrability

Continuity is sufficient for Riemann integrability, but it is not necessary. The natural question is how large a discontinuity set Riemann integration can tolerate. Darboux sums measure oscillation on intervals; the criterion below answers the question by asking whether discontinuities fit inside interval covers of arbitrarily small total length. This will make the Dirichlet–Thomae comparison decisive: density alone does not decide integrability. We use only countable interval covers, not a formal theory of Lebesgue measure.

Definition 9.40 (Oscillation and Riemann-negligible sets). Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded. For each $x \in [a, b]$, define

$$\omega_f(x) = \inf_{\delta > 0} \sup \{ |f(y) - f(z)| : y, z \in [a, b], |y - x| < \delta, |z - x| < \delta \}.$$

A set $E \subseteq [a, b]$ is **Riemann-negligible** if, for every

$$\eta > 0,$$

there are open intervals $I_1, I_2, \dots$ such that

$$E \subseteq \bigcup_{j=1}^{\infty} I_j \quad \text{and} \quad \sum_{j=1}^{\infty} \text{length}(I_j) < \eta.$$

Finite covers are allowed as a special case.

Remark 9.41 (Terminology). The temporary term **Riemann-negligible** refers only to the preceding covering property. In measure theory, these are precisely the sets of Lebesgue measure zero. No measure is being introduced or used here; the interval-covering property is all that is needed below.

Proposition 9.42 (Oscillation and continuity). Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded. Then

$$\omega_f(x) = 0$$

if and only if f is continuous at x .

Proof. If f is continuous at x , then, for a given $\varepsilon > 0$, all values in a sufficiently small relative neighborhood of x lie within $\varepsilon/2$ of $f(x)$. Their pairwise differences are therefore less than ε , so the defining infimum is zero. Conversely, if $\omega_f(x) = 0$, then the defining infimum supplies a $\delta > 0$ for which the pairwise oscillation in the relative δ -neighborhood of x is less than ε . Taking one of the two points to be x gives

$$|f(y) - f(x)| < \varepsilon$$

whenever $|y - x| < \delta$. □

Proposition 9.43 (Closed oscillation sets). Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded and let $\varepsilon > 0$. Then

$$D_\varepsilon = \{x \in [a, b] : \omega_f(x) \geq \varepsilon\}$$

is closed in $[a, b]$.

Proof. If $\omega_f(x) < \varepsilon$, then some neighborhood of x has pairwise oscillation less than ε . If this neighborhood has radius δ , then the relative $\delta/2$ -neighborhood of every sufficiently nearby point is contained in it. Thus

$$\omega_f(y) < \varepsilon$$

for every y sufficiently near x . Hence the complement of D_ε is open. □

Proposition 9.44 (Countable sets are Riemann-negligible). Every countable subset of

$$[a, b]$$

is Riemann-negligible.

Proof. Let

$$E \subseteq \{x_1, x_2, \dots\},$$

where repetitions are allowed if E is finite. Given $\eta > 0$, cover x_n by an open interval of length

$$\frac{\eta}{2^{n+1}}.$$

These intervals cover E , and their total length is less than η . □

Lemma 9.45 (Countable unions of negligible sets). If

$$E_1, E_2, \dots \subseteq [a, b]$$

are Riemann-negligible, then

$$\bigcup_{m=1}^{\infty} E_m$$

is Riemann-negligible.

Proof. Given $\eta > 0$, cover E_m by open intervals having total length less than

$$\frac{\eta}{2^{m+1}}.$$

The combined countable family covers the displayed union and has total length less than η . □

The criterion uses two error accounts. On good intervals the oscillation is small, even if their total length is large. On bad intervals the oscillation may be large, but their total length must be small.

$$\underbrace{\sum_{\text{good}} \text{osc}_I(f)|I|}_{\leq \eta(b-a)} + \underbrace{\sum_{\text{bad}} \text{osc}_I(f)|I|}_{\leq 2M \ell_{\text{bad}}}.$$

With $|f| \leq M$, choose $\eta(b-a) < \varepsilon/2$ and $2M\ell_{\text{bad}} < \varepsilon/2$. Large oscillation is acceptable when confined to intervals of sufficiently small combined length.

good: small oscillation

bad: short total length

Theorem 9.46 (Lebesgue criterion for Riemann integrability). Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded. The following are equivalent.

(a) f is Riemann integrable.

(b) For every

$$\varepsilon > 0,$$

the set

$$D_\varepsilon = \{x \in [a, b] : \omega_f(x) \geq \varepsilon\}$$

is Riemann-negligible.

(c) The discontinuity set of f is Riemann-negligible.

Remark 9.47 (Terminology). This result is commonly called **Lebesgue's criterion for Riemann integrability**. Some sources use a name such as the **Riemann–Lebesgue criterion**; it must not be confused with the Riemann–Lebesgue lemma from Fourier analysis.

Proof. Assume (a), and let $\varepsilon, \eta > 0$. Choose a partition P satisfying

$$U(f, P) - L(f, P) < \frac{\varepsilon\eta}{4}.$$

Except for the finitely many partition points, D_ε lies in partition intervals having oscillation at least ε : otherwise a small neighborhood of such a point contained in its partition interval would give oscillation less than ε . Since

$$U(f, P) - L(f, P)$$

is the sum of each partition interval's oscillation times its length, the total length of the indicated intervals is less than $\eta/4$. Enlarge the finitely many closed partition intervals in question to open intervals whose total length is less than $\eta/2$. Cover the finitely many partition points by open intervals having total length less than $\eta/2$. These two finite families form an open cover of D_ε of total length less than η . Thus (b) holds.

Conversely, let

$$|f(x)| \leq M \quad (x \in [a, b]),$$

where $M \geq 1$, and fix $\varepsilon > 0$. Put

$$\delta = \frac{\varepsilon}{2(b-a+1)}.$$

By (b), cover D_δ by open intervals whose total length is less than

$$\frac{\varepsilon}{8M+4}.$$

By [Proposition 9.43](#) and [Corollary 9.13](#), finitely many of these intervals still cover D_δ . Let O be their union. For every

$$x \in [a, b] \setminus O,$$

one has $\omega_f(x) < \delta$, so there is an open interval V_x containing x on which the oscillation of f is less than δ . The selected intervals together with all the intervals V_x form an open cover of $[a, b]$. Apply [Lemma 9.12](#), and choose a partition of mesh less than the resulting number λ .

Classify a partition interval as bad if it is contained in one of the selected intervals covering D_δ ; classify all other partition intervals as good. A bad interval has oscillation at most $2M$, and the sum of the lengths of all bad intervals is at most the sum of the lengths of the selected intervals. Hence the bad intervals contribute less than

$$2M \frac{\varepsilon}{8M+4} < \frac{\varepsilon}{2}$$

to the Darboux gap. Every good interval lies in one of the intervals V_x and therefore has oscillation less than δ . Thus the good intervals contribute less than

$$\delta(b-a) < \frac{\varepsilon}{2}.$$

The Darboux criterion proves (a).

By [Proposition 9.42](#), the discontinuity set is

$$\bigcup_{m=1}^{\infty} D_{1/m}.$$

By [Lemma 9.45](#), (b) implies (c). Conversely,

$$D_\varepsilon$$

is contained in the discontinuity set, so (c) implies (b). □

Example 9.48 (The Dirichlet function). For

$$\mathbf{1}_\mathbb{Q} : [0, 1] \rightarrow \mathbb{R},$$

every neighborhood contains rational and irrational points. Hence

$$\omega_{\mathbf{1}_\mathbb{Q}}(x) = 1 \quad (x \in [0, 1]),$$

so

$$D_{1/2} = [0, 1]$$

is not Riemann-negligible.

Example 9.49 (Thomae's function). For $x \in [0, 1]$, define

$$T(x) = \begin{cases} 1/q, & x = p/q \text{ is rational and written in lowest terms,} \\ 0, & x \text{ is irrational.} \end{cases}$$

This is commonly called **Thomae's function**, or the **popcorn function**; some texts call it Riemann's function, a name best avoided because it is ambiguous in analysis. The function T is continuous at every irrational point and discontinuous at every rational point. Consequently,

$$\text{Disc}(T) = \mathbb{Q} \cap [0, 1],$$

T is Riemann integrable, and

$$\int_0^1 T(x) \, dx = 0.$$

Proof. Let x be irrational, and fix $\varepsilon > 0$. There are only finitely many rational numbers in $[0, 1]$ that, in lowest terms, have denominator

$$q \leq \frac{1}{\varepsilon};$$

indeed, there are only finitely many possible numerators for each such denominator. Let F_ε be this finite set. Since

$$x \notin F_\varepsilon,$$

there is a neighborhood of x avoiding F_ε . At every point y in this neighborhood, either y is irrational and $T(y) = 0$, or its reduced denominator satisfies

$$q > \frac{1}{\varepsilon},$$

so

$$T(y) = \frac{1}{q} < \varepsilon.$$

Since $T(x) = 0$, this proves that T is continuous at x .

Because $(0, 1)$ is uncountable by [Theorem 4.4](#), whereas $\mathbb{Q}$ is countable, choose an irrational number

$$\alpha \in (0, 1).$$

If

$$x = p/q$$

is rational in lowest terms and $x < 1$, choose $N \in \mathbb{N}$ such that

$$\frac{\alpha}{N} < 1 - x,$$

and set

$$y_n = x + \frac{\alpha}{N + n}.$$

Then

$$y_n \in [0, 1] \setminus \mathbb{Q} \quad \text{and} \quad \lim_{n \rightarrow \infty} y_n = x.$$

If $x = 1$, the sequence

$$y_n = 1 - \frac{\alpha}{n + 1}$$

has the same two properties. In either formula, rationality of y_n would force α to be rational, so the asserted irrationality is justified. Thus, in every case, rational x is approached by irrational points y_n with

$$\lim_{n \rightarrow \infty} T(y_n) = 0 \neq \frac{1}{q} = T(x).$$

Thus T is discontinuous at every rational point, which proves the displayed description of its discontinuity set.

The set

$$\mathbb{Q} \cap [0, 1]$$

is countable and therefore Riemann-negligible by [Proposition 9.44](#). The criterion proves that T is integrable. Finally, every nondegenerate subinterval of $[0, 1]$ contains an irrational point: it is uncountable by an affine rescaling of [Theorem 4.4](#), while its rational points form a countable set. The infimum of T on it is therefore zero. Every lower Darboux sum is zero; since T is integrable, its integral is zero. $\square$

Remark 9.50 (The Dirichlet–Thomae contrast). Both functions are defined using the dense subset $\mathbb{Q}$ of $\mathbb{R}$, but density alone does not decide Riemann integrability:

$$\text{Dirichlet:} \quad \text{Disc}(\mathbf{1}_{\mathbb{Q}}) = [0, 1] \implies \text{not Riemann integrable,}$$

$$\text{Thomae:} \quad \text{Disc}(T) = \mathbb{Q} \cap [0, 1] \implies \text{Riemann integrable.}$$

The criterion measures the size of the discontinuity set in the covering sense, not whether that set is dense.

9.7 Limits of the Riemann Theory

Riemann integration is deliberately tied to partitions of intervals and to the size of a function's oscillation on each subinterval. This makes it excellent for continuous functions and for much of elementary calculus, but it also has sharp limitations. The contrast between [Example 9.48](#) and [Example 9.49](#) identifies the next obstacle precisely: dense sets can still be negligible, so a more systematic notion of the size of a subset of the real line is beginning to be required.

The Dirichlet function is not pathological from the viewpoint of total amount: the rational numbers are sparse, yet the Riemann criterion cannot ignore them because they occur in every interval. More generally, Riemann integration is awkward for functions with discontinuities distributed throughout an interval, for limits of functions, and for integrals on complicated sets. Improper integrals also force one to treat unbounded behavior and infinite domains by separate limiting conventions.

Remark 9.51 (Riemann–Stieltjes integration). Another classical extension replaces the ordinary Riemann sums

$$\sum_i f(t_i)(x_i - x_{i-1})$$

by sums of the form

$$\sum_i f(t_i)(\alpha(x_i) - \alpha(x_{i-1})),$$

whose limiting value, when it exists, is denoted by

$$\int_a^b f \, d\alpha.$$

The ordinary Riemann integral is recovered when

$$\alpha(x) = x.$$

This Riemann–Stieltjes integral records accumulation relative to changes in another function: increasing or step-function integrators can encode weighted or discrete contributions, and the theory leads naturally to functions of bounded variation. Under suitable regularity hypotheses, it has the later integration-by-parts identity

$$\int_a^b f \, d\alpha + \int_a^b \alpha \, df = f(b)\alpha(b) - f(a)\alpha(a),$$

which generalizes [Theorem 9.27](#). No Riemann–Stieltjes theory is developed here. The measure-theoretic viewpoint later subsumes many such integrals by associating suitable measures to increasing functions; it gives the broader framework previewed again in [Section 13.13](#).

Measure theory introduces a systematic notion of size for sets that is fine enough to distinguish a sparse set from an interval. The Lebesgue integral then organizes summation by the values of a function and by the sizes of their level sets, rather than solely by partitions of the domain. This gives stronger convergence theorems and integrates many functions beyond the Riemann class. We will not define measures or the Lebesgue integral in these notes; the point here is motivational: the Lebesgue criterion already shows that Riemann integrability is controlled by the size of the discontinuity set, while the limitations above indicate why a more flexible theory is valuable.

Exercises

Exercise 9.1. Use [Theorem 9.46](#) to prove that a bounded function with finitely many discontinuities is Riemann integrable.

Exercise 9.2. Let $f : [a, b] \rightarrow \mathbb{R}$ be monotone. Show that, for each

$$\varepsilon > 0,$$

the set

$$D_\varepsilon$$

is finite.

Exercise 9.3. Prove directly from the Darboux criterion that a function having only finitely many discontinuities on

$$[a, b]$$

and otherwise constant on finitely many intervals is Riemann integrable.

Exercise 9.4. Determine whether

$$\int_1^\infty \frac{1}{x^2} \, dx$$

and

$$\int_1^\infty \frac{1}{x} \, dx$$

converge. Use [Theorem 9.23](#), then compare the conclusion with the two corresponding series.

Exercise 9.5. Use [Theorem 9.25](#) to evaluate

$$\int_0^1 2t \exp(t^2) \, dt.$$

State the substitution and identify the endpoint values.

Exercise 9.6. Use [Theorem 9.27](#) to evaluate

$$\int_0^1 x \sin x \, dx.$$

Exercise 9.7. Explain why the function

$$\mathbf{1}_{\mathbb{Q}}$$

on

$$[0, 1]$$

has the same upper and lower sums for no partition. Which density facts from earlier chapters are being used?

Exercise 9.8. For $f_n(x) = x^n$ on $[0, 1]$, apply Arzelà bounded convergence to the pointwise limit and verify the integral conclusion directly. Explain why the uniform convergence theorem does not apply on this interval.

Exercise 9.9. Construct integrable nonnegative triangular spikes on $[0, 1]$ tending pointwise to zero with integrals equal to one. Identify exactly which Arzelà hypothesis fails.

Part II

Multivariable Calculus and Analysis

Chapter 10

Necessary Topology and Linear Algebra

In one variable, $|x - y|$ measures distance, one number $f'(a)$ describes the best linear approximation, and substitution rescales a short interval. In several variables these three familiar operations separate into several ideas. A point is now a vector, and proximity must make sense without privileging one coordinate. A derivative sends an input displacement to an output displacement: it is a linear transformation. Its Jacobian matrix is the table representing that transformation in chosen coordinates, not the transformation itself. Substitution must measure the volume distortion of a small box; the absolute determinant gives its magnitude, while the sign records orientation. Finally, forms evaluate several tangent directions at once, requiring dual spaces and alternating multilinear maps.

This chapter is not trying to reproduce full courses in topology and linear algebra. It develops the parts of those subjects needed for Chapters 11–13, but develops those parts carefully enough that they do not become hidden prerequisites. Our roadmap is

distance and neighborhoods $\longrightarrow$ compactness and continuity
 $\longrightarrow$ vectors and linear maps $\longrightarrow$ matrices and coordinates
 $\longrightarrow$ inner products and norms $\longrightarrow$ multilinearity
 $\longrightarrow$ determinants and orientation.

Keep asking two questions: what is the object, and which numbers describe it? A coordinate column, a matrix, and a covector row will answer different questions even when all three are arrays of real numbers.

10.1 Distances, Balls, and Neighborhoods

On $\mathbb{R}$, distance is $|x - y|$. In the plane the Pythagorean theorem gives

$$d_2(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}.$$

In $\mathbb{R}^n$ we use

$$d_2(x, y) = \left(\sum_{j=1}^n (x_j - y_j)^2 \right)^{1/2}.$$

Which properties of distance actually enter convergence and continuity arguments? Zero distance identifies a point, reversing the two points changes nothing, and an intermediate stop cannot shorten the trip. We abstract these properties, then verify that the Euclidean formula has them.

Definition 10.1 (Metric space). A metric on X is a map $d : X \times X \rightarrow [0, \infty)$ satisfying, for all $x, y, z \in X$,

$$\begin{aligned} d(x, y) = 0 &\iff x = y && \text{(identity of indiscernibles),} \\ d(x, y) &= d(y, x) && \text{(symmetry),} \\ d(x, z) &\leq d(x, y) + d(y, z) && \text{(triangle inequality).} \end{aligned}$$

The pair (X, d) is a metric space.

Theorem 10.2 (Cauchy–Schwarz inequality). Writing $x \cdot y = \sum_j x_j y_j$ and $\|x\|_2 = \sqrt{x \cdot x}$, we have $|x \cdot y| \leq \|x\|_2 \|y\|_2$. Consequently d_2 is a metric.

Proof. If $y = 0$ the inequality reads $0 \leq 0$. Otherwise substitute $t = (x \cdot y) / \|y\|_2^2$ into a sum of squares:

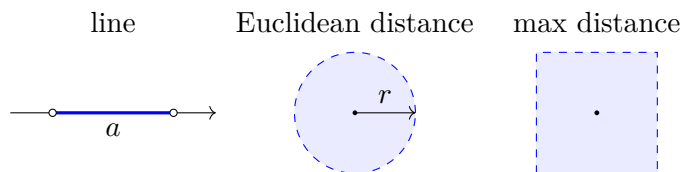
$$0 \leq \sum_j (x_j - ty_j)^2 = \|x\|_2^2 - \frac{(x \cdot y)^2}{\|y\|_2^2}.$$

Multiplication by $\|y\|_2^2$ and square roots give the assertion. For $u = x - y$ and $v = y - z$, it follows that

$$\|u + v\|_2^2 = \|u\|_2^2 + 2u \cdot v + \|v\|_2^2 \leq (\|u\|_2 + \|v\|_2)^2.$$

Taking square roots proves the triangle inequality. A sum of nonnegative squares is zero exactly when every coordinate difference is zero; changing the sign of these differences preserves their squares. These facts verify the other two metric axioms. $\square$

The points within distance $r > 0$ of a form $(a - r, a + r)$ on the line and an open disk in the plane. The limiting endpoints or circle are excluded.



Definition 10.3 (Balls and neighborhoods). The open ball of center a and radius $r > 0$ is

$$B_d(a, r) = \{x \in X : d(x, a) < r\}.$$

A neighborhood of a is any set containing some $B_d(a, r)$. The neighborhood itself need not be a ball, or even an open set.

Example 10.4 (A ball need not be round). On $\mathbb{R}^n$ set $d_\infty(x, y) = \max_j |x_j - y_j|$. Each coordinate distance is at most $d_\infty(x, z) + d_\infty(z, y)$, proving the triangle inequality after taking the maximum. The other axioms follow coordinatewise. Its balls in $\mathbb{R}^2$ are squares with sides parallel to the axes. Moreover,

$$d_\infty(x, y) \leq d_2(x, y) \leq \sqrt{n} d_\infty(x, y).$$

Thus a sufficiently small ball for either distance fits into any ball for the other. Neighborhood structure, rather than the roundness of balls, is what the topology will retain.

10.2 Open and Closed Sets in an Ambient Space

An interior point of an interval can move a little in either direction without leaving it. An interior point of a disk can move a little in every direction. In $\mathbb{R}$, the sets $(0, 1)$, $(2, \infty)$, and $(0, 1) \cup (2, 3)$ all have this property at every one of their points. For $x \in (0, 1)$, for instance, the radius $\min(x, 1-x)/2$ keeps the ball inside the interval. The set $[0, 1)$ fails this test at 0 when the ambient space is $\mathbb{R}$.

Definition 10.5 (Open and closed). A set $U \subseteq X$ is open if

$$\text{for every } x \in U \text{ there exists } r > 0 \text{ with } B_d(x, r) \subseteq U.$$

A set $F \subseteq X$ is closed if its complement $X \setminus F$ is open.

The set $[0, 1]$ is closed in $\mathbb{R}$, since its complement is $(-\infty, 0) \cup (1, \infty)$. Neither word is the negation of the other: $[0, 1]$ is neither open nor closed in $\mathbb{R}$, while $\emptyset$ and X are both. A ball is open: if $d(x, a) < r$, the positive margin $s = r - d(x, a)$ satisfies $B(x, s) \subseteq B(a, r)$ by the triangle inequality.

Proposition 10.6 (Metric open sets). The empty set and X are open. Arbitrary unions and finite intersections of open sets are open. Arbitrary intersections and finite unions of closed sets are closed.

Proof. There is no point to test in the empty set; every ball in X lies in X . If $x \in \bigcup_{\alpha} U_{\alpha}$, choose a member containing x and a ball inside that member. That same ball lies in the union. If $x \in U_1 \cap \cdots \cap U_m$, choose $r_i > 0$ with $B(x, r_i) \subseteq U_i$. The minimum of these finitely many positive radii is positive, and its ball lies in every U_i . The closed-set assertions follow by taking complements and applying De Morgan's laws. Finiteness matters: $\bigcap_{m \geq 1} (-1/m, 1/m) = \{0\}$ is not open in $\mathbb{R}$. $\square$

10.2.1 Subspaces: change the universe before testing openness

Restrict a metric on X to $Y \subseteq X$. Then $B_Y(y, r) = Y \cap B_X(y, r)$: only points allowed in the new universe remain. For $Y = [0, 1]$, a ball about 0 is $[0, r)$ when $0 < r \leq 1$. In particular $[0, 1/2)$ is open in Y : it equals $Y \cap (-1, 1/2)$. It is not open in $\mathbb{R}$, where every ball about 0 includes negative points.

Proposition 10.7 (Subspace open sets). A set $A \subseteq Y$ is open for the restricted metric if and only if $A = Y \cap U$ for some open $U \subseteq X$.

Proof. If $A = Y \cap U$, each $a \in A$ has an X -ball in U ; intersecting with Y gives a Y -ball in A . Conversely, for each $a \in A$ choose $r_a > 0$ with $B_Y(a, r_a) \subseteq A$. Put $U = \bigcup_{a \in A} B_X(a, r_a)$, which is open. Every $a \in A$ belongs to $Y \cap U$, and each point of $Y \cap U$ belongs to one of the chosen Y -balls, hence to A . Thus $Y \cap U = A$. $\square$

Closedness also depends on the ambient space: $[0, 1]$ is both open and closed in itself; $(0, 1)$ is closed in the space $(0, 1)$ but not in $\mathbb{R}$.

10.2.2 Interior, closure, and boundary

Three geometric questions give three different sets. Which points have a whole neighborhood inside E ? Which points can be approximated from E ? Which points are at the interface with the complement?

Definition 10.8 (Interior, closure, boundary). For $E \subseteq X$ define, with all balls taken in X ,

$$\begin{aligned} E^\circ &= \{x : \text{some } B(x, r) \text{ lies in } E\}, \\ \overline{E} &= \{x : \text{every } B(x, r) \text{ meets } E\}, \\ \partial E &= \{x : \text{every } B(x, r) \text{ meets both } E \text{ and } X \setminus E\}. \end{aligned}$$

These are the interior, closure, and boundary relative to X .

Proposition 10.9 (Closure characterizations). E° is the largest open subset of E , and $\overline{E}$ is the smallest closed superset. Also $\partial E = \overline{E} \setminus E^\circ$ and E is closed exactly when $\overline{E} = E$.

Proof. E° is the union of all open balls contained in E , so it is open; every open subset of E is included by the ball test. If $x \notin \overline{E}$, some ball about x misses E . Each point in that ball has a smaller ball still missing E , so $X \setminus \overline{E}$ is open. The closure contains E , because a ball about a point of E meets it at its center. If a closed F contains E and $x \notin F$, openness of $X \setminus F$ gives a ball missing E ; hence $x \notin \overline{E}$. This proves minimality. Finally, failure of the interior test means every ball meets the complement. Combining that test with the closure test proves the boundary formula. Minimality now gives the asserted criterion for closedness. $\square$

Example 10.10 (Computing all three sets). In $\mathbb{R}$, both $(0, 1)$ and $[0, 1]$ have interior $(0, 1)$, closure $[0, 1]$, and boundary $\{0, 1\}$. For $E = \{1/n : n \geq 1\}$ the interior is empty: no interval is contained in E . Its closure and boundary are both $E \cup \{0\}$. The point 0 is approached by $1/n$; any point outside this union has a small interval avoiding the finitely many nearby members and the tail near zero.

For the punctured disk $D = \{x \in \mathbb{R}^2 : 0 < \|x\|_2 < 1\}$, the interior is D , the closure is $\{x : \|x\|_2 \leq 1\}$, and the boundary is the unit circle together with the origin. If the ambient space is instead $X = \mathbb{R}^2 \setminus \{0\}$, the origin is not available: the closure is $\{x : 0 < \|x\|_2 \leq 1\}$ and the boundary is just the circle.

For $E = [0, 1/2)$ in $Y = [0, 1]$, its interior is E , its closure is $[0, 1/2]$, and its boundary is $\{1/2\}$. In $\mathbb{R}$ its interior is $(0, 1/2)$ and its boundary is $\{0, 1/2\}$. Always name the ambient space.

Definition 10.11 (Limit points and isolated points). A limit point of E is a point a such that

$$\text{for every } r > 0 \text{ there exists } x \in E \text{ with } 0 < d(x, a) < r.$$

An isolated point is $a \in E$ for which $B(a, r) \cap E = \{a\}$ for some $r > 0$. These are exactly the Chapter 5 tests with $d(x, a)$ replacing $|x - a|$. Closure allows the point itself; the limit-point test excludes it.

Thus every $1/n$ above is isolated, although it is also a boundary point. The only limit point of that set in $\mathbb{R}$ is 0.

10.3 Convergence, Completeness, and Metric Continuity

Definition 10.12 (Metric convergence). A sequence (x_n) in X converges to $x \in X$ if

$$\forall \varepsilon > 0 \exists N \forall n \geq N : d(x_n, x) < \varepsilon.$$

It is Cauchy if

$$\forall \varepsilon > 0 \exists N \forall m, n \geq N : d(x_m, x_n) < \varepsilon.$$

The space is complete if every Cauchy sequence has a limit in X .

A convergent sequence is Cauchy: put both terms within $\varepsilon/2$ of its limit and use the triangle inequality. Limits are unique: two limits would have distance at most the sum of two errors tending to zero.

Example 10.13 (Where must the limit live?). The sequence $1/n$ is Cauchy in $(0, 1]$, since $|1/m - 1/n| \leq 1/m + 1/n$, but its only possible real limit, 0, is missing. Thus $(0, 1]$ is not complete. The space $\mathbb{R}^n$ is complete: each coordinate of a Cauchy sequence is real Cauchy; the coordinate limits give a vector, and $\|x_n - x\|_2 \leq \sqrt{n} \max_j |x_{n,j} - x_j| \rightarrow 0$. In the discrete metric $d(x, y) = 1$ for $x \neq y$, a Cauchy sequence is eventually constant by the test $\varepsilon = 1/2$, so any such space is complete.

Definition 10.14 (Boundedness and diameter). $E \subseteq X$ is bounded if it is empty or lies in some ball $B(a, R)$. Its diameter is $\sup\{d(x, y) : x, y \in E\}$, possibly $+\infty$; the empty set has diameter zero. Boundedness alone gives no convergence: $(-1)^n$ is bounded but not Cauchy.

Theorem 10.15 (Sequential closure criterion). $x \in \overline{E}$ if and only if there is a sequence in E converging to x . Consequently a subset of a metric space is closed exactly when it contains every limit in X of every convergent sequence in that subset.

Proof. If $x \in \overline{E}$, each $E \cap B(x, 1/n)$ is nonempty. Choose x_n in that intersection. Then $d(x_n, x) < 1/n \rightarrow 0$. Conversely, if $x_n \in E$ and $x_n \rightarrow x$, every ball about x contains an eventual term and therefore meets E . Thus $x \in \overline{E}$. Apply [Proposition 10.9](#) to obtain the last assertion. $\square$

This gives a second way to prove closedness, alongside the open-complement test. For example limits of points satisfying $0 \leq x_n \leq 1$ still satisfy these inequalities. A closed subset of a complete metric space is complete: take the ambient limit of its Cauchy sequence and use closedness to keep that limit in the subset.

Definition 10.16 (Metric continuity). For metric spaces X, Y , a map $f : X \rightarrow Y$ is continuous at $a \in X$ if

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x \in X : d_X(x, a) < \delta \implies d_Y(f(x), f(a)) < \varepsilon.$$

It is continuous if this holds at every point of X .

The output tolerance is prescribed first; the input tolerance may depend on both it and the base point. For $f(x, y) = x + 2y$, $|f(x, y) - f(a, b)| \leq 3\|(x - a, y - b)\|_2$, so $\delta = \varepsilon/3$ works. For any metric, $|d(x, c) - d(a, c)| \leq d(x, a)$ by two applications of the triangle inequality. Hence distance to a fixed point is continuous.

Definition 10.17 (Limit of a map). If a is a limit point of $D \subseteq X$, then $\lim_{x \rightarrow a} f(x) = L$ for $f : D \rightarrow Y$ means

$$\forall \varepsilon > 0 \exists \delta > 0 : x \in D, 0 < d_X(x, a) < \delta \implies d_Y(f(x), L) < \varepsilon.$$

Example 10.18 (Paths disprove, estimates prove). The function $xy/(x^2 + y^2)$ has values $1/2$ along $y = x \neq 0$ and $-1/2$ along $y = -x \neq 0$, so has no limit at the origin. In contrast,

$$\left| \frac{x^2 y}{x^2 + y^2} \right| \leq |y| \leq \|(x, y)\|_2$$

proves a zero limit in every direction at once. Agreement along a few paths would not establish that all approaches are controlled.

Proposition 10.19 (Sequential continuity). f is continuous at a if and only if $x_n \rightarrow a$ implies $f(x_n) \rightarrow f(a)$.

Proof. For the forward direction choose δ for the prescribed ε ; eventually $d_X(x_n, a) < \delta$, so the output error is less than ε . For the converse the negation of continuity is

$$\exists \varepsilon_0 > 0 \forall \delta > 0 \exists x \in X : d_X(x, a) < \delta, \quad d_Y(f(x), f(a)) \geq \varepsilon_0.$$

One fixed ε_0 works at every scale. Set $\delta = 1/n$ and choose such an x_n . The inputs converge to a , but the output errors never fall below ε_0 , contradicting the sequential assertion. $\square$

10.4 Topological Spaces and Finite Products

The metric open-set rules allow us to retain neighborhoods without retaining a distance formula. This is useful for charts: two different coordinate descriptions can have the same open sets.

Definition 10.20 (Topological space). A topology $\mathcal{T}$ on X is a collection of subsets satisfying

$$\begin{aligned} \emptyset, X &\in \mathcal{T}, \\ U_\alpha \in \mathcal{T} \ (\alpha \in I) &\implies \bigcup_{\alpha \in I} U_\alpha \in \mathcal{T}, \\ U_1, \dots, U_m \in \mathcal{T} &\implies \bigcap_{i=1}^m U_i \in \mathcal{T}. \end{aligned}$$

Its members are called open. A topological space is a pair $(X, \mathcal{T})$. Closed means open complement; an open neighborhood is an open set containing the point. The subspace topology on $Y \subseteq X$ consists of $Y \cap U$ with U open in X .

[Propositions 10.6](#) and [10.7](#) explain why these definitions recover the metric notions. The subspace rules hold because intersections with Y commute with unions and finite intersections. In any topology define interior as the union of open subsets of E , closure as the intersection of closed supersets, and boundary as $\overline{E} \setminus E^\circ$. The proof of [Proposition 10.9](#), replacing balls by open neighborhoods, gives the same extremal and neighborhood characterizations. The sequential criterion is asserted here only for metric spaces.

Definition 10.21 (Topological continuity). A map $f : X \rightarrow Y$ is continuous if $f^{-1}(U)$ is open in X for every open $U \subseteq Y$.

Theorem 10.22 (The two definitions agree). For metric spaces this is equivalent to metric continuity at every point.

Proof. Suppose first that f is metric continuous. Given open $U \subseteq Y$, test a point $a \in f^{-1}(U)$. Openness gives $\varepsilon > 0$ with $B_Y(f(a), \varepsilon) \subseteq U$. Continuity gives $\delta > 0$ sending $B_X(a, \delta)$ into that ball. Thus $B_X(a, \delta) \subseteq f^{-1}(U)$. Every point of the inverse image passes the open-set test.

Conversely assume inverse images of open sets are open. Fix a and $\varepsilon > 0$. The open ball $B_Y(f(a), \varepsilon)$ has an open inverse image containing a , since its center belongs to it. Therefore some $B_X(a, \delta)$ lies in this inverse image. Unpacking membership gives exactly the metric continuity implication at a . $\square$

Composition is continuous: for open U , first $g^{-1}(U)$ is open and then $f^{-1}(g^{-1}(U)) = (g \circ f)^{-1}(U)$ is open.

Definition 10.23 (Homeomorphisms). A homeomorphism $h : X \rightarrow Y$ is a bijection such that h and h^{-1} are continuous. It identifies the open-set structures of the spaces.

For example $x \mapsto \tan(\pi(x - 1/2))$ gives a homeomorphism $(0, 1) \rightarrow \mathbb{R}$ with inverse $y \mapsto 1/2 + \arctan(y)/\pi$. It does not preserve lengths or boundedness.

10.4.1 Why rectangles suffice

To control a pair (x, y) , control each component: require $x \in U$ and $y \in V$. This describes the rectangle $U \times V$. Declare $G \subseteq X \times Y$ open if each of its points has a rectangle $U \times V$ containing it and contained in G , with U, V open. Equivalently, G is a union of such rectangles. This is a topology: unions retain the rectangles, and the intersection of two rectangles is $(U \cap U') \times (V \cap V')$, again a rectangle of open sets. The projections are $\pi_X(x, y) = x$ and $\pi_Y(x, y) = y$.

Proposition 10.24 (Product continuity). $f : Z \rightarrow X \times Y$ is continuous if and only if both components $\pi_X \circ f$ and $\pi_Y \circ f$ are continuous.

Proof. $\pi_X^{-1}(U) = U \times Y$ and $\pi_Y^{-1}(V) = X \times V$ are open, so the projections are continuous; composition proves one direction. Conversely write an arbitrary product-open set as $G = \bigcup_{\alpha} (U_{\alpha} \times V_{\alpha})$. Then

$$f^{-1}(G) = \bigcup_{\alpha} \left[(\pi_X \circ f)^{-1}(U_{\alpha}) \cap (\pi_Y \circ f)^{-1}(V_{\alpha}) \right].$$

Each bracket is an intersection of two open sets, and the union is open. This is why checking rectangles proves continuity for all open sets; no separate formalism about bases of topologies is needed. $\square$

For a finite product use rectangles $U_1 \times \cdots \times U_r$. The metric $d(x, y) = \max_j d_j(x_j, y_j)$ has balls equal to products of equal-radius balls. A rectangle about x contains such a ball: choose one radius for each component and take their positive minimum. Thus it induces the product topology. The Euclidean/max inequalities show this is also the Euclidean topology on $\mathbb{R}^n$. A map to $\mathbb{R}^n$ is therefore continuous exactly when each coordinate function is continuous.

10.5 Compactness: From Local Pieces to Global Control

A local estimate often comes with a neighborhood depending on its point. To obtain one global estimate we want only finitely many of those neighborhoods, so that their constants have a maximum and their positive scales have a positive minimum. Compactness solves this finite-selection problem:

$$\text{local information} \longrightarrow \text{finitely many local pieces} \longrightarrow \text{uniform control}.$$

For example, the relative intervals $U_t = [0, 1] \cap (t - 1/4, t + 1/4)$ for $t \in [0, 1]$ cover $[0, 1]$. The five with centers $0, 1/4, 1/2, 3/4, 1$ already cover it. This verifies one cover, whereas compactness makes a claim about every open cover. In contrast $(0, 1) = \bigcup_{n \geq 3} (1/n, 1 - 1/n)$ has no finite subcover: any finite selection lies in its largest member, leaving points near zero. Likewise $\mathbb{R} = \bigcup_{n \geq 1} (-n, n)$ has no finite subcover, since every finite selection is bounded.

Definition 10.25 (Compactness). An open cover of X is a family $(U_\alpha)_{\alpha \in I}$ of open sets with $X = \bigcup_\alpha U_\alpha$. A subcover retains enough of those same members to cover X . The space is compact if every open cover has a finite subcover $U_{\alpha_1}, \dots, U_{\alpha_m}$. A subset is compact in its subspace topology.

The empty space is compact (choose the empty subcover). The interval $[0, 1]$ is compact by the interval open-cover result of Chapter 9; the general Euclidean criterion will be proved below. The bounded interval $(0, 1)$ shows that boundedness is not a substitute for compactness.

Theorem 10.26 (Closed subsets of compact spaces). If X is compact and $F \subseteq X$ is closed, then F is compact.

Proof. Start with an arbitrary relatively open cover (A_α) of F . Write $A_\alpha = F \cap U_\alpha$, where U_α is open in X . The family consisting of all U_α together with $X \setminus F$ is an open cover of X : the first sets cover points of F , and the last set covers the other points. Compactness selects finitely many of them. List the selected members from the original indexed family as $U_{\alpha_1}, \dots, U_{\alpha_m}$. A point of F cannot be covered by $X \setminus F$, so belongs to one of these selected U 's. Intersecting back with F proves $F = \bigcup_{i=1}^m A_{\alpha_i}$, a finite subcover of the cover with which we began. Closedness made the adjoined set open. $\square$

Theorem 10.27 (Continuous images of compact spaces). If X is compact and $f : X \rightarrow Y$ is continuous, then $f(X)$ is compact.

Proof. Let (A_α) be a relatively open cover of $f(X)$ and write $A_\alpha = f(X) \cap U_\alpha$ with U_α open in Y . The sets $f^{-1}(U_\alpha)$ are open by continuity and cover X because every value $f(x)$ lies in some A_α . Select a finite subcover $f^{-1}(U_{\alpha_1}), \dots, f^{-1}(U_{\alpha_m})$. For any $y \in f(X)$ choose $x \in X$ with $f(x) = y$. That x lies in one selected inverse image, so $y \in U_{\alpha_i} \cap f(X) = A_{\alpha_i}$. The same selected indices give the required target subcover. $\square$

Definition 10.28 (Hausdorff space). X is Hausdorff if distinct points admit disjoint open neighborhoods. Every metric space is Hausdorff: balls of radius $d(x, y)/3$ about distinct x, y are disjoint by the triangle inequality.

Theorem 10.29 (Compact subsets are closed in Hausdorff spaces). A compact subset K of a Hausdorff space X is closed in X .

Proof. Fix $y \notin K$. For each $x \in K$, the Hausdorff hypothesis supplies open sets $x \in U_x$ and $y \in V_x$ with $U_x \cap V_x = \emptyset$. The $K \cap U_x$ cover K , so compactness supplies $x_1, \dots, x_m$ with $K \subseteq U_{x_1} \cup \dots \cup U_{x_m}$. Put

$$V = V_{x_1} \cap \dots \cap V_{x_m}.$$

It is open because the intersection is finite. It contains y because every factor does. It misses K : any $z \in K$ belongs to some U_{x_i} , whereas a point of V belongs to V_{x_i} , which is disjoint from that U_{x_i} . Thus every $y \notin K$ has an open neighborhood outside K . If K is empty use $V = X$. The complement is open, so K is closed. $\square$

This proof converts neighborhoods depending on x into one neighborhood of y . An infinite intersection of the V_x would not necessarily be open; the finite selection is the precise job of compactness.

10.6 Sequential Compactness, Finite Nets, and a Common Scale

Definition 10.30 (Sequential compactness and finite nets). A metric space is sequentially compact if every sequence has a subsequence converging to a point of that space. A finite ε -net consists of finitely many centers in the space whose open radius- ε balls cover it.

A net gives finitely many sample points. It does not yet make its balls fit inside a given cover. The next result also supplies one scale compatible with the entire cover; together the two tools produce a finite subcover.

Lemma 10.31 (Two consequences of sequential compactness). A sequentially compact metric space has a finite ε -net for every $\varepsilon > 0$. For every open cover there is $\lambda > 0$ such that each $B(x, \lambda)$ is contained in some member of that cover.

Proof. If no finite net exists, choose x_1 , then choose $x_{n+1} \notin \bigcup_{i=1}^n B(x_i, \varepsilon)$. This choice is possible precisely because the previous centers fail to be a net. Distinct selected points have distance at least ε . A convergent subsequence would be Cauchy, contradicting this separation.

If no common cover radius exists, for each n choose x_n such that $B(x_n, 1/n)$ lies in no cover member. Sequential compactness gives $x_{n_j} \rightarrow x$, where n_j increases strictly. Choose a cover member U containing x and $r > 0$ with $B(x, r) \subseteq U$. Eventually both $d(x_{n_j}, x) < r/2$ and $1/n_j < r/2$. For any $z \in B(x_{n_j}, 1/n_j)$,

$$d(z, x) \leq d(z, x_{n_j}) + d(x_{n_j}, x) < 1/n_j + r/2 < r.$$

Thus that entire ball lies in U , contradicting its selection. The empty space satisfies both assertions without choosing points. $\square$

Theorem 10.32 (Metric compactness). A metric space is compact if and only if it is sequentially compact.

Proof. For a sequence in a compact space use the closed sequence tails

$$F_N = \overline{\{x_n : n \geq N\}}.$$

They are nonempty and nested. If their intersection were empty, their open complements would cover the space. A finite subcover with indices $N_1, \dots, N_m$ would give $F_{\max N_i} = \emptyset$, a contradiction. Choose $x \in \bigcap_N F_N$. Start with $n_0 = 0$. At stage j the ball $B(x, 1/j)$ meets the tail with indices $n \geq n_{j-1} + 1$, because x belongs to its closure. Choose such an index n_j . Then $n_j > n_{j-1}$ and $d(x_{n_j}, x) < 1/j$, giving a convergent subsequence. The closures are essential: the limit need not occur among the terms.

Conversely, for a given open cover choose λ from the preceding lemma, then choose a finite $\lambda/2$ -net with centers $a_1, \dots, a_m$. For each center choose a cover member U_i containing $B(a_i, \lambda)$. Every point is within $\lambda/2$ of a center, so lies in one of these U_i . These are finitely many members of the original cover. For the empty space both properties hold directly. $\square$

Lemma 10.33 (Lebesgue number). For an open cover of a nonempty compact metric space there is $\lambda > 0$ such that every subset of diameter less than λ lies in one cover member.

Proof. Use the ball radius just proved. If E is nonempty, choose $a \in E$. For every $z \in E$, $d(z, a) \leq \text{diam}(E) < \lambda$, so $E \subseteq B(a, \lambda)$, which lies in a cover member. The empty set lies in any member; the cover has a member because the space is nonempty. $\square$

This is the scale needed when a fine grid must fit its cells into local neighborhoods. A finite net chooses points; a Lebesgue number controls pieces.

Theorem 10.34 (Compact extreme-value theorem). A continuous $f : K \rightarrow \mathbb{R}$ on a nonempty compact space attains its extrema.

Proof. By Theorems 10.27 and 10.29, the image is compact and closed in $\mathbb{R}$. Cover it by the intervals $(-m, m)$, $m \in \mathbb{N}$. Choose a finite subcover; its largest selected radius bounds the image. Let s be its supremum. For each n choose an image value $y_n > s - 1/n$, with $y_n \leq s$. Then $y_n \rightarrow s$, and closedness places s in the image. Applying the same argument to $-f$ gives its minimum. $\square$

Theorem 10.35 (Uniform continuity on compact metric spaces). A continuous map from a compact metric space to a metric space is uniformly continuous.

Proof. Failure would give one $\varepsilon_0 > 0$ and pairs x_n, y_n with $d_X(x_n, y_n) < 1/n$ but $d_Y(f(x_n), f(y_n)) \geq \varepsilon_0$. Extract $x_{n_j} \rightarrow x$ by metric compactness. Retain the same indices for the paired points, because only those pairs have the small-distance estimate:

$$d_X(y_{n_j}, x) \leq d_X(y_{n_j}, x_{n_j}) + d_X(x_{n_j}, x) < 1/n_j + d_X(x_{n_j}, x) \rightarrow 0.$$

Both image subsequences tend to $f(x)$ by continuity, so their mutual distance tends to zero by the triangle inequality, contradicting ε_0 . $\square$

Theorem 10.36 (Heine–Borel). A subset of Euclidean $\mathbb{R}^n$ is compact exactly when it is closed and bounded.

Proof. A compact subset is closed by Theorem 10.29. From the open cover $B(0, m)$ choose finitely many balls; their largest radius bounds the subset. Conversely let K be closed and bounded, and take a sequence in K . Apply the real Bolzano–Weierstrass theorem to its first coordinate. Within that subsequence extract one whose second coordinate converges; the first coordinate still converges. Continue through the finitely many coordinates. The final subsequence converges in every coordinate, hence in Euclidean distance by $\|x - y\|_2 \leq \sqrt{n} \max_i |x_i - y_i|$. Closedness puts the limit in K . Metric compactness now applies. $\square$

In particular the unit sphere is compact: it is bounded, and the reverse length estimate $|\|x\|_2 - \|y\|_2| \leq \|x - y\|_2$ makes it closed. A continuous positive function there has a positive minimum: attainment places the minimum at a point where the function is strictly positive. This observation will supply lower bounds in norm equivalence.

10.7 Connectedness and Paths

Can a space be split into two separated nonempty open pieces? The space $[0, 1] \cup [2, 3]$ can: each displayed interval is relatively open, using ambient open intervals that stop in the gap. A single interval has no such gap. A disk also has a geometric feature suggesting connectedness: one can travel along a straight segment between any two of its points without leaving it.

Definition 10.37 (Connectedness). A space is connected if it cannot be written $X = A \cup B$ with A, B disjoint, nonempty, and open in X . Such a pair is called a separation.

Definition 10.38 (Paths and convexity). A path from x to y is a continuous $\gamma : [0, 1] \rightarrow X$ with $\gamma(0) = x$ and $\gamma(1) = y$. A space is path connected if every two points can be joined by a path. A set $C \subseteq \mathbb{R}^n$ is convex if $(1 - t)x + ty \in C$ for $x, y \in C$ and $0 \leq t \leq 1$. Thus convex sets have the straight-line paths $\gamma(t) = (1 - t)x + ty$.

Proposition 10.39 (Connectedness). Continuous images of connected spaces are connected. Intervals in $\mathbb{R}$ are connected, every path-connected space is connected, and every connected subset of $\mathbb{R}$ is an interval (including the empty set and singletons).

Proof. Suppose A, B separate $f(X)$. Their inverse images under the continuous map into $f(X)$ are open in X . They cover X , are disjoint, and are both nonempty because every point of $f(X)$ has a preimage. This contradicts connectedness.

If an interval I had a separation A, B , define $h = 0$ on A and $h = 1$ on B . The relative openness of the pieces makes h locally constant, hence continuous. Choose one point from each piece. The closed interval between them lies in I , so the Intermediate Value Theorem would force the value $1/2$, impossible.

If a path-connected X had a separation, choose $x \in A, y \in B$ and a path joining them. The inverse images of A, B under the path would be a separation of $[0, 1]$: they are open, disjoint, cover, and contain $0, 1$ respectively. The preceding paragraph excludes this.

Finally, a set $E \subseteq \mathbb{R}$ that is not an interval has $a < c < b$ with $a, b \in E$ and $c \notin E$. The nonempty relatively open pieces $E \cap (-\infty, c)$ and $E \cap (c, \infty)$ separate it. $\square$

Corollary 10.40 (Constant sign). A continuous nonvanishing real function on a nonempty connected space has one constant sign.

Proof. Its connected image is an interval avoiding zero. If it contained both a negative and a positive value, the interval property would put zero in the image. Thus all its values have the same sign. $\square$

Later, continuity of a determinant, connectedness of a domain, and nonvanishing of that determinant will play these three separate roles.

10.8 Vector Spaces, Bases, and Dimension

Vectors describe directions; linear maps will describe first-order approximations. Coordinates let us calculate with these intrinsic objects. We develop the finite-dimensional real theory needed for that purpose.

10.8.1 What can be added and rescaled?

In $\mathbb{R}^2$, $(2, 1) + (1, -3) = (3, -2)$ and $2(2, 1) = (4, 2)$. In $\mathbb{R}^3$ the same operations describe displacements in three directions. But coordinate tuples are not essential: the polynomials $1 + t$ and t^2 can also be added and rescaled, as can continuous functions on $[a, b]$. Their sums remain objects of the same kind. The vector-space definition records the arithmetic laws common to these examples, without requiring that an object look like an arrow. Scalars are numbers; vectors are whatever objects the space contains. In the polynomial example, the zero vector is the zero polynomial, not a coordinate location.

Definition 10.41 (Real vector space). A real vector space is a set V with operations $+: V \times V \rightarrow V$ and $\mathbb{R} \times V \rightarrow V$, $(a, v) \mapsto av$. For all vectors u, v, w and scalars a, b , they satisfy

$$\begin{array}{ll}
 (u + v) + w = u + (v + w) & \text{(associativity of addition),} \\
 u + v = v + u & \text{(commutativity),} \\
 v + 0 = v & \text{(a zero vector exists),} \\
 v + (-v) = 0 & \text{(each vector has an additive inverse),} \\
 a(u + v) = au + av & \text{(distribution over vector addition),} \\
 (a + b)v = av + bv & \text{(distribution over scalar addition),} \\
 a(bv) = (ab)v & \text{(compatibility of scalar multiplication),} \\
 1v = v & \text{(scalar identity).}
 \end{array}$$

For another concrete example, the functions $a \cos t + b \sin t$ form a vector space under pointwise operations: addition adds the two coefficients and scalar multiplication rescales them. These are functions, not pairs; coefficients are a convenient description. More generally P_n denotes polynomials of degree at most n , including zero.

The model is $\mathbb{R}^n$ with componentwise operations. Another example is P_2 , the real polynomials of degree at most two (including the zero polynomial), with ordinary addition and scalar multiplication.

10.8.2 Subspaces

The plane $x + y + z = 0$ in $\mathbb{R}^3$ contains zero, and sums and multiples of solutions remain solutions. The parallel plane $x + y + z = 1$ lacks zero and is not a vector space under these operations. Polynomials with zero constant term form a subspace of P_n ; polynomials of degree exactly n do not, since zero is missing. The union of the two coordinate axes is not a subspace either: $(1, 0)$ and $(0, 1)$ belong to it but their sum does not.

Definition 10.42 (Subspace). A subspace U of V is a subset that is itself a vector space with the operations inherited from V .

Proposition 10.43 (Subspace criterion). A nonempty $U \subseteq V$ is a subspace exactly when $au + bv \in U$ for all $u, v \in U$ and all scalars a, b .

Proof. A subspace is closed under scaling and addition, giving the condition. Conversely choose $u \in U$. Taking $a = b = 0$ puts 0 in U ; taking $a = b = 1$ gives closure under addition, and taking $b = 0$ gives scalar closure. In particular $-u \in U$. All other axioms are identities already true in V , so they remain true for its subset under the inherited operations. $\square$

10.8.3 Linear combinations and span

Can $(3, 1)$ be produced from $u = (1, 1)$ and $v = (1, -1)$? The equation $au + bv = (3, 1)$ asks for $a + b = 3$ and $a - b = 1$, so $a = 2, b = 1$. For a general (x, y) the same equations give $a = (x + y)/2$, $b = (x - y)/2$. Thus this pair produces every vector in the plane.

Definition 10.44 (Linear combination). A linear combination of $v_1, \dots, v_r$ is a vector $a_1v_1 + \dots + a_rv_r$, where the a_i are scalars. The empty sum is zero.

Definition 10.45 (Span). For $S \subseteq V$, define

$$\text{span}(S) = \left\{ \sum_{j=1}^r a_j v_j : r \geq 0, v_j \in S, a_j \in \mathbb{R} \right\}.$$

Only finite sums are used, even if S is infinite.

Proposition 10.46 (What span constructs). $\text{span}(S)$ is a subspace, contains S , and is contained in every subspace containing S .

Proof. The empty sum gives zero. If $x = \sum_{i=1}^r a_i u_i$ and $y = \sum_{j=1}^s b_j v_j$ belong to the span, then $cx + dy = \sum_i (ca_i)u_i + \sum_j (db_j)v_j$ is another finite combination of members of S . The subspace criterion applies. Each $v \in S$ is the one-term combination $1v$, so S is contained. If U is any subspace containing S , scalar closure puts each $a_j v_j$ in U and repeated addition puts their finite sum in U . Hence the whole span lies in U . $\square$

Example 10.47 (Three kinds of span). In $\mathbb{R}^2$, $\text{span}(1, 2)$ is the line $(a, 2a)$. In $\mathbb{R}^3$, combinations of $u = (1, 0, 1)$ and $v = (0, 1, 1)$ are $(a, b, a + b)$, exactly the plane $z = x + y$. Thus $(2, -1, 1) = 2u - v$ lies in the span, while $(0, 0, 1)$ does not. In P_2 , combinations of $1 + t$ and $t + t^2$ have coefficients $(a, a + b, b)$, so $c + dt + et^2$ is in the span exactly when $d = c + e$.

Definition 10.48 (Spanning set). A set spans or generates V if its span is V . The question is whether every vector can be produced, regardless of whether there are redundant generators. A space is finite dimensional if it has a finite spanning set.

For example $(e_1, e_2, e_1 + e_2)$ spans $\mathbb{R}^2$, although the third vector adds nothing. In P_2 , $(1, t, t^2, 1 + t)$ is a redundant spanning list.

10.8.4 Independence means no redundancy

The list $u, v, u + v$ produces no more vectors than u, v . The relation $u + v - (u + v) = 0$ detects its redundancy with coefficients $1, 1, -1$. More generally a relation with some nonzero coefficient lets us solve for the corresponding vector and delete it from a generating list. The definition below says that such deletion is impossible. A basis must meet both requirements: it spans everything and contains no redundancy.

Example 10.49 (Solve the zero relation first). For $u = (1, 1)$ and $v = (1, -1)$, the relation $au + bv = 0$ gives $a + b = 0$, $a - b = 0$, hence $a = b = 0$. For the three vectors $(1, 0, 1), (0, 1, 1), (1, 1, 0)$ in $\mathbb{R}^3$, a zero relation gives

$$a + c = 0, \quad b + c = 0, \quad a + b = 0.$$

The first two imply $a = b = -c$, so the third gives $-2c = 0$; all three coefficients vanish. By contrast, replacing the third vector with $(1, 1, 2)$ gives the nontrivial relation $(1, 0, 1) + (0, 1, 1) - (1, 1, 2) = 0$. Whenever $\sum_i a_i v_i = 0$ has $a_j \neq 0$, solving gives $v_j = -\sum_{i \neq j} (a_i/a_j)v_i$. This is exactly a redundant vector.

Definition 10.50 (Linear independence). A list $(v_1, \dots, v_r)$ is linearly independent if

$$\sum_{j=1}^r a_j v_j = 0 \implies a_1 = \dots = a_r = 0.$$

Otherwise it is dependent. A set is independent if every finite list of distinct members is independent.

Definition 10.51 (Basis). A basis is an independent spanning set: it produces every vector and has no redundancy.

Spanning provides an expression for every vector; independence makes it unique, since subtracting two expressions gives a relation equal to zero. For instance $(1, t, t^2)$ is a basis of P_2 , since polynomial coefficients are unique. The empty list is a basis of the zero space.

Example 10.52 (Polynomial coordinates). The list $(1, 1+t, t^2)$ is a basis of P_2 . Indeed $a+b(1+t)+ct^2 = (a+b)+bt+ct^2$, so a zero combination has $c = b = a = 0$. Also $2+3t-t^2 = -1 \cdot 1 + 3(1+t) - t^2$. Every $c+dt+et^2$ is $(c-d) \cdot 1 + d(1+t) + et^2$, proving spanning as well. The same polynomial has columns $(2, 3, -1)^\top$ in $(1, t, t^2)$ and $(-1, 3, -1)^\top$ in $(1, 1+t, t^2)$. For contrast $(1+t, t+t^2, 1-t^2)$ is dependent because its first vector minus its second equals its third. Looking at three polynomials is not enough to conclude that they form a basis of a three-coordinate space.

10.8.5 Coordinates: existence and uniqueness

Fix an ordered basis $\mathcal{B} = (v_1, \dots, v_n)$. Spanning provides $v = \sum_j x_j v_j$. If also $v = \sum_j y_j v_j$, subtraction gives $\sum_j (x_j - y_j) v_j = 0$; independence gives $x_j = y_j$ for every j . Define $[v]_{\mathcal{B}} = (x_1, \dots, x_n)^\top$, its coordinate column. For $\mathcal{B} = ((1, 1), (1, -1))$, $[(3, 1)]_{\mathcal{B}} = (2, 1)^\top$. For $\mathcal{C} = ((1, 0, 1), (0, 1, 1), (0, 0, 1))$,

$$(x, y, z) = x(1, 0, 1) + y(0, 1, 1) + (z - x - y)(0, 0, 1),$$

so $[(2, -1, 4)]_{\mathcal{C}} = (2, -1, 3)^\top$. The polynomial example above supplies a third, non-geometric coordinate system.

Proposition 10.53 (Coordinate and synthesis maps). The maps

$$\kappa_{\mathcal{B}} : V \rightarrow \mathbb{R}^n, \quad v \mapsto [v]_{\mathcal{B}}, \quad s_{\mathcal{B}} : \mathbb{R}^n \rightarrow V, \quad x \mapsto \sum_j x_j v_j$$

preserve addition and scalar multiplication (they are linear), and are mutual inverses.

Proof. If $u = \sum_j x_j v_j$ and $v = \sum_j y_j v_j$, then $au + bv = \sum_j (ax_j + by_j) v_j$. Uniqueness of coordinates proves $\kappa_{\mathcal{B}}(au + bv) = a\kappa_{\mathcal{B}}(u) + b\kappa_{\mathcal{B}}(v)$. Distributing a finite sum proves $s_{\mathcal{B}}(ax + by) = as_{\mathcal{B}}(x) + bs_{\mathcal{B}}(y)$. For any $x \in \mathbb{R}^n$, the coefficients of $s_{\mathcal{B}}(x)$ are precisely x , so $\kappa_{\mathcal{B}} s_{\mathcal{B}}(x) = x$. For any $v \in V$, its coordinate expansion gives $s_{\mathcal{B}} \kappa_{\mathcal{B}}(v) = v$. Thus both composites are identities. $\square$

These explicit inverse maps will be called an isomorphism after linear maps are introduced formally. No dimension theorem was needed to construct them.

10.8.6 Why dimension is a theorem before it is a number

We would like to call the number of basis vectors the dimension. But that would be ambiguous if another basis could have a different size. The next three arguments remove that ambiguity. Reduction deletes redundant generators; exchange compares an independent list with a generating list; extension fills the missing directions.

Lemma 10.54 (Basis reduction). Every finite spanning set contains a basis.

Proof. Strategy. Remove one redundant vector at each step. If $a_1 v_1 + \dots + a_r v_r = 0$ and $a_j \neq 0$, then

$$v_j = -\sum_{i \neq j} \frac{a_i}{a_j} v_i.$$

In any combination of the original list, substitute this expression for v_j . The result uses only the remaining vectors, so deletion preserves span. Each deletion shortens the finite list by one. When no deletion is possible the list is independent and still spans. In the zero space the process ends with the empty list. $\square$

Example 10.55 (One replacement by hand). Insert $u = (2, 1)$ into the spanning list e_1, e_2 of $\mathbb{R}^2$. Since $u = 2e_1 + e_2$, the nonzero coefficient 2 allows us to solve $e_1 = (u - e_2)/2$. Thus u, e_2 still produces both old generators, and every new generator was already in the old span. The spans coincide. For example $(3, 4) = (3/2)u + (5/2)e_2$. In the general proof we must remove an old generator, never one of the independent vectors already inserted.

Theorem 10.56 (Replacement and exchange). If $(u_1, \dots, u_m)$ is independent and $(v_1, \dots, v_n)$ spans V , then $m \leq n$, and m members of the spanning list can be replaced by the u_i while preserving span. In particular, all finite bases of V have the same size.

Proof. Idea. Insert independent vectors one at a time, solving each time for a still-unused generator. After inserting $u_1, \dots, u_{k-1}$, write u_k in the current spanning list. Some coefficient of a remaining v_j must be nonzero, or u_k would lie in the span of its predecessors. Solving for that v_j lets us replace it by u_k while preserving span. There must be a remaining generator at every step, so $m \leq n$. Applying this inequality in both directions to two bases proves equality. More explicitly, at step k write

$$u_k = \sum_{i < k} a_i u_i + \sum_{j \in J} b_j v_j.$$

If all b_j vanished this would be a dependence of the u -list. Choose $b_{j_0} \neq 0$ and solve

$$v_{j_0} = b_{j_0}^{-1} \left(u_k - \sum_{i < k} a_i u_i - \sum_{j \in J \setminus \{j_0\}} b_j v_j \right).$$

This equation is the mechanism that preserves the old span. It never deletes an already inserted u_i , so the induction can continue. An infinite basis is impossible: it would contain an independent list of length $n + 1$. $\square$

Definition 10.57 (Dimension). The common size of the finite bases of V , now proved independent of the basis by exchange, is the dimension $\dim V$. In particular $\dim \mathbb{R}^n = n$ and $\dim P_n = n + 1$.

Lemma 10.58 (Subspaces and extensions). Every subspace of a finite-dimensional space has a finite basis. Every independent list extends to a basis.

Proof. Adjoin vectors outside the current span. The enlarged list remains independent: a relation with nonzero coefficient on the new vector would put it in the old span. Exchange bounds its length by $\dim V$, so the process stops with a spanning list. For a subspace start with the empty list and choose all new vectors in the subspace. Any independent set is already finite by exchange, so the extension assertion applies to sets too. $\square$

Example 10.59 (Extending a basis by finding a missing direction). Start in $\mathbb{R}^3$ with $u = (1, 0, 1)$ and $v = (0, 1, 1)$. Their span is $z = x + y$, so e_3 lies outside it. The formula above shows that (u, v, e_3) spans; the zero-combination equations $a = b = a + b + c = 0$ show independence. Extension keeps the original independent vectors. Reduction of $(u, v, u + v, e_3)$ instead deletes the redundant third vector. Both procedures give a basis, but answer different starting questions.

10.9 Finite Direct Sums

Sometimes a vector has several independent kinds of components. A polynomial in P_2 has a constant part and a part with zero constant term. A point in $\mathbb{R}^3$ has a component on the horizontal plane and one on the vertical axis. We need to distinguish building tuples from adding subspaces inside one space.

10.9.1 External sums: keep the components in separate positions

Definition 10.60 (External direct sum). For finitely many vector spaces $V_1, \dots, V_r$, their external direct sum is

$$V_1 \oplus \cdots \oplus V_r = \{(v_1, \dots, v_r) : v_i \in V_i\},$$

with addition and scalar multiplication performed componentwise. The injection $\iota_i : V_i \rightarrow \bigoplus_j V_j$ places a vector in slot i and zeros in the other slots; the projection $\pi_i : \bigoplus_j V_j \rightarrow V_i$ reads slot i .

The vector-space axioms hold in each component. Each injection and projection preserves both operations. Every tuple has the decomposition

$$(v_1, \dots, v_r) = \sum_i \iota_i(v_i), \quad \pi_i \iota_j = \begin{cases} \text{id}_{V_i} & i = j, \\ 0 & i \neq j. \end{cases}$$

Here the zero map when $i \neq j$ has source V_j and target V_i .

Proposition 10.61 (Basis of an external sum). If $\mathcal{B}_i$ is a basis of V_i , the vectors $\iota_i(b)$ with $b \in \mathcal{B}_i$ form a basis of the external sum. Consequently

$$\dim \bigoplus_{i=1}^r V_i = \sum_{i=1}^r \dim V_i.$$

Proof. Expand each v_i in $\mathcal{B}_i$ and insert these expansions into $(v_1, \dots, v_r) = \sum_i \iota_i(v_i)$. This proves spanning. For a zero relation among the proposed basis vectors, apply π_i . All terms from other slots vanish, leaving a zero relation in $\mathcal{B}_i$. Its coefficients are zero. Doing this for each i proves independence. Counting the basis vectors gives the dimension formula. $\square$

Proposition 10.62 (Assembling maps from the components). Given maps $L_i : V_i \rightarrow W$ preserving addition and scalar multiplication, there is a unique such map $L : \bigoplus_i V_i \rightarrow W$ with $L \iota_i = L_i$:

$$L(v_1, \dots, v_r) = \sum_i L_i(v_i).$$

Proof. The formula defines a map on tuples and preserves both operations by the corresponding identities for each L_i . On a tuple supported in slot i it gives $L_i(v_i)$, because $L_j(0) = 0$. Any other map with the prescribed restrictions must send $\sum_i \iota_i(v_i)$ to $\sum_i L_i(v_i)$ by additivity. Thus it equals L . This is the direct-sum construction and uniqueness rule. $\square$

10.9.2 Internal sums: uniqueness is an extra requirement

Definition 10.63 (Internal sum and directness). For subspaces $U_i \subseteq V$, define

$$U_1 + \cdots + U_r = \{u_1 + \cdots + u_r : u_i \in U_i\}.$$

This sum is direct if each of its vectors has exactly one such representation. We then write $U_1 \oplus \cdots \oplus U_r$ for the internal sum. Writing $V = \bigoplus_i U_i$ asserts also that the sum equals V .

The sum is a subspace by combining terms componentwise and applying the subspace criterion. The notation $\oplus$ asserts uniqueness; it does not make an arbitrary sum direct.

Theorem 10.64 (Tests for directness). The following conditions are equivalent:

- (a) The sum of the U_i is direct.
- (b) If $u_i \in U_i$ and $\sum_i u_i = 0$, then every $u_i = 0$.
- (c) For each i , $U_i \cap \sum_{j \neq i} U_j = \{0\}$.

In particular $V = U \oplus W$ exactly when $V = U + W$ and $U \cap W = \{0\}$.

Proof. Uniqueness applied to zero gives (b), since the all-zero representation already exists. If (b) holds and $\sum_i u_i = \sum_i u'_i$, then $\sum_i (u_i - u'_i) = 0$, so each difference is zero; this proves (a). For (b) $\Rightarrow$ (c), write a point in the intersection as $u_i = \sum_{j \neq i} u_j$. Move the right side to the left and use (b). For (c) $\Rightarrow$ (b), a zero sum gives $u_i = -\sum_{j \neq i} u_j$ in that intersection, so $u_i = 0$ for each i . With two subspaces, (c) is just the trivial-intersection condition. $\square$

Example 10.65 (A plane and a complementary line). Let $U = \{(x, y, z) : z = x + y\}$ and $W = \text{span}(0, 0, 1)$ in $\mathbb{R}^3$. The defining equation is preserved by linear combinations, so U is a subspace; W is a subspace by its span description. For every vector,

$$(x, y, z) = (x, y, x + y) + (0, 0, z - x - y).$$

The first term satisfies the equation of U , and the second is a multiple of $(0, 0, 1)$. This proves existence. For uniqueness, any decomposition $(x, y, z) = (a, b, a + b) + (0, 0, c)$ forces $a = x$, $b = y$, and $c = z - x - y$. Thus $\mathbb{R}^3 = U \oplus W$.

The component maps are the linear maps

$$P_U(x, y, z) = (x, y, x + y), \quad P_W(x, y, z) = (0, 0, z - x - y).$$

Their sum is the identity. The external summation map and its proposed inverse are

$$\Sigma : U \oplus_{\text{ext}} W \rightarrow \mathbb{R}^3, \quad (u, w) \mapsto u + w, \quad K(v) = (P_U v, P_W v).$$

Both maps are linear by their formulas. We have $\Sigma K(v) = v$. Conversely, for $u = (a, b, a + b)$ and $w = (0, 0, c)$,

$$K\Sigma(u, w) = K(a, b, a + b + c) = ((a, b, a + b), (0, 0, c)) = (u, w).$$

These are the two inverse identities. Directness means uniqueness of components. It imposes no condition on dot products: $(1, 0, 1) \in U$ has dot product 1 with $(0, 0, 1) \in W$.

Example 10.66 (Pairwise intersections do not suffice). In $\mathbb{R}^2$ let $U_1 = \text{span}(e_1)$, $U_2 = \text{span}(e_2)$, and $U_3 = \text{span}(e_1 + e_2)$. Any two meet only at zero. Nevertheless $e_1 + e_2 - (e_1 + e_2) = 0$ is a nonzero relation with one term in each space. Thus the sum of three subspaces is not direct. The correct test compares one subspace with the sum of all the others, not with each separately.

The summation map connects the two constructions:

$$\Sigma : U_1 \oplus_{\text{ext}} \cdots \oplus_{\text{ext}} U_r \rightarrow V, \quad \Sigma(u_1, \dots, u_r) = \sum_i u_i.$$

It preserves addition and scalar multiplication, reaches exactly the internal sum, and is injective exactly when the representation is unique. When the sum is all of V and direct, its reverse map sends a vector to its unique component tuple. Summing that tuple returns the vector, and decomposing a sum returns the original tuple. Thus both composites are identities.

Proposition 10.67 (Dimension of a sum). For finite-dimensional subspaces,

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

Proof. Choose a basis $(z_1, \dots, z_k)$ of $U \cap W$, extend it by $u_1, \dots, u_a$ to a basis of U , and by $w_1, \dots, w_b$ to a basis of W . The combined list (z_i, u_j, w_ℓ) spans $U + W$. If $\sum c_i z_i + \sum d_j u_j + \sum e_\ell w_\ell = 0$, then $\sum d_j u_j = -\sum c_i z_i - \sum e_\ell w_\ell$ lies in $U \cap W$. Express it in the z_i and use independence of the basis of U to get all $d_j = 0$. The remaining relation is in the basis of W , so all c_i, e_ℓ vanish. The combined list is a basis. Its size is $k + a + b = (k + a) + (k + b) - k$, proving the formula. $\square$

10.10 Linear Transformations, Kernels, and Images

A linear map preserves the two operations that define a vector space. The map $(x, y) \mapsto (x, 0)$ projects the plane onto its horizontal axis; $(x, y, z) \mapsto (x - y, y - z)$ records two differences; and differentiation $p \mapsto p'$ maps P_2 to P_1 . Each respects sums and scalar multiples. In contrast a translation $v \mapsto v + c$ with $c \neq 0$ cannot be linear, since a linear map sends zero to zero. The map $x \mapsto x^2$ sends zero to zero but fails additivity: $(1 + 1)^2 \neq 1^2 + 1^2$. Thus the zero test is a useful obstruction, not a sufficient test.

Definition 10.68 (Linear transformation). A map $T : V \rightarrow W$ is linear if

$$\begin{aligned} T(u + v) &= T(u) + T(v) && \text{(addition),} \\ T(av) &= aT(v) && \text{(scalar multiplication).} \end{aligned}$$

Equivalently $T(au + bv) = aT(u) + bT(v)$ for all vectors and scalars.

Indeed the two separate rules give the combined one; in the reverse direction take $a = b = 1$ for addition and $b = 0$ for scalar multiplication. For a linear map $T(0) = T(0 + 0) = T(0) + T(0)$, so subtraction gives $T(0) = 0$. Also $0 = T(v + (-v)) = T(v) + T(-v)$, giving $T(-v) = -T(v)$.

Theorem 10.69 (Constructing a linear map from basis values). Given a basis $(v_1, \dots, v_n)$ of V and arbitrary $w_1, \dots, w_n \in W$, there exists exactly one linear map $T : V \rightarrow W$ with $T(v_i) = w_i$.

Proof. For the unique coordinate expansion $v = \sum_i x_i v_i$, define $T(v) = \sum_i x_i w_i$. Uniqueness of the input coordinates makes this a function; the w_i need not be independent. Coordinates of $au + bv$ are the corresponding combinations of the coordinates of u, v , so distributing the finite output sum proves linearity. Substitution of a basis vector gives $T(v_i) = w_i$. If L is any other linear map with those basis values, then $L(v) = \sum_i x_i L(v_i) = \sum_i x_i w_i = T(v)$ for every v . This proves uniqueness after existence has been established. $\square$

Definition 10.70 (Kernel). The kernel $\ker T = \{v \in V : T(v) = 0\}$ is a subset of the source. It records the inputs whose information is lost.

Definition 10.71 (Image). The image $\operatorname{im} T = T(V) = \{T(v) : v \in V\}$ is a subset of the target. It records the outputs that can be reached.

Both are subspaces. The kernel contains zero, and if $T(u) = T(v) = 0$, then $T(au + bv) = 0$. The image contains zero, and a combination of image elements $aT(u) + bT(v)$ equals $T(au + bv)$, another image element.

Proposition 10.72 (Injectivity and surjectivity). A linear map is injective exactly when $\ker T = \{0\}$. It is surjective exactly when $\operatorname{im} T = W$.

Proof. If T is injective, $T(v) = 0 = T(0)$ forces $v = 0$. Conversely $T(u) = T(v)$ gives $T(u - v) = 0$; a zero kernel forces $u - v = 0$. The image equality is the assertion that every element of the target has a preimage. $\square$

Definition 10.73 (Rank and nullity). For a finite-dimensional source, the nullity is $\dim \ker T$ and the rank is $\dim \operatorname{im} T$. The image is finite dimensional because the images of a finite source basis span it.

Example 10.74 (A complete kernel and image calculation). For $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$, $T(x, y, z) = (x - y, y - z)$, the kernel equations are $x - y = 0$ and $y - z = 0$. Their solutions are (t, t, t) , so a kernel basis is $(1, 1, 1)$ and the nullity is 1. Given any target (a, b) , the source vector $(a, 0, -b)$ maps to it. Hence the image is all $\mathbb{R}^2$, with basis $(1, 0), (0, 1)$ and rank 2. Numerically $3 = 1 + 2$, anticipating rank-nullity. The map is surjective but not injective.

Example 10.75 (Differentiation loses one coefficient). For $D : P_2 \rightarrow P_1$, $D(a + bt + ct^2) = b + 2ct$. The kernel consists of constants, with basis (1) , and every target $u + vt$ is the derivative of $ut + (v/2)t^2$. Thus D is surjective but not injective. Its nullity is 1, its rank is 2, and their sum is 3. A map is injective when each output has at most one preimage, surjective when every target has a preimage, and bijective when both hold.

Theorem 10.76 (Rank-nullity). For a linear map $T : V \rightarrow W$ with V finite dimensional, $\dim V = \text{nullity } T + \text{rank } T$.

Proof. Choose a basis $(u_1, \dots, u_k)$ of $\ker T$ and extend it to a basis $(u_1, \dots, u_k, v_1, \dots, v_r)$ of V using Lemma 10.58. We prove that $T(v_1), \dots, T(v_r)$ is a basis of the image in two steps.

Spanning. For $y \in \operatorname{im} T$, choose $x \in V$ with $T(x) = y$. Write $x = \sum_i b_i u_i + \sum_j a_j v_j$. Applying T gives $y = \sum_j a_j T(v_j)$, because each $T(u_i) = 0$.

Independence. Suppose $\sum_j a_j T(v_j) = 0$. Then $\sum_j a_j v_j \in \ker T$, so $\sum_j a_j v_j = \sum_i b_i u_i$ for some b_i . Consequently

$$\sum_j a_j v_j - \sum_i b_i u_i = 0.$$

Independence of the full source basis forces every a_j and b_i to be zero. The image basis therefore has r elements. Since $k + r$ is the size of the source basis, $\dim V = k + r = \dim \ker T + \dim \operatorname{im} T$. $\square$

Corollary 10.77 (Linear bijectivity). For $\dim V = \dim W < \infty$, a linear $T : V \rightarrow W$ is injective if and only if it is surjective, hence if and only if it is bijective.

Proof. Rank-nullity makes zero nullity equivalent to rank $\dim W$. A subspace of W of this dimension is W , since its basis is already an independent list of maximal possible size. $\square$

Definition 10.78 (Isomorphism and endomorphism). A bijective linear map is an isomorphism. A linear map from V to itself is an endomorphism; it need not be bijective.

Proposition 10.79 (The inverse is linear). If $T : V \rightarrow W$ is a bijective linear map, its inverse $S : W \rightarrow V$ is linear.

Proof. For $y, z \in W$ and scalars a, b , compute

$$T(S(ay + bz)) = ay + bz, \quad T(aS(y) + bS(z)) = aT(S(y)) + bT(S(z)) = ay + bz.$$

Injectivity of T gives $S(ay + bz) = aS(y) + bS(z)$. This method proves equality of two vectors by applying an injective map to both of them. $\square$

The coordinate and synthesis maps of [Proposition 10.53](#) are therefore an explicitly constructed isomorphism. For the summation map of [Section 10.9](#), its image is the internal sum and its kernel is exactly the tuples $(u_1, \dots, u_r)$ with $\sum_i u_i = 0$. These are the linear relations among the components. Thus a direct internal sum is equivalent to injectivity of that map; if the sum is V , it is an isomorphism.

An isomorphism gives a reversible translation between vector spaces. The coordinate map $p \mapsto [p]_{(1,t,t^2)}$ is an isomorphism $P_2 \rightarrow \mathbb{R}^3$, although a polynomial and a column are different objects.

10.11 Matrices as Coordinate Representations

Example 10.80 (A square table from basis images). Take $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $T(x, y) = (2x - y, x + 3y)$, with source and target bases both $((1, 0), (0, 1))$. The basis images $T(1, 0) = (2, 1)$ and $T(0, 1) = (-1, 3)$ have target columns $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 3 \end{pmatrix}$. Assemble $A = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}$. Directly $T(3, 2) = (4, 9)$; the column calculation $3\begin{pmatrix} 2 \\ 1 \end{pmatrix} + 2\begin{pmatrix} -1 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ 9 \end{pmatrix}$ gives the same result.

Before using indices, consider differentiation $D : P_2 \rightarrow P_1$ in $\mathcal{B} = (1, t, t^2)$ and $\mathcal{C} = (1, t)$. Its three basis images are $0, 1, 2t$, whose target columns are $(0, 0)^T$, $(1, 0)^T$, and $(0, 2)^T$. Store them in that order:

$$[D]_{\mathcal{C} \leftarrow \mathcal{B}} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

For $p = 2 + 3t - t^2$, direct differentiation gives $3 - 2t$. Its output column is $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$, and the table acting on $(2, 3, -1)^T$ gives $2\begin{pmatrix} 0 \\ 0 \end{pmatrix} + 3\begin{pmatrix} 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$. There are two rows because outputs need two coordinates, and three columns because there are three input basis vectors. The matrix of $T(x, y, z) = (x - y, y - z)$ in standard bases is similarly $\begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}$. Here the source basis is (e_1, e_2, e_3) and the target basis is (e_1, e_2) . The basis images are $(1, 0), (-1, 1), (0, -1)$, already in target coordinates. For $(2, -1, 3)$, direct evaluation gives $(3, -4)$; the column sum $2(1, 0) - (-1, 1) + 3(0, -1) = (3, -4)$ agrees. These examples suggest the following general rule. Choose ordered bases $\mathcal{B} = (v_1, \dots, v_n)$ of V and $\mathcal{C} = (w_1, \dots, w_m)$ of W . To know T it suffices to know $T(v_j)$: linearity then determines $T(\sum_j x_j v_j)$. Conversely, any prescribed m -dimensional vectors $T(v_j)$ define one linear map by this formula, because the coordinates x_j are unique. Write each basis image in the target basis:

$$T(v_j) = \sum_{i=1}^m a_{ij} w_i.$$

Here j labels the input basis vector, hence the column. For that fixed j , i runs through the target basis and labels the row containing its coefficient. Thus a_{ij} means “coefficient of w_i in $T(v_j)$,” not the coefficient of the j th output in the i th input. An m -by- n **matrix** is a rectangular array of numbers; their space is denoted $M_{m,n}(\mathbb{R})$, or $M_n(\mathbb{R})$ when square. Define the representing matrix by its columns:

$$A = [T]_{\mathcal{C} \leftarrow \mathcal{B}} = \left(\begin{array}{c|ccc|c} & | & & & | \\ [T(v_1)]_{\mathcal{C}} & & \cdots & & [T(v_n)]_{\mathcal{C}} \\ & | & & & | \end{array} \right).$$

The notation $[T]_{\mathcal{C} \leftarrow \mathcal{B}}$ says: input coordinates are read in $\mathcal{B}$; output coordinates are written in $\mathcal{C}$. For $v = \sum_j x_j v_j$, apply linearity first and then collect coefficients:

$$T(v) = \sum_j x_j T(v_j) = \sum_i \left(\sum_j a_{ij} x_j \right) w_i, \quad [T(v)]_{\mathcal{C}} = \sum_j x_j [T(v_j)]_{\mathcal{C}}.$$

Thus the compatible definition of matrix-vector multiplication is $(Ax)_i = \sum_j a_{ij}x_j$. Matrix-vector multiplication is not an arbitrary convention. It is the unique coordinate rule compatible with linearity and the fact that the columns record the images of basis vectors.

Example 10.81 (A rectangular multiplication, two views). Let $A = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}$ and $x = (2, -1, 3)^T$. By columns,

$$Ax = 2 \begin{pmatrix} 1 \\ 0 \end{pmatrix} - \begin{pmatrix} -1 \\ 1 \end{pmatrix} + 3 \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}.$$

By rows, $(Ax)_1 = 1(2) + (-1)(-1) + 0(3) = 3$ and $(Ax)_2 = 0(2) + 1(-1) + (-1)(3) = -4$. The column view combines basis images; the row view gives one formula for each output coordinate. They are the same calculation grouped differently.

Example 10.82 (Recovering a map with a nonstandard target basis). Suppose a map $H : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ has matrix

$$[H]_{\mathcal{C} \leftarrow \text{std}} = \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix}, \quad \mathcal{C} = ((1, 1), (1, -1)).$$

The columns are target coordinates, not the standard output vectors. They say $H(e_1) = (1, 1) - (1, -1) = (0, 2)$ and $H(e_2) = 2(1, 1) + (1, -1) = (3, 1)$. The basis-value theorem therefore constructs $H(x, y) = (3y, 2x + y)$. For input $(2, -1)$, matrix multiplication gives the $\mathcal{C}$ -column $(0, -3)^T$. Synthesizing the output gives $0(1, 1) - 3(1, -1) = (-3, 3)$, agreeing with the intrinsic calculation.

Example 10.83 (Coordinates through a nonstandard input basis). Let $\mathcal{B} = ((1, 1), (1, -1))$ and consider the map $H(x, y) = (x + y, y, 2x)$. The two basis images are $(2, 1, 2)$ and $(0, -1, 2)$, so

$$[H]_{\text{std} \leftarrow \mathcal{B}} = \begin{pmatrix} 2 & 0 \\ 1 & -1 \\ 2 & 2 \end{pmatrix}.$$

The vector $(3, 1)$ has $\mathcal{B}$ -column $(2, 1)^T$, not $(3, 1)^T$. Multiplying the correct column gives $2(2, 1, 2) + (0, -1, 2) = (4, 1, 6) = H(3, 1)$.

10.12 Matrix Multiplication Comes from Composition

First compose two concrete maps. For $T(x, y) = (x + y, y)$ and $S(u, v) = (2u - v, u)$, substitution gives $S(T(x, y)) = (2x + y, x + y)$. The composite sends e_1 to $(2, 1)$ and e_2 to $(1, 1)$, so its table is $\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$. The tables of T and S are respectively $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix}$. Applying B to each column of A produces exactly those composite columns. Now let $S : W \rightarrow X$ have matrix $B = [S]_{\mathcal{D} \leftarrow \mathcal{C}}$. The middle coordinates must match, since

$$[S(T(v))]_{\mathcal{D}} = B[T(v)]_{\mathcal{C}} = B(A[v]_{\mathcal{B}}).$$

The j th column of the composite is B applied to the j th column of A , and therefore has entries $\sum_k b_{ik}a_{kj}$. This forces:

Proposition 10.84 (Coordinate rules and composition). For the preceding choices,

$$[T(v)]_{\mathcal{C}} = A[v]_{\mathcal{B}}, \quad [S \circ T]_{\mathcal{D} \leftarrow \mathcal{B}} = BA, \quad (BA)_{ij} = \sum_{k=1}^m b_{ik}a_{kj}.$$

The types can be read from

$$\mathbb{R}^n \xrightarrow{A^{(m \times n)}} \mathbb{R}^m \xrightarrow{B^{(\ell \times m)}} \mathbb{R}^\ell.$$

Here A is m -by- n , B is ℓ -by- m , and BA is ℓ -by- n .

Proof. The preceding derivation computes every output coordinate and every basis image of the composite, which uniquely determine its matrix. $\square$

Matrix multiplication is the unique multiplication rule compatible with composition of linear maps. The number of columns of B must equal the number of rows of A , the dimension of the intermediate space. “First A , then B ” gives BA , just as $S \circ T$ applies T first. Associativity follows from associativity of composition and uniqueness of representing matrices. The identity map has the matrix I_n with diagonal entries 1 and all other entries zero. An inverse matrix represents the inverse linear map and satisfies $A^{-1}A = AA^{-1} = I_n$.

Example 10.85 (Why entrywise multiplication fails). Recall the opening maps $T(x, y) = (x + y, y)$ and $S(u, v) = (2u - v, u)$. Their composite sends e_2 to $(1, 1)$. Multiplying the entries in their two tables position by position would instead give

$$\begin{pmatrix} 2 & -1 \\ 0 & 0 \end{pmatrix} e_2 = \begin{pmatrix} -1 \\ 0 \end{pmatrix}.$$

This fails the basis-image test. Matrix multiplication must sum over the intermediate coordinate; multiplying matching entries loses that information.

Example 10.86 (A rectangular composition). Let $T(x, y, z) = (x - y, y - z)$ and $S(u, v) = (u + v, 2u)$. Then

$$A = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 \\ 2 & 0 \end{pmatrix}, \quad BA = \begin{pmatrix} 1 & 0 & -1 \\ 2 & -2 & 0 \end{pmatrix}.$$

For instance entry $(2, 2)$ is $2(-1) + 0(1) = -2$: the second input basis vector goes to $(-1, 1)$, and the second coordinate of its image under S is -2 . Directly, $S(T(x, y, z)) = (x - z, 2x - 2y)$. AB is not defined: S produces two coordinates, while T needs three.

10.13 Changing Coordinates while Keeping the Map

Before introducing a conversion matrix, solve a conversion. For $\mathcal{B} = ((1, 1), (1, -1))$ and the standard basis $\mathcal{C}$, a column $(a, b)^\top$ in $\mathcal{B}$ describes $(a + b, a - b)$. Thus the conversion is $(a, b) \mapsto (a + b, a - b)$, with inverse $(x, y) \mapsto ((x + y)/2, (x - y)/2)$. Both preserve linear combinations. For example $(2, 1)^\top$ converts to $(3, 1)^\top$. Coordinate conversion is itself a linear map on columns.

The vector v is one object. The columns $[v]_{\mathcal{B}}$ and $[v]_{\mathcal{C}}$ are two descriptions of it. To translate between descriptions, the intrinsic operation is simply $v \mapsto v$; only its coordinate table changes. This is why an identity map can have a nonidentity matrix when its source and target bases differ. For two bases of the same space, define

$$P_{\mathcal{C} \leftarrow \mathcal{B}} = [\text{id}_V]_{\mathcal{C} \leftarrow \mathcal{B}}, \quad [v]_{\mathcal{C}} = P_{\mathcal{C} \leftarrow \mathcal{B}} [v]_{\mathcal{B}}.$$

The columns are the old basis vectors expressed in the new basis. Composing the identity coordinate changes in both directions gives $P_{\mathcal{B} \leftarrow \mathcal{C}} = P_{\mathcal{C} \leftarrow \mathcal{B}}^{-1}$. For $T : V \rightarrow W$, change the input coordinates, apply the old matrix, then change the output coordinates. The composition rule yields

$$[T]_{\mathcal{C}' \leftarrow \mathcal{B}'} = P_{\mathcal{C}' \leftarrow \mathcal{C}} [T]_{\mathcal{C} \leftarrow \mathcal{B}} P_{\mathcal{B} \leftarrow \mathcal{B}'}.$$

For an endomorphism using the same basis on both sides, abbreviate its matrix by $[T]_{\mathcal{B}}$. Setting $P = P_{\mathcal{B} \leftarrow \mathcal{B}'}$ gives $[T]_{\mathcal{B}'} = P^{-1}[T]_{\mathcal{B}}P$.

Example 10.87 (A complete change of basis). Take $T(x, y) = (2x + y, y)$, $\mathcal{B} = (e_1, e_2)$, and $\mathcal{B}' = ((1, 1), (0, 1))$. Then

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}, \quad P = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad P^{-1} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}, \quad P^{-1}AP = \begin{pmatrix} 3 & 1 \\ -2 & 0 \end{pmatrix}.$$

To check the columns directly, $T(1, 1) = (3, 1) = 3(1, 1) - 2(0, 1)$, and $T(0, 1) = (1, 1)$. For $v = (2, 3)$ the old coordinates are $(2, 3)$, the new ones are $P^{-1}(2, 3)^T = (2, 1)^T$, and the new matrix gives $(7, -4)^T$. Converting back gives $P(7, -4)^T = (7, 3)^T = T(v)$.

The linear map is intrinsic; its matrix depends on the coordinate systems. In Chapter 11, $Df(a)$ will be the derivative itself and the Jacobian its matrix in standard coordinates.

$$\begin{array}{ccccc} \mathbb{R}^n & \xrightarrow{P_{\mathcal{B} \leftarrow \mathcal{B}'}} & \mathbb{R}^n & \xrightarrow{[T]_{\mathcal{C} \leftarrow \mathcal{B}}} & \mathbb{R}^m \\ [T]_{\mathcal{C}' \leftarrow \mathcal{B}'} \downarrow & & & & \downarrow P_{\mathcal{C}' \leftarrow \mathcal{C}} \\ \mathbb{R}^m & \xlongequal{\hspace{1.5cm}} & & & \mathbb{R}^m \end{array}$$

Read the top route in order: translate the input, perform the old calculation, translate the output. The equality of routes derives the transformation law. The expression $P^{-1}AP$ is called a similarity transformation here only in this coordinate sense.

Example 10.88 (A second plane calculation, with different bases on each side). Let $T(x, y) = (x - y, 2y)$, $\mathcal{B} = ((1, 1), (0, 1))$, and $\mathcal{C} = ((1, 0), (1, 1))$. The standard-coordinate matrices of these bases and the map are

$$P = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad Q = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}.$$

Thus $[T]_{\mathcal{C} \leftarrow \mathcal{B}} = Q^{-1}AP = \begin{pmatrix} -2 & -3 \\ 2 & 2 \end{pmatrix}$. Directly $T(1, 1) = (0, 2) = -2(1, 0) + 2(1, 1)$ and $T(0, 1) = (-1, 2) = -3(1, 0) + 2(1, 1)$, verifying both columns. For the identity map the transition is $P_{\mathcal{C} \leftarrow \mathcal{B}} = Q^{-1}P = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$, whose inverse is $\begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}$.

Example 10.89 (Three-dimensional coordinates). Take $T(x, y, z) = (x + y, y + z, z)$ and $\mathcal{B}' = ((1, 0, 0), (1, 1, 0), (0, 0, 1))$. Relative to the standard basis,

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad P = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad P^{-1} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Multiplication gives

$$P^{-1}AP = \begin{pmatrix} 1 & 1 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

The images of the new basis vectors are $(1, 0, 0)$, $(2, 1, 0)$, $(0, 1, 1)$. Writing

$$(x, y, z) = (x - y)(1, 0, 0) + y(1, 1, 0) + z(0, 0, 1)$$

gives their columns $(1, 0, 0)$, $(1, 1, 0)$, $(-1, 1, 1)$, exactly as above.

Matrices have entrywise vector-space operations:

$$(A + B)_{ij} = a_{ij} + b_{ij}, \quad (cA)_{ij} = ca_{ij}.$$

The zero matrix and entrywise negatives satisfy the vector-space axioms because real numbers do. For linear maps $S, T : V \rightarrow W$, the j th column of the matrix of $aS + bT$ records $(aS + bT)(v_j) = aS(v_j) + bT(v_j)$. Linearity of target coordinates gives

$$[aS + bT]_{C \leftarrow B} = a[S]_{C \leftarrow B} + b[T]_{C \leftarrow B}.$$

This proves the linearity in the next identification.

Definition 10.90 (The space of linear maps). Write $\mathcal{L}(V, W)$ for the vector space of linear maps $V \rightarrow W$, with $(S + T)(v) = S(v) + T(v)$ and $(cT)(v) = cT(v)$. The vector-space axioms hold pointwise. Choosing bases identifies it linearly and bijectively with $M_{m,n}(\mathbb{R})$; the matrices with one entry 1 and all others 0 form a basis. Hence $\dim \mathcal{L}(V, W) = mn$. The special case $\mathcal{L}(V, \mathbb{R})$ is the **dual space** V^* , developed after the norm estimates.

10.14 Inner Products and Norms

The dot product measures how two displacements interact. On polynomials, another meaningful interaction is the average product over an interval. Both lead to a length by pairing a vector with itself.

Definition 10.91 (Real inner product). An inner product is a map $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ satisfying, for all vectors and real scalars,

$$\begin{aligned} \langle au + bv, w \rangle &= a\langle u, w \rangle + b\langle v, w \rangle && \text{(linearity),} \\ \langle u, v \rangle &= \langle v, u \rangle && \text{(symmetry),} \\ \langle v, v \rangle &\geq 0, \quad \langle v, v \rangle = 0 \iff v = 0 && \text{(positive definiteness).} \end{aligned}$$

Symmetry also gives linearity in the second variable: $\langle u, av + bw \rangle = a\langle u, v \rangle + b\langle u, w \rangle$. Set $\|v\| = \sqrt{\langle v, v \rangle}$; the next theorem proves its length laws.

Example 10.92 (Two inner products). On $\mathbb{R}^n$ use $\sum_i x_i y_i$. On P_2 use $\langle p, q \rangle = \int_0^1 p(t)q(t) dt$. Linearity of the integral proves bilinearity, and $pq = qp$ proves symmetry. If p is nonzero, it cannot vanish at every point of $[0, 1]$; a polynomial of degree at most two with three distinct roots is zero by successive factorization. At a point where $p \neq 0$, continuity supplies a subinterval of positive length on which p^2 is bounded below by a positive constant. Hence $\int_0^1 p^2 > 0$. For example $\langle 1, t \rangle = 1/2$, $\|t\| = 1/\sqrt{3}$, and $\|1 + t\|^2 = \int_0^1 (1 + 2t + t^2) dt = 7/3$.

The goal is to show that the induced length satisfies the following three axioms, which also make sense without an inner product.

Definition 10.93 (Norm). A **norm** on a real vector space V is a function

$$\|\cdot\| : V \longrightarrow [0, \infty)$$

such that

$$\begin{aligned} \|v\| &= 0 \iff v = 0 && \text{(definiteness),} \\ \|av\| &= |a|\|v\| && \text{(absolute homogeneity),} \\ \|v + w\| &\leq \|v\| + \|w\| && \text{(triangle inequality).} \end{aligned}$$

A vector space equipped with a norm is a **normed vector space**. The associated metric is

$$d(v, w) = \|v - w\|.$$

Theorem 10.94 (Inner-product Cauchy–Schwarz). For a real inner product, $|\langle u, v \rangle| \leq \|u\|\|v\|$. The induced length is a norm.

Proof. Strategy. Use the nonnegative squared length of $u - tv$ and choose the scalar that removes the linear term. If $v = 0$ the inequality is immediate. Otherwise set $t = \langle u, v \rangle / \langle v, v \rangle$. Bilinearity and symmetry give

$$0 \leq \langle u - tv, u - tv \rangle = \|u\|^2 - \frac{\langle u, v \rangle^2}{\|v\|^2}.$$

Multiply by $\|v\|^2$ and take square roots. Consequently

$$\|u + v\|^2 = \|u\|^2 + 2\langle u, v \rangle + \|v\|^2 \leq (\|u\| + \|v\|)^2,$$

which proves the triangle inequality. Positive definiteness gives zero length only at zero, and bilinearity gives $\|av\| = |a|\|v\|$. $\square$

Thus an inner product gives a norm, and a norm gives a metric. A general norm need only preserve the length properties needed for estimates; it need not come from an inner product. Norms quantify vector size and approximation errors. Operator norms measure the largest amplification of those errors by a linear map. Compactness will make these bounds finite in the spaces used throughout Part II.

Norms measure vector size and therefore quantify errors in linear approximations. On $\mathbb{R}^2$, the unit balls of $\|x\|_1 = |x_1| + |x_2|$, $\|x\|_2$, and $\|x\|_\infty$ are, respectively, a diamond, a circle, and a square. They look different, but in finite dimensions they lead to the same convergence and continuity theory; the next theorem explains why this is important before derivatives are defined.

For a map r with values in a normed space, the notation

$$r(h) = o(\|h\|) \quad (h \rightarrow 0)$$

means precisely

$$\frac{\|r(h)\|}{\|h\|} \longrightarrow 0.$$

Likewise, $r(h) = O(\|h\|)$ means that $\|r(h)\| \leq C\|h\|$ for all sufficiently small h . This is the norm-valued version of the asymptotic notation introduced in Chapter 5; it will describe approximation errors in Chapter 11.

Proposition 10.95 (Reverse triangle inequality). Every norm satisfies

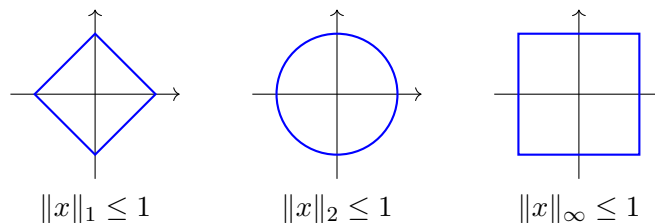
$$|\|v\| - \|w\|| \leq \|v - w\|.$$

Hence its norm map is continuous.

Proof. The triangle inequality gives

$$\|v\| \leq \|v - w\| + \|w\|.$$

Interchange v, w and combine the two inequalities. $\square$



Different unit balls do not mean different convergent sequences. In $\mathbb{R}^n$, $\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1 \leq \sqrt{n}\|x\|_2$. The first two inequalities compare the entries directly; the last is Cauchy–Schwarz applied to $(|x_i|)$ and (1) . The following proof follows the same pattern for any norm: coordinate upper bound, continuity, compact sphere, positive minimum, then homogeneous rescaling to obtain a lower bound.

Theorem 10.96 (Equivalence of norms). If V is finite dimensional and

$$\|\cdot\|_a, \quad \|\cdot\|_b$$

are norms on V , then there are $c, C > 0$ such that

$$c\|v\|_a \leq \|v\|_b \leq C\|v\|_a$$

for every $v \in V$. Thus the two norms have the same open sets, convergent and Cauchy sequences, continuous maps, and compact subsets.

Proof. Idea. First bound a norm above in coordinates, then obtain a uniform lower bound on the compact Euclidean sphere. If $V = \{0\}$ any $c, C > 0$ work. Otherwise choose a basis and identify V with $\mathbb{R}^n$. For any norm on $\mathbb{R}^n$,

$$\|x\|_b \leq \sum_{j=1}^n |x_j| \|e_j\|_b \leq C\|x\|_2$$

by Cauchy–Schwarz. Together with [Proposition 10.95](#), this gives

$$|\|x\|_b - \|y\|_b| \leq C\|x - y\|_2,$$

so $x \mapsto \|x\|_b$ is Euclidean-continuous. It has a positive minimum c on the Euclidean unit sphere, which is compact by [Theorem 10.36](#); [Theorem 10.34](#) supplies the attained minimum. It is positive because every sphere point is nonzero. Homogeneity gives

$$c\|x\|_2 \leq \|x\|_b.$$

Comparing each of the two given norms with the Euclidean norm yields the claim. The final statements follow directly from the two-sided estimate. $\square$

Corollary 10.97 (Completeness in finite dimensions). Every finite-dimensional normed vector space is complete, and its compact subsets are exactly its closed bounded subsets.

Proof. Coordinates reduce the assertion to Euclidean space, where completeness follows coordinatewise from [Theorem 3.33](#) and compactness from [Theorem 10.36](#); use [Theorem 10.96](#). $\square$

Completeness is not merely formal here. It ensures that Cauchy approximations have limits, which is the key step in the contraction theorem and in the nonlinear existence arguments of Chapter 11.

10.15 Operator Norms and Finite-Dimensional Estimates

Proposition 10.98 (Continuity of linear maps). Every linear map between finite-dimensional normed spaces is continuous.

Proof. Strategy. Bound each output coordinate, then add their squares. For matrix $A = (a_{ij})$, Cauchy–Schwarz gives

$$\sum_i \left| \sum_j a_{ij} x_j \right|^2 \leq \sum_i \left(\sum_j a_{ij}^2 \right) \left(\sum_j x_j^2 \right).$$

Thus $\|Ax\|_2 \leq (\sum_{i,j} a_{ij}^2)^{1/2} \|x\|_2$. Norm equivalence transfers this estimate to arbitrary norms, which proves continuity at zero and hence everywhere by linearity. $\square$

Definition 10.99 (Operator norm). For a linear map $L : V \rightarrow W$ between finite-dimensional normed spaces, define

$$\|L\|_{\text{op}} = \sup \{ \|Lv\|_W : \|v\|_V \leq 1 \}.$$

This number is finite in finite dimensions, and

$$\|Lv\|_W \leq \|L\|_{\text{op}} \|v\|_V.$$

Proposition 10.100 (Operator-norm estimates). The operator norm is a norm on the space of linear maps, and $\|SL\|_{\text{op}} \leq \|S\|_{\text{op}} \|L\|_{\text{op}}$.

Proof. The closed unit ball is compact by [Corollary 10.97](#). The function $v \mapsto \|Lv\|$ is continuous by [Propositions 10.95](#) and [10.98](#), so [Theorem 10.34](#) makes the supremum an attained finite maximum. For $v \neq 0$, apply the defining supremum to $v/\|v\|$ to obtain $\|Lv\| \leq \|L\|_{\text{op}} \|v\|$; the case $v = 0$ is immediate. Homogeneity and the triangle inequality follow pointwise and then by taking suprema. If the norm is zero, the displayed bound gives $Lv = 0$ for every v , so $L = 0$. Finally $\|SLv\| \leq \|S\|_{\text{op}} \|Lv\|$; take the supremum over $\|v\| \leq 1$. $\square$

The estimate $\|T(v)\| \leq C\|v\|$ says quantitatively that a linear map cannot amplify a small input error without bound. Every finite-dimensional linear map has such a bound; this is the prototype for the operator-norm estimates on derivatives used later.

Example 10.101 (A diagonal operator norm). For $L(x, y) = (3x, -2y)$ in Euclidean norm,

$$\|L(x, y)\|_2^2 = 9x^2 + 4y^2 \leq 9(x^2 + y^2).$$

Equality on $(1, 0)$ proves $\|L\|_{\text{op}} = 3$. More generally a diagonal map has Euclidean operator norm $\max_i |a_{ii}|$, by the same bound and a maximizing coordinate vector.

Example 10.102 (An operator norm depends on the chosen lengths). Let $T(x, y) = (x + 2y, 0)$. With the max norm on both spaces,

$$\|T(x, y)\|_{\infty} = |x + 2y| \leq 3\|(x, y)\|_{\infty}.$$

The max-unit vector $(1, 1)$ attains 3, so $\|T\|_{\infty \rightarrow \infty} = 3$. With the sum norm on both spaces,

$$\|T(x, y)\|_1 = |x + 2y| \leq |x| + 2|y| \leq 2\|(x, y)\|_1.$$

The sum-unit vector $(0, 1)$ attains 2, so $\|T\|_{1 \rightarrow 1} = 2$. The map is unchanged but the exact norms differ. An upper bound alone is not automatically the exact operator norm: equality requires an input attaining the bound, or inputs approaching it.

Example 10.103 (An estimate transferred to a polynomial space). On P_2 use the coefficient norm $\|a + bt + ct^2\|_c = |a| + |b| + |c|$, and on P_1 use the analogous norm. Differentiation satisfies $\|D(a + bt + ct^2)\|_c = |b| + 2|c| \leq 2\|a + bt + ct^2\|_c$. Equality at t^2 gives operator norm 2. If the spaces instead use the integral inner-product norms, this number need not remain 2, but norm equivalence still guarantees a finite bound. The intrinsic map has not changed, while a numerical error-amplification constant depends on how the two spaces measure error.

10.16 Dual Spaces and Multilinear Maps

A scalar-valued derivative measures how a direction changes a scalar output. This calls for a linear functional, rather than another direction. Recall $V^* = \mathcal{L}(V, \mathbb{R})$; its elements are **covectors**. For a basis $(e_1, \dots, e_n)$, define the coordinate functionals e^i by

$$e^i \left(\sum_j x_j e_j \right) = x_i, \quad e^i(e_j) = \delta_j^i,$$

where δ_j^i is 1 for $i = j$ and 0 otherwise. These form the **dual basis**: for any $\lambda \in V^*$, $\lambda = \sum_i \lambda(e_i) e^i$, and evaluation on each e_j proves uniqueness. Thus $\dim V^* = n$. In standard coordinates write $dx_i = e^i$. For example, $\lambda = 2dx_1 - dx_2$ evaluates on $(3, 4)$ as 2. Its coordinate row is $(2, -1)$, whereas a vector's coordinate column has a different role. For scalar-valued f in Chapter 11, $Df(a) \in V^*$. Its Euclidean vector representative is $\nabla f(a)$, determined by $Df(a)h = \nabla f(a) \cdot h$.

Example 10.104 (Finding a dual basis, rather than guessing it). For $b_1 = (1, 1), b_2 = (1, -1)$, seek $b^1(x, y) = ax + by$ satisfying $a + b = 1$ and $a - b = 0$. Thus $b^1 = (dx + dy)/2$. Similarly $b^2 = (dx - dy)/2$. Each extracts the coefficient of its own basis vector: $(x, y) = b^1(x, y)b_1 + b^2(x, y)b_2$. For $\lambda(x, y) = 3x - y$, its values on b_1, b_2 are 2, 4, so $\lambda = 2b^1 + 4b^2$. The coordinate row changes from $(3, -1)$ to $(2, 4)$ while the functional stays fixed.

The gradient is an inner-product representative, not the covector itself. For example the functional $\lambda(x, y) = x$ is represented by $(1, 0)$ for the usual dot product, but by $(1/2, 0)$ for $\langle u, v \rangle = 2u_1v_1 + u_2v_2$. Both pairings return x . No such choice is needed to evaluate λ or to pull it back.

Definition 10.105 (Pullback of a covector). For linear $T : V \rightarrow W$, define $T^* : W^* \rightarrow V^*$ by $T^*(\lambda) = \lambda \circ T$. Vectors are sent forward by T ; covectors are pulled back by composition. This map is linear because composition distributes over sums and scalar multiples of functionals.

For $T(x, y) = (x + 2y, 3y)$ and $\lambda(u, v) = 2u - v$, one obtains $T^*\lambda(x, y) = 2x + y$. In coordinates the row $(2, -1)$ is multiplied on the right by the matrix of T .

Definition 10.106 (Multilinear maps). A map $A : V_1 \times \dots \times V_k \rightarrow W$ is k -multilinear if each slot r satisfies the following conditions, with all other inputs fixed:

$$\begin{aligned} A(\dots, u + v, \dots) &= A(\dots, u, \dots) + A(\dots, v, \dots), \\ A(\dots, au, \dots) &= aA(\dots, u, \dots). \end{aligned}$$

The vectors u, v belong to V_r and $a \in \mathbb{R}$. For $k = 2$ the map is called bilinear. A single scaling of the whole input tuple is not the condition; one input varies at a time.

Examples are $x \cdot y$, $x^T A y$ for a fixed matrix A , and the signed-area expression $v_1 w_2 - v_2 w_1$ in the plane. A bilinear map is generally *not* linear on the product vector space: $B(2x, 2y) = 4B(x, y)$, whereas product-space linearity would require $2B(x, y)$. Ordinary multiplication $B(x, y) = xy$ already demonstrates this.

Expanding one argument at a time shows that a multilinear map is determined by its values on tuples of basis vectors. In the bilinear case,

$$B\left(\sum_i x_i e_i, \sum_j y_j f_j\right) = \sum_i \sum_j x_i y_j B(e_i, f_j).$$

Repeating this for k slots gives a sum of products of k coordinates times the corresponding basis-tuple value. This also shows the space of such maps is finite dimensional. In finite-dimensional normed spaces every multilinear map is bounded: coordinate bounds and the triangle inequality give a constant C such that $\|A(v_1, \dots, v_k)\| \leq C \prod_j \|v_j\|$. For bilinear maps, for example, bound each $|x_i|$ and $|y_j|$ by fixed multiples of $\|x\|$ and $\|y\|$ and sum the finitely many $\|B(e_i, f_j)\|$. This bound also proves continuity, by telescoping the difference of two evaluations one slot at a time.

Example 10.107 (Basis values and a complete bilinear estimate). On $\mathbb{R}^2$, suppose $B(e_1, e_1) = 1$, $B(e_1, e_2) = 2$, $B(e_2, e_1) = -1$, and $B(e_2, e_2) = 3$. Then

$$B(x, y) = x_1 y_1 + 2x_1 y_2 - x_2 y_1 + 3x_2 y_2.$$

The four basis values force this expression. Since every coordinate magnitude is at most the Euclidean norm, $|B(x, y)| \leq 7\|x\|_2\|y\|_2$. This is an upper bound, not a claim that 7 is optimal. To compare nearby input pairs, write the exact two-slot telescoping identity

$$B(x, y) - B(x', y') = B(x - x', y) + B(x', y - y').$$

Near a fixed pair the factors $\|y\|$ and $\|x'\|$ stay bounded, so the displayed estimate proves continuity. For more slots replace one argument at a time and add the resulting differences. This explains the multilinear continuity mechanism used for higher derivatives.

Definition 10.108 (Alternating maps). An alternating k -multilinear map $A : V^k \rightarrow W$ is one that vanishes whenever two arguments are equal:

$$v_i = v_j \text{ for some } i \neq j \implies A(v_1, \dots, v_k) = 0.$$

The repeated-argument-zero condition is the definition; a sign rule will be a consequence.

Proposition 10.109 (Alternation). Swapping two arguments of an alternating map changes its sign. An alternating map vanishes on linearly dependent arguments.

Proof. Put $u + v$ in the two slots and expand the value 0. The terms with two u 's or two v 's vanish, leaving $A(\dots, u, \dots, v, \dots) + A(\dots, v, \dots, u, \dots) = 0$. For dependent arguments, express one as a combination of the others; every term then repeats an argument. A zero argument gives zero directly by linearity, including the one-slot case. $\square$

Proposition 10.110 (Norm-valued remainder rules). As $h \rightarrow 0$, sums of $O(\|h\|)$ terms remain $O(\|h\|)$ and sums of $o(\|h\|)$ terms remain $o(\|h\|)$. If L is bounded linear and $r(h) = o(\|h\|)$, then $Lr(h) = o(\|h\|)$. If $k(h) = O(\|h\|)$ and $r(k) = o(\|k\|)$ with $r(0) = 0$, then $r(k(h)) = o(\|h\|)$, whenever the composition is defined near 0. For bounded bilinear B , if $u(h), v(h) = O(\|h\|)$, then $B(u(h), v(h)) = O(\|h\|^2) = o(\|h\|)$.

Proof. Idea. Replace the symbols by norm inequalities, including the case of a zero intermediate increment. Triangle inequalities prove the sum rules. The bound $\|Lr(h)\| \leq \|L\|_{\text{op}}\|r(h)\|$ proves the linear rule after division by $\|h\|$. For composition, choose $C \geq 1$ with $\|k(h)\| \leq C\|h\|$. Given $\varepsilon > 0$, the estimate $\|r(k)\| \leq (\varepsilon/C)\|k\|$ holds for sufficiently small nonzero k , and for $k = 0$ because $r(0) = 0$. Since $k(h) \rightarrow 0$, it gives $\|r(k(h))\| \leq \varepsilon\|h\|$ eventually. Finally the bilinear bound gives $\|B(u(h), v(h))\| \leq C_B C_u C_v \|h\|^2$; division by $\|h\|$ makes this tend to zero. $\square$

The condition $r(0) = 0$ is essential if $k(h)$ sometimes vanishes; the punctured little-o statement alone imposes no value at zero. Derivative remainders have this value by their definition. These estimates justify the product and chain-rule manipulations in Chapter 11.

10.17 Determinants and Orientation

In $\mathbb{R}^2$, $v_1 w_2 - v_2 w_1$ measures the signed area of the ordered parallelogram spanned by v, w . Doubling one side doubles it; replacing a side by a sum adds the signed contributions; equal sides give zero. The standard ordered unit square has value 1. These suggest three properties of signed volume in higher dimensions: multilinearity, alternation, and normalization on the standard basis. Volume magnitude and orientation sign are related but distinct; Chapter 12 proves the precise volume-scaling theorem.

Definition 10.111 (Normalized determinant form). A normalized determinant form on $\mathbb{R}^n$ is an alternating n -linear scalar-valued map whose value on $(e_1, \dots, e_n)$ is 1.

The sign-change lemma is [Proposition 10.109](#): putting $u + v$ in two slots and expanding its four terms proves the minus sign for an interchange. This lemma supplies the effect of a swap on a form; the elementary parity facts supply how the signs of successive swaps combine. It remains to show that the structural requirements are consistent and determine exactly one form.

Theorem 10.112 (The normalized determinant form). There is a unique alternating n -linear map $\det : (\mathbb{R}^n)^n \rightarrow \mathbb{R}$ with $\det(e_1, \dots, e_n) = 1$. More generally, every alternating n -linear form F on $\mathbb{R}^n$ is determined by its one value $c = F(e_1, \dots, e_n)$ and satisfies $F = c \det$.

Proof. Idea. Expand basis coordinates one slot at a time. Alternation removes repeated choices; the remaining basis tuples can be reordered to the standard one. We separate uniqueness from existence. Write $x_j = \sum_i a_{ij} e_i$. The slot number j specifies which input is being expanded. The index i specifies which standard basis vector occurs in that input. Thus a_{ij} lies in row i , column j of the matrix of input columns. The separate indices $i_1, i_2, \dots$ below are independent choices: i_2 is not constrained by the choice already made for i_1 . Expanding the first slot gives

$$F(x_1, x_2, \dots, x_n) = \sum_{i_1=1}^n a_{i_1,1} F(e_{i_1}, x_2, \dots, x_n).$$

For $n \geq 2$, expand the second slot of *each* resulting term:

$$F(x_1, \dots, x_n) = \sum_{i_1=1}^n \sum_{i_2=1}^n a_{i_1,1} a_{i_2,2} F(e_{i_1}, e_{i_2}, x_3, \dots, x_n).$$

Continuing through all slots gives

$$F(x_1, \dots, x_n) = \sum_{i_1, \dots, i_n=1}^n a_{i_1,1} \cdots a_{i_n,n} F(e_{i_1}, \dots, e_{i_n}).$$

Each term chooses one basis vector from each input, with the product of the chosen coefficients. There are n^n choices. Terms with repeated indices vanish. In each remaining term every index occurs once, and a sequence of swaps puts the arguments into $(e_1, \dots, e_n)$, determining its value from c . Thus two forms with the same c agree on every tuple. A multi-index $(i_1, \dots, i_n)$ records one such choice for every slot. If $i_r = i_s$, the basis tuple repeats a vector, so alternation kills that whole term even if its coefficient is nonzero. Without repetitions, n choices from n labels use every label exactly once: these are exactly the $n!$ permutations. Multilinearity produces all choices; alternation removes repeated choices; permutations are exactly the choices that remain.

For existence, assign each distinct-index tuple the value $+1$ or -1 according to the parity of its reordering, and assign repeated-index tuples zero. Extend by the displayed coordinate expansion. We use the elementary permutation facts that every reordering is a product of swaps and that the parity of the number of swaps is independent of the expression. Appendix A proves these facts without group theory. The constructed expression is linear in each column of coordinates. When two inputs agree, pair each distinct-index term with the term exchanging their selected indices. The coefficients agree and the assigned signs are opposite, so each pair cancels. Thus the expression is alternating. On the standard basis only the identity choice remains and has value 1. This proves existence. The same uniqueness argument applied to F and $c \det$ proves the final assertion, including $c = 0$. $\square$

The parity construction and its formal verification are expanded in Appendix A; the main text uses only those stated elementary facts, not the additional constructions of that appendix.

Remark 10.113 (Why this proof has so many indices). This first complete multilinear expansion is unusually index-heavy because every input is simultaneously expressed in a basis. Later arguments may say “expand by multilinearity” after identifying the slots being expanded; they usually work directly with the given vectors and do not need this full bookkeeping. The underlying operation is always the same distribution of one sum at a time.

Example 10.114 (From 27 choices to six). For $n = 3$ the expansion has terms $a_{i1}a_{j2}a_{k3}F(e_i, e_j, e_k)$ for $1 \leq i, j, k \leq 3$. The term $(1, 1, 2)$ vanishes. The choices $(1, 2, 3)$ and $(2, 1, 3)$ contribute $a_{11}a_{22}a_{33}c$ and $-a_{21}a_{12}a_{33}c$. Of the 27 formal terms, only the six orderings of $(1, 2, 3)$ can survive; some of these may still have zero coefficients. This is the mechanism behind “expand by multilinearity,” not an extra determinant rule.

For a square matrix A with columns $A_1, \dots, A_n$, define $\det A = \det(A_1, \dots, A_n)$. The preceding proof adapts the slot-by-slot argument in the author’s algebra notes, Theorem 5.12.3.

Theorem 10.115 (Determinant laws). Swapping columns negates the determinant, scaling a column scales it by the same factor, and adding a multiple of another column leaves it unchanged. Dependent columns give zero. Moreover

$$\det I_n = 1, \quad \det(AB) = \det A \det B, \quad \det A \neq 0 \iff A \text{ is invertible.}$$

An upper or lower triangular matrix has determinant $\prod_i a_{ii}$.

Proof. Idea. Use alternation for column operations and uniqueness for composition. Scaling is multilinearity; swaps and dependent columns were proved in Proposition 10.109. Adding cA_i to a different column adds a determinant with two equal columns, which is zero. To measure what a fixed map A does to every parallelepiped, feed its transformed edge vectors into the volume

form. This motivates $F(v_1, \dots, v_n) = \det(Av_1, \dots, Av_n)$ is alternating and n -linear. Its value at the standard basis is $\det A$. Uniqueness gives

$$\det(Av_1, \dots, Av_n) = (\det A) \det(v_1, \dots, v_n).$$

Taking the columns of B as inputs proves multiplicativity. If A is invertible, $1 = \det(AA^{-1}) = \det A \det A^{-1}$. If it is not, its columns are dependent by the basis and bijectivity criteria, so its determinant is zero. For upper triangular A , the first column is $a_{11}e_1$. In the expansion of column two only $a_{22}e_2$ can survive, since the e_1 choice repeats. Continuing through the columns leaves precisely the diagonal product. For lower triangular matrices run this argument from the last column to the first. $\square$

The coordinate formula follows from the structure

A **permutation** σ is a bijection of $\{1, \dots, n\}$ with itself; S_n denotes their set. Its sign is $(-1)^r$ if it is obtained by r swaps, well defined by the parity proof in Appendix A. Every ordering is obtainable by swaps, by placing each entry in its desired position successively. Concatenating swap expressions proves

$$\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}(\sigma) \operatorname{sgn}(\tau), \quad \operatorname{sgn}(\sigma^{-1}) = \operatorname{sgn}(\sigma). \quad (10.1)$$

The determinant proof now reads as the forced **Leibniz formula**:

$$\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{j=1}^n a_{\sigma(j), j}.$$

Multilinearity produces all choices; alternation removes repeated choices; permutations are exactly the choices that remain. Each product uses one entry from each column and each row. For $\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$ the six terms are

$$aei + bfg + cdh - ceg - bdi - afh.$$

Example 10.116 (The six Leibniz terms). In the table, the list is $(\sigma(1), \sigma(2), \sigma(3))$. It selects one row in column one, one in column two, and one in column three. Use the matrix

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}.$$

list	parity/sign	selected positions	product	contribution
123	even/+	(1, 1), (2, 2), (3, 3)	aei	$+aei$
132	odd/-	(1, 1), (3, 2), (2, 3)	ahf	$-ahf$
213	odd/-	(2, 1), (1, 2), (3, 3)	dbi	$-dbi$
231	even/+	(2, 1), (3, 2), (1, 3)	dhc	$+dhc$
312	even/+	(3, 1), (1, 2), (2, 3)	gbf	$+gbf$
321	odd/-	(3, 1), (2, 2), (1, 3)	gec	$-gec$

For $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 8 \\ 1 & 3 & 6 \end{pmatrix}$ the six contributions in this order are 30, -24, -24, 18, 16, -15. Their sum is 1. Only for three-by-three matrices does this short six-term mnemonic apply; a four-by-four determinant has 24 terms.

Proposition 10.117 (Transpose and intrinsic determinant). The transpose $A^\top$, obtained by exchanging rows and columns, has the same determinant as A . For an endomorphism $T : V \rightarrow V$, $\det[T]_{\mathcal{B}}$ is independent of the basis $\mathcal{B}$; we may therefore denote this scalar by $\det T$.

Proof. In the Leibniz sum for $A^\top$, replace σ by σ^{-1} and reorder the scalar factors. The products and signs become those for A . A basis change gives $P^{-1}[T]_{\mathcal{B}}P$; multiplicativity and $\det(P^{-1})\det P = 1$ cancel the coordinate factors. $\square$

Transpose invariance gives all the column rules for rows as well. For maps between different spaces a determinant requires choices of volume normalizations; a scalar independent of both bases is not asserted.

Minors, cofactors, and efficient computation

The structural laws are best for theory. Jacobians also require actual numbers, and cofactor expansion reduces a determinant to smaller ones.

Definition 10.118 (Minors and cofactors). For $A \in M_n(\mathbb{R})$, let $M_{ij}(A)$ be the determinant obtained by deleting row i and column j , preserving the remaining order. The **cofactor** is $C_{ij}(A) = (-1)^{i+j} M_{ij}(A)$. The empty 0-by-0 determinant is 1, so these conventions cover $n = 1$.

Theorem 10.119 (Laplace expansion). Along any fixed column j or fixed row i , respectively,

$$\det A = \sum_{i=1}^n a_{ij} C_{ij}(A), \quad \det A = \sum_{j=1}^n a_{ij} C_{ij}(A).$$

Proof. Idea. Expand a column in the standard basis, and identify the coefficient of each entry by moving that basis vector to the corner. Multilinearity in column j gives

$$\det A = \sum_i a_{ij} \det(A_1, \dots, e_i, \dots, A_n).$$

Move this column to the first position using $j - 1$ swaps, then move row i to the first position using $i - 1$ swaps. The resulting first column is e_1 ; the remaining rows and columns have their original relative order. Expanding the other columns, any choice of e_1 repeats the first argument and vanishes. The remaining expression is the normalized $(n - 1)$ -dimensional determinant of the deleted matrix, by uniqueness. Undoing the swaps gives $(-1)^{i+j-2} M_{ij} = C_{ij}$. This proves column expansion; applying it to $A^\top$ proves row expansion, since deleting its row j and column i transposes the minor. $\square$

Example 10.120 (Choose the zeros first). For

$$A = \begin{pmatrix} 2 & 0 & 1 \\ 3 & 0 & 4 \\ 5 & 6 & 7 \end{pmatrix},$$

column two has only one nonzero entry. Expanding there gives

$$\det A = 6(-1)^{3+2} \det \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix} = -6(8 - 3) = -30.$$

As a check, row one gives $2(0 \cdot 7 - 4 \cdot 6) + (3 \cdot 6 - 0 \cdot 5) = -48 + 18 = -30$, requiring two minors instead of one. For an even greater saving take

$$D = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 5 & 0 \\ 2 & 1 & 0 & 3 \\ 4 & 0 & 1 & 2 \end{pmatrix}.$$

Row two leaves a single minor:

$$\begin{aligned} \det D &= -5 \det \begin{pmatrix} 1 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 0 & 2 \end{pmatrix} \\ &= -5\{1(2) - 2(4 - 12) + 4(0 - 4)\} = -10. \end{aligned}$$

Expanding the original first row would require four 3-by-3 minors.

A determinant may be expanded along *any row or any column*. For hand computation normally choose the one with the most zeros, since each zero entry removes a term. Cofactors are useful for small matrices, sparse rows or columns, and recursive proofs. For larger dense matrices, elimination is usually more efficient: record row swaps and row scalings, use row additions freely, and finish with the triangular diagonal product. Thus structural properties, cofactor expansion, and elimination serve different purposes within the same determinant theory.

Example 10.121 (Elimination with no determinant change). For $A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 3 \\ 3 & 2 & 2 \end{pmatrix}$, subtract twice row one from row two and three times row one from row three. This gives $\begin{pmatrix} 1 & 2 & 1 \\ 0 & -3 & 1 \\ 0 & -4 & -1 \end{pmatrix}$. Subtract $4/3$ times row two from row three to obtain diagonal entries $1, -3, -7/3$. All operations were row additions, so $\det A = 7$.

The determinant exists algebraically without order; its *sign* uses the ordered real field. Over $\mathbb{R}$, $|\det|$ records volume magnitude, while sign distinguishes the two orientations. These facts lead to the following definition using the coordinate changes already constructed.

In the plane, an ordered coordinate pair can be counterclockwise or clockwise; in three dimensions this is the familiar distinction between right-handed and left-handed coordinates. In arbitrary finite dimension, the sign of a change-of-basis determinant is the algebraic version of this handedness. Orientation matters because scalar volume forgets the sign via $|\det|$, while oriented integration and Stokes' theorem retain it.

Definition 10.122 (Orientation). For $n \geq 1$, two ordered bases of an n -dimensional real vector space are **equivalently oriented** if the change-of-coordinate matrix between them has positive determinant. An **orientation** is one of the two equivalence classes of ordered bases. The class of the standard basis is the standard orientation of $\mathbb{R}^n$.

In $\mathbb{R}^2$, the ordered basis

$$(e_1, e_2)$$

has the standard orientation, whereas

$$(e_2, e_1)$$

has the opposite orientation because the change-of-basis determinant is -1 . The basis $(e_1 + e_2, e_2)$ has the standard orientation because its change-of-basis determinant is positive.

Proposition 10.123 (Orientation). Equivalent orientation is an equivalence relation. The sign of the determinant of an invertible map between oriented spaces does not depend on the positively oriented bases chosen.

Proof. Fix one ordered basis. Every other basis has either positive or negative transition determinant relative to it. Multiplicativity shows bases of the same sign are equivalent, and bases of opposite sign are not. Both signs occur by negating the first basis vector, so there are exactly two classes. The identity has determinant 1, inverses of positive-determinant matrices have positive determinant, and products of such matrices have positive determinant. A change to another positively oriented basis multiplies the matrix of a map on the left or right by a positive-determinant matrix, so multiplicativity preserves its sign. $\square$

Lemma 10.124 (Elementary factorization by elimination). An **elementary matrix** is obtained from the identity by one row operation; left multiplication performs that operation on any matrix. Every invertible real square matrix is a product of elementary matrices: row interchanges, nonzero row scalings, and additions of a multiple of one row to another. Their determinants are respectively -1 , the scaling factor, and 1 .

Proof. Idea. Reduce to the identity one pivot at a time and reverse the operations. The determinant rules give the three stated values. Each operation is invertible, with inverse of the same kind. In the first column of an invertible matrix some entry is nonzero (otherwise its first basis vector is in the kernel). Swap that row to the top, scale its entry to one, and subtract multiples of it from the other rows. The result is $\begin{pmatrix} 1 & b \\ 0 & C \end{pmatrix}$ with C invertible: if $Cv = 0$, then $(-bv, v)$ is in the full kernel, so $v = 0$; use the equal-dimension injectivity criterion. Induct on the size to reduce C to the identity. Then subtract multiples of the lower rows from the first row to clear b . This gives $E_k \cdots E_1 A = I$, hence $A = E_1^{-1} \cdots E_k^{-1}$ as claimed. $\square$

Example 10.125 (Stretch, shear, reflection). The maps $(x, y) \mapsto (3x, y)$, $(x, y) \mapsto (x + 2y, y)$, and $(x, y) \mapsto (-x, y)$ have determinants $3, 1, -1$. The first stretches a unit square, the second tilts its sides into a parallelogram, and the third reflects it. The determinant sign records orientation; its absolute value will be proved to give area scaling in Chapter 12.

10.18 Perspective and Transition

Before continuing, check that you can move between a linear map and its matrix, multiply a matrix by a vector, and compute composition both ways. Use $\|h\|$ to quantify an input error and explain how compactness turns local estimates into uniform ones. Distinguish $\det L$, $|\det L|$, and the sign of $\det L$: they answer related but different questions.

Exercise 10.1. For $L(x, y) = (2x - y, y)$ and $S(u, v) = (u, u + v)$, compute SL as maps and matrices. Bound $\|Lh\|_2$ by a constant times $\|h\|_2$, and determine whether each map preserves orientation.

The derivative in Chapter 11 will be a linear map between finite-dimensional normed spaces. The equivalence of norms makes differentiability independent of a coordinate norm. Compactness supplies finite local control, while determinants and orientation distinguish the two local senses of direction needed in change of variables and Stokes' theorem.

Exercises

Exercise 10.2. Prove that every open ball in a metric space is open and every singleton is closed.

Exercise 10.3. Prove that, for $E \subseteq X$,

$$\partial E = \overline{E} \cap \overline{X \setminus E}.$$

Exercise 10.4. Which of the sets

$$[0, 1/2), \quad (0, 1/2), \quad [0, 1/2]$$

are open in the subspace $[0, 1] \subseteq \mathbb{R}$?

Exercise 10.5. Prove directly from epsilon–delta definitions that a composition of continuous maps between metric spaces is continuous.

Exercise 10.6. For $f(t) = (t^2, \sin t)$, use coordinate continuity to prove continuity into $\mathbb{R}^2$. Explain why an estimate for the max metric also controls Euclidean continuity.

Exercise 10.7. Use [Theorem 10.32](#) to prove that a compact metric space is complete.

Exercise 10.8. Which of the punctured plane, two disjoint open disks, and a closed triangle are connected? Give paths or a separation as appropriate.

Exercise 10.9. Show that the open unit ball in $\mathbb{R}^n$ is not compact.

Exercise 10.10. Extend $(1, 1, 0), (0, 1, 1)$ to a basis of $\mathbb{R}^3$ using standard basis vectors. Identify a generator that can be replaced when inserting each given vector into the standard spanning list.

Exercise 10.11. For

$$T(x, y, z) = (x + 2y - z, 2x + 4y - 2z),$$

find bases of the kernel and image, and verify rank–nullity. Explain why the explicit target–lift argument for the earlier surjective example cannot work for an arbitrary target here.

Exercise 10.12. Let $T : V \rightarrow V$ be a linear endomorphism of a finite-dimensional real vector space and suppose that

$$T^2 = T$$

Show that

$$\ker T \cap T(V) = \{0\},$$

and that every $v \in V$ can be written uniquely in the form

$$v = u + w, \quad u \in \ker T, \quad w \in T(V).$$

Exercise 10.13. Give maps between finite-dimensional spaces that are injective but not surjective, and surjective but not injective.

Exercise 10.14. For

$$\|x\|_1 = \sum_{j=1}^n |x_j|, \quad \|x\|_\infty = \max_{1 \leq j \leq n} |x_j|,$$

prove

$$\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1 \leq \sqrt{n} \|x\|_2.$$

Exercise 10.15. Give an incomplete normed vector space, for example by considering polynomials on $[0, 1]$ with the supremum norm.

Exercise 10.16. Suppose square matrices A, B satisfy $AB = I$. First use linear bijectivity to prove $BA = I$, then determine $\det B$ from $\det A$. Explain why the first implication can fail for rectangular matrices, and give an example.

Exercise 10.17. Show that

$$(x_1, \dots, x_n) \mapsto (-x_1, x_2, \dots, x_n)$$

reverses the standard orientation of $\mathbb{R}^n$.

Exercise 10.18. In P_2 , decide whether $(1 + t, t + t^2, 1 - t^2)$ is independent and find a basis of its span. Give a polynomial outside that span.

Exercise 10.19. Let $\mathcal{B} = ((1, 1), (1, -1))$ and let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ satisfy $T(1, 1) = (1, 2, 0)$, $T(1, -1) = (0, 1, 3)$. Find its matrix from $\mathcal{B}$ to the standard basis. Compute $T(3, 1)$ first by expressing the input in $\mathcal{B}$ and then by the corresponding column sum.

Exercise 10.20. For $T(x, y) = (x + 2y, y)$ and $S(u, v) = (u - v, 2u)$, compute the matrix of $S \circ T$ and explain each column using basis images. Compare $T \circ S$.

Exercise 10.21. For $\mathcal{B} = ((1, 1), (0, 1))$ and $\mathcal{C} = ((1, 0), (1, 1))$, find both change-of-coordinate matrices. Transform the standard matrix of $T(x, y) = (x - y, 2y)$ to $[T]_{\mathcal{C} \leftarrow \mathcal{B}}$ and verify its two columns directly. Then compute $[T]_{\mathcal{B}}$.

Exercise 10.22. For $\lambda = 3dx_1 - 2dx_2$ and $T(x, y) = (x + y, 2x)$, compute $\lambda(2, 1)$ and $T^*\lambda$. Distinguish the input vector, the covector, and their coordinate column and row.

Exercise 10.23. Show $B(x, y) = x_1y_2 + x_2y_1$ is bilinear and symmetric but not alternating. Show it is not linear as a function on $\mathbb{R}^2 \times \mathbb{R}^2$.

Exercise 10.24. Compute the determinant of $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 8 \\ 1 & 3 & 6 \end{pmatrix}$ using row additions and a triangular matrix. Record which operations preserve the determinant.

Exercise 10.25. Compute $\det \begin{pmatrix} 2 & 0 & 3 \\ 1 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix}$ by two different cofactor expansions. Which row or column gives the shortest calculation, and why?

Exercise 10.26. If $A(t)$ is a continuous square-matrix-valued function on an interval and every $A(t)$ is invertible, prove that $\det A(t)$ has constant sign. State exactly where continuity, connectedness, and invertibility are used.

Exercise 10.27. For $E = \{1/n : n \geq 1\} \cup \{2\}$ in $\mathbb{R}$, compute interior, closure, boundary, isolated points, and accumulation points. Repeat with ambient space $\overline{E}$, identifying which answers change.

Exercise 10.28. On P_1 with $\langle p, q \rangle = \int_0^1 pq$, compute $\|1 - t\|$ and $\langle 1 + t, 1 - t \rangle$. Verify Cauchy-Schwarz for these two polynomials. Does the coordinate sum norm arise from this pairing?

Exercise 10.29. Find the dual basis of $((1, 0, 1), (0, 1, 1), (0, 0, 1))$ in $\mathbb{R}^3$. Express $2dx - dy + 3dz$ in that basis and check its value on $(1, 2, 3)$ using both descriptions.

Exercise 10.30. For $T(x, y) = (x + y, x - y)$ find its operator norm from the sum norm to the max norm, then from the max norm to the sum norm. Give an upper bound and a maximizing input in each case.

Exercise 10.31. A student identifies the matrix of the identity in two different bases with I . Supply a concrete counterexample and explain when the claim is valid. Then explain why $[T]_{\mathcal{B}'} = P^{-1}[T]_{\mathcal{B}}P$ preserves the determinant but need not preserve any particular entry.

Exercise 10.32. Reconstruct the proof that a compact subset of a Hausdorff space is closed. Label the step using Hausdorff separation, the finite selection using compactness, and the step that would fail for an infinite intersection. Explain why boundedness alone supplies none of that finite selection.

Exercise 10.33. Negate metric continuity at a with all quantifiers displayed. Construct a sequence witnessing the failure, and explain why its output error uses one fixed positive number, not a number depending on the sequence index.

Exercise 10.34. For subspaces U_1, U_2, U_3 , prove that uniqueness of a representation in $U_1 + U_2 + U_3$ is equivalent to the zero-relation criterion. Test both the criterion and pairwise intersections on three distinct lines in $\mathbb{R}^2$.

Exercise 10.35. Keep $U = \{z = x + y\} \subset \mathbb{R}^3$ but replace the complementary line in [Example 10.65](#) by $W_t = \text{span}(1, 0, t)$. For which real t is $\mathbb{R}^3 = U \oplus W_t$? In those cases construct both component maps. At the exceptional value, identify separately what fails for existence and for uniqueness.

Exercise 10.36. Let $\mathcal{C} = ((1, 1), (1, -1))$, and suppose the matrices of $S, T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ from the standard basis to $\mathcal{C}$ are $A = \begin{pmatrix} 0 & 1 \\ 2 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -2 \\ 0 & 3 \end{pmatrix}$. Construct $2S - T$ from its basis images and compare its matrix with $2A - B$. Explain why adding a matrix with a different target basis would not be valid until its output coordinates had been converted.

Chapter 11

Differentiation in Finite-Dimensional Spaces

11.1 Differentiability and the Derivative

With the little-o notation introduced in Chapter 5 and extended to norm-valued remainders in Chapter 10, one-variable differentiability at a may be written as

$$f(a + h) = f(a) + f'(a)h + o(|h|).$$

The remainder is negligible relative to the increment: after subtracting the best linear approximation, its size divided by $|h|$ tends to zero. Multiplication by the scalar $f'(a)$ is already a linear map of the increment h . For a map between higher-dimensional spaces, the correct replacement is a linear map $Df(a)$, and the derivative remains the best linear approximation to the change in f .

Definition 11.1 (Fréchet derivative). Let V, W be finite-dimensional normed real vector spaces, let $U \subseteq V$ be open, and let

$$f : U \longrightarrow W.$$

The function f is **differentiable at** $a \in U$ if there is a linear map

$$L : V \longrightarrow W$$

such that

$$\lim_{h \rightarrow 0} \frac{\|f(a + h) - f(a) - Lh\|_W}{\|h\|_V} = 0.$$

The map L , if it exists, is denoted by

$$Df(a)$$

and is called the **derivative** of f at a .

Example 11.2 (A derivative with its error visible). For $F(x, y) = (x^2 + y, xy)$ at $a = (1, 1)$, put $h = (u, v)$. Then

$$F(a + h) - F(a) = (2u + v + u^2, u + v + uv) = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} + (u^2, uv).$$

The remainder satisfies $\|(u^2, uv)\|_2 = |u|\sqrt{u^2 + v^2} \leq \|h\|_2^2$. For $h \neq 0$ its relative size is at most $\|h\|_2$, which tends to zero. Thus the displayed matrix represents $DF(1, 1)$; it is the Jacobian, and the estimate proves the approximation for every small increment at once.

Proposition 11.3 (The one-dimensional case). For a function $f : U \subseteq \mathbb{R} \rightarrow \mathbb{R}$, Fréchet differentiability at $a \in U$ is equivalent to differentiability in the sense of Chapter 8. Under this equivalence,

$$Df(a)h = f'(a)h.$$

Proof. Every linear map $L : \mathbb{R} \rightarrow \mathbb{R}$ has the form

$$Lh = ch$$

with $c = L(1)$. The Fréchet remainder condition is therefore exactly

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = c,$$

after separating the harmless case $h = 0$. This is the ordinary derivative definition, and the asserted formula follows. $\square$

Remark 11.4. Equivalently,

$$f(a+h) = f(a) + Df(a)h + r(h), \quad \frac{\|r(h)\|_W}{\|h\|_V} \rightarrow 0.$$

By [Theorem 10.96](#), this definition is independent of the norms chosen on finite-dimensional spaces.

Proposition 11.5 (Uniqueness and continuity). If f is differentiable at a , then $Df(a)$ is unique and f is continuous at a .

Proof. If L, M both have the required remainder property, then, for $h \neq 0$,

$$\frac{\|(L-M)h\|_W}{\|h\|_V} \leq \frac{\|f(a+h) - f(a) - Lh\|_W}{\|h\|_V} + \frac{\|f(a+h) - f(a) - Mh\|_W}{\|h\|_V}.$$

Put

$$h = tv$$

and let $t \rightarrow 0$ for each fixed nonzero v . This gives

$$(L-M)v = 0,$$

so $L = M$. The remainder formula, together with the continuity of a linear map from [Proposition 10.98](#), proves continuity of f . $\square$

Proposition 11.6 (Algebra and the chain rule). Let $f, g : U \rightarrow W$ be differentiable at a and let $c \in \mathbb{R}$. Then

$$D(f+g)(a) = Df(a) + Dg(a), \quad D(cf)(a) = cDf(a).$$

For real-valued f, g ,

$$D(fg)(a) = f(a)Dg(a) + g(a)Df(a).$$

If $g(a) \neq 0$, then

$$D\left(\frac{f}{g}\right)(a) = \frac{g(a)Df(a) - f(a)Dg(a)}{g(a)^2}.$$

If $f : U \rightarrow W$ is differentiable at a and $g : Z \rightarrow X$ is differentiable at $f(a)$, with $f(a) \in Z$ and Z open, then

$$D(g \circ f)(a) = Dg(f(a)) \circ Df(a).$$

Proof. Adding and scaling the remainder formulas proves the first two identities. We use the norm-valued estimates of [Proposition 10.110](#); each derivative remainder is defined to be zero at increment zero. For products, expand

$$f(a+h)g(a+h) - f(a)g(a)$$

by adding and subtracting

$$f(a)g(a+h).$$

The product of the two increments is

$$o(\|h\|_V)$$

because each differentiable increment is $O(\|h\|)$. To establish the reciprocal, set $c = g(a) \neq 0$ and $\Delta = g(a+h) - c = Dg(a)h + o(\|h\|)$. Continuity bounds $|c + \Delta|$ away from zero. The exact identity

$$\frac{1}{c + \Delta} - \frac{1}{c} = -\frac{\Delta}{c^2} + \frac{\Delta^2}{c^2(c + \Delta)}$$

has final term $O(\|h\|^2)$, so the reciprocal has derivative $-Dg(a)/c^2$. Applying the already proved product rule now gives the quotient.

For the chain rule set $A = Df(a)$ and $B = Dg(f(a))$. Write both remainder formulas with their own input variables:

$$f(a+h) = f(a) + Ah + r(h), \quad g(f(a)+k) = g(f(a)) + Bk + s(k),$$

where $r(h) = o(\|h\|)$ and $s(k) = o(\|k\|)$. Substituting $k = Ah + r(h)$ gives the complete expression

$$g(f(a+h)) = g(f(a)) + BAh + Br(h) + s(Ah + r(h)).$$

The candidate derivative is the linear map BA . Its remainder is $Br(h) + s(Ah + r(h))$. Boundedness of A and the first remainder bound give $\|Ah + r(h)\| = O(\|h\|)$. Boundedness of B gives $Br(h) = o(\|h\|)$, and the composition remainder rule of [Proposition 10.110](#) gives $s(Ah + r(h)) = o(\|h\|)$. Their sum is therefore $o(\|h\|)$, proving the asserted derivative. $\square$

Remark 11.7. The chain rule says that local linear approximations compose:

$$f(a+h) \approx f(a) + Df(a)h$$

and then

$$g(f(a)+k) \approx g(f(a)) + Dg(f(a))k.$$

Thus the linear part of $g \circ f$ is the composite

$$Dg(f(a))Df(a).$$

After coordinates are chosen, composition of linear maps is matrix multiplication, which is why that operation appears in the Jacobian formula.

Definition 11.8 (Directional derivatives and Jacobian). Let $U \subseteq \mathbb{R}^n$ be open and let $f : U \rightarrow \mathbb{R}^m$. The **directional derivative** at a in direction v is

$$D_v f(a) = \lim_{t \rightarrow 0} \frac{f(a + tv) - f(a)}{t},$$

when this limit exists. The partial derivative with respect to x_j is

$$\frac{\partial f}{\partial x_j}(a) = D_{e_j} f(a).$$

The **Jacobian matrix** of a differentiable f is the standard matrix of $Df(a)$:

$$J_f(a) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(a) & \cdots & \frac{\partial f_1}{\partial x_n}(a) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(a) & \cdots & \frac{\partial f_m}{\partial x_n}(a) \end{pmatrix}.$$

Proposition 11.9 (Derivative and directional derivatives). If f is differentiable at a , then

$$D_v f(a) = Df(a)v$$

for every $v \in \mathbb{R}^n$. In particular, its Jacobian is the matrix displayed above.

Proof. Put

$$h = tv$$

in the defining remainder formula and divide by t . The remainder divided by t tends to zero. Taking $v = e_j$ identifies the columns of the matrix. $\square$

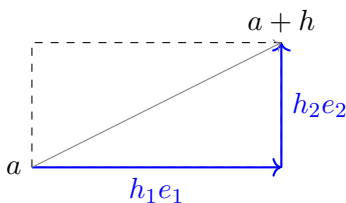
Example 11.10 (Partial derivatives need not give differentiability). Define

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Both partial derivatives at the origin exist and equal zero, since f vanishes on both coordinate axes. But along the line $y = x$,

$$f(x, x) = \frac{1}{2}$$

for $x \neq 0$; hence f is not even continuous at the origin and cannot be differentiable there. Continuous partial derivatives, as in the next theorem, are sufficient because they control all nearby coordinate increments.



The continuous-partials proof walks from a to $a + h$ one coordinate at a time. Each segment admits the one-variable mean value theorem. All intermediate points remain within $\|h\|_2$ of a , so continuity of the finitely many partials supplies one error bound for all segments. That common neighborhood is what upgrades coordinate information to a uniform first-order approximation.

Theorem 11.11 (Continuous partial derivatives). Let $U \subseteq \mathbb{R}^n$ be open. If all partial derivatives of

$$f : U \longrightarrow \mathbb{R}^m$$

exist on a neighborhood of a and are continuous at a , then f is differentiable at a and its derivative is $J_f(a)$.

Proof. It is enough to consider a real-valued component. For small h , telescope the increment from a to $a + h$ along the coordinate polygonal path. Put

$$p_0 = a, \quad p_j = a + h_1 e_1 + \cdots + h_j e_j \quad (1 \leq j \leq n).$$

For sufficiently small h , every segment from p_{j-1} to p_j lies in the neighborhood on which the partial derivatives exist. Applying the one-variable Mean Value Theorem to the function that varies only its j th coordinate gives a point

$$a_j = p_{j-1} + t_j h_j e_j \quad (0 < t_j < 1)$$

such that

$$f(a + h) - f(a) = \sum_{j=1}^n \frac{\partial f}{\partial x_j}(a_j) h_j.$$

Each a_j tends to a with h . Subtract the same sum with a_j replaced by a . Given $\varepsilon > 0$, continuity of the partial derivatives makes the absolute value of the difference at most

$$\varepsilon \sum_{j=1}^n |h_j| \leq \varepsilon \sqrt{n} \|h\|_2.$$

This is the required remainder estimate. Apply it componentwise. □

Definition 11.12 (Class C^1). A differentiable map on an open set is of class

$$C^1$$

when its derivative is continuous as a map into the finite-dimensional space of linear maps.

Remark 11.13 (A practical differentiability workflow). First compute the only possible derivative: if differentiability holds, its matrix must have the partial derivatives as columns. If the partial derivatives exist in a neighborhood and are continuous at the point, apply [Theorem 11.11](#). Otherwise return to the definition, subtract the candidate linear map, and estimate the remainder divided by $\|h\|$. Partial or directional derivatives inspect individual lines; total differentiability requires one linear approximation that controls all small increments simultaneously.

Example 11.14 (A direct remainder estimate). Define $F(0, 0) = 0$ and, away from the origin,

$$F(x, y) = \frac{x^2 y^2}{x^2 + y^2}.$$

The candidate derivative at the origin is zero. Since $4x^2 y^2 \leq (x^2 + y^2)^2$,

$$\frac{|F(x, y)|}{\|(x, y)\|_2} \leq \frac{1}{4} \|(x, y)\|_2 \longrightarrow 0.$$

Hence F is differentiable at the origin by the definition, even without first proving continuity of its nearby partial derivatives.

The counterexamples form a hierarchy of increasingly demanding tests:

$$\begin{aligned} & \text{partials} < \text{all directional derivatives} \\ & < \text{linear directional data} < \text{uniform linear approximation.} \end{aligned}$$

Here each step means a stronger requirement, not a numerical inequality. Partial derivatives test coordinate lines only. Directional derivatives test each fixed line, but their values might not depend linearly on direction. Even linear directional data can fail to control directions that change with the size of the increment. Differentiability requires one remainder estimate valid for all sufficiently small increments simultaneously.

Example 11.15 (Directional derivatives are still not enough). Set $G(0,0) = 0$ and

$$G(x,y) = \frac{x^3}{x^2 + y^2}$$

elsewhere. For every $v \neq 0$,

$$D_v G(0) = \frac{v_1^3}{v_1^2 + v_2^2}.$$

All directional derivatives therefore exist, but their dependence on v is not linear. They cannot equal $DG(0)v$ for a linear map, so G is not differentiable at the origin. Existence of all partial derivatives is an even weaker condition.

Example 11.16 (Even linear directional data can fail). Let $H(0,0) = 0$ and $H(x,y) = x^3y/(x^4 + y^2)$ elsewhere. Since $2|x^2y| \leq x^4 + y^2$, $|H(x,y)| \leq |x|/2$, so H is continuous at zero. For $v = (a,b)$ with $b \neq 0$,

$$\frac{H(ta, tb)}{t} = \frac{ta^3b}{t^2a^4 + b^2} \rightarrow 0;$$

for $b = 0$ it is identically zero. Every directional derivative is therefore the zero linear map. But $H(x, x^2) = x/2$ for $x \neq 0$, and $|H(x, x^2)|/\|(x, x^2)\|_2 \rightarrow 1/2$. Total differentiability requires uniform first-order control over all increments, including curved approaches.

11.2 Higher Derivatives and Local Estimates

The one-variable Mean Value and Taylor theorems describe a function by its successive local derivatives. The same mechanism works along a line segment in $\mathbb{R}^n$: restrict the function to that segment, apply the Part I theorem, and interpret the resulting coefficients as multilinear maps.

For vector-valued functions there need not be a single intermediate point giving the scalar mean-value equality. For $c(t) = (\cos t, \sin t)$ on $[0, 2\pi]$, the endpoint difference is zero but $c'(t)$ is never zero. The norm estimate below gives the useful control that survives.

Theorem 11.17 (Mean-value estimate). Let $U \subseteq \mathbb{R}^n$ be open, let $f : U \rightarrow \mathbb{R}^m$ be differentiable, and suppose the segment from x to y lies in U . If

$$\|Df((1-t)x + ty)\|_{\text{op}} \leq M \quad (0 \leq t \leq 1),$$

then

$$\|f(y) - f(x)\|_2 \leq M\|y - x\|_2.$$

Proof. If $f(y) \neq f(x)$, put

$$u = \frac{f(y) - f(x)}{\|f(y) - f(x)\|_2}$$

The choice of this unit vector is made so that

$$u \cdot (f(y) - f(x)) = \frac{\|f(y) - f(x)\|_2^2}{\|f(y) - f(x)\|_2} = \|f(y) - f(x)\|_2.$$

Thus one scalar projection captures the full length of the difference. Keep u fixed while applying the one-variable Mean Value Theorem to

$$t \mapsto u \cdot f((1-t)x + ty).$$

For some $c \in (0, 1)$,

$$\|f(y) - f(x)\|_2 = u \cdot Df((1-c)x + cy)(y-x).$$

Cauchy–Schwarz and the operator-norm bound prove the assertion. The zero case is immediate. $\square$

Definition 11.18 (Higher derivatives). For a differentiable map on an open subset of a finite-dimensional space, define inductively

$$D^0 f = f, \quad D^{k+1} f(a) = D(D^k f)(a),$$

when the right side exists. Here is the identification explicitly. Recall $\mathcal{L}(V, W)$ from Definition 10.90. Since $Df : U \rightarrow \mathcal{L}(V, W)$, its derivative satisfies

$$D(Df)(a) : V \rightarrow \mathcal{L}(V, W), \quad D^2 f(a)[h, k] = (D(Df)(a)h)(k).$$

This is linear in h because $D(Df)(a)$ is linear, and in k because each output is itself a linear map. Thus it is bilinear in the sense of Section 10.16. Inductively, evaluate a linear-map-valued derivative on its first argument and then on the remaining arguments to obtain $D^k f(a) : V^k \rightarrow W$. Its coefficients in bases form a finite array; continuity and differentiation of that array use the equivalent finite-dimensional norms. The map is of class

$$C^k$$

when all derivatives through order k exist and are continuous.

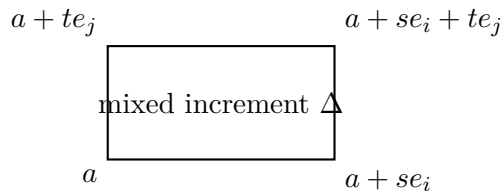
To unpack the second derivative, first let the base point move by h :

$$Df(a+h) = Df(a) + D(Df)(a)h + R(h), \quad \frac{\|R(h)\|_{\text{op}}}{\|h\|} \rightarrow 0.$$

Every term is a linear map $V \rightarrow W$. Applying these maps to a second vector k gives

$$Df(a+h)k = Df(a)k + D^2 f(a)[h, k] + R(h)k, \quad \|R(h)k\| \leq \|R(h)\|_{\text{op}}\|k\|.$$

Thus h describes motion of the base point, while k is the direction on which the changing derivative is tested. Setting $k = h$ produces a quadratic value; it does not turn the two-input object into a linear map.



Theorem 11.19 (Equality of mixed partials). Let $U \subseteq \mathbb{R}^n$ be open and let $f : U \rightarrow \mathbb{R}$ have continuous second partial derivatives. Then

$$\frac{\partial^2 f}{\partial x_i \partial x_j}(a) = \frac{\partial^2 f}{\partial x_j \partial x_i}(a).$$

Proof. Strategy. Compute the same rectangular increment in both orders. For $i = j$ there is nothing to prove. Otherwise choose small positive s, t so the entire coordinate rectangle is in U , and write

$$\Delta = f(a + se_i + te_j) - f(a + se_i) - f(a + te_j) + f(a).$$

Apply the mean value theorem to $u \mapsto f(a + ue_i + te_j) - f(a + ue_i)$ on $[0, s]$. For some $\xi \in (0, s)$,

$$\Delta = s\{\partial_i f(a + \xi e_i + te_j) - \partial_i f(a + \xi e_i)\}.$$

Apply it again, now to $v \mapsto \partial_i f(a + \xi e_i + ve_j)$ on $[0, t]$. For some $\eta \in (0, t)$ this yields $\Delta/(st) = \partial_j \partial_i f(a + \xi e_i + \eta e_j)$. Reversing the two initial differences instead gives, for possibly different ξ', η' , $\Delta/(st) = \partial_i \partial_j f(a + \xi' e_i + \eta' e_j)$. Both points tend to a as $s, t \rightarrow 0^+$. Continuity of the two mixed partials equates their limits. A single choice of positive shrinking rectangles suffices to compare their values at a . $\square$

Definition 11.20 (Gradient, Hessian, and critical points). Let $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable at a . Its **gradient** is the coordinate vector

$$\nabla f(a) = \left(\frac{\partial f}{\partial x_1}(a), \dots, \frac{\partial f}{\partial x_n}(a) \right).$$

Thus the invariant derivative has the coordinate form

$$Df(a)h = \nabla f(a) \cdot h.$$

If f is of class C^2 , its **Hessian** at a is the matrix

$$H_f(a) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}(a) \right)_{i,j=1}^n.$$

Under the standard coordinate identification,

$$D^2 f(a)[h, k] = h^T H_f(a) k.$$

By [Theorem 11.19](#), the Hessian is symmetric. A point $a \in U$ is a **critical point** of a differentiable scalar-valued function when

$$Df(a) = 0,$$

or equivalently when $\nabla f(a) = 0$.

The derivative $Df(a)$ is a linear functional, independent of coordinates. Its Jacobian is a row matrix after bases are chosen. The gradient is a vector representing that functional through the Euclidean inner product: $Df(a)h = \langle \nabla f(a), h \rangle$. Changing the inner product changes this vector representation, not the underlying functional. This distinction was introduced through dual spaces in Chapter 10 and will govern differential forms in Chapter 13.

Example 11.21 (Two directions versus a repeated direction). For $f(x, y) = x^2 + 3xy$, the second derivative at every a is

$$D^2 f(a)[h, k] = 2h_1 k_1 + 3h_1 k_2 + 3h_2 k_1.$$

This is bilinear and symmetric: interchanging h, k preserves its value. For $h = (1, 0)$ and $k = (0, 1)$ it gives 3, whereas $D^2 f(a)[h, h] = 2$. The diagonal specialization is the quadratic expression $2h_1^2 + 6h_1 h_2$ in one direction; it is not the two-input bilinear map.

By equality of mixed partials, $D^2f(a)[h, k] = D^2f(a)[k, h]$ for C^2 maps (componentwise for vector-valued maps). More generally C^k maps have symmetric k -linear derivatives: their coordinate coefficients are iterated partials, and adjacent differentiation orders can be interchanged by the mixed-partial theorem applied to the appropriate lower derivative. Adjacent swaps produce any ordering. Thus higher derivatives naturally produce *symmetric* multilinear maps, while determinants and the differential forms of Chapter 13 produce *alternating* maps.

Theorem 11.22 (Taylor's theorem along a segment). Let $U \subseteq \mathbb{R}^n$ be open, let $f : U \rightarrow \mathbb{R}^m$ be of class

$$C^{k+1},$$

and suppose the segment from a to $a + h$ is in U . If $m = 1$, there is

$$\theta \in (0, 1)$$

such that

$$f(a + h) = \sum_{j=0}^k \frac{1}{j!} D^j f(a)[h, \dots, h] + \frac{1}{(k+1)!} D^{k+1} f(a + \theta h)[h, \dots, h].$$

For vector-valued f , this formula holds componentwise, possibly with a different value of θ for each component.

Proof. For a scalar component define

$$g(t) = f(a + th).$$

Repeated application of the chain rule gives

$$g^{(j)}(t) = D^j f(a + th)[h, \dots, h].$$

Apply the one-variable Taylor theorem [Theorem 8.22](#) to g at zero and evaluate at one. Do this for each component. $\square$

Corollary 11.23 (Second-order Taylor–Peano formula). For $f : U \rightarrow \mathbb{R}^m$ of class C^2 near a ,

$$f(a + h) = f(a) + Df(a)h + \frac{1}{2} D^2 f(a)[h, h] + o(\|h\|^2).$$

Proof. For each scalar component apply the segment Taylor theorem with $k = 1$. The remainder after subtracting $\frac{1}{2} D^2 f(a)[h, h]$ is $\frac{1}{2} (D^2 f(a + \theta h) - D^2 f(a))[h, h]$. Its norm is bounded by one half the maximum operator-norm change of the second derivative on the segment, times $\|h\|^2$. Continuity at a makes that maximum tend to zero: every point of the segment is within $\|h\|$ of a . In finitely many components take a common bound. No third derivative is required. $\square$

Corollary 11.24 (Second-order test). Let $f : U \rightarrow \mathbb{R}$ be of class

$$C^2$$

and satisfy

$$Df(a) = 0.$$

If

$$D^2 f(a)[h, h] > 0 \quad (h \neq 0),$$

then a is a strict local minimum. If the displayed inequality is reversed, then a is a strict local maximum.

Proof. On the Euclidean unit sphere, the continuous quadratic form has a positive minimum by [Theorem 10.36](#). Hence

$$D^2f(a)[h, h] \geq c\|h\|_2^2$$

for some $c > 0$. The sphere minimum is attained at a nonzero vector, so it is strictly positive; pointwise positivity alone would not yield this uniform constant without compactness. By [Corollary 11.23](#), choose $\|h\|$ small enough that the absolute remainder is at most $c\|h\|^2/4$. Then $f(a+h) - f(a) \geq (c/2 - c/4)\|h\|^2 > 0$ for $h \neq 0$. Apply this to $-f$ for the maximum assertion. $\square$

Remark 11.25 (Saddles and inconclusive Hessians). At a critical point, an indefinite Hessian gives directions u, v with $D^2f(a)[u, u] > 0 > D^2f(a)[v, v]$. Taylor expansion along tu and tv then gives values above and below $f(a)$ arbitrarily nearby: a saddle. For example $x^2 - y^2$ has this behavior at zero. A semidefinite Hessian is generally inconclusive: both $x^4 + y^4$ and $x^4 - y^4$ have zero Hessian at zero, but the first has a strict minimum and the second a saddle.

11.3 The Contraction Principle

Definition 11.26 (Contraction). A map

$$T : X \longrightarrow X$$

on a metric space is a **contraction** if

$$d(Tx, Ty) \leq qd(x, y)$$

for all $x, y \in X$ and some

$$0 \leq q < 1.$$

Theorem 11.27 (Contraction mapping theorem). A contraction of a nonempty complete metric space has a unique fixed point. For every x_0 , the sequence

$$x_{n+1} = T(x_n)$$

converges to it.

Proof. The contraction inequality gives

$$d(x_{n+1}, x_n) \leq q^n d(x_1, x_0).$$

Thus, for $m > n$,

$$d(x_m, x_n) \leq \frac{q^n}{1-q} d(x_1, x_0).$$

The sequence is Cauchy and has a limit x . Continuity of T implies

$$T(x) = x.$$

If x, y are fixed, then

$$d(x, y) \leq qd(x, y),$$

so $x = y$. $\square$

Example 11.28 (Solving $x = \cos x$ by iteration). On $X = [1/2, 1]$ with the usual metric, $T(x) = \cos x$ maps X into itself. Indeed, the elementary-function theory gives $1/2 < \cos 1 \leq \cos x < 1$ there. The Mean Value Theorem and $|\sin x| \leq \sin 1 < 1$ give

$$|T(x) - T(y)| \leq (\sin 1)|x - y|.$$

The closed interval is complete, so the contraction theorem gives a unique solution in X , and the numerical iteration $x_{n+1} = \cos x_n$ converges to it from every starting point in X . Completeness is used precisely when the Cauchy iterates are asserted to have a limit. The same mechanism later underlies existence arguments for differential equations, integral equations, and problems in functional analysis.

What each contraction hypothesis contributes. The self-map hypothesis keeps every iterate in the space. The factor $q < 1$ makes successive steps shrink geometrically. Completeness places the Cauchy limit in that same space. For $T(x) = x/2$ on $(0, 1)$, the first two properties hold but the fixed point 0 is missing, so completeness cannot be omitted. On $\mathbb{R}$, $T(x) = x + 1$ shows why a Lipschitz factor 1 is insufficient even in a complete space.

The geometric tail gives usable stopping bounds. For $x_n = T^n x_0$ and the fixed point p ,

$$d(x_n, p) \leq \frac{q^n}{1 - q} d(x_1, x_0), \quad d(x_n, p) \leq \frac{q}{1 - q} d(x_n, x_{n-1}) \quad (n \geq 1).$$

Indeed every future step is bounded by a successive power of q times the first remaining step; sum that geometric series and let the far endpoint tend to p . The second bound uses the observed last step, so it can certify the accuracy during an iteration.

Proposition 11.29 (Perturbation of an invertible map). Let $A, B : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be linear and let A be invertible. If

$$\|I - A^{-1}B\|_{\text{op}} < 1,$$

then B is invertible.

Proof. Strategy. Build the inverse by a geometric series in the operator norm. Set $T = I - A^{-1}B$ and $q = \|T\|_{\text{op}} < 1$. Let

$$S_N = I + T + \cdots + T^N.$$

Submultiplicativity gives $\|T^k\|_{\text{op}} \leq q^k$, so $\|S_m - S_N\|_{\text{op}} \leq q^{N+1}/(1 - q)$ for $m > N$. The finite-dimensional space of linear maps is complete; hence $S_N \rightarrow S$. Multiplying the finite sums and cancelling consecutive powers gives

$$(I - T)S_N = S_N(I - T) = I - T^{N+1}.$$

Since $\|T^{N+1}\| \leq q^{N+1} \rightarrow 0$, continuity of multiplication (again from submultiplicativity) gives $(I - T)S = S(I - T) = I$. Thus $B = A(I - T)$ has inverse SA^{-1} . $\square$

11.4 Inverse and Implicit Functions

For an invertible linear map $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$, invertibility is equivalent to $\det T \neq 0$. In one variable, the corresponding hypothesis $f'(a) \neq 0$ says that the linear approximation $h \mapsto f'(a)h$ can be inverted. For a map

$$\mathbb{R}^n \longrightarrow \mathbb{R}^n,$$

the exact replacement is that $Df(a)$ is an invertible linear map. Since

$$f(a+h) = f(a) + Df(a)h + o(\|h\|),$$

the natural question is whether an invertible best linear approximation forces f itself to be invertible nearby. The inverse function theorem answers yes, locally: close to a , the map neither folds space nor collapses a direction, and nearby points can be recovered uniquely from their images.

Example 11.30 (Local is not global). The map

$$P(u, v) = (e^u \cos v, e^u \sin v)$$

has

$$\det DP(u, v) = e^{2u} > 0$$

at every point. It is therefore locally invertible everywhere, but it is not globally one-to-one because $P(u, v) = P(u, v + 2\pi)$. A nonzero determinant at a point, or even at every point, is a local conclusion unless additional global information is supplied.

Theorem 11.31 (Inverse function theorem). Let $U \subseteq \mathbb{R}^n$ be open and let

$$f : U \longrightarrow \mathbb{R}^n$$

be of class

$$C^1.$$

If

$$Df(a)$$

is invertible, then there are open neighborhoods

$$U_0 \subseteq U, \quad V_0 \subseteq \mathbb{R}^n$$

of a and $f(a)$ such that

$$f : U_0 \longrightarrow V_0$$

is bijective. Its inverse g is of class

$$C^1$$

and

$$Dg(y) = (Df(g(y)))^{-1}.$$

Proof. Roadmap. Convert $f(x) = y$ into a fixed-point equation, prove existence and uniqueness by contraction, control variation of the inverse, differentiate it using that control, and prove continuity of its derivative. The map $T_y(x) = x - Df(a)^{-1}(f(x) - y)$ corrects the output error using the inverse linear approximation, like a Newton step with a fixed derivative.

1. Construct nearby solutions. Put

$$A = Df(a)^{-1}.$$

Choose q with

$$0 < q < 1.$$

Because U is open and Df is continuous at a , there is $r > 0$ such that the closed ball

$$K = \{x \in \mathbb{R}^n : \|x - a\|_2 \leq r\}$$

is contained in U and

$$\|I - ADf(x)\|_{\text{op}} \leq q \quad (x \in K).$$

The set K is complete because it is closed in the complete space $\mathbb{R}^n$. Define

$$G(x) = x - Af(x).$$

Since

$$DG(x) = I - ADf(x),$$

the segment between any two points $x, z \in K$ lies in the convex set K , and [Theorem 11.17](#) gives

$$\|G(x) - G(z)\|_2 \leq q\|x - z\|_2.$$

For $y \in \mathbb{R}^n$, define

$$T_y(x) = x - A(f(x) - y).$$

Then

$$T_y(x) - T_y(z) = G(x) - G(z),$$

so T_y is a q -contraction on K . Moreover, for $x \in K$,

$$\begin{aligned} \|T_y(x) - a\|_2 &= \|[x - Af(x)] - [a - Af(a)] + A(y - f(a))\|_2 \\ &\leq q\|x - a\|_2 + \|A\|_{\text{op}}\|y - f(a)\|_2. \end{aligned}$$

Set

$$\rho = \frac{(1 - q)r}{\|A\|_{\text{op}}}.$$

If $\|y - f(a)\|_2 < \rho$, the last expression is strictly smaller than r . Thus $T_y(K) \subseteq K$, and [Theorem 11.27](#) gives a unique fixed point $g(y) \in K$. The fixed-point equation is equivalent to

$$f(g(y)) = y.$$

If $x, z \in K$ and $f(x) = f(z)$, then

$$\|x - z\|_2 = \|[x - Af(x)] - [z - Af(z)]\|_2 \leq q\|x - z\|_2.$$

Since $q < 1$, this forces $x = z$, so f is injective on K .

Take

$$V_0 = B_2\left(f(a), \frac{\rho}{2}\right)$$

and set

$$U_0 = B_2(a, r) \cap f^{-1}(V_0).$$

The set V_0 is open, contains $f(a)$, and its closure lies in the range of y 's for which the contraction argument applies. Since f is continuous, U_0 is open and contains a . For every $y \in V_0$, the self-map estimate with $\|y - f(a)\|_2 < \rho/2$ puts the fixed point $g(y)$ in the open ball $B_2(a, r)$, so $g(y) \in U_0$. Thus $f(U_0) = V_0$, and injectivity on K shows that

$$f : U_0 \longrightarrow V_0$$

is bijective with inverse g .

2. Control variation of the inverse. Subtracting the fixed-point equations

$$g(y) = G(g(y)) + Ay, \quad g(z) = G(g(z)) + Az$$

and using the Lipschitz estimate for G gives

$$(1 - q)\|g(y) - g(z)\|_2 \leq \|A\|_{\text{op}}\|y - z\|_2,$$

and hence

$$\|g(y) - g(z)\|_2 \leq \frac{\|A\|_{\text{op}}}{1 - q}\|y - z\|_2.$$

In particular, g is continuous. For $x \in K$,

$$\|I - Df(a)^{-1}Df(x)\|_{\text{op}} = \|I - ADf(x)\|_{\text{op}} \leq q < 1.$$

Thus [Proposition 11.29](#), with its invertible map equal to $Df(a)$ and its perturbed map equal to $Df(x)$, shows that every $Df(x)$ on K is invertible.

3. Differentiate the inverse. The Lipschitz estimate converts errors small relative to the input increment into errors small relative to the output increment. Fix $y \in V_0$, put $x = g(y)$, and, for h small enough that $y + h \in V_0$, set

$$\Delta x = g(y + h) - g(y).$$

Differentiability of f at x gives

$$h = f(x + \Delta x) - f(x) = Df(x)\Delta x + r(\Delta x),$$

where

$$\frac{\|r(\Delta x)\|_2}{\|\Delta x\|_2} \rightarrow 0 \quad (\Delta x \rightarrow 0).$$

The Lipschitz estimate for g gives

$$\|\Delta x\|_2 \leq \frac{\|A\|_{\text{op}}}{1 - q}\|h\|_2.$$

Consequently,

$$\frac{\|r(\Delta x)\|_2}{\|h\|_2} \leq \frac{\|A\|_{\text{op}}}{1 - q} \frac{\|r(\Delta x)\|_2}{\|\Delta x\|_2} \rightarrow 0,$$

where for $h \neq 0$ one has $\Delta x \neq 0$ because g is injective. Therefore $r(\Delta x) = o(\|h\|_2)$. Multiplying by $Df(x)^{-1}$ yields

$$\Delta x = Df(x)^{-1}h + o(\|h\|_2),$$

and therefore

$$Dg(y) = Df(g(y))^{-1}.$$

4. Continuity of the derivative. Fix an invertible matrix C . If

$$\|C^{-1}(B - C)\|_{\text{op}} < \frac{1}{2},$$

then the geometric-series proof of [Proposition 11.29](#) gives

$$\|B^{-1}\|_{\text{op}} \leq 2\|C^{-1}\|_{\text{op}}.$$

Together with

$$B^{-1} - C^{-1} = B^{-1}(C - B)C^{-1},$$

this proves that inversion is continuous at C . Since g and Df are continuous and $Df(g(y))$ is invertible throughout V_0 , the formula for Dg is continuous. Thus g is of class C^1 . $\square$

Lemma 11.32 (Smooth matrix inversion). Matrix inversion is smooth on $\text{GL}_n(\mathbb{R}) = \{A : \det A \neq 0\}$.

Proof. Let $\text{adj}(A)$ be the transpose of the cofactor matrix from Chapter 10. The (i, j) entry of $A \text{adj}(A)$ is $\sum_k a_{ik} C_{jk}$. If $i = j$ this is the row- j cofactor expansion of $\det A$. If $i \neq j$ it is the determinant obtained by replacing row j with row i , because the minors deleting row j are unchanged; the resulting two equal rows give zero. Column expansion proves $\text{adj}(A)A = (\det A)I$ as well. Hence

$$A^{-1} = \frac{\text{adj}(A)}{\det A} \quad (\det A \neq 0).$$

Each numerator and the denominator are polynomials in the matrix entries. The denominator is nonzero on the open set $\text{GL}_n(\mathbb{R})$, so the scalar quotient and chain rules, iterated, prove smoothness of every entry. $\square$

Corollary 11.33 (Higher regularity of local inverses). If f is C^r , $1 \leq r \leq \infty$, the local inverse supplied by the inverse function theorem is C^r . For $r = \infty$ this means smooth.

Proof. The theorem already gives $g \in C^1$ and $Dg(y) = [Df(g(y))]^{-1}$. Suppose g is C^j and f is C^{j+1} . Then Df is C^j , so composition with g and smooth matrix inversion make Dg a C^j function. Thus g is C^{j+1} . Induction reaches every order available for f , and every finite order when f is smooth. $\square$

Example 11.34 (Using the inverse function theorem). Let

$$f(u, v) = (u + v^2, u^2 + v).$$

Its Jacobian is

$$Df(u, v) = \begin{pmatrix} 1 & 2v \\ 2u & 1 \end{pmatrix}, \quad \det Df(u, v) = 1 - 4uv.$$

At $(0, 0)$ the determinant is 1, so the theorem supplies neighborhoods of $(0, 0)$ on which f has a unique C^1 inverse. In ordinary language, every output sufficiently close to $(0, 0)$ comes from exactly one nearby input, and that input varies differentiably with the output. The theorem does not make a global claim; on the curve $uv = 1/4$ the derivative is singular, so the Inverse Function Theorem gives no conclusion about local invertibility there.

The derivative formula is exactly what formal differentiation predicts: from $f^{-1} \circ f = \text{id}$, the chain rule suggests

$$D(f^{-1})(f(a))Df(a) = I,$$

and hence $D(f^{-1})(f(a)) = Df(a)^{-1}$. The theorem proves that this formal calculation applies to an actual local inverse.

Definition 11.35 (Diffeomorphisms). Let $U \subseteq \mathbb{R}^n$ and $V \subseteq \mathbb{R}^m$ be open. A bijection

$$f : U \longrightarrow V$$

is a C^r **diffeomorphism** if both f and f^{-1} are of class C^r . When $r = \infty$, it is a **smooth diffeomorphism**. Thus the inverse function theorem says, in this language, that a C^1 map with invertible derivative at a point is a C^1 diffeomorphism after restriction to sufficiently small open neighborhoods.

Remark 11.36 (From an equation to a local graph). The circle

$$x^2 + y^2 = 1$$

is not the graph of one function $y = g(x)$ globally: most x values have two points on the circle. Yet near $(0, 1)$, the equation

$$F(x, y) = x^2 + y^2 - 1 = 0$$

can be solved for y because $F_y(0, 1) = 2 \neq 0$. Near $(1, 0)$ it can instead be solved for x , since $F_x(1, 0) = 2 \neq 0$. The Implicit Function Theorem is the general existence-and-local-parametrization principle behind this familiar calculus picture: it solves selected variables locally even when there is no useful explicit algebraic formula.

Conceptually it is another form of local invertibility. For $F : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^m$, the auxiliary map

$$H(x, y) = (x, F(x, y))$$

keeps the prospective independent variables and records the equations as the remaining output coordinates. Invertibility of $D_y F$ makes DH invertible, so a local inverse for H recovers the desired dependent variables.

Theorem 11.37 (Implicit function theorem). Let $U \subseteq \mathbb{R}^n \times \mathbb{R}^m$ be open and let

$$F : U \longrightarrow \mathbb{R}^m$$

be of class

$$C^1.$$

Suppose

$$F(a, b) = 0$$

and

$$D_y F(a, b)$$

is invertible. Then there are open neighborhoods

$$A \subseteq \mathbb{R}^n, \quad B \subseteq \mathbb{R}^m$$

of a and b , respectively, and a unique C^1 map

$$\varphi : A \longrightarrow B$$

such that

$$\varphi(a) = b, \quad F(x, \varphi(x)) = 0,$$

and $A \times B \subseteq U$ with the local graph equality

$$\{(x, y) \in A \times B : F(x, y) = 0\} = \{(x, \varphi(x)) : x \in A\}.$$

Moreover,

$$D\varphi(x) = -(D_y F(x, \varphi(x)))^{-1} D_x F(x, \varphi(x)).$$

Proof. Geometrically, the theorem solves the equation

$$F(x, y) = 0$$

near (a, b) for the y -variables as functions of the x -variables. To convert this into an inverse-function problem, retain x as an output coordinate and define

$$H(x, y) = (x, F(x, y)).$$

Its derivative at (a, b) is

$$DH(a, b)(u, v) = (u, D_x F(a, b)u + D_y F(a, b)v).$$

It is invertible because, given (p, q) , its inverse is

$$(p, q) \mapsto \left(p, (D_y F(a, b))^{-1}(q - D_x F(a, b)p) \right).$$

Continuity of $D_y F$ and [Proposition 11.29](#) give an open product neighborhood $P \times R$ of (a, b) , contained in U , on which every $D_y F(x, y)$ is invertible: shrink the product until

$$\|I - D_y F(a, b)^{-1} D_y F(x, y)\|_{\text{op}} < 1.$$

Apply [Theorem 11.31](#) to H . There are open neighborhoods W of (a, b) and Z of $(a, 0)$ such that

$$H : W \longrightarrow Z$$

is a C^1 diffeomorphism. Intersect W with $P \times R$ and replace Z by the image of this intersection; because $H : W \rightarrow Z$ is a homeomorphism, these are still open neighborhoods and the restricted map is still a C^1 diffeomorphism. Thus assume

$$W \subseteq P \times R.$$

Because W is open, choose open neighborhoods A_1 of a and B of b such that

$$A_1 \times B \subseteq W.$$

Because Z is open at $(a, 0)$, choose an open neighborhood A_2 of a such that

$$A_2 \times \{0\} \subseteq Z.$$

Continuity of H^{-1} at $(a, 0)$ permits A_2 to be shrunk so that

$$H^{-1}(A_2 \times \{0\}) \subseteq \mathbb{R}^n \times B.$$

Set

$$A = A_1 \cap A_2.$$

Then $A \times B \subseteq W$ and $A \times \{0\} \subseteq Z$.

For $x \in A$, define

$$\varphi(x) = \pi_2(H^{-1}(x, 0)).$$

The preceding inclusion shows that $\varphi(x) \in B$. Since H^{-1} , the map $x \mapsto (x, 0)$, and the projection π_2 are C^1 , so is φ . If

$$H^{-1}(x, 0) = (x', y'),$$

then

$$H(x', y') = (x, 0).$$

The first component forces $x' = x$, and the second gives $F(x, y') = 0$. Therefore

$$F(x, \varphi(x)) = 0.$$

If $x \in A$, $y \in B$, and $F(x, y) = 0$, then both (x, y) and $(x, \varphi(x))$ lie in W , and

$$H(x, y) = (x, 0) = H(x, \varphi(x)).$$

Injectivity of H on W gives $y = \varphi(x)$, proving the stated local uniqueness. Moreover, $D_y F(x, \varphi(x))$ is invertible for every $x \in A$, because the graph lies in $W \subseteq P \times R$.

Differentiate

$$F(x, \varphi(x)) = 0$$

using the chain rule. As linear maps from $\mathbb{R}^n$ to $\mathbb{R}^m$, this gives

$$D_x F(x, \varphi(x)) + D_y F(x, \varphi(x)) D\varphi(x) = 0.$$

Here $D_x F$ is an m -by- n map, $D_y F$ is an invertible m -by- m map, and $D\varphi$ is an m -by- n map. Multiplication on the left by $D_y F(x, \varphi(x))^{-1}$ proves

$$D\varphi(x) = -D_y F(x, \varphi(x))^{-1} D_x F(x, \varphi(x)).$$

□

Example 11.38 (An equation without an explicit solution). For $F(x, y) = y + \sin y - x$, $F(0, 0) = 0$ and $F_y(0, 0) = 2$. Thus near zero there is a unique $y = \varphi(x)$ with $\varphi(x) + \sin \varphi(x) = x$, and $\varphi'(x) = 1/(1 + \cos \varphi(x))$, in particular $\varphi'(0) = 1/2$. The theorem supplies existence and derivatives without an explicit inverse formula.

Example 11.39 (Implicit differentiation justified). For $F(x, y) = x^2 + y^2 - 1$, take $(a, b) = (0, 1)$. Then

$$F(0, 1) = 0, \quad F_y(0, 1) = 2 \neq 0.$$

The Implicit Function Theorem therefore gives a unique differentiable function $y = g(x)$ near 0 with $g(0) = 1$ and $x^2 + g(x)^2 = 1$. Its derivative is

$$g'(x) = -\frac{F_x(x, g(x))}{F_y(x, g(x))} = -\frac{x}{g(x)}.$$

This is the familiar implicit-differentiation rule from calculus, now backed by a theorem that first supplies the local function being differentiated. At $(0, -1)$ one obtains the lower local graph; at $(\pm 1, 0)$ one must solve locally for x as a function of y instead.

Corollary 11.40 (Higher regularity of implicit functions). In the implicit function theorem, C^r input data give a C^r implicit function, including the smooth case.

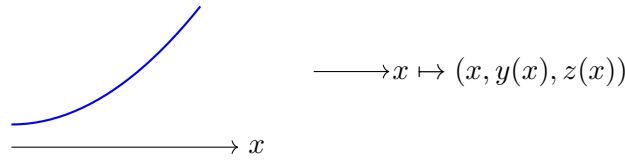
Proof. The proof used $H(x, y) = (x, F(x, y))$, which is C^r when F is. Its local inverse is C^r by [Corollary 11.33](#). The implicit function is the dependent-variable component of that inverse restricted to $(x, 0)$, so is C^r by composition with the inclusion and projection. □

Example 11.41 (Two dependent variables in an implicit system). Consider

$$F(x, y, z) = (y + z - x, y - z + x^2).$$

At $(0, 0, 0)$ the matrix of derivatives in (y, z) is $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, of determinant -2 . Thus both y and z can be solved as smooth functions of the one independent variable x . Directly adding and subtracting the equations gives $y(x) = (x - x^2)/2$, $z(x) = (x + x^2)/2$. At zero the implicit derivative formula yields $(y'(0), z'(0))^T = (1/2, 1/2)^T$, which agrees with the explicit solution. The invertible object is the entire dependent-variable block, not one selected partial derivative.

constraint locally becomes a graph



Theorem 11.42 (Lagrange multipliers for one constraint). Let $f, g : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ be of class C^1 . Suppose that a is a local extremum of f subject to

$$g(x) = c,$$

with $g(a) = c$, and that $\nabla g(a) \neq 0$. Then there is $\lambda \in \mathbb{R}$ such that

$$\nabla f(a) = \lambda \nabla g(a).$$

Proof. The Implicit Function Theorem makes the level set $g = c$ near a a C^1 graph. Its tangent directions form

$$T = \ker Dg(a).$$

Indeed, after relabeling coordinates so that a partial derivative of g is nonzero, the theorem writes the level set as a local graph, whose derivative has precisely this tangent range. Indeed, writing $a = (a', a_n)$ and the graph as $(x', \varphi(x'))$, its velocities are $(u, D\varphi(a')u)$. Differentiating the constraint puts them in $\ker Dg(a)$; conversely the equation $D_{x'}g(a)u + g_{x_n}(a)v = 0$ forces $v = D\varphi(a')u$. Every such velocity is realized by the graph curve $(a' + tu, \varphi(a' + tu))$. Restricting f to that graph gives a local unconstrained extremum, so

$$Df(a)v = 0 \quad (v \in T).$$

Thus $Df(a)$ vanishes on $\ker Dg(a)$. Write $\ell = Dg(a)$ and $m = Df(a)$. Choose w with $\ell(w) = 1$; this is possible by rescaling a vector where the nonzero functional ℓ does not vanish. Every vector splits as

$$v = (v - \ell(v)w) + \ell(v)w, \quad v - \ell(v)w \in \ker \ell.$$

Applying m kills the first summand, giving $m(v) = m(w)\ell(v)$. Thus $m = m(w)\ell$ as functionals. Geometrically the allowed tangent directions lie in the constraint kernel; a constrained extremum has no first-order change in any such direction. With the Euclidean inner product the two representing gradients are parallel. Since $Dg(a)$ is a nonzero linear functional, any linear functional vanishing on this kernel is a scalar multiple of $Dg(a)$. Hence $Df(a) = \lambda Dg(a)$, which is the displayed gradient equation. $\square$

Example 11.43 (Extrema of xy on the unit circle). To maximize or minimize

$$f(x, y) = xy$$

subject to $g(x, y) = x^2 + y^2 = 1$, observe that

$$\nabla f = (y, x), \quad \nabla g = (2x, 2y).$$

The Lagrange equations are

$$(y, x) = \lambda(2x, 2y), \quad x^2 + y^2 = 1.$$

They give $\lambda = 1/2$ and $y = x$, or $\lambda = -1/2$ and $y = -x$. Thus $f = 1/2$ at

$$\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \quad \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right),$$

and $f = -1/2$ at the two points with $y = -x$. These are respectively the maximum and minimum, as also follows from $2|xy| \leq x^2 + y^2 = 1$.

Remark 11.44 (How to recognize which theorem to use). For a map $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$, a question about a local inverse calls for computing $Df(a)$ and checking that it is invertible (equivalently, that its determinant is nonzero). For equations $F(x, y) = 0$ and a request to solve for the y variables, compute the block $D_y F(a, b)$ instead. In either case, the result concerns neighborhoods of the specified point, not a global inverse or a global parametrization.

Exercises

Exercise 11.1. For

$$f(x, y) = (x^2y, \sin(x + y)),$$

compute $Df(x, y)$ and $Df(x, y)(u, v)$.

Exercise 11.2. Give a function from $\mathbb{R}^2$ to $\mathbb{R}$ with both partial derivatives at the origin but which is not differentiable there.

Exercise 11.3. Prove that a linear map between finite-dimensional normed spaces is differentiable everywhere and is its own derivative.

Exercise 11.4. Prove directly from the definition that differentiability at a implies continuity at a .

Exercise 11.5. Use [Theorem 11.17](#) to prove that a C^1 map with zero derivative on a convex open set is constant.

Exercise 11.6. Compute the Hessian of

$$x^3 + xy^2 - 3x$$

and classify its critical points.

Exercise 11.7. Apply Taylor's theorem to find the quadratic approximation of

$$e^x \cos y$$

at the origin.

Exercise 11.8. Give a function with unequal mixed partial derivatives at the origin and explain why [Theorem 11.19](#) does not apply.

Exercise 11.9. Use the contraction mapping theorem to prove that

$$x = \cos x$$

has a unique solution in

$$[0, 1].$$

Exercise 11.10. Let A be a matrix with

$$\|I - A\|_{\text{op}} < 1.$$

Show that

$$x_{k+1} = (I - A)x_k + b$$

converges to the unique solution of

$$Ax = b.$$

Exercise 11.11. Apply the inverse function theorem to

$$f(x, y) = (e^x \cos y, e^x \sin y)$$

at $(0, 0)$ and compute the derivative of its local inverse at $(1, 0)$.

Exercise 11.12. Near $(0, 1)$, use the implicit function theorem on

$$x^2 + y^2 + xy = 1$$

to solve for y as a C^1 function of x and find its derivative at zero.

Exercise 11.13. Explain why

$$y^2 = x^3$$

does not define y as a C^1 function of x near the origin.

11.5 Perspective and Transition

Chapter 12 uses differentiability, determinants, and local inverses to study transformations of domains. Chapter 13 reinterprets derivatives as pullbacks of alternating multilinear forms.

Exercise 11.14. For $f(x, y) = e^x \cos y$, compute $Df(0)$ and $D^2f(0)$ and write the second-order Peano formula. Distinguish its quadratic term from the bilinear second derivative evaluated on two different vectors.

Exercise 11.15. Iterate $T(x) = \cos x/2$ on $[0, 1]$. Check self-map, contraction, and completeness, then give an a priori error bound after ten steps starting at zero. Which bound uses the last observed step?

Exercise 11.16. For $F(x, y, z) = (y+z-x, y-z+x^2)$ calculate the implicit derivative at $x = 1$ from the block formula and the explicit solution. Identify its source dimension and target dimension.

Exercise 11.17. At the origin compare the second-order tests for $x^2 + 2y^2$, $x^2 - y^2$, and $x^2 + y^4$. In the last case, explain why the theorem is inconclusive even though the definition decides the minimum.

Chapter 12

Riemann Integration in Several Variables

12.1 Rectangles, Partitions, and Darboux Sums

This is the direct extension of Chapter 9: intervals become rectangles, length becomes volume, and one partitions each coordinate interval at once. The Darboux definition is unchanged in spirit. Chapter 10 supplies the compactness and uniform-continuity facts that make continuous functions integrable on these higher-dimensional compact rectangles.

Definition 12.1 (Rectangles and volume). A **rectangle** in $\mathbb{R}^n$ is a product

$$Q = [a_1, b_1] \times \cdots \times [a_n, b_n],$$

where

$$a_j < b_j \quad (1 \leq j \leq n).$$

Its **volume** is

$$\text{vol}(Q) = \prod_{j=1}^n (b_j - a_j).$$

Definition 12.2 (Grid partitions and Darboux sums). Let Q be a rectangle. A **grid partition** of Q is obtained by choosing a finite partition of each interval $[a_j, b_j]$ and taking all products of the resulting subintervals. Its members are **subrectangles**. For bounded

$$f : Q \longrightarrow \mathbb{R}$$

and a grid partition $\mathcal{P}$, put

$$U(f, \mathcal{P}) = \sum_{R \in \mathcal{P}} \sup_R f \, \text{vol}(R), \quad L(f, \mathcal{P}) = \sum_{R \in \mathcal{P}} \inf_R f \, \text{vol}(R).$$

The function f is **Riemann integrable** on Q if

$$\inf_{\mathcal{P}} U(f, \mathcal{P}) = \sup_{\mathcal{P}} L(f, \mathcal{P}).$$

The common value is denoted by

$$\int_Q f(x) \, dx.$$

Proposition 12.3 (Darboux criterion). A bounded function $f : Q \rightarrow \mathbb{R}$ is Riemann integrable if and only if, for every

$$\varepsilon > 0,$$

there is a grid partition $\mathcal{P}$ such that

$$U(f, \mathcal{P}) - L(f, \mathcal{P}) < \varepsilon.$$

Proof. Refining a grid partition decreases upper sums and increases lower sums. Thus every lower sum is at most every upper sum. If the two extremal values are equal, choose partitions whose upper and lower sums are within

$$\varepsilon/2$$

of that common value and take their common refinement. Conversely, the displayed condition forces the difference of the two extremal values to be smaller than every positive ε , hence zero by [Lemma 4.5](#). $\square$

Theorem 12.4 (Continuous functions are integrable). Every continuous function

$$f : Q \longrightarrow \mathbb{R}$$

is Riemann integrable.

Proof. The rectangle Q is compact by [Theorem 10.36](#), so [Theorem 10.35](#) makes f uniformly continuous. Choose a grid whose subrectangles have Euclidean diameter less than δ . On each subrectangle R ,

$$\sup_R f - \inf_R f < \frac{\varepsilon}{\text{vol}(Q)}.$$

Therefore

$$U(f, \mathcal{P}) - L(f, \mathcal{P}) < \varepsilon,$$

and [Proposition 12.3](#) applies. $\square$

Example 12.5 (Continuity succeeds; boundedness alone fails). On $Q = [0, 1]^2$, the function $f(x, y) = x^2 + y^2$ is integrable by [Theorem 12.4](#). In contrast, define

$$g(x, y) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Every nondegenerate subrectangle contains points of both kinds, so its oscillation is 1. Consequently every grid partition satisfies

$$U(g, \mathcal{P}) - L(g, \mathcal{P}) = \text{vol}(Q) = 1,$$

and g is not Riemann integrable. The example separates boundedness from integrability in several variables just as the Dirichlet function does in one variable.

Theorem 12.6 (Basic properties). If $f, g : Q \rightarrow \mathbb{R}$ are integrable and $c \in \mathbb{R}$, then

$$\int_Q (f + g) \, dx = \int_Q f \, dx + \int_Q g \, dx, \quad \int_Q cf \, dx = c \int_Q f \, dx.$$

If

$$f \leq g,$$

then their integrals have the same order. Also,

$$\left| \int_Q f \, dx \right| \leq \int_Q |f| \, dx,$$

and

$$\left| \int_Q f \, dx \right| \leq M \operatorname{vol}(Q)$$

whenever

$$|f(x)| \leq M \quad (x \in Q).$$

Proof. The upper and lower sum inequalities used in the proof of [Theorem 9.7](#) hold verbatim for subrectangles:

$$U(f + g, \mathcal{P}) \leq U(f, \mathcal{P}) + U(g, \mathcal{P}),$$

$$L(f + g, \mathcal{P}) \geq L(f, \mathcal{P}) + L(g, \mathcal{P}).$$

Taking a common refinement of partitions that make the two individual gaps small proves integrability of the sum. The same inequalities, now compared with the integrals, prove equality of its integral with the sum of the integrals. Scalar multiplication and order preservation are identical to the one-variable arguments.

The inequality

$$||u| - |v|| \leq |u - v|$$

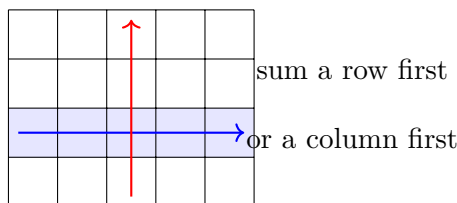
shows from a common partition that $|f|$ is integrable. Since

$$-|f| \leq f \leq |f|,$$

order preservation gives the first estimate. The second follows by integrating the constant functions $-M$ and M . $\square$

12.2 Iterated Integration

The equality of an integral over a rectangle with repeated one-variable integrals is not a new definition. It is a theorem saying that the same finite rectangular approximations can be grouped in either order. The continuity hypothesis keeps the oscillation on every sufficiently small product rectangle under control.



A finite rectangular grid has the same total whether its terms are added row by row or column by column. Riemann–Fubini passes this finite regrouping to a limit; the estimates ensure both limiting operations describe the same integral.

Theorem 12.7 (Riemann–Fubini theorem for continuous functions). Let

$$Q = A \times B,$$

where $A \subseteq \mathbb{R}^p$ and $B \subseteq \mathbb{R}^q$ are rectangles, and let

$$f : Q \longrightarrow \mathbb{R}$$

be continuous. Then the functions

$$x \longmapsto \int_B f(x, y) \, dy, \quad y \longmapsto \int_A f(x, y) \, dx$$

are continuous, and

$$\int_{A \times B} f(x, y) \, d(x, y) = \int_A \left(\int_B f(x, y) \, dy \right) dx = \int_B \left(\int_A f(x, y) \, dx \right) dy.$$

Proof. Uniform continuity of f on the compact rectangle implies that, when

$$x' \rightarrow x,$$

the functions

$$y \longmapsto f(x', y)$$

converge uniformly to

$$y \longmapsto f(x, y).$$

The bounded-function estimate from Chapter 9 therefore proves continuity of the first displayed function; the other is similar.

Let $\eta > 0$. Choose product grids in A and B so fine that the oscillation of f on every product subrectangle is less than

$$\frac{\eta}{\text{vol}(A) \text{vol}(B)}.$$

Choose one point (x_R, y_S) in each product cell $R \times S$, and form the common finite sum

$$\Sigma = \sum_{R, S} f(x_R, y_S) \text{vol}(R) \text{vol}(S).$$

The upper and lower Darboux sums on the product grid differ by less than η , so Σ differs from

$$\int_{A \times B} f$$

by less than η . For each fixed R , the same oscillation estimate shows that

$$\left| \int_B f(x_R, y) \, dy - \sum_S f(x_R, y_S) \text{vol}(S) \right| < \frac{\eta}{\text{vol}(A)}.$$

Multiplying by $\text{vol}(R)$, summing over R , and then using the analogous one-variable-in- x estimate shows that Σ differs from

$$\int_A \left(\int_B f(x, y) \, dy \right) dx$$

by at most 2η . Since η is arbitrary, the first equality follows. Interchanging A and B proves the second. $\square$

Example 12.8 (A double integral on a rectangle). For the continuous function $f(x, y) = x + 2y$ on $[0, 1] \times [0, 2]$, Riemann–Fubini gives

$$\begin{aligned} \int_{[0,1] \times [0,2]} (x + 2y) \, d(x, y) &= \int_0^1 \left(\int_0^2 (x + 2y) \, dy \right) dx \\ &= \int_0^1 (2x + 4) \, dx = 5. \end{aligned}$$

Here the rectangle makes iterated integration natural; no coordinate change is needed.

Theorem 12.9 (Coordinatewise integration by parts). Let

$$Q = [a_1, b_1] \times \cdots \times [a_n, b_n],$$

let u, v be of class C^1 on an open neighborhood of Q , and fix

$$j \in \{1, \dots, n\}.$$

Write

$$Q_{\hat{j}} = \prod_{\substack{1 \leq i \leq n \\ i \neq j}} [a_i, b_i], \quad dx_{\hat{j}} = dx_1 \cdots dx_{j-1} dx_{j+1} \cdots dx_n.$$

Then

$$\int_Q u(x) \frac{\partial v}{\partial x_j}(x) \, dx = \int_{Q_{\hat{j}}} [u(x)v(x)]_{x_j=a_j}^{x_j=b_j} dx_{\hat{j}} - \int_Q v(x) \frac{\partial u}{\partial x_j}(x) \, dx.$$

In the middle integral, the coordinates other than x_j are held fixed while the bracket is evaluated at the two endpoints.

Proof. For fixed $x_{\hat{j}} \in Q_{\hat{j}}$, apply [Theorem 9.27](#) to the C^1 functions of x_j obtained from u and v . The resulting identity has continuous dependence on $x_{\hat{j}}$. Integrating it over $Q_{\hat{j}}$ and using [Theorem 12.7](#) gives the displayed formula. $\square$

Remark 12.10 (Rectangular and intrinsic integration by parts). The preceding formula uses only one-variable integration by parts and Fubini, so it is naturally a statement about rectangles and their coordinate faces. At this point no intrinsic integral over a general boundary has been developed. Chapter 13 replaces the coordinate-face term by an integral over the oriented boundary of a manifold, obtained from the graded Leibniz rule and Stokes' theorem.

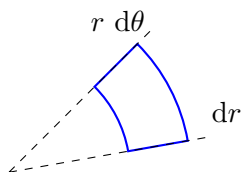
12.3 Jordan Sets and Integration over Bounded Regions

A triangular region is not a rectangle, so our existing definition does not yet integrate over it. Place it inside a rectangle and extend the integrand by zero outside the region. The ordinary rectangular integral then counts exactly the desired portion. The new question is whether the introduced jump along the boundary obstructs Riemann integrability. Jordan measurability isolates the case where it does not. For area itself the integrand is 1 inside, 0 outside: the indicator function.

An early polar preview. For the familiar parameter order (r, θ) , the map

$$(r, \theta) \longmapsto (r \cos \theta, r \sin \theta)$$

takes a tiny parameter rectangle to an annular sector. The radial side has length about dr , the angular side about $r \, d\theta$, so its area is about $r \, dr \, d\theta$. This motivates the factor r before the change-of-variables theorem justifies it. The later rigorous calculation removes the origin and an angular seam, then estimates the omitted pieces.



Definition 12.11 (Jordan measurability). Let E be a bounded subset of $\mathbb{R}^n$ and choose a rectangle Q containing E . The set E is **Jordan measurable** if its indicator

$$\mathbf{1}_E(x) = \begin{cases} 1, & x \in E, \\ 0, & x \notin E \end{cases}$$

is Riemann integrable on Q . Its **Jordan volume** is

$$\text{vol}(E) = \int_Q \mathbf{1}_E(x) \, dx.$$

Definition 12.12 (Jordan-null set). A bounded set $N \subseteq \mathbb{R}^n$ is **Jordan-null** if, for every

$$\varepsilon > 0,$$

it can be covered by finitely many rectangles whose total volume is less than ε .

The word “negligible” has two different meanings in our two integration chapters. Chapter 9 allows countably many covering intervals; Jordan-nullness requires finitely many rectangles. The rationals in $[0, 1]$ are negligible in the former sense, by [Proposition 9.44](#), but are not Jordan-null. A finite union of closed intervals covering this dense set is closed, so it covers its closure $[0, 1]$ and has total length at least one. Countably many small intervals need not have this property.

Lemma 12.13 (Fine grids near a compact set). Let K be compact and let G be an open neighborhood of K in $\mathbb{R}^n$. There is $d > 0$ such that every closed grid cell of diameter less than d meeting K lies in G . If G is a finite union of open rectangles of combined volume v , the total volume of such cells in any fixed finite grid is at most v .

Proof. For each $x \in K$ choose $r_x > 0$ with $B(x, 2r_x) \subset G$. Finitely many balls $B(x, r_x)$ cover K ; let d be the minimum of their radii. If a cell contains $y \in K \cap B(x, r_x)$, then for every point z in the cell, $|z - x| < d + r_x \leq 2r_x$, so $z \in G$. For the volume bound, subdivide at all grid and covering-rectangle faces. Each resulting cell is counted at most once in the grid union and at least once among the covering rectangles. Adding its elementary volume gives the inequality. This is a finite volume comparison, not an assumption that arbitrary neighborhoods can be refined by their faces. $\square$

Proposition 12.14 (Boundary criterion for Jordan measurability). A bounded set $E \subseteq \mathbb{R}^n$ is Jordan measurable if and only if its boundary is Jordan-null.

Proof. Strategy. Put all cells meeting the boundary inside an open cover of small volume, using its positive margin. Choose Q with $\overline{E}$ in its interior. The boundary is compact because it is closed and bounded. If it is Jordan-null, enlarge a finite rectangular cover slightly to make it open while keeping total volume below ε . By [Lemma 12.13](#), a sufficiently fine grid has all its boundary cells inside this cover. A cell missing the boundary is either wholly in E° or wholly in the interior of the complement: otherwise these two relatively open pieces would disconnect the convex cell. The indicator is constant there. Its Darboux gap is consequently at most the combined volume of the boundary cells, less than ε .

Conversely choose a partition whose indicator gap is less than $\varepsilon/2$. Its mixed cells, where the indicator takes both values, have combined volume less than $\varepsilon/2$. Any boundary point in the interior of a grid cell makes that cell mixed by the definition of boundary. The other boundary points lie on finitely many grid faces, coverable by thin boxes of combined volume less than $\varepsilon/2$. Together these rectangles cover the boundary with total volume less than ε . $\square$

Proposition 12.15 (Elementary algebra of Jordan sets). If $E, F \subseteq \mathbb{R}^n$ are Jordan measurable, so are $E \cup F$, $E \cap F$, and $E \setminus F$.

Proof. All three sets are bounded. If $x \notin \partial E \cup \partial F$, choose one ball about x on which membership in E is constant and another on which membership in F is constant. On the smaller ball, membership in each of the three displayed sets is constant. Therefore

$$\partial(E \cup F), \partial(E \cap F), \partial(E \setminus F) \subseteq \partial E \cup \partial F.$$

The right side is Jordan-null: combine finite rectangle covers of its two parts with total volumes below $\varepsilon/2$ each. Each boundary on the left is thus Jordan-null, and [Proposition 12.14](#) proves the claims. $\square$

Proposition 12.16 (Continuous functions on Jordan sets). Let $E \subseteq \mathbb{R}^n$ be bounded with Jordan-null boundary, and let

$$f : E \longrightarrow \mathbb{R}$$

be bounded and continuous for the relative topology on E . Then its zero extension to a containing rectangle is Riemann integrable.

Proof. Strategy. Separate a small boundary neighborhood from a compact region on which the zero extension is continuous. Write $\tilde{f}$ for that extension and choose $M \geq 1$ bounding $|f|$. Cover ∂E by a finite union G of open rectangles of total volume less than $\varepsilon/(4M)$. Choose an open neighborhood H of the boundary with $\overline{H} \subset G$: finitely many small balls with doubled closed balls in G provide one. On $K = Q \setminus H$ the zero extension is continuous, because every point avoids ∂E and locally lies entirely in E° or outside $\overline{E}$. The compact set K therefore has a finite neighborhood cover on which the relative oscillation is less than $\eta = \varepsilon/(2(1 + \text{vol } Q))$. Its Lebesgue number ensures that sufficiently small subsets of K lie in one such neighborhood.

Take a grid fine enough for this number and for the margin between $\overline{H}$ and the complement of G . Every cell meeting $\overline{H}$ lies in G , by [Lemma 12.13](#); these cells contribute at most $2M \text{vol}(G) < \varepsilon/2$ to the gap. Each other cell lies in K and has oscillation below η , so their contribution is at most $\eta \text{vol}(Q) < \varepsilon/2$. The Darboux criterion proves the claim. If the boundary is empty the bounded set is empty, and the zero extension is identically zero. $\square$

Proposition 12.17 (Independence and elementary Jordan sets). The Jordan volume is independent of the containing rectangle. Every rectangle, and every finite union of rectangles with pairwise disjoint interiors, is Jordan measurable; its volume is the sum of the volumes of its rectangles.

Proof. Extending an indicator by zero from one containing rectangle to a larger one changes no upper or lower sum except on a finite family of adjoining rectangles, where the function is identically zero. Common refinements therefore give the same integral.

The indicator of a rectangle is constant except on its finitely many faces. Given

$$\varepsilon > 0,$$

cover these faces by finitely many thin rectangles of total volume less than

$$\varepsilon.$$

Outside that cover, use a grid refining the rectangle's faces; there the upper and lower sums agree. The Darboux criterion proves integrability. The same argument works for a finite union, and additivity gives the volume formula. $\square$

Proposition 12.18 (Null modifications and finite additivity). Let $N \subseteq \mathbb{R}^n$ be bounded and Jordan-null.

- (a) The set N is Jordan measurable and has volume zero.
- (b) A bounded function supported on N , extended by zero to a containing rectangle, is Riemann integrable with integral zero. Consequently, changing a bounded Riemann-integrable function on N does not change its integral.
- (c) Let $E_1, \dots, E_m$ be Jordan measurable and suppose every $E_i \cap E_j$, $i \neq j$, is Jordan-null. Then their union is Jordan measurable and

$$\text{vol} \left(\bigcup_{i=1}^m E_i \right) = \sum_{i=1}^m \text{vol}(E_i).$$

More generally, if h is bounded on the union and the zero extension of $h|_{E_i}$ is integrable for every i , then the zero extension of h from the union is integrable and

$$\int_{\bigcup_i E_i} h = \sum_{i=1}^m \int_{E_i} h.$$

Proof. Given $\varepsilon > 0$, cover N by finitely many closed rectangles of combined volume less than $\varepsilon/2$. Their union also contains $\overline{N}$. Enlarge them to open rectangles with combined volume less than ε . The compact set $\overline{N}$ has a positive margin inside this open union, so [Lemma 12.13](#) puts every sufficiently fine grid cell meeting N inside it. All other cells have indicator zero. The upper sum is at most ε , and lower sums are nonnegative. Hence the indicator is integrable with integral zero.

If $|h| \leq M$ and h is supported on N , then

$$|h| \leq M \mathbf{1}_N.$$

The same grid argument proves integrability, and order preservation gives zero integral. The difference of two functions which agree off N is of this form, proving the modification assertion.

Put

$$N_0 = \bigcup_{i < j} (E_i \cap E_j).$$

If $m \leq 1$, there is nothing to prove. If $m \geq 2$, this finite union is Jordan-null: combine finite covers whose total volumes are each less than $\varepsilon/\binom{m}{2}$. Away from N_0 ,

$$\mathbf{1}_{\cup_i E_i} = \sum_i \mathbf{1}_{E_i}.$$

The right side is integrable, so the null-modification result proves integrability of the left side and the volume formula. The same argument applied to the zero extensions of $h|_{E_i}$ proves the final assertion. $\square$

Definition 12.19 (Integration over a Jordan set). Let $E \subseteq \mathbb{R}^n$ be Jordan measurable and let $f : E \rightarrow \mathbb{R}$ be bounded. If the extension

$$\tilde{f}(x) = \begin{cases} f(x), & x \in E, \\ 0, & x \notin E \end{cases}$$

is integrable on a containing rectangle Q , define

$$\int_E f(x) \, dx = \int_Q \tilde{f}(x) \, dx.$$

Remark 12.20. This definition is intentionally more restrictive than later Lebesgue integration. A bounded function on a Jordan-measurable set need not have an integrable zero extension. The qualification in the definition is therefore mathematical content, not merely a technicality.

12.3.1 Simple planar regions and changing the order

Lemma 12.21 (Riemann–Fubini with integrable sections). Let g be Riemann integrable on $A \times B$, and suppose every section $y \mapsto g(x, y)$ is integrable on B . Then $G(x) = \int_B g(x, y) \, dy$ is integrable on A and $\int_A G = \int_{A \times B} g$.

Proof. For product grids with cells $R \times S$, put $m_{RS} = \inf_{R \times S} g$ and $M_{RS} = \sup_{R \times S} g$. For every $x \in R$,

$$\sum_S m_{RS} \operatorname{vol}(S) \leq G(x) \leq \sum_S M_{RS} \operatorname{vol}(S).$$

Taking infima and suprema on R and summing gives $L(g) \leq L(G) \leq U(G) \leq U(g)$ for these grids. Choose product grids with $U(g) - L(g)$ arbitrarily small. This proves integrability of G and equality of the integrals, since each lies between the same lower and upper sums. $\square$

Theorem 12.22 (Integration between continuous graphs). Let $\alpha, \beta : [a, b] \rightarrow \mathbb{R}$ be continuous, $\alpha \leq \beta$, and

$$D = \{(x, y) : a \leq x \leq b, \alpha(x) \leq y \leq \beta(x)\}.$$

Then D is Jordan measurable, and for continuous $f : D \rightarrow \mathbb{R}$,

$$\int_D f = \int_a^b \left(\int_{\alpha(x)}^{\beta(x)} f(x, y) \, dy \right) dx.$$

The analogous assertion holds with x and y interchanged.

Proof. The boundary of D lies in the two graphs and the two vertical end segments. To cover a continuous graph, partition $[a, b]$ so that its oscillation in each cell is less than η . Rectangles of height at most 2η cover these pieces, with total area at most $2\eta(b - a)$; end segments have covers of arbitrarily small area. Uniform continuity and the boundary criterion give Jordan measurability. Compactness of D and Proposition 12.16 make the zero extension of f integrable on a containing rectangle. Each vertical section is continuous on $[\alpha(x), \beta(x)]$ and zero outside, with at most two junction discontinuities, hence integrable by the one-variable Darboux criterion. Apply Lemma 12.21 to this extension. We have not applied the continuous-rectangle theorem to a discontinuous function. $\square$

Example 12.23 (The same triangle, two orders). For $D = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 1 - x\}$,

$$\int_D x = \int_0^1 \int_0^{1-x} x \, dy \, dx = \frac{1}{6} = \int_0^1 \int_0^{1-y} x \, dx \, dy.$$

Reversing the order means describing the same points with horizontal sections; it does not mean merely exchanging two differential symbols.

Example 12.24 (A triangular zero extension and a split order). For $D = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq x\}$ and $f = x + y$, the zero extension on $[0, 1]^2$ has jumps only along the three boundary segments. The graph-region theorem gives

$$\int_D (x + y) = \int_0^1 \int_0^x (x + y) \, dy \, dx = \int_0^1 \frac{3}{2}x^2 \, dx = \frac{1}{2}.$$

Reversing the order gives $\int_0^1 \int_y^1 (x + y) \, dx \, dy = 1/2$. For a region requiring a split, take $E = \{(x, y) : 0 \leq x \leq 2, 0 \leq y \leq \min(x, 2 - x)\}$. Horizontal slices have $0 \leq y \leq 1, y \leq x \leq 2 - y$; vertical slices require $[0, 1]$ with upper edge x and $[1, 2]$ with upper edge $2 - x$. Thus its area is both

$$\int_0^1 (2 - 2y) \, dy = 1 \quad \text{and} \quad \int_0^1 x \, dx + \int_1^2 (2 - x) \, dx = 1.$$

The switch in the upper boundary at $x = 1$ is why one iterated description must be split.

Lemma 12.25 (Lipschitz images of boxes and faces). If F is L -Lipschitz on a subset of an n -cube of side h , its image fits in a box of side $2L\sqrt{n}h$ (or is a singleton when $L = 0$). The image of a compact $(n - 1)$ -dimensional rectangular face under a Lipschitz map into $\mathbb{R}^n$ is Jordan-null.

Proof. Choose one point of the subset. Every other point has distance at most $\sqrt{n}h$ from it, so every image coordinate differs by at most $L\sqrt{n}h$. This proves the box bound. Partition a face into at most $Ch^{-(n-1)}$ pieces of side at most h . The covering image boxes have total volume at most $C(2L\sqrt{n})^n h$, which tends to zero. For $n = 1$ a face is a point and a small interval covers its image. The same argument handles a finite union of faces. $\square$

12.4 Linear Changes of Variables

The one-variable substitution rule says that $|\phi'(t)|$ measures the local stretching of length. In several variables, the corresponding linear fact comes first: an invertible linear map L sends a small box to a parallelepiped and scales its volume by $|\det L|$. A coordinate scaling by c has factor $|c|$, a diagonal map has the product of its coordinate scalings, a shear has factor 1, and a coordinate reflection preserves volume. Thus the determinant is the natural volume-scaling number, not a formal correction inserted after the fact.

Example 12.26 (A rectangle becomes a parallelogram). Let

$$L(s, t) = (2s + t, s + 3t).$$

The unit square is sent to the parallelogram spanned by $(2, 1)$ and $(1, 3)$. Since

$$\det L = \det \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} = 5,$$

its area is 5. More generally, the image of a rectangle of area A has area $5A$. The linear change-of-variables theorem below turns this geometric calculation into an integral formula.

Lemma 12.27 (Volume under an invertible linear map). Let

$$L : \mathbb{R}^n \longrightarrow \mathbb{R}^n$$

be invertible and let Q be a rectangle. Then $L(Q)$ is Jordan measurable and

$$\text{vol}(L(Q)) = |\det L| \text{vol}(Q).$$

More generally, if $E \subseteq \mathbb{R}^n$ is Jordan measurable, then $L(E)$ is Jordan measurable and

$$\text{vol}(L(E)) = |\det L| \text{vol}(E).$$

Proof. We first record the effect of the three elementary invertible linear maps on Jordan volume. A coordinate permutation merely permutes the side intervals of a rectangle and preserves its volume. Multiplication of one coordinate by $c \neq 0$ takes a rectangle to a rectangle and multiplies its volume by $|c|$.

For the remaining operation, fix $i \neq j$ and consider the shear

$$S(x)_i = x_i + cx_j, \quad S(x)_k = x_k \quad (k \neq i).$$

Let $R = \prod_k [a_k, b_k]$, put $d_i = b_i - a_i$, and write z for the coordinates other than x_i . Above $z \in R_{\widehat{i}}$, the x_i -fiber of $S(R)$ is

$$[\ell(z), \ell(z) + d_i], \quad \ell(z) = a_i + cz_j.$$

Partition $R_{\widehat{i}}$ into finitely many rectangles P on each of which the oscillation $\omega_P(\ell)$ is less than a prescribed γ , where $0 < \gamma < d_i$. Over each P , the fibered set is contained in the rectangular prism

$$P \times [\inf_P \ell, d_i + \sup_P \ell]$$

and contains the prism

$$P \times [\sup_P \ell, d_i + \inf_P \ell].$$

The outer and inner prisms have respective fiber lengths

$$d_i + \omega_P(\ell) \quad \text{and} \quad d_i - \omega_P(\ell).$$

After summing over P , their total volume difference is at most

$$2\gamma \text{vol}(R_{\widehat{i}}).$$

The upper and lower Jordan integrals of $\mathbf{1}_{S(R)}$ are therefore squeezed to $\text{vol}(R)$ as $\gamma \rightarrow 0$. Hence $S(R)$ is Jordan measurable and

$$\text{vol}(S(R)) = \text{vol}(R).$$

These conclusions extend from rectangles to every Jordan-measurable set. Indeed, for such a set E , the Darboux criterion applied to $\mathbf{1}_E$ gives finite unions I and O of grid rectangles with pairwise disjoint interiors such that

$$I \subseteq E \subseteq O, \quad \text{vol}(O) - \text{vol}(I) < \varepsilon.$$

Images of distinct grid rectangles under an elementary map overlap only in images of their common faces. If M is the operator norm of the elementary map, a fixed face can be covered by at most $C_0 h^{-(n-1)}$ face-cubes of side at most h . Each image face-cube lies in an axis-parallel box of side $2M\sqrt{n}h$, so these boxes have total volume at most

$$C_0(2M\sqrt{n})^n h,$$

which tends to zero with h . Thus these overlaps are Jordan-null, and [Proposition 12.18](#) shows that the volumes of the images of I and O are the sums of the volumes of their pieces. The inclusions

$$S(I) \subseteq S(E) \subseteq S(O)$$

then make the upper and lower Jordan integrals of $\mathbf{1}_{S(E)}$ differ by less than ε . The same argument applies to coordinate permutations and coordinate scalings. Consequently, every elementary map sends Jordan-measurable sets to Jordan-measurable sets and multiplies their volumes by the absolute value of its determinant.

Let $K = [0, 1]^n$. [Lemma 10.124](#) supplies elementary matrices $E_1, \dots, E_k$ such that

$$E_k \cdots E_1 L = I.$$

Applying the preceding volume rule successively to the intermediate Jordan sets gives

$$1 = \text{vol}(K) = \left(\prod_{\nu=1}^k |\det E_\nu| \right) \text{vol}(L(K)).$$

Taking determinants in the elimination identity gives

$$\left(\prod_{\nu=1}^k |\det E_\nu| \right) |\det L| = 1.$$

Combining the two equalities yields

$$\text{vol}(L(K)) = |\det L|.$$

The same calculation applies to an arbitrary Jordan-measurable set E . The elementary-map result already proved ensures that every intermediate image is Jordan measurable, and successive application of that result gives

$$\text{vol}(E) = \left(\prod_{\nu=1}^k |\det E_\nu| \right) \text{vol}(L(E)).$$

The determinant identity therefore gives

$$\text{vol}(L(E)) = |\det L| \text{vol}(E).$$

For a general rectangle $Q = t + DK$, where D is diagonal with positive diagonal entries equal to the side lengths of Q , translation preserves Jordan volume and

$$L(Q) = Lt + (LD)(K).$$

Thus

$$\text{vol}(L(Q)) = |\det(LD)| = |\det L| \text{vol}(Q).$$

For completeness, measurability can also be read directly from the boundary. Since L is a homeomorphism,

$$\partial L(Q) = L(\partial Q).$$

Each face of Q can be covered by at most $C_0 h^{-(n-1)}$ face-cubes of side at most h . If $M = \|L\|_{\text{op}}$, the image of each such piece has diameter at most $M\sqrt{n}h$, and hence lies in an axis-parallel n -box of side $2M\sqrt{n}h$. The total volume of these boxes is at most

$$C_0(2M\sqrt{n})^n h,$$

which tends to zero. A finite union over the faces proves that $\partial L(Q)$ is Jordan-null, in agreement with [Proposition 12.14](#). $\square$

Theorem 12.28 (Linear change of variables). Let $L : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be invertible, let E be Jordan measurable, and let

$$f : L(E) \rightarrow \mathbb{R}$$

have an integrable zero extension. Then $L(E)$ is Jordan measurable and

$$\int_{L(E)} f(y) \, dy = \int_E f(Lx) |\det L| \, dx.$$

Proof. Let R be a rectangle and let

$$s = \sum_{\nu=1}^N c_\nu \mathbf{1}_{R_\nu}$$

be a finite sum for the closed cells of a grid partition of R . On shared faces this formula sums all incident terms. This is harmless for its integral, but we do not use those values as pointwise bounds for another function. More generally we call a bounded function a grid step function if it is constant on each cell interior, allowing arbitrary bounded values on the union G of grid faces. Such a function differs from the displayed sum only on the Jordan-null set G , so has the same integral by [Proposition 12.18](#). Its transform differs only on $L(G)$, also Jordan-null by [Lemma 12.25](#). The sets $L(R_\nu)$ are Jordan measurable by [Lemma 12.27](#). Two of them intersect only in the image of a common grid face, which is Jordan-null by [Lemma 12.25](#). Hence [Proposition 12.18](#) gives

$$\int_{L(R)} s(L^{-1}y) \, dy = \sum_{\nu=1}^N c_\nu \text{vol}(L(R_\nu)) = |\det L| \int_R s(x) \, dx.$$

Thus the formula is established first for rectangular step functions; their transforms are finite sums of indicators of the Jordan sets $L(R_\nu)$, not rectangular step functions in the y -coordinates.

Now let $h : R \rightarrow \mathbb{R}$ be Riemann integrable. For every $\varepsilon > 0$, choose a grid whose Darboux gap is less than ε . On the interior of its cell R_ν , put $\ell = \inf_{R_\nu} h$ and $u = \sup_{R_\nu} h$. On the entire face set G , instead put $\ell = -M$ and $u = M$, where $M \geq \sup_R |h|$. These are grid step functions in the preceding sense. The bounds now hold even on shared and exterior faces. Null modification leaves their integrals equal to the lower and upper Darboux sums, so

$$\ell \leq h \leq u, \quad \int_R (u - \ell) < \varepsilon.$$

Their transforms are integrable on $L(R)$, and

$$\ell(L^{-1}y) \leq h(L^{-1}y) \leq u(L^{-1}y).$$

The gap between the transformed upper and lower integrals is

$$|\det L| \int_R (u - \ell) < |\det L| \varepsilon.$$

To make the integrability conclusion explicit, choose Darboux partitions for the two transformed integrable step functions whose upper–lower gaps have sum less than an arbitrary $\delta > 0$, and pass to a common refinement. On that refinement, the upper sum of $h \circ L^{-1}$ is at most the upper sum of $u \circ L^{-1}$, while its lower sum is at least the lower sum of $\ell \circ L^{-1}$. Its Darboux gap is consequently less than

$$|\det L| \varepsilon + \delta.$$

Since ε and δ are arbitrary, $h \circ L^{-1}$ is integrable on $L(R)$. Squeezing its integral between the two transformed step-function integrals proves

$$\int_{L(R)} h(L^{-1}y) \, dy = |\det L| \int_R h(x) \, dx.$$

All integrations here may be read through zero extensions to containing rectangles, as in the definition of integration over a Jordan set.

Apply the same lower–upper construction to $\mathbf{1}_E$ on a rectangle containing E . The transformed indicators squeeze $\mathbf{1}_{L(E)}$, with integral gap multiplied by $|\det L|$. Hence $L(E)$ is Jordan measurable before any integral over it is used.

Finally, choose a rectangle P containing $L(E)$, and let $\tilde{f}$ be the integrable zero extension of f to P . Apply the preceding argument to L^{-1} and $\tilde{f}$. It proves that

$$x \longmapsto \tilde{f}(Lx)$$

has an integrable zero extension from $L^{-1}(P)$, and

$$\int_P \tilde{f}(y) \, dy = |\det L| \int_{L^{-1}(P)} \tilde{f}(Lx) \, dx.$$

The integrands vanish, respectively, outside $L(E)$ and outside E . Using the definition of integration over Jordan sets therefore turns this identity into the asserted formula. $\square$

12.5 Nonlinear Changes of Variables

For a differentiable nonlinear map,

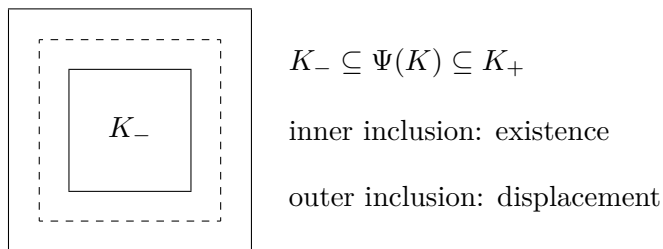
$$\Phi(a + h) = \Phi(a) + D\Phi(a)h + o(\|h\|).$$

Consequently, a sufficiently small box near a behaves approximately like the parallelepiped produced by the linear map $D\Phi(a)$. The progression is therefore

linear volume scaling $\longrightarrow$ local linearization $\longrightarrow |\det D\Phi(a)|$ as local volume distortion.

The following lemma makes this geometric principle quantitative; the theorem then adds the Riemann-sum bookkeeping needed to integrate over a whole region.

Proof architecture. Linear maps scale volume by $|\det L|$. Normalize a nonlinear map by its derivative so that it is close to the identity. A displacement estimate puts the image inside a slightly larger cube; a contraction argument puts a smaller cube inside the image, ruling out holes. Compactness makes these estimates uniform, after which we sum over cells and control thin remainders.



The dashed square is K , the outer square is K_+ , and the image is trapped between the two solid squares.

Lemma 12.29 (Local volume comparison on cubes). Let $U \subseteq \mathbb{R}^n$ be open, let

$$\Phi : U \longrightarrow \mathbb{R}^n$$

be of class C^1 , and suppose that $D\Phi(a)$ is invertible. For every

$$\varepsilon > 0,$$

there is a neighborhood N of a such that, for every closed cube

$$C \subseteq N$$

on which Φ is injective,

$$|\text{vol}(\Phi(C)) - |\det D\Phi(a)| \text{vol}(C)| \leq \varepsilon \text{vol}(C).$$

Moreover, $\Phi(C)$ is Jordan measurable.

Proof. Put

$$A = D\Phi(a).$$

Choose $\delta > 0$ so small that

$$0 < \delta < \min \left\{ 1, \frac{1}{\sqrt{n}} \right\}$$

and

$$|\det A| \max \{ (1 + \delta\sqrt{n})^n - 1, 1 - (1 - \delta\sqrt{n})^n \} < \varepsilon.$$

Continuity of $D\Phi$ permits us to choose an open convex ball N about a , with closure contained in U , such that

$$\|A^{-1}D\Phi(x) - I\|_{\text{op}} < \delta \quad (x \in N).$$

Let $C = c + sK$, where

$$K = [-1/2, 1/2]^n.$$

Define

$$\Psi(u) = s^{-1}A^{-1}(\Phi(c + su) - \Phi(c)) \quad (u \in K).$$

Because $C \subseteq N$, every segment used below lies in C , hence in N . The chain rule gives

$$D\Psi(u) = A^{-1}D\Phi(c + su).$$

Put $g(u) = \Psi(u) - u$. Then

$$\Psi(u) = u + g(u), \quad \|Dg(u)\|_{\text{op}} < \delta.$$

The mean-value estimate in the Euclidean norm gives

$$\|g(u) - g(v)\|_2 \leq \delta \|u - v\|_2.$$

Since

$$\|w\|_{\infty} \leq \|w\|_2 \leq \sqrt{n} \|w\|_{\infty},$$

we also have

$$\|g(u) - g(v)\|_{\infty} \leq \delta\sqrt{n} \|u - v\|_{\infty}.$$

Because $\Psi(0) = 0$, one has $g(0) = 0$, and hence

$$\|g(u)\|_{\infty} \leq \frac{\delta\sqrt{n}}{2} \quad (u \in K).$$

It follows that

$$\Psi(K) \subseteq \left\{ y : \|y\|_{\infty} \leq \frac{1 + \delta\sqrt{n}}{2} \right\}.$$

Conversely, suppose

$$\|y\|_{\infty} \leq \frac{1 - \delta\sqrt{n}}{2}$$

and define $T_y(u) = y - g(u)$. For $u \in K$,

$$\|T_y(u)\|_{\infty} \leq \|y\|_{\infty} + \|g(u)\|_{\infty} \leq \frac{1}{2},$$

so $T_y(K) \subseteq K$. Moreover,

$$\|T_y(u) - T_y(v)\|_2 \leq \delta \|u - v\|_2.$$

Thus T_y is a contraction in the Euclidean norm. The closed cube K is complete, so [Theorem 11.27](#) gives a fixed point $u \in K$, and the fixed-point equation is $\Psi(u) = y$. Therefore

$$\left\{ y : \|y\|_\infty \leq \frac{1 - \delta\sqrt{n}}{2} \right\} \subseteq \Psi(K).$$

The estimate

$$\|\Psi(u) - \Psi(v)\|_2 \geq (1 - \delta)\|u - v\|_2$$

also proves directly that Ψ , and hence Φ , is injective on C .

We next justify Jordan measurability. For $x \in N$,

$$\|I - A^{-1}D\Phi(x)\|_{\text{op}} < 1,$$

so [Proposition 11.29](#), with invertible map A and perturbed map $D\Phi(x)$, shows that $D\Phi(x)$ is invertible. The continuous bijection $\Phi|_C : C \rightarrow \Phi(C)$ is a homeomorphism: a closed subset of the compact set C is compact, its image is compact, and compact subsets of the Hausdorff space $\mathbb{R}^n$ are closed. If $x \in C^\circ$, the inverse function theorem gives a neighborhood $O \subseteq C^\circ$ for which $\Phi(O)$ is open. Thus $\Phi(x)$ is interior to $\Phi(C)$. Since $\Phi(C)$ is compact and therefore closed, this proves

$$\partial\Phi(C) \subseteq \Phi(\partial C).$$

Because $\overline{N}$ is compact and contained in U , continuity of $D\Phi$ gives

$$L = \max_{x \in \overline{N}} \|D\Phi(x)\|_{\text{op}} < \infty.$$

The mean-value estimate makes Φ L -Lipschitz on N . Each face of C can be covered by at most

$$C_0(s/h)^{n-1}$$

face-cubes of side at most h . The image of each lies in an axis-parallel box of side $2L\sqrt{n}h$, so the total volume of these boxes is at most

$$C_0(2L\sqrt{n})^n s^{n-1} h.$$

Letting $h \rightarrow 0$ proves that $\Phi(\partial C)$ is Jordan-null, and [Proposition 12.14](#) shows that $\Phi(C)$ is Jordan measurable.

Finally,

$$\Phi(C) = \Phi(c) + sA\Psi(K).$$

Translation invariance and [Lemma 12.27](#) give

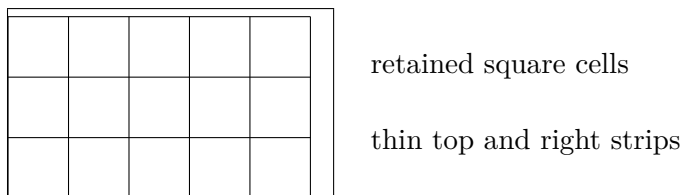
$$\text{vol}(\Phi(C)) = |\det A| s^n \text{vol}(\Psi(K)).$$

The inner and outer cube containments, together with $\text{vol}(C) = s^n$, yield

$$|\det A|(1 - \delta\sqrt{n})^n \text{vol}(C) \leq \text{vol}(\Phi(C)) \leq |\det A|(1 + \delta\sqrt{n})^n \text{vol}(C).$$

The choice of δ proves the estimate. □

In the full theorem keep four errors separate: variation of f on image cells, error in local volume scaling, the source Riemann-sum error, and the contribution of discarded thin strips. Each estimate below pays for one of these errors.



Theorem 12.30 (Change of variables). Let $Q \subseteq \mathbb{R}^n$ be a rectangle, let U be an open neighborhood of Q , and let

$$\Phi : U \longrightarrow \mathbb{R}^n$$

be of class C^1 . Assume that Φ is injective on Q and that

$$\det D\Phi(x) \neq 0 \quad (x \in Q).$$

Then $\Phi(Q)$ is Jordan measurable. If

$$f : \Phi(Q) \longrightarrow \mathbb{R}$$

is continuous, then

$$\int_{\Phi(Q)} f(y) \, dy = \int_Q f(\Phi(x)) |\det D\Phi(x)| \, dx.$$

Proof. Write

$$J(x) = |\det D\Phi(x)|.$$

The restriction $\Phi|_Q : Q \rightarrow \Phi(Q)$ is a homeomorphism. Indeed, it is a continuous bijection, Q is compact, and $\Phi(Q) \subseteq \mathbb{R}^n$ is Hausdorff. More explicitly, a closed subset of Q is compact, its image is compact, and compact subsets of $\mathbb{R}^n$ are closed; hence the inverse is continuous. If $x \in Q^\circ$, the inverse function theorem gives an open neighborhood $O \subseteq Q^\circ$ such that $\Phi(O)$ is open. Therefore $\Phi(x)$ is interior to $\Phi(Q)$, and

$$\partial\Phi(Q) \subseteq \Phi(\partial Q).$$

We shall use one uniform Lipschitz constant. Since Q is compact and U is open, there is $\tau > 0$ such that the closed τ -neighborhood of Q lies in U . Put

$$K_\tau = \{x \in \mathbb{R}^n : \text{dist}(x, Q) \leq \tau\}.$$

This compact set is contained in U . The open τ -neighborhood W is convex because Q is a rectangle, and $W \subseteq K_\tau$. Put

$$L = \max_{x \in K_\tau} \|D\Phi(x)\|_{\text{op}}.$$

Then [Theorem 11.17](#) shows that Φ is L -Lipschitz on W . Every face of Q can be covered by at most $C_0 h^{-(n-1)}$ face-cubes of side at most h , where C_0 is independent of h . Their images lie in n -boxes of side $2L\sqrt{n}h$, whose total volume is at most

$$C_0(2L\sqrt{n})^n h.$$

After summing over the finitely many faces and letting $h \rightarrow 0$, $\Phi(\partial Q)$ is Jordan-null. By the boundary inclusion just proved and [Proposition 12.14](#), $\Phi(Q)$ is Jordan measurable before any integral over it is used.

We also need a uniform estimate for images of thin rectangles. If $R \subseteq Q$ is a nondegenerate rectangle, cover it, for arbitrarily small $d > 0$, by grid cubes of side d . Taking d smaller than every

side length of R and smaller than $\tau/\sqrt{n}$, the covering cubes lie in W , and the sum of their volumes is at most

$$2^n \operatorname{vol}(R).$$

Each image cube lies in an axis-parallel box of side $2L\sqrt{n}d$. The same boundary argument as above shows that $\Phi(R)$ is Jordan measurable, and the covering estimate gives

$$\operatorname{vol}(\Phi(R)) \leq C_L \operatorname{vol}(R), \quad C_L = 2^n (2L\sqrt{n})^n.$$

Degenerate rectangles have Jordan-null images by the face estimate.

The functions J and

$$H(x) = f(\Phi(x))J(x)$$

are continuous on Q . Let

$$B = \max_{y \in \Phi(Q)} |f(y)|, \quad M_H = \max_{x \in Q} |H(x)|.$$

For $t \geq 0$, let $\omega_f(t)$ and $\omega_H(t)$ be the suprema of $|f(y) - f(z)|$ and $|H(x) - H(w)|$, respectively, over pairs at distance at most t in their compact domains. Uniform continuity gives

$$\lim_{t \rightarrow 0} \omega_f(t) = 0, \quad \lim_{t \rightarrow 0} \omega_H(t) = 0.$$

Fix $\varepsilon > 0$. Choose $\eta > 0$ so small that

$$B\eta \operatorname{vol}(Q) < \frac{\varepsilon}{4}.$$

If $B = 0$, any positive η satisfies this requirement. For each $a \in Q$, apply [Lemma 12.29](#) with error $\eta/2$, and then shrink its neighborhood $N_a \subseteq W$ so that

$$|J(x) - J(a)| < \frac{\eta}{2} \quad (x \in N_a \cap Q).$$

Choose finitely many of the sets $N_a \cap Q$ which cover Q . By [Lemma 10.33](#), there is $\lambda > 0$ such that every subset of Q of diameter less than λ lies in one member of this finite cover. Hence, if $C \subseteq Q$ is a cube with $\operatorname{diam}(C) < \lambda$, and $x_C \in C$, then for a suitable center a ,

$$\begin{aligned} |\operatorname{vol}(\Phi(C)) - J(x_C) \operatorname{vol}(C)| &\leq |\operatorname{vol}(\Phi(C)) - J(a) \operatorname{vol}(C)| \\ &\quad + |J(a) - J(x_C)| \operatorname{vol}(C) \\ &\leq \eta \operatorname{vol}(C). \end{aligned}$$

Here Φ is injective on C because it is injective on Q .

Write

$$Q = \prod_{i=1}^n [a_i, b_i], \quad d_i = b_i - a_i.$$

Choose $h > 0$, smaller than every d_i , and put

$$m_i = \left\lfloor \frac{d_i}{h} \right\rfloor, \quad Q_h = \prod_{i=1}^n [a_i, a_i + m_i h].$$

The rectangle Q_h is the union of finitely many closed cubes of side h , called the retained cubes. For $1 \leq i \leq n$, set

$$R_{h,i} = \left(\prod_{k < i} [a_k, a_k + m_k h] \right) \times [a_i + m_i h, b_i] \times \left(\prod_{k > i} [a_k, b_k] \right),$$

and put

$$R_h = \bigcup_{i=1}^n R_{h,i}.$$

We have $Q_h \cup R_h = Q$, and $Q_h \cap R_h$ is contained in finitely many coordinate faces. If d_i is an exact multiple of h , the corresponding remainder rectangle can have zero width; it is still Jordan-null and does not affect these assertions or the estimates below. The rectangles $R_{h,i}$ have pairwise Jordan-null overlaps: two distinct pieces can meet only where a coordinate equals an endpoint of a shortened interval. A telescoping product estimate gives

$$\text{vol}(R_h) = \text{vol}(Q) - \text{vol}(Q_h) \leq C_Q h, \quad C_Q = \sum_{i=1}^n \prod_{k \neq i} d_k.$$

By the rectangle-image bound, finite additivity, and the nullity of the overlaps,

$$\text{vol}(\Phi(R_h)) \leq C_L \text{vol}(R_h).$$

Choose h small enough that

$$\sqrt{n} h < \lambda, \quad \omega_f(L\sqrt{n} h) \text{vol}(\Phi(Q)) < \frac{\varepsilon}{4}, \quad \omega_H(\sqrt{n} h) \text{vol}(Q) < \frac{\varepsilon}{4},$$

and

$$(BC_L + M_H)C_Q h < \frac{\varepsilon}{4}.$$

For each retained cube C , choose a tag $x_C \in C$ and put $J_C = J(x_C)$. Then

$$\begin{aligned} & \int_{\Phi(C)} f(y) \, dy - f(\Phi(x_C)) J_C \text{vol}(C) \\ &= \int_{\Phi(C)} (f(y) - f(\Phi(x_C))) \, dy \\ & \quad + f(\Phi(x_C)) (\text{vol}(\Phi(C)) - J_C \text{vol}(C)). \end{aligned}$$

Since

$$\text{diam}(\Phi(C)) \leq L \text{diam}(C) \leq L\sqrt{n} h,$$

the bounded-function estimate and the local tag estimate above imply

$$\begin{aligned} & \left| \int_{\Phi(C)} f(y) \, dy - f(\Phi(x_C)) J_C \text{vol}(C) \right| \\ & \leq \omega_f(L\sqrt{n} h) \text{vol}(\Phi(C)) + B\eta \text{vol}(C). \end{aligned}$$

Distinct retained cubes meet only on faces. Injectivity gives

$$\Phi(C) \cap \Phi(C') = \Phi(C \cap C'),$$

and the Lipschitz face estimate makes this intersection Jordan-null. Therefore [Proposition 12.18](#) gives

$$\int_{\Phi(Q_h)} f = \sum_C \int_{\Phi(C)} f, \quad \text{vol}(\Phi(Q_h)) = \sum_C \text{vol}(\Phi(C)).$$

Summing this local estimate yields

$$\begin{aligned} \left| \int_{\Phi(Q_h)} f - \sum_C f(\Phi(x_C)) J_C \text{vol}(C) \right| \\ \leq \omega_f(L\sqrt{n}h) \text{vol}(\Phi(Q)) + B\eta \text{vol}(Q). \end{aligned}$$

Similarly, continuity of H on each retained cube gives

$$\left| \sum_C H(x_C) \text{vol}(C) - \int_{Q_h} H(x) \, dx \right| \leq \omega_H(\sqrt{n}h) \text{vol}(Q).$$

Finally, $Q = Q_h \cup R_h$, and their overlap is Jordan-null. Their images also have Jordan-null overlap, by injectivity and the Lipschitz face estimate. Finite additivity and the bounded-function estimate give

$$\left| \int_{\Phi(R_h)} f \right| \leq B \text{vol}(\Phi(R_h)) \leq BC_L \text{vol}(R_h)$$

and

$$\left| \int_{R_h} H \right| \leq M_H \text{vol}(R_h).$$

Combining these two estimates with the summed local estimate and the Riemann-sum estimate, and observing that $H(x_C) = f(\Phi(x_C))J_C$, gives

$$\begin{aligned} \left| \int_{\Phi(Q)} f - \int_Q H \right| &\leq \omega_f(L\sqrt{n}h) \text{vol}(\Phi(Q)) + B\eta \text{vol}(Q) \\ &\quad + \omega_H(\sqrt{n}h) \text{vol}(Q) + (BC_L + M_H) \text{vol}(R_h) \\ &< \varepsilon \end{aligned}$$

by the choices of η and h , together with the strip-volume bound. Since $\varepsilon > 0$ was arbitrary, the change-of-variables formula follows. $\square$

Remark 12.31 (A working strategy for multiple integrals). First identify the domain. On a rectangle, iterated integration is often the simplest computational tool. For a nonrectangular region, ask whether it decomposes into simple pieces or suggests coordinates adapted to its geometry. For a proposed change of variables, compute $D\Phi$ and $|\det D\Phi|$, transform both the integrand and the domain, and verify the injectivity and nonvanishing-Jacobian hypotheses (allowing only carefully controlled boundary exceptions). Keep two tasks distinct: continuity, Darboux estimates, and Jordan measurability address existence of an integral; iterated integration and change of variables usually address its calculation.

Corollary 12.32 (Change of variables for compact Jordan regions). If $\Phi : U \rightarrow V$ is a C^1 diffeomorphism of open sets, $E \subset U$ is compact and Jordan measurable, and f is continuous on $\Phi(E)$, then $\Phi(E)$ is Jordan measurable and

$$\int_{\Phi(E)} f = \int_E f(\Phi(x)) |\det D\Phi(x)| \, dx.$$

Proof. 1. Control the image boundary. If $E = \emptyset$ both integrals are zero. Otherwise choose $\delta > 0$ such that the closed δ -neighborhood of E is compact and lies in U . This is possible by a finite ball cover of the compact set E . Continuity of $D\Phi$ gives a finite bound L there, enlarged to be positive. On any sufficiently small cube meeting E , every segment in the cube stays in that neighborhood, so the mean-value estimate bounds image distances by L times source distances. The image of a cube of side h lies in a cube of side $2L\sqrt{n}h$. Thus finitely many such source cubes of total volume v have images with outer volume at most Cv , where $C = (2L\sqrt{n})^n$.

The compact boundary ∂E is Jordan-null. Given $\eta > 0$, cover it by finitely many open rectangles of total volume less than η . Compactness gives a positive distance from ∂E to the complement of their union (if necessary intersect with a fixed bounded neighborhood). For a sufficiently fine grid, every cell meeting ∂E lies in that union. Their total volume v_h is at most η by the finite grid comparison, so $v_h \rightarrow 0$. The preceding image estimate covers $\Phi(\partial E)$ by boxes of total volume at most $Cv_h \rightarrow 0$. A homeomorphism between open sets preserves relative interior and closure; since E and $\Phi(E)$ are compact subsets of those open sets, their relative boundaries equal their ambient boundaries. Hence $\partial\Phi(E) = \Phi(\partial E)$. The boundary criterion proves that $\Phi(E)$ is Jordan measurable.

2. Separate retained cells from boundary cells. Take a finite grid in a containing rectangle. Retain every closed cell lying wholly in E and let A_h be their union. A cell meeting E but not retained has a point of E and a point outside E ; along the connecting segment it must meet ∂E (closedness of E supplies the last point still in E). Thus $E \setminus A_h$ is contained in the boundary cells and has volume at most v_h . Its image has outer volume at most Cv_h . The retained cells meet only on faces; injectivity and the Lipschitz face estimate make their image overlaps Jordan-null. Their unions and differences with E and $\Phi(E)$ are Jordan measurable by [Proposition 12.15](#).

3. Add local formulas and bound the discarded pieces. On each retained cell apply the rectangular change-of-variables theorem, and add using the null-overlap rule. With $g(x) = f(\Phi(x))|\det D\Phi(x)|$, this gives

$$\int_{\Phi(A_h)} f = \int_{A_h} g.$$

The function f is bounded on $\Phi(E)$, say by M_f , and g is continuous and bounded on E , say by M_g . Their integrals exist by continuous integrability on compact Jordan regions. The error estimates are

$$\left| \int_{\Phi(E)} f - \int_{\Phi(A_h)} f \right| \leq M_f Cv_h, \quad \left| \int_E g - \int_{A_h} g \right| \leq M_g v_h.$$

Both errors tend to zero. Taking limits in the finite sum identity proves the corollary without extending f outside its given domain. $\square$

Corollary 12.33 (Change of variables on open sets with compact support). Let $\Phi : U \rightarrow V$ be a C^1 diffeomorphism of open subsets of $\mathbb{R}^n$ and let $f : V \rightarrow \mathbb{R}$ be continuous with compact support in V . Integrals are defined by continuous zero extension to any containing rectangle. Then

$$\int_V f(y) \, dy = \int_U f(\Phi(x)) |\det D\Phi(x)| \, dx.$$

If $\det D\Phi > 0$, the absolute value may be omitted; if it is negative everywhere, omitting it reverses the sign.

Proof. Idea. Only a compact part of each open set matters, so finitely many small closed cubes reduce the assertion to the rectangular theorem. Let $K = \text{supp } f$. Its inverse image $L = \Phi^{-1}(K)$ is compact in U . Choose a sufficiently fine grid so that the finitely many closed cubes meeting L

lie in U ; their union E contains a neighborhood of L . Distinct cubes meet only on faces. On a compact neighborhood of E , the derivative is bounded, so their image faces are Jordan-null by the face-cover estimate in the change-of-variables proof. Each cube and its image obey [Theorem 12.30](#); add these finitely many identities. Both integrands vanish outside E and $\Phi(E)$ respectively, so finite additivity identifies the sums with the open-set integrals. This also proves independence of the containing rectangles. The signed cases follow directly. $\square$

Corollary 12.34 (The half-space version). The same formula holds for a C^1 diffeomorphism between relatively open subsets of $\mathbb{H}^n = \{x_n \geq 0\}$, smooth across the boundary locally, with smooth inverse and preserving the boundary face. A continuous compactly supported integrand is integrated over a containing half-rectangle, extending by zero only across artificial chart boundaries.

Proof. Such an extension is continuous within the half-space, so its integral over a half-rectangle exists. For $\delta > 0$ let $\chi_\delta(t) = \min\{1, \max\{0, (t - \delta)/\delta\}\}$. Apply the open-set corollary on the interiors to the target function $f(y)\chi_\delta((\Phi^{-1}(y))_n)$. It has compact support away from the boundary. The source error is supported in a fixed bounded box intersected with $0 \leq x_n \leq 2\delta$, and is bounded, hence its integral tends to zero. The target error is likewise bounded on $K = \text{supp } f$. By compactness of $\Phi^{-1}(K)$ and boundary preservation, its support lies in $0 \leq y_n \leq \eta(\delta)$ with $\eta(\delta) \rightarrow 0$. Indeed a sequence contradicting this would have a subsequence converging to a source boundary point with target height bounded below. The target error therefore also has integral tending to zero by the bounded-integral estimate. Pass to the limit. No zero extension across the genuine boundary is required to be continuous. $\square$

12.5.1 Alternative Viewpoint: Pullbacks and Manifold Integration

The proof of [Theorem 12.30](#) is necessarily detailed because it establishes an analytic fact in the Riemann/Jordan setting. Its geometric core is simple. Differentiability gives

$$\Phi(x + h) = \Phi(x) + D\Phi(x)h + o(\|h\|),$$

so $D\Phi(x)$ is the infinitesimal linear approximation to Φ . For a linear map, $\det L$ records oriented volume scaling and $|\det L|$ records unsigned volume scaling; see the determinant properties in [Theorem 10.115](#) and the linear volume result [Lemma 12.27](#). Thus

$$|\det D\Phi(x)|$$

is not an arbitrary correction factor: it is the local unsigned volume distortion of the derivative. The proof above makes this statement quantitative by comparing small cubes with their linearized images and then passing to Riemann sums.

Accordingly, the scalar formula just proved has the schematic form

$$\int_V f(y) \, dy = \int_U f(\Phi(x)) |\det D\Phi(x)| \, dx.$$

The absolute value occurs because ordinary scalar volume is unsigned. Chapter 13 gives a complementary interpretation. For a top-degree form on V ,

$$\omega = f(y) \, dy_1 \wedge \cdots \wedge dy_n,$$

the pullback by a smooth map $\Phi : U \rightarrow V$ is

$$\Phi^*\omega = f(\Phi(x)) \det D\Phi(x) \, dx_1 \wedge \cdots \wedge dx_n.$$

Indeed,

$$\Phi^*(dy_j) = \sum_{i=1}^n \frac{\partial \Phi^j}{\partial x_i} dx_i,$$

and the alternating multilinearity of the wedge product produces the determinant. In this language, the determinant appears automatically when a top-degree form is transported to the parameter domain.

The sign is now meaningful. For an orientation-preserving diffeomorphism, the oriented formula is

$$\int_V \omega = \int_U \Phi^* \omega;$$

an orientation-reversing diffeomorphism changes the sign. This distinction between $\det D\Phi$ and $|\det D\Phi|$ is why orientation is part of the natural language of integration on manifolds. One can formulate unsigned integration intrinsically using densities, but that theory is beyond the scope of these notes.

This pullback identity is the form of change of variables that survives on a smooth manifold, where no single global coordinate system need exist. Chapter 13 develops pullbacks and top-degree forms in [Section 13.7](#), and its change-of-parametrization result [Proposition 13.42](#) uses the present theorem to establish coordinate independence. The logical order is essential:

$$\begin{aligned} & \text{Euclidean change of variables} \\ \implies & \text{coordinate invariance of form integration} \\ \implies & \text{well-defined integration on manifolds.} \end{aligned}$$

Thus the differential-form discussion is a conceptual, coordinate-invariant reformulation, not an independent proof of the Euclidean analytic theorem.

Finally, Lebesgue integration later provides a substantially more flexible change-of-variables theory for measurable sets and functions under weaker hypotheses. It replaces much of the present Jordan-boundary and Riemann-sum bookkeeping by measurable sets, null sets, Lebesgue integration, and countable decompositions. Measure theory is not introduced here; see [Section 13.13](#) for its place among the natural continuations of these notes.

Example 12.35 (Polar coordinates and the unit disk). Consider

$$\Phi(\theta, r) = (r \cos \theta, r \sin \theta).$$

Its derivative and Jacobian are

$$D\Phi(\theta, r) = \begin{pmatrix} -r \sin \theta & \cos \theta \\ r \cos \theta & \sin \theta \end{pmatrix}, \quad \det D\Phi(\theta, r) = -r, \quad |\det D\Phi(\theta, r)| = r.$$

The order (θ, r) reverses orientation, while the scalar area factor remains positive. The factor r is geometric: an angular change $d\theta$ at radius r sweeps an arc of length approximately $r d\theta$, while the radial change has length approximately dr .

To use the theorem without hiding its hypotheses, first take

$$Q_{\varepsilon, \eta} = [\eta, 2\pi - \eta] \times [\varepsilon, 1]$$

with $\varepsilon, \eta > 0$. On this rectangle the map is injective and its Jacobian never vanishes. Its image is an annular sector. For example, since $x^2 + y^2 = r^2$,

$$\begin{aligned} \int_{\Phi(Q_{\varepsilon, \eta})} (x^2 + y^2) \, d(x, y) &= \int_{\eta}^{2\pi - \eta} \int_{\varepsilon}^1 r^2 r \, dr \, d\theta \\ &= \frac{(2\pi - 2\eta)(1 - \varepsilon^4)}{4}. \end{aligned}$$

Letting $\varepsilon, \eta \rightarrow 0^+$ gives

$$\int_{x^2 + y^2 \leq 1} (x^2 + y^2) \, d(x, y) = \frac{\pi}{2}.$$

These discarded-area bounds must not use the disk-area formula we are about to derive. The central disk lies in $[-\varepsilon, \varepsilon]^2$, so its area is at most $4\varepsilon^2$. For $0 < \eta < \pi/2$, the omitted angular seam has $0 \leq x \leq 1$ and $|y| \leq \sin \eta \leq \eta$; it lies in $[0, 1] \times [-\eta, \eta]$, of area 2η . Their combined outer area is at most $4\varepsilon^2 + 2\eta \rightarrow 0$. The integrand is bounded by 1, so the discarded integrals obey the same bound. The unit circle is Jordan-null independently, by the Lipschitz image-of-an-interval estimate applied to $(\cos t, \sin t)$ on $[0, 2\pi]$; thus the disk's Jordan measurability also precedes its area calculation. The same calculation with integrand 1 gives the area of the unit disk:

$$\int_{x^2 + y^2 \leq 1} 1 \, d(x, y) = \pi.$$

12.6 Uniform Limits and Multiple Integration

The same principle from Chapter 9 survives unchanged on a rectangle: a uniform error in the integrand is at most that error times the volume of the domain after integration. Pointwise convergence alone is not the Riemann theory's general interchange principle.

Theorem 12.36 (Uniform interchange for multiple Riemann integrals). Let $Q \subseteq \mathbb{R}^n$ be a rectangle. Suppose that every function

$$f_k : Q \longrightarrow \mathbb{R}$$

is Riemann integrable and that f_k converges uniformly to f on Q . Then f is Riemann integrable and

$$\lim_{k \rightarrow \infty} \int_Q f_k(x) \, dx = \int_Q f(x) \, dx.$$

Moreover,

$$\lim_{k \rightarrow \infty} \int_Q |f_k(x) - f(x)| \, dx = 0.$$

Proof. The oscillation argument in the proof of [Theorem 9.11](#) applies to grid subrectangles, using [Proposition 12.3](#). Once integrability of f is known, [Theorem 12.6](#) gives

$$\left| \int_Q f_k \, dx - \int_Q f \, dx \right| \leq \int_Q |f_k - f| \, dx \leq \text{vol}(Q) \sup_{x \in Q} |f_k(x) - f(x)|.$$

The right-hand side tends to zero, and the same estimate gives the final claim. $\square$

Remark 12.37 (Uniform convergence and bounded convergence). This section records the clean uniform theorem in several variables: uniform convergence both guarantees integrability of the limit and permits interchange with integration. Chapter 9 also proved the one-dimensional Arzelà

theorem, [Theorem 9.16](#): uniformly bounded pointwise convergence gives integral convergence when the limit is separately assumed Riemann integrable. Uniform convergence is therefore not the only Riemann interchange mechanism proved in these notes. We do not assert a multivariable Arzelà theorem here.

A nonnegative integrable dominating function does not by itself supply the hypotheses of the uniform theorem above. The Lebesgue Dominated Convergence Theorem later handles almost-everywhere convergence under integrable domination and also establishes integrability of the limit in its measurable setting. That theorem requires a different integration theory.

As in one variable, pointwise convergence alone does not control a Riemann integral. On a compact rectangle, continuity and monotonicity provide the extra structure that can force uniform convergence.

Theorem 12.38 (Dini's theorem). Let $Q \subseteq \mathbb{R}^n$ be a rectangle, and let

$$f_k : Q \longrightarrow \mathbb{R}$$

be continuous functions satisfying

$$f_k(x) \leq f_{k+1}(x) \quad (x \in Q, k \in \mathbb{N}).$$

If f_k converges pointwise on Q to a continuous function f , then the convergence is uniform and

$$\lim_{k \rightarrow \infty} \int_Q f_k(x) \, dx = \int_Q f(x) \, dx.$$

The same conclusion holds for a pointwise decreasing sequence.

Proof. Put $h_k = f - f_k$. If h_k did not converge uniformly to zero, there would be $\varepsilon > 0$ and $x_k \in Q$ with

$$h_k(x_k) \geq \varepsilon.$$

By compactness of Q and [Theorem 10.32](#), some subsequence x_{k_j} tends to a point $x \in Q$. For every fixed m , monotonicity gives

$$h_m(x_{k_j}) \geq h_{k_j}(x_{k_j}) \geq \varepsilon$$

for all sufficiently large j . Continuity yields $h_m(x) \geq \varepsilon$ for every m , contradicting $h_m(x) \rightarrow 0$. Thus the convergence is uniform, and [Theorem 12.36](#) gives the formula. The decreasing case follows by negation. $\square$

Remark 12.39 (Dini versus Lebesgue monotone convergence). This is Dini's theorem on a compact rectangle: continuity, compactness, and monotonicity upgrade pointwise convergence to uniform convergence, after which [Theorem 12.36](#) gives the integral formula. The full multivariable Lebesgue Monotone Convergence Theorem instead assumes nonnegative measurable functions and does not require continuity or uniform convergence; it lies beyond the present Riemann framework.

Exercises

Exercise 12.1. Prove that replacing a grid partition by a refinement decreases its upper sum and increases its lower sum.

Exercise 12.2. Compute

$$\int_{\{0 \leq y \leq 1, y \leq x \leq y+2\}} (x + 2y) \, d(x, y)$$

directly from iterated one-variable integrals.

Exercise 12.3. Use [Theorem 12.9](#) with $j = 1$ to evaluate

$$\int_{[0,1]^2} x \frac{\partial}{\partial x} (x^2 y) \, d(x, y).$$

Exercise 12.4. Prove that a constant function c on Q has integral

$$c \operatorname{vol}(Q).$$

Exercise 12.5. Show that a finite set in $\mathbb{R}^n$ has Jordan volume zero.

Exercise 12.6. Prove that the open unit ball in $\mathbb{R}^n$ is Jordan measurable, assuming that its boundary can be covered by finitely many boxes of arbitrarily small total volume.

Exercise 12.7. Give a bounded subset of $[0, 1]$ whose indicator is not Riemann integrable.

Exercise 12.8. Use [Theorem 12.28](#) to compute the area of the parallelogram spanned by two vectors in $\mathbb{R}^2$.

Exercise 12.9. Verify the determinant computation in the polar-coordinate example.

Exercise 12.10. For $a, b > 0$, use a linear map and polar coordinates to compute

$$\int_{\{x^2/a^2 + y^2/b^2 \leq 1\}} \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} \right) d(x, y).$$

Transform the domain and both scaling factors. Justify the exceptional origin and angular seam by truncation and bounds, as in the disk example.

12.7 Limits of the Riemann Theory

The Riemann integral handles continuous functions and geometrically regular domains very effectively. Its limitations become sharper in several variables: indicators of sets with complicated boundaries may fail to be integrable, iterated integrals need not be available for arbitrary bounded functions, and the present change-of-variables theorem requires continuity and a globally injective parametrization. Lebesgue measure and integration replace Jordan volume by a much more flexible notion of size and later permit theorems under substantially weaker hypotheses. Those extensions are not developed in these notes.

Chapter 13

Differential Forms, Stokes' Theorem, and Calculus on Manifolds

13.1 Parametrized Curves and Line Integrals

Integration can accumulate mass along a wire or work along a path. The same geometric curve supports both questions, but they require different integrands: mass uses length, while work records direction. We start with these familiar quantities before introducing the objects that unify their boundary formulas.

13.1.1 A curve is a traversal, not just a set

The maps $t \mapsto (\cos t, \sin t)$ on $[0, 2\pi]$ and $t \mapsto (\cos 2t, \sin 2t)$ on the same interval have the same image, but the second traverses it twice. Integrals can distinguish these maps. The velocity $\gamma'(t)$ is a tangent vector; its norm is speed. For a short time Δt , the displacement is approximately $\gamma'(t)\Delta t$ and the distance traveled is approximately $\|\gamma'(t)\|\Delta t$. This suggests length as accumulated speed.

For a scalar density f , multiply each small length by f to obtain a scalar line integral. For a force field F , multiply displacement by the component of force in its direction: $F \cdot \gamma'\Delta t$. This gives work. These two constructions use different information, although both are computed by a one-variable integral.

Definition 13.1 (Parametrized curve and velocity). A C^1 parametrized curve is a map $\gamma : [a, b] \rightarrow \mathbb{R}^n$ with continuous derivative, using one-sided derivatives at endpoints. Its velocity at time t is $\gamma'(t)$, and it is **regular** if this vector is never zero. Its orientation is the direction of increasing parameter. Its image is the set $\gamma([a, b])$, which does not record the speed or multiplicity of traversal.

For $\gamma(t) = (t, t^2)$ the velocity is $(1, 2t)$ and the speed $\sqrt{1 + 4t^2}$ varies with time. A tangent vector is a displacement direction; speed is a nonnegative scalar.

Definition 13.2 (Smooth-curve length). We define the length of a C^1 curve by

$$\ell(\gamma) = \int_a^b \|\gamma'(t)\|_2 \, dt.$$

For continuous piecewise C^1 curves, add the lengths of the finitely many smooth pieces. Refining that subdivision does not change the sum, by additivity of the one-variable integral. No equivalence with a polygonal-supremum definition is assumed here.

To integrate a scalar function along the curve, the small length is weighted by the scalar value at the visited point.

Definition 13.3 (Scalar line integral). For continuous f along the image of γ , set

$$\int_{\gamma} f \, ds = \int_a^b f(\gamma(t)) \|\gamma'(t)\|_2 \, dt.$$

Taking $f = 1$ recovers length. Negative values of f contribute negatively.

Work instead pairs a vector field with the directed displacement.

Definition 13.4 (Work integral). For a continuous vector field F along the image, set

$$\int_{\gamma} F \cdot dr = \int_a^b F(\gamma(t)) \cdot \gamma'(t) \, dt.$$

Both kinds of line integral extend to piecewise C^1 curves by finite addition over their smooth pieces.

Proposition 13.5 (Reparametrization and reversal). Let $h : [c, d] \rightarrow [a, b]$ be a C^1 diffeomorphism and set $\tilde{\gamma} = \gamma \circ h$. The scalar line integral and length are unchanged. The work integral is unchanged if h is increasing and changes sign if h is decreasing.

Proof. The chain rule gives $\tilde{\gamma}' = \gamma'(h)h'$, and hence $\|\tilde{\gamma}'\| = \|\gamma'(h)\||h'|$. If h is increasing, substitution in Chapter 9 gives both assertions. If it is decreasing, substitution with h' changes the limits from b to a . For work this contributes a minus sign. For the scalar integral, $|h'| = -h'$ contributes a second minus sign, cancelling the first. $\square$

Thus $\int_C f \, ds$ is independent of orientation, but it need not be positive: if $f = -1$ on a non-constant curve, it equals minus the length. In contrast, $\int_C F \cdot dr$ is signed. The notation C presupposes a specified traversal up to these reparametrizations: tracing a curve twice generally doubles its integral and is not a diffeomorphic reparametrization.

Example 13.6 (Changing speed and reversing direction). Let $\gamma(t) = (t, t^2)$ for $0 \leq t \leq 1$ and $h(s) = (s + s^2)/2$ on $[0, 1]$. Then $h'(s) = (1 + 2s)/2 > 0$, so h is an increasing diffeomorphism with the same endpoint values. The new velocity is $(h'(s), 2h(s)h'(s))$; the traversal speeds differ, but substitution shows both lengths equal $\int_0^1 \sqrt{1 + 4t^2} \, dt$. For $F = (1, 0)$, both work integrals are 1: in the new parameter the integral is $\int_0^1 h'(s) \, ds = h(1) - h(0)$. Using $h(s) = 1 - s$ instead reverses direction and gives work -1 . The scalar integral of $f = -1$ stays the same negative length.

Example 13.7 (One running circulation). For $F(x, y) = (-y, x)$ and the counterclockwise unit circle $\gamma(t) = (\cos t, \sin t)$, $0 \leq t \leq 2\pi$, the velocity is $F(\gamma(t))$. Thus its length and circulation are both 2π . Reversal keeps the length and negates the circulation. The expression $x \, dy - y \, dx$ will reappear as a differential form later in this chapter.

Theorem 13.8 (Fundamental theorem for line integrals). If f is C^1 near a piecewise C^1 curve γ , then

$$\int_{\gamma} \nabla f \cdot dr = f(\gamma(b)) - f(\gamma(a)).$$

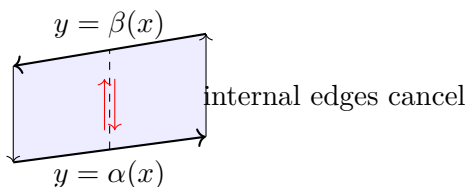
Proof. On each smooth piece the chain rule gives $(f \circ \gamma)' = Df(\gamma)\gamma' = \nabla f(\gamma) \cdot \gamma'$. Integrate by the one-variable FTC. Intermediate endpoint values cancel when the pieces are added. $\square$

This is our first boundary/interior formula: integrating a derivative along the interior leaves only the oriented endpoint values.

Exercise 13.1. Integrate $F = (2x, y)$ along the segment from $(0, 0)$ to $(1, 2)$, directly and using a potential. Compute $\int_C y \, ds$ as well and explain how the two answers change under reversal.

13.2 Green's Theorem in the Plane

The next formula turns circulation around a planar boundary into accumulated local rotation. We first prove it for a precise elementary class of regions.



Integrating a vertical derivative first subtracts its lower boundary value from its upper value. Positive boundary traversal runs rightward on the lower edge and leftward on the upper edge, explaining the minus sign in the P term. The horizontal calculation explains the Q term. After subdivision, a shared edge is traversed in opposite directions by its two adjacent pieces, leaving only the outside boundary.

Theorem 13.9 (Green for elementary planar regions). Suppose D has both a vertical and a horizontal description

$$\begin{aligned} D &= \{(x, y) : a \leq x \leq b, \alpha(x) \leq y \leq \beta(x)\}, \\ D &= \{(x, y) : c \leq y \leq d, \lambda(y) \leq x \leq \mu(y)\}, \end{aligned}$$

where the four bounding functions are continuous and piecewise C^1 , with strict lower/upper inequalities in the interiors of their intervals. Give its piecewise C^1 boundary the positive direction, with the region on the left. If P, Q are C^1 on a neighborhood of D , then

$$\int_{\partial D} P \, dx + Q \, dy = \int_D (Q_x - P_y) \, d(x, y).$$

The formula also holds for a finite union of such regions with disjoint interiors and shared boundary arcs that cancel in opposite directions.

Proof. Idea. Integrate P_y vertically and Q_x horizontally; the FTC produces exactly the upper/lower and right/left boundary contributions. By [Theorem 12.22](#) and the FTC,

$$-\int_D P_y = \int_a^b (P(x, \alpha(x)) - P(x, \beta(x))) \, dx.$$

The lower boundary goes left to right and the upper one right to left; vertical boundary segments contribute nothing to $P \, dx$. This is therefore $\int_{\partial D} P \, dx$. Similarly,

$$\int_D Q_x = \int_c^d (Q(\mu(y), y) - Q(\lambda(y), y)) \, dy = \int_{\partial D} Q \, dy.$$

The right boundary goes upward and the left downward. Add the identities. For a finite decomposition, piecewise C^1 arcs are Jordan-null: bounded derivatives cover each arc by $O(1/h)$ squares of area $O(h^2)$. Thus area integrals add without overlap contributions. Each shared arc is traversed twice in opposite directions, so its line integrals cancel. $\square$

Rectangles and triangles belong to this class. The theorem does not assert a decomposition for every conceivable piecewise smooth boundary. Later manifold Stokes will supply the global smooth-boundary result.

Example 13.10 (Area from circulation). For $P = -y/2$, $Q = x/2$, one has $Q_x - P_y = 1$. Thus for these regions

$$\text{area}(D) = \frac{1}{2} \int_{\partial D} (x \, dy - y \, dx).$$

On the triangle with vertices $(0,0), (1,0), (0,1)$, the two coordinate edges contribute zero. On $(x,y) = (1-t,t)$, $0 \leq t \leq 1$, the integrand is dt , yielding area $1/2$. Internal-edge cancellation is already a concrete model of the later passage from local formulas to a global one.

Exercise 13.2. Verify Green for $P = xy$, $Q = x^2$ on the unit square by computing all four edge integrals. Then split the square along its diagonal and identify the two cancelling contributions.

13.3 Parametrized Surfaces and Surface Integrals

A surface needs two independent tangent directions. A parametrization turns a small parameter rectangle into an approximately planar parallelogram; its area factor is the two-dimensional counterpart of $\|\gamma'\|$.

Definition 13.11 (Cross product). For $a, b \in \mathbb{R}^3$, define

$$a \times b = (a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1).$$

Proposition 13.12 (Cross-product identities). For $a, b, c \in \mathbb{R}^3$,

$$(a \times b) \cdot c = \det(a, b, c), \quad \|a \times b\|^2 = \|a\|^2\|b\|^2 - (a \cdot b)^2.$$

The cross product is orthogonal to a, b (meaning that its Euclidean dot product with each is zero). It is nonzero exactly when a, b are independent, and its norm is their parallelogram area.

Proof. The dot product expands as $(a_2b_3 - a_3b_2)c_1 + (a_3b_1 - a_1b_3)c_2 + (a_1b_2 - a_2b_1)c_3$, the determinant expansion in the third column. Taking $c = a$ or $c = b$ gives zero by alternation. For the squared norm, expansion gives

$$\sum_{i < j} (a_i b_j - a_j b_i)^2 = \sum_{i \neq j} a_i^2 b_j^2 - 2 \sum_{i < j} a_i a_j b_i b_j = \|a\|^2 \|b\|^2 - (a \cdot b)^2.$$

If $a \neq 0$, write $b = ta + h$ with $t = (a \cdot b)/\|a\|^2$ and $a \cdot h = 0$. Substitution in the last identity gives $\|a \times b\|^2 = \|a\|^2 \|h\|^2$. It vanishes exactly when b is a multiple of a ; otherwise $\|a\|$ and $\|h\|$ are the base and perpendicular height. If $a = 0$, both the cross product and the area are zero, and the pair is dependent. $\square$

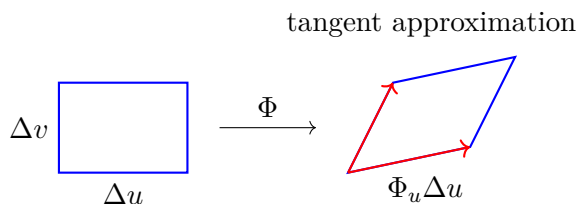
Definition 13.13 (Regular surface patch and its orientation). Let $D \subset \mathbb{R}^2$ be a compact Jordan region equal to the closure of its interior. A regular C^1 surface patch is a C^1 map Φ on an open neighborhood of D , injective on D , with independent derivatives Φ_u, Φ_v throughout D . Its tangent plane at $\Phi(u, v)$ is the span of these vectors, translated to that point. The parameter order (u, v) chooses the unit normal

$$n = \frac{\Phi_u \times \Phi_v}{\|\Phi_u \times \Phi_v\|}.$$

The opposite normal gives the reverse orientation.

13.3.1 A small parameter rectangle becomes a tangent parallelogram

At (u, v) the two edge increments are approximately $\Phi_u \Delta u$ and $\Phi_v \Delta v$. Their parallelogram has area $\|\Phi_u \times \Phi_v\| \Delta u \Delta v$. The scalar $\|\Phi_u \times \Phi_v\|$ measures area per unit parameter area. The vector $\Phi_u \times \Phi_v$ contains the same magnitude together with an oriented normal direction. Taking its norm loses precisely the information that flux needs.



Definition 13.14 (Area, scalar surface integral, and flux). For $S = \Phi(D)$ and continuous f, F along S , define

$$\begin{aligned} \text{area}(S) &= \int_D \|\Phi_u \times \Phi_v\| \, d(u, v), & \int_S f \, dS &= \int_D f(\Phi) \|\Phi_u \times \Phi_v\| \, d(u, v), \\ \int_S F \cdot n \, dS &= \int_D F(\Phi) \cdot (\Phi_u \times \Phi_v) \, d(u, v). \end{aligned}$$

These are classical smooth parametrized-surface integrals. We are not constructing an independent general surface-measure theory.

Proposition 13.15 (Surface reparametrization). If $\Psi : D' \rightarrow D$ is induced by a C^1 diffeomorphism of neighborhoods, then scalar surface integrals are unchanged. Flux is unchanged for a positive determinant and changes sign for a negative determinant.

Proof. The chain rule and bilinearity of the cross product give

$$(\Phi \circ \Psi)_s \times (\Phi \circ \Psi)_t = (\det D\Psi)(\Phi_u \times \Phi_v) \circ \Psi.$$

Taking norms produces $|\det D\Psi|$ for scalar integrals; retaining the vector produces $\det D\Psi$ for flux. Apply [Corollary 12.32](#) to obtain both assertions. $\square$

Example 13.16 (A graph and a simple flux). For $\Phi(u, v) = (u, v, h(u, v))$,

$$\Phi_u = (1, 0, h_u), \quad \Phi_v = (0, 1, h_v), \quad \Phi_u \times \Phi_v = (-h_u, -h_v, 1).$$

The upward area factor is $\sqrt{1 + h_u^2 + h_v^2}$. The upward flux of $(0, 0, 1)$ is $\int_D 1 = \text{area}(D)$, although the graph's surface area is generally larger. Reversing orientation negates the flux and leaves the scalar area unchanged.

Exercise 13.3. For $\Phi(u, v) = (u, v, u + 2v)$ on $[0, 1]^2$, compute surface area and upward flux of $F = (0, 0, z)$. Explain the effect of interchanging the parameter order.

Example 13.17 (Area and flux on a curved paraboloid). Let $\Phi(u, v) = (u, v, u^2 + v^2)$ for $u^2 + v^2 \leq 1$. The map is injective because its first two coordinates recover (u, v) ; its cross product $(-2u, -2v, 1)$ never vanishes and points upward. Thus the area is

$$\int_0^{2\pi} \int_0^1 \sqrt{1 + 4r^2} \, r \, dr \, d\theta = \frac{\pi}{6} (5^{3/2} - 1).$$

Here substitution $s = 1 + 4r^2$, $ds = 8r \, dr$ evaluates the inner integral. For $F = (0, 0, z)$ the upward flux is instead

$$\int_{u^2+v^2 \leq 1} (u^2 + v^2) \, d(u, v) = 2\pi \int_0^1 r^3 \, dr = \frac{\pi}{2}.$$

The normalizing factor cancels against the scalar area factor in the flux formula. Reversing the surface orientation negates only the flux.

$$\begin{array}{ccc} D & \xrightarrow{\Phi} & \Phi(D) \\ \uparrow & & \uparrow \\ \partial D & \xrightarrow{\Phi|_{\partial D}} & \Phi(\partial D) \end{array}$$

A parameter-plane velocity (u', v') becomes $\Phi_u u' + \Phi_v v'$ on the surface. Taking its dot product with $F(\Phi)$ gives $Au' + Bv'$ with $A = F(\Phi) \cdot \Phi_u$ and $B = F(\Phi) \cdot \Phi_v$. These coefficients are forced by the work integrand. Differentiating them produces two kinds of terms: derivatives of F , through the chain rule, and second derivatives of Φ . In $B_u - A_v$ the latter cancel because $\Phi_{vu} = \Phi_{uv}$. The surviving force-field derivatives combine into the curl. This is the plan behind the component calculation.

13.4 Classical Stokes for a Parametrized Surface Patch

Theorem 13.18 (Parametrized-patch Stokes theorem). Let D satisfy [Theorem 13.9](#), let Φ be a regular C^2 patch on D , and let $F = (F_1, F_2, F_3)$ be C^1 near $S = \Phi(D)$. Orient S by $\Phi_u \times \Phi_v$, and traverse its boundary by the image of the positive boundary of D . Then

$$\int_{\partial S} F \cdot dr = \int_S (\nabla \times F) \cdot n \, dS,$$

where $\nabla \times F = (\partial_y F_3 - \partial_z F_2, \partial_z F_1 - \partial_x F_3, \partial_x F_2 - \partial_y F_1)$.

Proof. Idea. Transport work to the parameter plane and apply Green. Put $A = F(\Phi) \cdot \Phi_u$ and $B = F(\Phi) \cdot \Phi_v$. The chain rule along every boundary piece gives $\int_{\partial S} F \cdot dr = \int_{\partial D} A \, du + B \, dv$. Write $a = \Phi_u$, $b = \Phi_v$. Differentiating, the terms $F \cdot \Phi_{vu}$ and $F \cdot \Phi_{uv}$ cancel by equality of mixed partials. The remaining terms, with derivatives evaluated at Φ , are

$$\begin{aligned} B_u - A_v &= \sum_{i,j} (\partial_j F_i) (a_j b_i - b_j a_i) \\ &= (\partial_2 F_3 - \partial_3 F_2) (a_2 b_3 - a_3 b_2) \\ &\quad + (\partial_3 F_1 - \partial_1 F_3) (a_3 b_1 - a_1 b_3) \\ &\quad + (\partial_1 F_2 - \partial_2 F_1) (a_1 b_2 - a_2 b_1) \\ &= (\nabla \times F)(\Phi) \cdot (\Phi_u \times \Phi_v). \end{aligned}$$

Green gives the desired surface integral. Our boundary orientation is precisely the one induced by the ordered tangent pair; reversing it and the surface orientation negates both sides. $\square$

This proves Stokes for one patch. An arbitrary compact oriented surface may require several charts, and proving the global statement is the purpose of the manifold machinery below.

Example 13.19 (Stokes on a tilted patch). Take $\Phi(u, v) = (u, v, u+v)$ on the unit square and $F = (-y/2, x/2, 0)$. Its curl is $(0, 0, 1)$, so the flux is 1. Boundary work pulls back to $(-v \, du + u \, dv)/2$; the bottom and left edges contribute zero and the other two contribute $1/2$ each.

Proposition 13.20 (Divergence on a box). For a C^1 vector field near a box $Q \subset \mathbb{R}^3$, $\int_Q \operatorname{div} F = \int_{\partial Q} F \cdot n \, dS$, where the right side is the sum of the outward fluxes across its six faces and $\operatorname{div} F = \partial_x F_1 + \partial_y F_2 + \partial_z F_3$.

Proof. Integrate each $\partial_j F_j$ first in the j th coordinate by FTC and then in the other two by Fubini. The endpoint difference is exactly the flux through the positive and negative j -faces. Add the three identities. $\square$

The curved-boundary divergence theorem will follow from general Stokes.

13.5 Why Differential Forms?

Work, Green, and patch Stokes all relate an interior derivative to an oriented boundary integral. We want an integrand whose transformation law already remembers directions and signs. A 1-form such as $\omega = P \, dx + Q \, dy + R \, dz$ evaluates a tangent vector v as $Pv_1 + Qv_2 + Rv_3$. Along a curve this becomes $F \cdot \gamma' \, dt$. An oriented area element must instead accept two directions, change sign when they are interchanged, and vanish when they are dependent. That is why alternating multilinear maps, rather than scalar functions alone, are the natural integrands on oriented objects of higher dimension.

The flux form associated with F is

$$\omega_F = F_1 \, dy \wedge dz + F_2 \, dz \wedge dx + F_3 \, dx \wedge dy.$$

The wedge notation, defined next, makes its value on a, b equal to $F \cdot (a \times b)$. Thus on a surface patch it becomes $F(\Phi) \cdot (\Phi_u \times \Phi_v) \, du \wedge dv$. Line and flux integrals are already examples of form integrals. The next sections build the algebra that makes this observation precise, then use charts to extend it beyond a single parametrization.

13.6 Alternating Multilinear Algebra

Recall dual spaces, multilinear maps, and alternation from [Section 10.16](#). The determinant is our prototype: an alternating form measures an oriented volume from an ordered list of directions. We now organize all degrees of such forms without using tensor products. The symmetric higher derivatives in Chapter 11 and these alternating forms share multilinearity, but have different behavior under interchange. For standard coordinates continue to write $dx_i = e^i$; these symbols denote the coordinate covectors $dx_i(v) = v_i$.

Definition 13.21 (Exterior powers of the dual). Set $\Lambda^k(V^*)$ equal to the vector space of alternating k -linear maps $V^k \rightarrow \mathbb{R}$, with pointwise operations. Its members are **k -covectors**. By convention $\Lambda^0(V^*) = \mathbb{R}$, and $\Lambda^1(V^*) = V^*$. If $k > \dim V$, every list is dependent, so $\Lambda^k(V^*) = \{0\}$ by [Proposition 10.109](#).

Definition 13.22 (Elementary wedges). For covectors $\alpha_1, \dots, \alpha_k$, their **elementary wedge** is the alternating form

$$(\alpha_1 \wedge \dots \wedge \alpha_k)(v_1, \dots, v_k) = \det(\alpha_i(v_j))_{i,j=1}^k.$$

It is multilinear in the input vectors by column multilinearity of the determinant, and vanishes on repeated inputs because columns then agree. It is also multilinear in its covectors, by the corresponding row rules.

In particular, $(dx_1 \wedge \cdots \wedge dx_n)(v_1, \dots, v_n) = \det[v_1 \cdots v_n]$. For two covectors, $(\alpha \wedge \beta)(v, w) = \alpha(v)\beta(w) - \alpha(w)\beta(v)$. Thus $dx \wedge dy$ is signed area, and

$$(dx + 2dy) \wedge (3dx - dy) = -7dx \wedge dy.$$

For example $dx \wedge dy$ evaluates on $((1, 2), (3, 1))$ as -5 .

Theorem 13.23 (Basis of alternating covectors). For a dual basis $(e^1, \dots, e^n)$, the elementary wedges

$$e^{i_1} \wedge \cdots \wedge e^{i_k}, \quad i_1 < \cdots < i_k,$$

form a basis of $\Lambda^k(V^*)$. Consequently its dimension is $\binom{n}{k}$ for $0 \leq k \leq n$.

Proof. Idea. Values on increasing basis tuples are exactly the free coefficients. Expanding the inputs in a basis reduces every value to basis tuples. Repeated tuples vanish; other tuples can be sorted with the appropriate sign. Thus the increasing-tuple values determine a form. The elementary wedge indexed by $I = (i_1, \dots, i_k)$ evaluates to 1 on $(e_{i_1}, \dots, e_{i_k})$ and to 0 on every other increasing tuple: its evaluation matrix is the identity in the first case, and has a zero row in the second. Hence every form A equals

$$\sum_{i_1 < \cdots < i_k} A(e_{i_1}, \dots, e_{i_k}) e^{i_1} \wedge \cdots \wedge e^{i_k},$$

and evaluation on increasing tuples proves independence. For $k = 0$ the basis consists of the scalar 1. $\square$

Elementary wedges span, but a general covector of degree k need not itself be one elementary wedge. A concrete degree-two example will follow the product laws, where it can be checked in one calculation.

Before general shuffle notation, take a one-covector α and an alternating two-covector β . On three vectors there are exactly three ways to assign an input to α , keeping the other two in their original order:

$$\begin{aligned} (\alpha \wedge \beta)(v_1, v_2, v_3) &= \alpha(v_1)\beta(v_2, v_3) - \alpha(v_2)\beta(v_1, v_3) \\ &\quad + \alpha(v_3)\beta(v_1, v_2). \end{aligned}$$

The corresponding lists are 123, 213, 312, with signs $+, -, +$. For $\alpha = dx$, $\beta = dy \wedge dz$, and $v_1 = (1, 1, 0)$, $v_2 = (2, 0, 1)$, $v_3 = (0, 1, 1)$, the three contributions are $-1, -2, 0$, totaling -3 . This is also the determinant of the three vector columns. A shuffle is precisely this selection of two internally ordered blocks, generalized to degrees p and q .

Definition 13.24 (Wedge product by shuffles). For $\alpha \in \Lambda^p(V^*)$, $\beta \in \Lambda^q(V^*)$, define

$$\begin{aligned} (\alpha \wedge \beta)(v_1, \dots, v_{p+q}) &= \sum_{\sigma \in \text{Sh}(p, q)} \text{sgn}(\sigma) \alpha(v_{\sigma(1)}, \dots, v_{\sigma(p)}) \\ &\quad \cdot \beta(v_{\sigma(p+1)}, \dots, v_{\sigma(p+q)}). \end{aligned}$$

Here a (p, q) -**shuffle** is a permutation with $\sigma(1) < \dots < \sigma(p)$ and $\sigma(p+1) < \dots < \sigma(p+q)$: choose the first increasing block, and the remaining inputs form the second increasing block. Degree-zero factors act by scalar multiplication. Equivalently, over $\mathbb{R}$ the full-permutation formula is

$$(\alpha \wedge \beta)(v_1, \dots, v_{p+q}) = \frac{1}{p!q!} \sum_{\sigma \in S_{p+q}} \text{sgn}(\sigma) \alpha(v_{\sigma(1)}, \dots, v_{\sigma(p)}) \cdot \beta(v_{\sigma(p+1)}, \dots, v_{\sigma(p+q)}).$$

Indeed, the $p!q!$ internal reorderings of each shuffle give the same summand: the form signs cancel the permutation signs. The full sum is alternating by reindexing after an input interchange; when two inputs agree, terms pair with opposite signs. Thus the product belongs to $\Lambda^{p+q}(V^*)$.

For elementary wedges, this product agrees with concatenating their covector lists. Expand the evaluation determinant according to the set of columns used by its first p rows. Each choice is a shuffle term, with the remaining rows supplying the second determinant. Equivalently, expand both determinants into permutations: each full permutation occurs once. Thus the earlier notation for elementary wedges is consistent.

Proposition 13.25 (Exterior-algebra rules). The wedge product is bilinear, associative, and graded commutative: $\alpha \wedge \beta = (-1)^{pq} \beta \wedge \alpha$ for degrees p, q .

Proof. Idea. Track increasing blocks in the shuffle formula. Bilinearity follows by distributing each summand. Exchanging the two blocks moves p inputs past q inputs and changes the sign by $(-1)^{pq}$. For associativity, both $(\alpha \wedge \beta) \wedge \gamma$ and $\alpha \wedge (\beta \wedge \gamma)$ sum once over all ordered partitions of the inputs into increasing blocks of sizes p, q, r . The first groups the first two blocks initially; the second groups the last two. Both have the sign of the permutation listing the three blocks, because permutation signs multiply under composition. Their summands therefore coincide. $\square$

For a 1-covector α , $\alpha \wedge \alpha = 0$, also directly from the determinant with repeated rows. For even degree the square need not vanish. In dimension four, put $\omega = e^1 \wedge e^2 + e^3 \wedge e^4$. Then

$$\omega \wedge \omega = 2e^1 \wedge e^2 \wedge e^3 \wedge e^4 \neq 0.$$

If ω were one elementary wedge $\alpha \wedge \beta$, its square would have repeated factors and be zero. Thus this also proves that a general 2-covector need not be elementary.

Definition 13.26 (The exterior algebra). Using the finite direct sum constructed in [Section 10.9](#), set

$$\Lambda^*(V^*) = \bigoplus_{k=0}^n \Lambda^k(V^*).$$

Every element has a unique component in each degree. Distribute the wedge product over these components: the degree- r part is $\sum_{p+q=r} \alpha_p \wedge \beta_q$. An element supported in degree p is homogeneous of degree p . The direct-sum basis and dimension facts are those already proved in Chapter 10.

The optional tensor construction in [Appendix B](#) gives an algebraic explanation of these spaces; none of the following analysis requires it.

13.7 Differential Forms, Pullbacks, and Exterior Derivatives

For a smooth function $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$, Chapter 11 gives

$$Df(x) : \mathbb{R}^n \longrightarrow \mathbb{R}.$$

Thus $Df(x)$ is a 1-covector. The notation $df_x = Df(x)$ turns the familiar one-variable identity

$$df = f'(x) dx$$

into a coordinate-free statement that continues to make sense in every dimension. In coordinates it becomes

$$df = \sum_{i=1}^n \frac{\partial f}{\partial x_i} dx_i.$$

This is the direct chain from the Part I derivative, through the Fréchet derivative of Chapter 11, to a differential 1-form.

Definition 13.27 (Differential forms). Let $U \subseteq \mathbb{R}^n$ be open and let

$$0 \leq k \leq n.$$

A **smooth k -form** on U is a function that assigns to each

$$x \in U$$

an element of

$$\Lambda^k((\mathbb{R}^n)^*)$$

whose coordinate coefficients are smooth. Thus it has a unique expression

$$\omega = \sum_{1 \leq i_1 < \dots < i_k \leq n} a_{i_1 \dots i_k}(x) dx_{i_1} \wedge \dots \wedge dx_{i_k}.$$

Here the sum is over the increasing multi-indices

$$I = (i_1, \dots, i_k), \quad 1 \leq i_1 < \dots < i_k \leq n,$$

and we abbreviate

$$dx_I = dx_{i_1} \wedge \dots \wedge dx_{i_k}.$$

By convention,

$$\Lambda^0((\mathbb{R}^n)^*) = \mathbb{R},$$

so a smooth 0-form is simply a smooth real-valued function. The integer

$$k$$

is the **degree** of a k -form. Write $\Omega^k(U)$ for the vector space of smooth k -forms, with pointwise operations. For $k > n$ set $\Omega^k(U) = \{0\}$, consistent with the empty increasing-index basis. Thus the derivative of a top-degree form and a two-form pulled back to a curve have a specified zero target.

Proposition 13.28 (Differential of a function). For a smooth function $f : U \rightarrow \mathbb{R}$,

$$df = \sum_{i=1}^n \frac{\partial f}{\partial x_i} dx_i.$$

At each $x \in U$, the form df_x is the derivative $Df(x)$.

Proof. Both sides are linear functionals. Evaluating the displayed sum on the standard basis vector e_i gives

$$\frac{\partial f}{\partial x_i}(x) = Df(x)e_i.$$

The coordinate formula for a linear functional proves equality. $\square$

Definition 13.29 (Pullback). Let

$$\Phi : U \longrightarrow V$$

be a smooth map, where

$$U \subseteq \mathbb{R}^n, \quad V \subseteq \mathbb{R}^m$$

are open. For a k -form ω on V , its **pullback** is the smooth k -form

$$\Phi^*\omega$$

on U defined by

$$(\Phi^*\omega)_x(v_1, \dots, v_k) = \omega_{\Phi(x)}(D\Phi(x)v_1, \dots, D\Phi(x)v_k).$$

Its smoothness follows from the smoothness of the coefficients of ω and of Φ .

Example 13.30 (Pullback as substitution and directions). For $\omega = y \, dx + x \, dy$ and $\gamma(t) = (t, t^2)$,

$$\gamma^*x = t, \quad \gamma^*y = t^2, \quad \gamma^*(dx) = dt, \quad \gamma^*(dy) = 2t \, dt.$$

Substitution into the coefficients and application of $D\gamma$ to directions therefore give

$$\gamma^*\omega = t^2 \, dt + t(2t \, dt) = 3t^2 \, dt, \quad \int_0^1 3t^2 \, dt = 1.$$

The output is a 1-form on the parameter line, not a form on the plane. For the running form $\omega_0 = x \, dy - y \, dx$, the circle gives $\gamma^*\omega_0 = dt$. If using injective curve patches, split the circle into the two semicircles $[0, \pi]$ and $[\pi, 2\pi]$, each injective, and add their integrals to obtain 2π .

Exercise 13.4. If $\Phi : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is smooth and η is a 2-form on $\mathbb{R}^3$, what are the degree and domain of $\Phi^*\eta$? What happens for a 3-form? Compute the pullback of the flux form of $(0, 0, z)$ under $\Phi(u, v) = (u, v, u + 2v)$ and compare with the earlier flux exercise.

Remark 13.31. Pullback is the form-theoretic version of substitution. For the standard volume form on $\mathbb{R}^n$,

$$\Phi^*(dy_1 \wedge \dots \wedge dy_n) = \det D\Phi(x) \, dx_1 \wedge \dots \wedge dx_n.$$

Thus the determinant in Chapter 12 is exactly the coefficient produced by pulling a volume form back to the parameter domain.

Proposition 13.32 (Pullback rules). For smooth maps

$$U \xrightarrow{\Phi} V \xrightarrow{\Psi} W,$$

and forms of compatible degree,

$$(\Psi \circ \Phi)^* = \Phi^* \circ \Psi^*, \quad \Phi^*(\alpha \wedge \beta) = \Phi^*\alpha \wedge \Phi^*\beta.$$

Proof. The chain rule from [Proposition 11.6](#) gives

$$D(\Psi \circ \Phi)(x) = D\Psi(\Phi(x))D\Phi(x).$$

For an arbitrary k -form ω on W , evaluate both proposed pullbacks at $x \in U$ on $v_1, \dots, v_k$. The left side gives $\omega_{\Psi(\Phi(x))}(D(\Psi \circ \Phi)(x)v_1, \dots, D(\Psi \circ \Phi)(x)v_k)$. The chain rule makes this the same scalar as the right side:

$$\omega_{\Psi(\Phi(x))}(D\Psi(\Phi(x))D\Phi(x)v_1, \dots, D\Psi(\Phi(x))D\Phi(x)v_k).$$

Both forms already exist by the pullback definition; equality on every tuple proves their equality. The reversal of order comes from first evaluating $\Psi^*\omega$ on the vectors $D\Phi(x)v_i$. For the wedge rule, its alternating finite sum has the same terms on both sides after each argument is replaced by $D\Phi(x)v_i$, proving the second identity. $\square$

Definition 13.33 (Exterior derivative). For a smooth k -form

$$\omega = \sum_I a_I dx_I$$

on an open subset of $\mathbb{R}^n$, define its **exterior derivative** by

$$d\omega = \sum_I \sum_{j=1}^n \frac{\partial a_I}{\partial x_j} dx_j \wedge dx_I.$$

Remark 13.34. The exterior derivative raises degree by one:

$$\text{functions} \longrightarrow \text{1-forms} \longrightarrow \text{2-forms} \longrightarrow \dots$$

It generalizes ordinary differentiation while retaining the product rule. The identities $d^2 = 0$ and pullback-commutation below are the two algebraic facts that allow a local integral identity to pass through coordinates and then to global manifolds.

For our running example,

$$d(x dy - y dx) = dx \wedge dy - dy \wedge dx = 2 dx \wedge dy.$$

Green's area formula is thus already the statement that the boundary integral of ω_0 equals the interior integral of $d\omega_0$.

Theorem 13.35 (Fundamental exterior-derivative rules). For smooth forms α, β of fixed degrees,

$$d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^{\deg \alpha} \alpha \wedge d\beta,$$

$$d(d\alpha) = 0, \quad \Phi^*(d\omega) = d(\Phi^*\omega).$$

Proof. For elementary coordinate forms $\alpha = a dx_I$ and $\beta = b dx_J$ with $|I| = p$, the coefficient product rule gives

$$d(\alpha \wedge \beta) = d(ab) \wedge dx_I \wedge dx_J = b da \wedge dx_I \wedge dx_J + a db \wedge dx_I \wedge dx_J.$$

The first term is $d\alpha \wedge \beta$. To identify the second, move the one-form db past the p one-forms in dx_I : $dx_I \wedge db = (-1)^p db \wedge dx_I$. Thus $(-1)^p \alpha \wedge d\beta$ is precisely the second term, since $(-1)^{2p} = 1$. If indices repeat both expressions still obey the same alternating identities. Every form is a finite

sum of elementary coordinate forms; bilinearity and linearity of d extend the rule to all forms. For the second identity write

$$\alpha = \sum_I a_I dx_I.$$

Applying the definition twice gives the double sum

$$d(d\alpha) = \sum_I \sum_{j,k} \frac{\partial^2 a_I}{\partial x_k \partial x_j} dx_k \wedge dx_j \wedge dx_I.$$

If a differential is repeated, the corresponding term vanishes because

$$dx_j \wedge dx_j = 0.$$

For distinct indices, the terms indexed by (j, k) and (k, j) pair: their coefficients agree by [Theorem 11.19](#), while

$$dx_k \wedge dx_j = -dx_j \wedge dx_k.$$

They therefore cancel, proving $d^2 = 0$.

For a smooth function f , the multivariable chain rule gives

$$\Phi^*(df) = d(f \circ \Phi).$$

In particular, pullback commutes with d on coordinate functions. For an elementary term $\omega = f dy_{i_1} \wedge \cdots \wedge dy_{i_k}$, its pullback is $(f \circ \Phi) d\Phi^{i_1} \wedge \cdots \wedge d\Phi^{i_k}$. The Leibniz rule and $d^2\Phi^i = 0$ show that its derivative is $d(f \circ \Phi) \wedge d\Phi^{i_1} \wedge \cdots \wedge d\Phi^{i_k}$, exactly the pullback of $d\omega$. Add the finitely many elementary terms. $\square$

Example 13.36. In $\mathbb{R}^3$, write

$$\omega = P dx + Q dy + R dz.$$

Then

$$d\omega = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx \wedge dy + \left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z} \right) dx \wedge dz + \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) dy \wedge dz.$$

The three coefficients encode the usual curl.

13.7.1 One form through all its types

Keep $\omega = x dy - y dx$ as a running example. The form ω is a field of covectors on the plane. At $p = (2, 3)$, its value is the covector $\omega_p = 2 dy|_p - 3 dx|_p$. On $v = (4, -1) \in T_p\mathbb{R}^2$ it returns the scalar $\omega_p(v) = 2(-1) - 3(4) = -14$. Along $\gamma(t) = (\cos t, \sin t)$,

$$(\gamma^*\omega)_t = \omega_{\gamma(t)}(\gamma'(t)) dt = (\cos^2 t + \sin^2 t) dt = dt.$$

Its exterior derivative is $d\omega = dx \wedge dy - dy \wedge dx = 2 dx \wedge dy$. Thus ω , ω_p , and $\omega_p(v)$ are respectively a form, a covector at a point, and a number. Exterior differentiation acts on the varying form, not on an isolated numerical evaluation.

13.7.2 Pullback as a workflow, including a surface calculation

For $\Phi : U \rightarrow V$, the evaluation rule reads

$$\begin{array}{ccc} (h_1, \dots, h_k) \text{ at } u & \xrightarrow{D\Phi(u)} & (D\Phi(u)h_1, \dots, D\Phi(u)h_k) \text{ at } \Phi(u) \\ & & \downarrow \omega_{\Phi(u)} \\ & & (\Phi^*\omega)_u(h_1, \dots, h_k) \in \mathbb{R} \end{array}$$

First move the source tangent vectors forward, then evaluate the target form. The resulting rule is a covector on the source. The degree k is preserved and the base domain changes. Invertibility is unnecessary. If the source dimension is smaller than k , all k input vectors are dependent and the pulled-back k -form is zero.

Example 13.37 (Pulling a two-form onto a graph). Let $\Phi(u, v) = (u, v, u^2 + v^2)$ and take $\eta = z \, dx \wedge dy + x \, dy \wedge dz$ on $\mathbb{R}^3$. Substitute both coefficients and covectors:

$$\Phi^*z = u^2 + v^2, \quad \Phi^*x = u, \quad \Phi^*dx = du, \quad \Phi^*dy = dv, \quad \Phi^*dz = 2u \, du + 2v \, dv.$$

Consequently

$$\Phi^*\eta = (u^2 + v^2) \, du \wedge dv + u \, dv \wedge (2u \, du + 2v \, dv) = (v^2 - u^2) \, du \wedge dv.$$

Over $[0, 1]^2$ its integral is $1/3 - 1/3 = 0$. The map from a plane into space is not invertible, yet every step of the pullback is defined. As a second example, pulling $dx \wedge dy$ onto any curve gives zero, a degree-two form on a one-dimensional source.

13.8 Integration and the Local Stokes Theorem

Before the higher-dimensional statement, recall its one-dimensional model. For a smooth function on an oriented interval,

$$\int_{[a,b]} df = f(b) - f(a) = \int_{\partial[a,b]} f.$$

The local theorem is the same assertion on a rectangle: its proof applies the Part I Fundamental Theorem of Calculus one coordinate at a time.

Definition 13.38 (Integration of top-degree forms). Let $Q \subseteq \mathbb{R}^n$ be a rectangle with the standard orientation. For a continuous n -form

$$\omega = f \, dx_1 \wedge \dots \wedge dx_n,$$

define

$$\int_Q \omega = \int_Q f(x) \, dx.$$

Reversing the orientation changes the sign of the integral.

Integration over an open set. Let $V \subseteq \mathbb{R}^n$ be open, and let

$$\omega = f \, dx_1 \wedge \cdots \wedge dx_n$$

be a continuous top-degree form whose support

$$K = \text{supp}(\omega) = \overline{\{x \in V : f(x) \neq 0\}}$$

is compact and contained in V , where the closure is taken in V . Because a compact subset of the open set V is separated from $\mathbb{R}^n \setminus V$ (by a positive distance when $K \neq \emptyset$), there is a neighborhood of $\mathbb{R}^n \setminus V$ disjoint from K . The function f vanishes on the part of that neighborhood lying in V . Consequently, extending f by zero outside V gives a continuous function $\tilde{f}$. Choose any rectangle Q whose interior contains K , and define

$$\int_V \omega = \int_Q \tilde{f}(x) \, dx.$$

This is independent of Q . Indeed, after enlarging two such rectangles to a common containing rectangle, finite additivity of the Riemann integral and the vanishing of $\tilde{f}$ off K show that both integrals equal the integral over the larger rectangle. If ω is smooth, then the extended coefficients are identically zero on a neighborhood of the artificial boundary of V , so the same zero extension is smooth.

Definition 13.39 (Boundary orientation). The boundary of the standard rectangle is oriented by the **outward-normal-first rule**: a basis of a boundary face is positive when adjoining an outward normal before it gives the orientation of the rectangle. The integral over the boundary is the signed sum over its faces. Restriction means pullback by inclusion: if $\iota : \partial Q \rightarrow Q$ is the inclusion, $\int_{\partial Q} \omega$ abbreviates $\int_{\partial Q} \iota^* \omega$, interpreted face by face here.

For the standard square the outward-normal-first calculation gives: bottom edge $(-e_2, e_1)$ is positive, right edge (e_1, e_2) is positive, top edge $(e_2, -e_1)$ is positive, and left edge $(-e_1, -e_2)$ is positive. Thus the boundary goes right, up, left, down: counterclockwise. Rectangles and cubes have corners. The theorem below is a facewise result; they are not automatically smooth-boundary examples under our later definitions.

Theorem 13.40 (Stokes theorem on a rectangle). Let $Q \subseteq \mathbb{R}^n$ be a rectangle and let ω be a smooth

$$(n-1)\text{-form}$$

on an open neighborhood of Q . Then

$$\int_Q d\omega = \int_{\partial Q} \omega.$$

Proof. Write

$$\omega = \sum_{j=1}^n (-1)^{j-1} f_j \, dx_1 \wedge \cdots \wedge \widehat{dx_j} \wedge \cdots \wedge dx_n.$$

Only

$$\frac{\partial f_j}{\partial x_j}$$

contributes to the coefficient of

$$dx_1 \wedge \cdots \wedge dx_n$$

in $d\omega$. Apply the one-variable Fundamental Theorem of Calculus in the x_j variable and then [Theorem 12.7](#) to integrate the remaining variables. To verify the signs, put $E_j = (e_1, \dots, \widehat{e}_j, \dots, e_n)$, the standard ordered face frame. The ordered frame (e_j, E_j) has ambient sign $(-1)^{j-1}$, since moving e_j into position j uses $j-1$ swaps. On the upper face $x_j = b_j$ the outward normal is e_j , so its induced face orientation has sign $(-1)^{j-1}$ relative to E_j . The coefficient of the j th term of ω has this same sign; multiplying gives $+f_j$. On the lower face $x_j = a_j$ the outward normal is $-e_j$, reversing the induced face sign to $(-1)^j$. Multiplication by the coefficient $(-1)^{j-1}$ gives $-f_j$. Every other term restricts to zero on these faces because it contains dx_j . The paired face integral is therefore

$$\int_{\prod_{i \neq j} [a_i, b_i]} (f_j(\dots, b_j, \dots) - f_j(\dots, a_j, \dots)) dx_1 \cdots \widehat{dx}_j \cdots dx_n,$$

exactly the endpoint difference from the FTC. Summing over j proves the formula. $\square$

Definition 13.41 (Parametrized integration). Let $D \subseteq \mathbb{R}^n$ be a compact rectangle and let

$$\Phi : D \longrightarrow \mathbb{R}^m$$

be a smooth injective parametrization with injective derivative on the interior. For a continuous n -form ω on a neighborhood of

$$\Phi(D),$$

define

$$\int_{\Phi(D)} \omega = \int_D \Phi^* \omega,$$

where the image carries the orientation induced by Φ .

Proposition 13.42 (Change of parametrization). Let $D, D' \subseteq \mathbb{R}^n$ be compact rectangles, and suppose that $\Psi : D' \rightarrow D$ is the restriction of a smooth diffeomorphism between open neighborhoods of these rectangles, with $\Psi(D') = D$. Replacing Φ by $\Phi \circ \Psi$ leaves the integral unchanged when $\det D\Psi > 0$ on D' , and changes its sign when $\det D\Psi < 0$ on D' .

Proof. Pullback functoriality gives

$$(\Phi \circ \Psi)^* \omega = \Psi^*(\Phi^* \omega).$$

Write $\Phi^* \omega = g(x) dx_1 \wedge \cdots \wedge dx_n$. Then

$$\Psi^*(\Phi^* \omega) = g(\Psi(u)) \det D\Psi(u) du_1 \wedge \cdots \wedge du_n.$$

Because D' is connected and the determinant is continuous and nowhere zero, its sign is constant. The scalar change-of-variables theorem [Theorem 12.30](#) therefore gives

$$\int_{D'} g(\Psi(u)) \det D\Psi(u) du = \begin{cases} \int_D g(x) dx, & \det D\Psi > 0, \\ -\int_D g(x) dx, & \det D\Psi < 0. \end{cases}$$

These are precisely the asserted orientation-preserving and orientation-reversing cases. $\square$

Proposition 13.43 (Invariance on chart overlaps). For an orientation-preserving smooth diffeomorphism $\Psi : V' \rightarrow V$ and a continuous compactly supported top-degree form η on V , $\int_{V'} \Psi^* \eta = \int_V \eta$. This holds both for open Euclidean sets and for relatively open half-space sets with boundary-preserving transitions.

Proof. Write $\eta = f dx_1 \wedge \cdots \wedge dx_n$. The pullback coefficient is $f(\Psi) \det D\Psi$, and the determinant is positive. Apply [Corollary 12.33](#) or [Corollary 12.34](#). These statements cover arbitrary compact support on an overlap, unlike the rectangular reparametrization proposition by itself. $\square$

13.9 Smooth Manifolds, Tangent Spaces, and Orientation

The local Stokes theorem works inside one coordinate region. A curved global space generally needs many overlapping charts, so a global integral and boundary formula require coordinate invariance together with a mechanism by which artificial internal boundaries cancel. The manifold and refinement machinery introduced next supplies exactly that passage from local Euclidean calculations to one coherent global theorem.

Example 13.44 (A graph supplies actual coordinates). For a smooth graph $M = \{(x, y, h(x, y)) : (x, y) \in U\}$, with U open,

$$\varphi : M \rightarrow U, \quad (x, y, h(x, y)) \mapsto (x, y), \quad \varphi^{-1}(u, v) = (u, v, h(u, v)).$$

The chart goes from the surface to coordinates; its inverse is the parametrization. Its tangent vectors are $(1, 0, h_u)$ and $(0, 1, h_v)$. For the upper circle arc in [Example 11.39](#), the chart is $(x, y) \mapsto x$ on $y > 0$, with inverse $t \mapsto (t, \sqrt{1-t^2})$, $-1 < t < 1$. Near the rightmost point use y instead. The implicit function theorem supplies these local graphs; it does not make the whole circle one graph.

Topology terminology. The following modest amount of general-topology language will be used only to state the refinement theorem below. A topological space is **second countable** if there is a countable collection of open sets such that every open set is a union of members of that collection. A family

$$\{U_\alpha : \alpha \in I\}$$

is **locally finite** if every point has a neighborhood meeting only finitely many of its members. A cover

$$\{V_j : j \in J\}$$

refines a cover $\{U_\alpha : \alpha \in I\}$ if every V_j is contained in some U_α . A subset E is **precompact** if its closure is compact. Finally, a topological space is **paracompact** if every open cover admits a locally finite open refinement. These definitions use the notions of closure and compactness from Chapter 10.

Supports. If f is a function on a topological space X , its **support** is

$$\text{supp}(f) = \overline{\{x \in X : f(x) \neq 0\}}.$$

A function has **compact support** when this closed set is compact. This definition makes precise the condition that a function vanish outside a compact region.

13.9.1 Charts undo parametrizations

On the upper semicircle, the coordinate map is $\varphi(x, y) = x$, with image $(-1, 1)$; its inverse $u \mapsto (u, \sqrt{1-u^2})$ is a parametrization. A chart sends geometric points to numbers. A parametrization sends coordinate numbers to geometric points. A graph surface $(u, v) \mapsto (u, v, h(u, v))$ has the inverse chart $(x, y, z) \mapsto (x, y)$. For $[a, b]$, an endpoint requires coordinates in a half-line, whereas an interior point has coordinates in an open interval.

$$\begin{array}{ccc} & \varphi \text{ (chart)} & \\ & \curvearrowright & \\ U \subset M & & \varphi(U) \subset \mathbb{R}^n \\ & \curvearrowleft & \\ & \varphi^{-1} \text{ (parametrization)} & \end{array}$$

Example 13.45 (Computing an actual chart transition). On the upper semicircle use the x chart above; on the right semicircle use $\psi(x, y) = y$. Their overlap is the first-quadrant arc, on which

$$(\psi \circ \varphi^{-1})(u) = \sqrt{1 - u^2}, \quad 0 < u < 1.$$

Its derivative is $-u/\sqrt{1 - u^2}$, nonzero throughout the overlap. The inverse has the same square-root formula with the coordinate renamed. Both are smooth on their open interval. The negative derivative changes coordinate orientation; it does not obstruct smooth compatibility. Multiple charts are needed to cover the circle because these graph charts exclude the points where their chosen coordinate ceases to work.

An atlas is a collection of charts covering every point. Compatibility requires that overlapping coordinate descriptions translate smoothly in both directions. A smooth structure includes every chart compatible with the chosen atlas. These are successive choices, not one unexplained package. The formal definition also specifies the mild global topological conditions needed later for locally finite localization.

Definition 13.46 (Coordinate chart). A chart of dimension n on a topological space M is a homeomorphism $\varphi : U \rightarrow V$, where U is open in M and V is open in $\mathbb{R}^n$. Its inverse is a parametrization of U .

The homeomorphism requirement says that both geometric points and their coordinates vary continuously with one another. It does not supply a global coordinate system.

Definition 13.47 (Atlas and compatibility). An **atlas** is a family of charts whose domains cover M . Two charts are **smoothly compatible** if the transition $\psi \circ \varphi^{-1}$ and its inverse are smooth on their open domains. A **smooth atlas** has pairwise compatible charts.

Compatibility allows a calculation in one chart to be translated into another by the ordinary chain rule. The semicircle transition above is an explicit example of this test.

Definition 13.48 (Manifold and smooth structure). A topological n -manifold is a Hausdorff, second-countable topological space with an atlas of dimension n . A **smooth structure** is a maximal smooth atlas: all charts compatible with it are included. Such a space with a smooth structure is a **smooth manifold**.

Hausdorffness separates distinct points. Second countability and the local Euclidean structure supply the countable, locally finite refinement used later. These global requirements do not alter the coordinate computations on any individual chart.

Definition 13.49 (Manifold with boundary). Allow charts into relatively open subsets of $\mathbb{H}^n = \{(x_1, \dots, x_n) : x_n \geq 0\}$, requiring smooth transition maps with smooth inverses. Smooth here means locally the restriction of a smooth map on a Euclidean neighborhood, including at $x_n = 0$. With the same global topological conditions this defines a smooth manifold with boundary. Its boundary consists of points whose last coordinate is zero in a boundary chart.

An interior chart remains possible wherever the last coordinate is positive. The boundary designation is consistent across charts, as the following argument proves.

Boundary points are independent of chart. To see the analytic reason, suppose a transition sent a boundary point to an interior point. Its inverse would be smooth on an open ball and take values in the half-space, with last coordinate zero at an interior point of that ball. That coordinate has a local minimum, so its derivative is zero there. But transition and inverse derivatives are inverses,

by the chain rule (also at boundary points by limits from the interior). A matrix with a zero row cannot be invertible. The contradiction proves boundary preservation.

Hausdorffness gives the expected separation and uniqueness behavior for points, while second countability excludes excessively large pathological spaces and is one ingredient behind the refinement and partition-of-unity theory used later. Every smooth atlas determines a unique maximal smooth atlas: adjoin every chart smoothly compatible with it. Thus maximality adds all allowable coordinate systems without changing the underlying geometry.

13.9.2 Deriving the tangent-coordinate rule from a moving point

At an interior point p , let $\gamma(0) = p$ and suppose two charts φ, ψ are available. Put $v = (\varphi \circ \gamma)'(0)$ and $w = (\psi \circ \gamma)'(0)$. Since $\psi \circ \gamma = (\psi \circ \varphi^{-1}) \circ (\varphi \circ \gamma)$, the chain rule forces

$$w = D(\psi \circ \varphi^{-1})(\varphi(p))v.$$

The geometric motion has not changed, only its velocity coordinates. This equation suggests declaring chart-vector pairs equivalent when they represent the same direction by exactly this rule. The equivalence-class construction below preserves that interpretation without choosing an ambient Euclidean embedding. It also gives a full vector space at boundary points, where two-sided curves inside the manifold would be too restrictive.

Definition 13.50 (Smooth manifold map). A map $F : M \rightarrow N$ is smooth if, for every $p \in M$, there are charts $\varphi : U \rightarrow \varphi(U)$ and $\psi : V \rightarrow \psi(V)$ with $p \in U$, $F(p) \in V$, and $F(U) \subseteq V$, such that

$$\psi \circ F \circ \varphi^{-1} : \varphi(U) \rightarrow \psi(V)$$

is smooth. The chart neighborhood U may be shrunk to obtain this containment. For boundary charts the domain is relatively open in the appropriate half-space; smooth means locally extendible to a smooth map on a Euclidean open neighborhood, as in the chart definition.

This condition implies continuity: on each U the map is $\psi^{-1} \circ (\psi F \varphi^{-1}) \circ \varphi$, a composition of continuous maps, and these neighborhoods cover M . For other charts $\varphi' : U' \rightarrow \varphi'(U')$ and $\psi' : V' \rightarrow \psi'(V')$ at the same points, restrict to the open neighborhood $U \cap U' \cap F^{-1}(V \cap V')$. The new coordinate representative is

$$\psi' F (\varphi')^{-1} = (\psi' \psi^{-1}) \circ (\psi F \varphi^{-1}) \circ (\varphi (\varphi')^{-1})$$

on the image of that neighborhood under φ' . Smooth transition maps and composition preserve smoothness, including local extensions at boundary points. Thus the definition does not privilege the initial charts. We construct tangent vectors by declaring when two coordinate vectors mean the same direction. For charts φ, ψ at p , define the linear map

$$C_{\psi\varphi}(p) = D(\psi \circ \varphi^{-1})(\varphi(p)) : \mathbb{R}^n \rightarrow \mathbb{R}^n.$$

The rightmost chart is the input coordinate system; the leftmost is the output. At boundary points derivatives are those of local smooth extensions. They are unique: extensions agree on interior points, hence their derivatives agree at the boundary by continuity. The chain rule identities below therefore hold at boundary points as well.

Lemma 13.51 (Transition identities). For charts at a common point,

$$C_{\varphi\varphi} = I, \quad C_{\varphi\psi} = C_{\psi\varphi}^{-1}, \quad C_{\chi\varphi} = C_{\chi\psi} C_{\psi\varphi}.$$

Proof. The first transition is the identity in coordinates, so its derivative is I . The maps $\varphi \circ \psi^{-1}$ and $\psi \circ \varphi^{-1}$ are inverse on the overlap. Differentiating their two identity composites gives $C_{\varphi\psi}C_{\psi\varphi} = I$ and $C_{\psi\varphi}C_{\varphi\psi} = I$, proving the inverse assertion. Finally factor the actual transition as $\chi \circ \varphi^{-1} = (\chi \circ \psi^{-1}) \circ (\psi \circ \varphi^{-1})$. The intermediate evaluation point is $\psi(p)$; the chain rule at $\varphi(p)$ gives $C_{\chi\varphi} = C_{\chi\psi}C_{\psi\varphi}$. $\square$

Definition 13.52 (Tangent-vector relation). For chart-vector pairs (φ, v) with $v \in \mathbb{R}^n$, declare

$$(\varphi, v) \sim (\psi, w) \iff w = C_{\psi\varphi}(p)v.$$

Proposition 13.53 (The relation is an equivalence relation). Its classes form a set denoted T_pM ; a class is a tangent vector at p .

Proof. Reflexivity follows from $C_{\varphi\varphi}v = v$. If $w = C_{\psi\varphi}v$, applying the inverse gives $v = C_{\varphi\psi}w$, proving symmetry. If also $z = C_{\chi\psi}w$, then $z = C_{\chi\psi}C_{\psi\varphi}v = C_{\chi\varphi}v$, proving transitivity. These are precisely the three identities of the lemma. $\square$

Theorem 13.54 (Vector-space structure of the tangent space). Fix one chart φ at p . Represent two classes in it as $[\varphi, v]$ and $[\varphi, w]$, and define

$$[\varphi, v] + [\varphi, w] = [\varphi, v + w], \quad a[\varphi, v] = [\varphi, av].$$

These operations are independent of chart and make T_pM an n -dimensional real vector space.

Proof. Every class has a representative in φ : replace (ψ, z) by $(\varphi, C_{\varphi\psi}z)$. It has only one representative in that chart, since $[\varphi, v] = [\varphi, v']$ forces $v' = Iv = v$. Thus the displayed operations specify outputs once φ is fixed. If we use ψ instead, the input representatives become $C_{\psi\varphi}v$ and $C_{\psi\varphi}w$. Linearity gives

$$C_{\psi\varphi}(v + w) = C_{\psi\varphi}v + C_{\psi\varphi}w, \quad C_{\psi\varphi}(av) = aC_{\psi\varphi}v.$$

The outputs from the two charts are therefore equivalent pairs. This proves well-definedness before any vector-space conclusion is asserted. The zero vector is $[\varphi, 0]$, and the negative of $[\varphi, v]$ is $[\varphi, -v]$. All vector-space axioms can now be checked in this one chart, where they are exactly those of $\mathbb{R}^n$.

Explicitly the coordinate map $\kappa_\varphi : T_pM \rightarrow \mathbb{R}^n$ sends a class to its unique φ representative vector; its reverse is $s_\varphi(v) = [\varphi, v]$. The definitions of operations make both maps linear. We have $\kappa_\varphi s_\varphi(v) = v$ and $s_\varphi \kappa_\varphi(\xi) = \xi$. They are inverse isomorphisms, so the classes $[\varphi, e_i]$ form a basis and $\dim T_pM = n$. $\square$

13.9.3 The differential: intrinsic map, changing coordinate matrices

For smooth $F : M \rightarrow N$, with charts φ at p and ψ at $F(p)$, define the coordinate derivative

$$A_{\psi\varphi} = D(\psi \circ F \circ \varphi^{-1})(\varphi(p)) : \mathbb{R}^{\dim M} \rightarrow \mathbb{R}^{\dim N}.$$

Theorem 13.55 (Well-defined differential). The rule $dF_p[\varphi, v] = [\psi, A_{\psi\varphi}v]$ defines one intrinsic linear map $dF_p : T_pM \rightarrow T_{F(p)}N$, independent of both charts.

Proof. For other charts φ', ψ' , the actual coordinate maps factor as

$$\psi' F (\varphi')^{-1} = (\psi' \psi^{-1}) \circ (\psi F \varphi^{-1}) \circ (\varphi (\varphi')^{-1}).$$

Differentiating at $\varphi'(p)$ gives

$$A_{\psi'\varphi'} = C_{\psi'\psi}(F(p))A_{\psi\varphi}C_{\varphi\varphi'}(p).$$

If $v' = C_{\varphi'\varphi}(p)v$ is the new representative of the same input tangent vector, then

$$A_{\psi'\varphi'}v' = C_{\psi'\psi}A_{\psi\varphi}C_{\varphi\varphi'}C_{\varphi'\varphi}v = C_{\psi'\psi}A_{\psi\varphi}v.$$

These are exactly equivalent output tangent representatives. Thus the rule is well defined as a map on classes, with target $T_{F(p)}N$. Fixing charts, linearity of $A_{\psi\varphi}$ proves addition and scalar preservation; the already proved chart independence makes this intrinsic linearity. $\square$

This is the Chapter 10 change-of-basis formula with transition derivatives as coordinate conversions. The map dF_p stays fixed while the source and target coordinate descriptions change. In those coordinates the chain rule also gives $d(G \circ F)_p = dG_{F(p)} \circ dF_p$ and $d(\text{id}_M)_p = \text{id}_{T_pM}$.

At an interior point, every coordinate vector v is the velocity of $t \mapsto \varphi^{-1}(\varphi(p) + tv)$ for small positive and negative t . At a boundary point this interpretation needs care. A two-sided curve into $[0, \infty)$ through zero has derivative zero there, since zero is a minimum. Yet $T_0[0, \infty) = \mathbb{R}$: the tangent space contains inward and outward vectors. Coordinate vectors, rather than two-sided curves constrained to remain inside the manifold, give the correct full tangent space. Derivatives of local smooth extensions in half-space charts are unique: they agree on interior points and hence at boundary points by continuity.

Remark 13.56 (Two different boundaries). The circle in $\mathbb{R}^2$ is its own ambient topological boundary (it is closed with empty planar interior), but its manifold boundary is empty: every point has an interval chart. A closed disk has the circle as its manifold boundary. The genuine boundary is recognized in half-space charts, not by taking the boundary in an arbitrary ambient Euclidean space.

Definition 13.57 (Forms and intrinsic orientation). A smooth k -form assigns to each $p \in M$ an alternating covector $\omega_p \in \Lambda^k(T_p^*M)$, with smooth coefficients in charts. Here coordinate coefficients are computed using the tangent-coordinate isomorphisms of [Theorem 13.54](#); at boundary points they have local smooth extensions. Write $\Omega^k(M)$ for these forms, with pointwise operations, and put $\Omega^k(M) = \{0\}$ for $k > \dim M$. Degree zero consists of smooth real-valued functions. The transition rule for the coordinate forms is verified below. An **orientation** chooses one of the two orientation classes of ordered bases in each T_pM , varying continuously: locally there is a smooth frame (a smoothly varying basis of tangent vectors) in the chosen class at every point. Such frames are called positive frames. For zero-dimensional manifolds an orientation is a choice of sign at each point. The boundary uses the outward-normal-first convention.

In an interior chart we may choose coordinates whose coordinate frame is positive; transitions between such choices have positive determinants. In a boundary chart into the fixed model $\mathbb{H}^n$, instead record the locally constant sign $s_\varphi = \pm 1$ of the coordinate frame relative to the selected tangent orientation. Shrink chart components if needed so this sign is constant. On overlaps,

$$\text{sgn det } D(\psi \circ \varphi^{-1}) = s_\psi s_\varphi.$$

One may equivalently orient each coordinate model with its recorded sign; the chart is then an oriented chart relative to that model. We do not require every chart into one fixed half-space to have sign $+1$.

Example 13.58 (Why the oriented interval needs chart signs). Orient $[a, b]$ by increasing x . Near a the half-line chart is $u = x - a$, whose coordinate frame is $+\partial_x$, so $s = +1$. Near b the half-line chart is $v = b - x$, whose frame is $-\partial_x$, so $s = -1$. The manifold remains consistently oriented.

The outward vector at a is $-\partial_x$, and at b it is ∂_x . Outward-normal-first therefore assigns the point signs -1 at a , $+1$ at b . Explicitly,

$$\partial[a, b] = \{b\} - \{a\}, \quad \int_{\partial[a, b]} f = f(b) - f(a).$$

Requiring both endpoint coordinates to be positive relative to the fixed half-line would incorrectly exclude this basic example.

Here “outward normal” can mean any outward transverse tangent vector; no metric on the manifold is needed. In half-space coordinates an outward vector has negative last component. Replacing it by another outward vector multiplies its transverse component by a positive factor and adds a boundary tangent vector, which does not change the determinant sign. Transitions preserve the inward side, so oriented coordinate changes give the same boundary orientation. On ∂M the first $n - 1$ coordinates form charts, and their transition derivatives are invertible; thus the boundary is itself a smooth $(n - 1)$ -manifold without boundary.

For dimension zero the integral is the finite sum of values weighted by the chosen signs of the points. In particular, the oriented interval has boundary $\{b\} - \{a\}$, so $\int_{\partial[a, b]} f = f(b) - f(a)$.

13.9.4 Pullbacks, wedge products, and exterior derivatives on manifolds

The preceding definition specifies the values of a form. We now construct the operations used in manifold integration and Stokes, starting from the Euclidean operations in [Section 13.7](#).

For a smooth $F : M \rightarrow N$, define

$$F^* : \Omega^k(N) \rightarrow \Omega^k(M), \quad (F^*\omega)_p(v_1, \dots, v_k) = \omega_{F(p)}(dF_p v_1, \dots, dF_p v_k).$$

Initially this is a pointwise rule. The intrinsic linear map dF_p from [Theorem 13.55](#) makes it an alternating covector at p , independent of tangent representatives. For $k = 0$ the rule is $F^*f = f \circ F$. To check smoothness, choose charts with $F(U) \subseteq V$ and write $f = \psi F \varphi^{-1}$. If ω_ψ is the expression of ω in target coordinates, evaluation on coordinate vectors gives

$$(F^*\omega)_\varphi = f^*\omega_\psi.$$

Indeed both sides apply $\omega_\psi(f(x))$ to $Df(x)a_1, \dots, Df(x)a_k$. The right side is the already constructed Euclidean pullback and has smooth coefficients. This verifies the claimed target space, and proves agreement with Euclidean pullback. Linearity follows by evaluation at every point and tuple. The chain rule similarly gives $(G \circ F)^* = F^*G^*$ and $\text{id}^* = \text{id}$.

For $\alpha \in \Omega^p(M)$ and $\beta \in \Omega^q(M)$, define

$$(\alpha \wedge \beta)_x = \alpha_x \wedge \beta_x.$$

The right side is the alternating-covector wedge already defined in [Section 13.6](#). In coordinates its coefficients are finite sums of signed products of the coefficients of α and β ; they are smooth. Thus this defines a bilinear map into $\Omega^{p+q}(M)$. Pointwise pullback preserves it by the Euclidean shuffle formula.

Proposition 13.59 (The manifold exterior derivative). There is a unique operator $d : \Omega^k(M) \rightarrow \Omega^{k+1}(M)$ whose expression in every chart is the Euclidean exterior derivative.

Proof. For $\varphi : U \rightarrow \varphi(U)$ write $\omega_\varphi = (\varphi^{-1})^*(\omega|_U)$; this notation is now legitimate by the pullback construction. On an overlap with ψ , put $\chi = \psi\varphi^{-1}$. The identity $\varphi^{-1} = \psi^{-1} \circ \chi$ and pullback composition give

$$\omega_\varphi = \chi^* \omega_\psi.$$

Both sides have the same values on every tuple of coordinate vectors. Apply the already proved Euclidean identity in [Theorem 13.35](#) to obtain

$$d\omega_\varphi = d(\chi^* \omega_\psi) = \chi^*(d\omega_\psi).$$

Define the candidate on U as $\varphi^*(d\omega_\varphi)$. The displayed identity makes this equal to $\varphi^* \chi^*(d\omega_\psi) = \psi^*(d\omega_\psi)$ on the overlap. The local candidates therefore define a single global form with smooth coefficients. This is existence and chart independence. Any operator with the prescribed coordinate rule has these same local values, so is unique.

For a relatively open half-space domain, use local smooth extensions of the coefficients. Two extensions agree on interior points; all their derivatives agree there and then agree at boundary points by continuity. Thus the derivative is independent of the extension, and the Euclidean pullback identity above extends to the boundary by the same argument. This also justifies all preceding coordinate checks in boundary charts. $\square$

Equality of forms can be checked in charts. Consequently [Theorem 13.35](#) gives the intrinsic identities

$$d^2 = 0, \quad d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^p \alpha \wedge d\beta, \quad F^*(d\omega) = d(F^*\omega).$$

The last identity uses the local representative $f = \psi F \varphi^{-1}$ and the Euclidean identity for f , so it applies between different manifolds as well. Degrees above the dimension use the zero-space convention.

Support of a differential form. For a differential form ω on M , define

$$\text{supp}(\omega) = \overline{\{p \in M : \omega_p \neq 0\}}.$$

The form has **compact support** when this closed set is compact.

13.10 Partitions of Unity and Integration on Manifolds

Lemma 13.60 (Working with local finiteness). For a locally finite family of subsets (A_i) , every compact set meets only finitely many members. A locally finite family of smooth functions (meaning their supports form such a family) has a smooth pointwise sum. A locally finite union of closed subsets is closed.

Proof. For each x of a compact set K , choose an open neighborhood N_x meeting only finitely many A_i . Select finitely many $N_{x_1}, \dots, N_{x_m}$ covering K . Any A_i meeting K meets one selected neighborhood, so its index belongs to the union of their finitely many finite index lists. Only finitely many indices occur.

Near any point, only finitely many supports of the smooth functions meet one chosen neighborhood. All other functions vanish on that neighborhood, so the sum there is an ordinary finite smooth sum, and is smooth.

For closed A_i , let $x \notin \bigcup_i A_i$. Choose an open neighborhood N of x meeting only $A_{i_1}, \dots, A_{i_m}$. The set $N \setminus (A_{i_1} \cup \dots \cup A_{i_m})$ is open, contains x , and misses every A_i . Thus the complement of the full union is open. $\square$

A form may not be supported in one chart. We want to decompose it into smoothly varying chart-supported pieces without changing the total form:

$$\omega = \sum_{\alpha} \rho_{\alpha} \omega, \quad \sum_{\alpha} \rho_{\alpha} = 1.$$

Differentiating the weights introduces extra terms, but they cancel:

$$\sum_{\alpha} d\rho_{\alpha} = d\left(\sum_{\alpha} \rho_{\alpha}\right) = d1 = 0.$$

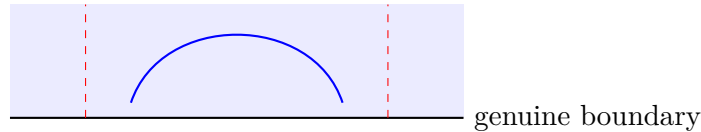
Local finiteness makes these operations finite near every point. The next theorem is our external topological input. The subsequent construction of smooth weights from the cutoff in [Theorem 8.37](#) is proved here.

Example 13.61 (Two weights before a locally finite family). Suppose two nonnegative smooth functions χ_1, χ_2 are supported in their respective chart domains and at least one is positive at each point under consideration. Then $\chi_1 + \chi_2 > 0$, so

$$\rho_1 = \frac{\chi_1}{\chi_1 + \chi_2}, \quad \rho_2 = \frac{\chi_2}{\chi_1 + \chi_2}$$

are smooth and sum to one. If the unnormalized values in an overlap are 2 and 3, the weights are $2/5$ and $3/5$: each chart gets its share and their sum counts the original form exactly once. For a concrete line example use the smooth transition s constructed from θ in Chapter 8, put $\chi_1 = 1 - s(x)$, $\chi_2 = s(x)$, and take chart domains $(-\infty, 2)$ and $(-1, \infty)$. Their supports lie in these domains and their sum is already one.

support stays away from artificial chart edges



A chart-supported function can vanish near its artificial edges while remaining nonzero on the genuine boundary. Zero extension across the former is smooth; zero extension across the latter generally is not. Local finiteness in the general construction makes the two-weight arithmetic finite in a neighborhood of each point.

Theorem 13.62 (External refinement theorem (used without proof)). Every second-countable Hausdorff smooth manifold, possibly with boundary, is paracompact. Given an open cover of M , there are countable locally finite families of open coordinate neighborhoods V_j and W_j , with the V_j covering M , such that $\overline{V_j}$ is compact and

$$\overline{V_j} \subseteq W_j \subseteq U_{\alpha(j)}$$

for a suitable index $\alpha(j) \in I$.

We use this standard topological refinement theorem without proof; see, for example, [5, Chapters 1–2]. Its proof belongs to general topology rather than to the Euclidean calculus developed here. The smooth bump-function construction and normalization required for partitions of unity are proved below.

Theorem 13.63 (Partition of unity). Let M be a smooth manifold, possibly with boundary, and let

$$\{U_\alpha : \alpha \in I\}$$

be an open cover. There are smooth functions

$$\rho_\alpha : M \rightarrow [0, 1]$$

such that the family is locally finite,

$$\text{supp}(\rho_\alpha) \subseteq U_\alpha,$$

and

$$\sum_{\alpha \in I} \rho_\alpha = 1.$$

In this situation, the partition of unity is said to be **subordinate** to the cover: its support condition says precisely that each

$$\text{supp}(\rho_\alpha)$$

lies in the corresponding U_α .

Proof. Apply the external refinement theorem [Theorem 13.62](#) to obtain

$$\overline{V_j} \subseteq W_j \subseteq U_{\alpha(j)}.$$

For a manifold with boundary, use boundary charts in this construction; the smooth-extension convention in the definition of a manifold with boundary makes the same Euclidean bump-function construction applicable.

For each j , let $\varphi_j : W_j \rightarrow O_j$ be its chart, with O_j open in $\mathbb{R}^n$ or relatively open in $\mathbb{H}^n$. Write $O_j = \tilde{O}_j \cap \mathbb{H}^n$ in the boundary case, with $\tilde{O}_j$ open in $\mathbb{R}^n$; in the interior case use $\tilde{O}_j = O_j$. The compact coordinate set $K_j = \varphi_j(\overline{V_j})$ is covered by finitely many open Euclidean balls whose closed balls lie in $\tilde{O}_j$. For their centers c_ℓ and radii r_ℓ , put

$$b_j(x) = \sum_{\ell} \theta(1 - \|x - c_\ell\|_2^2 / r_\ell^2).$$

By [Theorem 8.37](#) and the chain rule this is smooth, nonnegative, positive on K_j , and supported in the union of those closed balls. Restrict to O_j , transfer by φ_j , and extend by zero outside W_j in M , obtaining χ_j . Its support is compact inside W_j , so this extension is smooth across artificial chart boundaries. At the genuine boundary we retain the smooth Euclidean bump extension, which need not be zero on the other side. Thus

$$\chi_j \geq 0, \quad \chi_j > 0 \text{ on } V_j, \quad \text{supp } \chi_j \subset W_j.$$

Because the family (W_j) is locally finite and $\text{supp}(\chi_j) \subseteq W_j$, the family of supports $(\text{supp}(\chi_j))$ is locally finite. Therefore the sum

$$\chi = \sum_{j=1}^{\infty} \chi_j$$

is smooth and well defined by [Lemma 13.60](#); it is positive because the V_j cover M . Set

$$\rho_j = \frac{\chi_j}{\chi}.$$

The denominator is positive, and local finiteness makes χ smooth and each quotient smooth. The family (ρ_j) is locally finite, satisfies

$$\text{supp}(\rho_j) \subseteq \text{supp}(\chi_j) \subseteq W_j \subseteq U_{\alpha(j)},$$

and sums to one. For each original index α , define

$$\rho_\alpha = \sum_{\{j:\alpha(j)=\alpha\}} \rho_j.$$

These sums are locally finite, so the ρ_α are smooth and the regrouped family remains locally finite. Moreover, by [Lemma 13.60](#), a locally finite union of closed sets is closed, and therefore

$$\text{supp}(\rho_\alpha) \subseteq \bigcup_{\{j:\alpha(j)=\alpha\}} \text{supp}(\rho_j) \subseteq U_\alpha.$$

Regrouping the locally finite sum also gives $\sum_\alpha \rho_\alpha = 1$. Thus the partition is subordinate to the original cover. $\square$

Integrating in a boundary chart. If O is relatively open in $\mathbb{H}^n$ and $\eta = f \, dx_1 \wedge \cdots \wedge dx_n$ has compact support in O , extend f by zero within $\mathbb{H}^n$ outside O . It is continuous on the half-space. Integrate over a containing rectangle $[a_1, b_1] \times \cdots \times [a_{n-1}, b_{n-1}] \times [0, b_n]$. Finite additivity proves independence of this choice. Smooth coefficients extend locally across $x_n = 0$ by the definition of smoothness; that extension is generally not zero. Zero extension is used only across artificial chart boundaries, where compact support supplies a neighborhood of vanishing. This is the half-space integral in [Corollary 12.34](#).

Definition 13.64 (Integration on an oriented manifold). Let M be an oriented n -manifold and let ω be a smooth n -form with compact support. Choose a partition of unity subordinate to oriented coordinate charts and define

$$\int_M \omega = \sum_\alpha s_\alpha \int_{\varphi_\alpha(U_\alpha)} (\varphi_\alpha^{-1})^* (\rho_\alpha \omega).$$

Here s_α is the coordinate-frame sign defined above; each written coordinate integral uses the standard orientation of its Euclidean or half-space model. Equivalently integrate without the explicit sign in the correspondingly oriented model. Only finitely many supports $\text{supp}(\rho_\alpha)$ meet the compact set $\text{supp}(\omega)$. Furthermore,

$$\text{supp}(\rho_\alpha \omega) \subseteq \text{supp}(\rho_\alpha) \cap \text{supp}(\omega) \subseteq U_\alpha.$$

This support is a closed subset of the compact set $\text{supp}(\omega)$, so it is compact. Its image under φ_α is therefore compact and contained in the chart image, open in Euclidean space or relatively open in the half-space. Thus every nonzero summand is the appropriate compactly supported coordinate integral defined above, and the displayed sum is finite.

Proposition 13.65 (Well-definedness). The preceding integral is independent of the oriented charts and subordinate partition of unity.

Proof. To compare two choices, compare both to a common refinement. Let (ρ_α) and (σ_β) be partitions subordinate to chart families $(U_\alpha, \varphi_\alpha)$ and (V_β, ψ_β) . Let $K = \text{supp}(\omega)$, which is compact.

By Lemma 13.60, only finitely many supports in either family meet K . Thus only finitely many products below can be nonzero. Define $\eta_{\alpha\beta} = \rho_\alpha \sigma_\beta \omega$. Its support is closed and is contained in

$$K \cap \text{supp}(\rho_\alpha) \cap \text{supp}(\sigma_\beta) \subseteq U_\alpha \cap V_\beta.$$

It is therefore compact and lies within the chart overlap, so either chart can integrate this same localized form.

For a chart-supported form θ , write its signed coordinate integral as

$$I_\alpha(\theta) = s_\alpha \int_{\varphi_\alpha(U_\alpha)} (\varphi_\alpha^{-1})^* \theta, \quad J_\beta(\theta) = s_\beta \int_{\psi_\beta(V_\beta)} (\psi_\beta^{-1})^* \theta.$$

On the overlap the transition determinant has sign $s_\beta s_\alpha$. The signed change-of-coordinates formula (Proposition 13.43 and Corollary 12.34) therefore gives $I_\alpha(\eta_{\alpha\beta}) = J_\beta(\eta_{\alpha\beta})$, including opposite coordinate signs. Now display the full comparison:

$$\begin{aligned} \sum_\alpha I_\alpha(\rho_\alpha \omega) &= \sum_{\alpha, \beta} I_\alpha(\eta_{\alpha\beta}) \\ &= \sum_{\alpha, \beta} J_\beta(\eta_{\alpha\beta}) \\ &= \sum_\beta J_\beta(\sigma_\beta \omega). \end{aligned}$$

The first equality uses $\sum_\beta \sigma_\beta = 1$ and linearity in a fixed chart. The middle equality is the overlap formula just verified. The last uses $\sum_\alpha \rho_\alpha = 1$. Compact support and local finiteness make this a finite double sum, so no infinite rearrangement is involved. Both choices give the same integral. $\square$

Example 13.66 (The circle integral using two chart-supported pieces). Orient S^1 counterclockwise and restrict $\omega = -y \, dx + x \, dy$ from the plane. The angle parametrization $\gamma(t) = (\cos t, \sin t)$ has $\gamma^* \omega = dt$, but no single angle chart covers the circle. Use $U_+ = S^1 \setminus \{(-1, 0)\}$ with angles $t \in (-\pi, \pi)$ and $U_- = S^1 \setminus \{(1, 0)\}$ with angles $t \in (0, 2\pi)$. Both angle coordinate frames have positive orientation.

Let θ be the smooth function from Chapter 8 that is zero on $(-\infty, 0]$ and $e^{-1/s}$ for $s > 0$. For $(x, y) \in S^1$, define

$$\rho_+(x, y) = \frac{\theta(x + 1/2)}{\theta(x + 1/2) + \theta(1/2 - x)}, \quad \rho_- = 1 - \rho_+.$$

The denominator is positive: the two arguments cannot both be nonpositive. These are smooth nonnegative weights summing to one. The support of ρ_+ lies in $\{x \geq -1/2\} \subset U_+$, and that of ρ_- lies in $\{x \leq 1/2\} \subset U_-$. Both supports are compact. Thus this is an explicit subordinate partition of unity. Put $r_\pm(t) = \rho_\pm(\gamma(t))$, which are 2π -periodic. The definition of manifold integration gives the two localized coordinate integrals

$$\int_{S^1} \omega = \int_{-\pi}^{\pi} r_+(t) \, dt + \int_0^{2\pi} r_-(t) \, dt.$$

Their integrands vanish near the artificial endpoints of their respective charts, as required by chart-supported integration. Split the first integral at zero and substitute $s = t + 2\pi$ on $[-\pi, 0]$. Periodicity changes it to $\int_0^{2\pi} r_+(s) \, ds$. Consequently

$$\int_{S^1} \omega = \int_0^{2\pi} (r_+ + r_-)(t) \, dt = \int_0^{2\pi} 1 \, dt = 2\pi.$$

The global definition reproduces the earlier line integral by adding two chart-supported pieces; it never assumes a global angle coordinate.

13.11 The General Stokes Theorem

For the boundary inclusion $\iota : \partial M \rightarrow M$, the notation $\int_{\partial M} \omega$ always means $\int_{\partial M} \iota^* \omega$. The pullback restricts the form to tangent directions of the boundary.

The line-integral FTC, Green, and patch Stokes already exhibit the formula. The global proof now has four visible steps: localize the form, apply the Euclidean or half-space theorem, cancel artificial terms, and retain only the genuine manifold boundary.

Theorem 13.67 (General Stokes theorem). Let M be an oriented smooth n -manifold with boundary, $n \geq 1$, and let ω be a smooth

$$(n-1)\text{-form}$$

with compact support. Then

$$\int_M d\omega = \int_{\partial M} \omega.$$

Proof. Choose a partition of unity subordinate to oriented manifold-with-boundary charts that carry M locally into $\mathbb{H}^n$ and carry ∂M into $\{x_n = 0\}$. Compactness of $\text{supp}(\omega)$ and local finiteness show that only finitely many forms

$$\rho_\alpha \omega$$

are nonzero. Each has compact support contained in the corresponding chart, because $\text{supp}(\rho_\alpha \omega) \subseteq \text{supp}(\rho_\alpha) \cap \text{supp}(\omega)$. By the product rule,

$$d(\rho_\alpha \omega) = d\rho_\alpha \wedge \omega + \rho_\alpha d\omega.$$

After summing, the first terms cancel because

$$\sum_\alpha d\rho_\alpha = d\left(\sum_\alpha \rho_\alpha\right) = 0.$$

This identity is legitimate because the partition is locally finite: near each point both sums are finite, so exterior differentiation commutes with the displayed sum. In each local Stokes calculation multiply both sides by the chart sign s_α . Outward-normal-first transfers the same sign to its genuine boundary integral, so the local formula remains valid for either orientation of the coordinate model. It is therefore enough to prove the formula for a form supported in one chart, where “supported in” means that its closure-based support lies in the chart domain. In an interior chart, the coordinate support is compact in the open chart image. Hence the coordinate representative vanishes on a neighborhood of the artificial boundary of that image and extends smoothly by zero to $\mathbb{R}^n$. Choose a rectangle whose interior contains the extended support. The extension vanishes near the rectangle boundary, so [Theorem 13.40](#) gives the formula with no manifold boundary contribution. In a boundary chart, the half-space definition of smoothness gives a smooth (not necessarily zero) extension of the coordinate representative across

$$x_n = 0$$

to a neighborhood in $\mathbb{R}^n$. Compact coordinate support permits zero extension across the remaining artificial boundary of the coordinate image; no zero extension is made across the genuine face $x_n = 0$. Choose a rectangle

$$[a_1, b_1] \times \cdots \times [a_{n-1}, b_{n-1}] \times [0, b_n]$$

whose relative interior contains its coordinate support and whose artificial faces are disjoint from it. Using a cutoff which is one near that support, retain the smooth extension across $x_n = 0$ and

extend by zero only across the artificial faces; the resulting form vanishes near all of those faces. Apply [Theorem 13.40](#) to the smooth extension. Its only surviving boundary term is the face $x_n = 0$, which is the coordinate representative of the genuine manifold boundary. The outward-normal-first convention gives exactly the induced boundary sign. Pullback compatibility from [Theorem 13.35](#) transfers each coordinate formula back to the chart. Summing the finitely many contributions proves the result. $\square$

Corollary 13.68 (Integration by parts for differential forms). Let M be an oriented smooth n -manifold with boundary, let

$$0 \leq p \leq n - 1,$$

let α be a smooth p -form with compact support, and let β be a smooth

$$(n - p - 1)\text{-form}.$$

Then

$$\int_M d\alpha \wedge \beta = \int_{\partial M} \alpha \wedge \beta - (-1)^p \int_M \alpha \wedge d\beta.$$

Proof. The form

$$\alpha \wedge \beta$$

has degree $n - 1$ and compact support, so [Theorem 13.67](#) applies. The graded Leibniz rule from [Theorem 13.35](#) gives

$$d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^p \alpha \wedge d\beta.$$

Integrate this identity over M , apply Stokes to its left side, and rearrange. $\square$

Remark 13.69 (Three levels of integration by parts). The same structural identity appears at three increasing levels of generality:

$$\begin{aligned} & \text{product rule plus the Fundamental Theorem of Calculus} \\ & \implies \text{one-variable integration by parts,} \\ & \text{one-variable integration by parts plus Fubini} \\ & \implies \text{the coordinatewise rectangular formula,} \\ & \text{graded Leibniz rule plus Stokes} \\ & \implies \text{integration by parts for forms.} \end{aligned}$$

The final formula is coordinate-free. Green's identities and weak formulations of partial differential equations are important later descendants, but they require additional analytic machinery.

13.12 Classical Theorems Revisited

These are now instances of one theorem. The form degree specifies the dimension of integration; the exterior derivative specifies the interior quantity. In three dimensions the flux form is the one already calculated on surface patches. The general theorem additionally joins the chart pieces consistently and handles arbitrary compact smooth oriented surfaces.

Corollary 13.70 (Fundamental theorem of calculus). For a smooth function f on

$$[a, b],$$

$$\int_a^b f'(x) \, dx = f(b) - f(a).$$

Proof. Apply [Theorem 13.67](#) to the oriented interval and the 0-form f . $\square$

Corollary 13.71 (Green's theorem). Let $D \subseteq \mathbb{R}^2$ be a compact smooth 2-manifold with boundary, equipped with the orientation induced by the standard orientation of $\mathbb{R}^2$, and let P, Q be smooth on a neighborhood of D . The induced boundary orientation is the usual positive orientation. Then

$$\int_{\partial D} P \, dx + Q \, dy = \int_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx \, dy.$$

Proof. Apply [Theorem 13.67](#) to

$$\omega = P \, dx + Q \, dy.$$

$\square$

Corollary 13.72 (Classical Stokes theorem). Let S be an oriented smooth compact surface with boundary in $\mathbb{R}^3$ and let

$$F = (P, Q, R)$$

be smooth near S . Then

$$\int_{\partial S} P \, dx + Q \, dy + R \, dz = \int_S (\nabla \times F) \cdot n \, dS.$$

Proof. Apply [Theorem 13.67](#) to the 1-form

$$P \, dx + Q \, dy + R \, dz.$$

The coordinate calculation of the exterior derivative identifies it with the curl flux form, as in the parametrized-patch theorem. $\square$

Definition 13.73 (Flux in arbitrary dimension). For a smooth vector field $F = (F_1, \dots, F_n)$ on a Euclidean open set, its flux form is the $(n-1)$ -form

$$\omega_F = \sum_{j=1}^n (-1)^{j-1} F_j \, dx_1 \wedge \cdots \wedge \widehat{dx_j} \wedge \cdots \wedge dx_n.$$

For an oriented boundary with inclusion ι , the notation

$$\int_{\partial \Omega} F \cdot n \, dS \quad \text{means} \quad \int_{\partial \Omega} \iota^* \omega_F.$$

This defines the shorthand through form integration; no separate hypersurface-measure construction is required here.

For $n = 3$ and an oriented patch $\Phi(u, v)$, evaluation gives $\omega_F(\Phi_u, \Phi_v) = \det(F, \Phi_u, \Phi_v) = F \cdot (\Phi_u \times \Phi_v)$ by [Proposition 13.12](#). Thus the definition agrees with the earlier unit-normal flux formula. For $n = 1$, the flux form is the function F_1 and the signed endpoint integral gives its value at the right endpoint minus its value at the left.

Corollary 13.74 (Divergence theorem). Let $\Omega \subseteq \mathbb{R}^n$ be a compact smooth n -manifold with boundary, equipped with the orientation induced by the standard orientation of $\mathbb{R}^n$, and let

$$F = (F_1, \dots, F_n)$$

be smooth near Ω . The induced boundary orientation is the usual outward orientation. Then

$$\int_{\partial \Omega} F \cdot n \, dS = \int_{\Omega} \left(\sum_{j=1}^n \frac{\partial F_j}{\partial x_j} \right) dx.$$

Proof. Apply [Theorem 13.67](#) to the flux form ω_F of [Definition 13.73](#). Its exterior derivative is $(\sum_j \partial F_j / \partial x_j) dx_1 \wedge \cdots \wedge dx_n$: only the j th derivative of F_j survives, and moving dx_j past $j-1$ factors cancels the coefficient sign. Its boundary integral is the outward flux by definition. $\square$

Exercises

Exercise 13.5. Prove that

$$\dim \Lambda^k((\mathbb{R}^n)^*) = \binom{n}{k}.$$

Exercise 13.6. Let α, β be 1-covectors. Prove

$$\alpha \wedge \beta = -\beta \wedge \alpha$$

and give an example in which

$$\alpha \wedge \beta \neq 0.$$

Exercise 13.7. For

$$\omega = x \, dy + y \, dz + z \, dx,$$

compute

$$d\omega.$$

Exercise 13.8. Let

$$\Phi(u, v) = (u^2 - v^2, 2uv).$$

Compute

$$\Phi^*(x \, dy - y \, dx).$$

Exercise 13.9. Use [Theorem 13.40](#) in dimension one to recover the Fundamental Theorem of Calculus.

Exercise 13.10. Apply [Corollary 13.68](#) to the oriented interval

$$[a, b]$$

and the forms $\alpha = u$ and $\beta = v$ to recover the one-variable integration-by-parts formula for smooth functions u and v .

Exercise 13.11. For the unit square with its standard orientation, evaluate

$$\int_{\partial Q} x \, dy$$

and compare it with

$$\int_Q d(x \, dy).$$

Exercise 13.12. Use Green's theorem to compute the area enclosed by a positively oriented smooth simple closed plane curve.

Exercise 13.13. Use the divergence theorem to compute the outward flux of

$$F(x, y, z) = (x, y, z)$$

across the unit sphere.

Exercise 13.14. Explain why

$$d^2 = 0$$

is compatible with the boundary-of-boundary principle reflected in Stokes' theorem. Distinguish this compatibility from a chain-level theorem, which has not been developed here.

Exercise 13.15 (Hypotheses and synthesis). Test the scope of the preceding constructions.

- (a) Why does the full circular parametrization on $[0, 2\pi]$ fail the injective-patch hypothesis? Describe a valid subdivision.
- (b) Why can a two-sided curve in $[0, \infty)$ through zero not represent the outward tangent vector there?
- (c) If $\rho_1 + \rho_2 = 1$ near the support of ω , expand $d(\rho_1\omega) + d(\rho_2\omega)$ and identify the cancelling terms.
- (d) For $\Phi(u, v) = (u, v, u^2 + v^2)$ on a triangle and $F = (-y, x, 0)$, verify patch Stokes by pulling back work to the triangle and computing curl flux. State exactly why this does not by itself prove Stokes on a sphere.

Exercise 13.16. For $\gamma(t) = (t, t^2)$ on $[0, 1]$, compute the work of $F = (y, x)$. Repeat with $h(s) = (s + s^2)/2$ by substitution and under reversal. Explain why doubling the traversal is a different operation.

Exercise 13.17. Pull back $z \, dx \wedge dy$ by $(u, v) \mapsto (u, v, uv)$ and integrate over $[0, 1]^2$. Interpret the result as an oriented flux.

Exercise 13.18. For $\omega = x \, dy - y \, dx$, compute $\omega_{(1,2)}(3, 4)$, its pullback by $t \mapsto (t, t^2)$, and $d\omega$. State the type of each answer and why a two-form pulled back to this curve is zero.

Exercise 13.19. Use the upper- and right-semicircle charts to transform a tangent coordinate at $(1/\sqrt{2}, 1/\sqrt{2})$. Check the answer by differentiating the corresponding parametrizations in the ambient plane.

Exercise 13.20. For a positively oriented interval verify its two endpoint chart signs and Stokes for $f(x) = x^2$ on $[1, 3]$. Explain why the endpoint at 3 can have a negative chart sign and a positive boundary sign.

Exercise 13.21. Construct two smooth weights on an overlapping cover of an interval by normalizing cutoff functions. Check positivity of the denominator and show directly that $d\rho_1 + d\rho_2 = 0$.

Exercise 13.22. For a disk covered by two overlapping surface charts, explain which chart edges disappear in the Stokes proof and which boundary remains. Distinguish cancellation of derivatives of weights from zero extension near an artificial chart edge.

Exercise 13.23. For two circle angle charts differing by 2π on an overlap component, compute their transition derivative and check the tangent-vector equivalence relation. Then replace an angle coordinate by twice that coordinate and compute how a tangent-vector component changes without changing the vector.

Exercise 13.24. Use the coordinate transformation formula for dF_p to compare two source and two target charts for $F : S^1 \rightarrow S^1$, $F(\cos t, \sin t) = (\cos 2t, \sin 2t)$ locally. Check explicitly that equivalent source representatives give equivalent outputs.

Exercise 13.25. Reconstruct the common-refinement proof for manifold integration. Identify which support assumption permits each overlap integral and which hypothesis makes the double sum finite.

Exercise 13.26. Let $F : S^1 \rightarrow S^1$ be $F(x, y) = (x^2 - y^2, 2xy)$, and let $f(x, y) = x$ on S^1 . In angle coordinates compute $F^*(df)$ and $d(F^*f)$. Repeat near the same point with source coordinate $s = 2t$ and explain why the two coordinate answers describe the same one-form.

Exercise 13.27. For the inclusion $\iota : S^1 \rightarrow \mathbb{R}^2$ and $\alpha = x \, dy$, compute $\iota^*\alpha$ in angle coordinates and verify $d(\iota^*\alpha) = \iota^*(d\alpha) = 0$. Explain the role of $\Omega^2(S^1) = \{0\}$. Then reconstruct the overlap identity that makes the manifold exterior derivative well defined.

13.13 Perspective and Further Directions

Differential forms separate the three ingredients hidden in vector-calculus formulas: differentiation is the exterior derivative, orientation controls sign, and integration pairs top-degree forms with domains. Stokes' theorem is the single compatibility law joining them. The familiar one-, two-, and three-dimensional formulas differ only in the notation used to identify forms with scalar and vector fields.

Measure Theory and Lebesgue Integration. The limitations of the Riemann and Jordan theories noted in Chapters 9 and 12 lead first to measure theory and Lebesgue integration. There one develops sigma-algebras, measures, measurable functions, and a more flexible integral. The genuine Monotone Convergence Theorem, Dominated Convergence Theorem, Fatou's lemma, and the Tonelli–Fubini theorems then hold under hypotheses beyond the Riemann results available here. Uniform convergence guarantees integrability of the limit and convergence of integrals; the one-dimensional Arzelà theorem also permits uniformly bounded pointwise convergence, but assumes integrability of the limit separately. Lebesgue convergence theorems address measurable limits and domination in a broader setting.

General Topology. General topology studies the open-set language of Chapter 10 beyond metric and finite-dimensional spaces: bases and subbases, quotient and product spaces, separation axioms, compactness, connectedness, paracompactness, and metrization questions. The external refinement theorem used above is a small but important example of this broader theory.

Functional Analysis. Functional analysis extends normed-space methods to Banach and Hilbert spaces, bounded linear operators, infinite-dimensional differentiation, contraction principles, spectral theory, and weak convergence. In infinite dimensions, the equivalence of norms and the compactness behavior familiar from Chapter 10 no longer persist automatically.

Complex Analysis. The power-series, differentiation, and integration methods also open the way to complex analysis. Complex differentiability for maps

$$f : \mathbb{C} \longrightarrow \mathbb{C}$$

is far more restrictive than real differentiability and leads to holomorphic functions, contour integrals, Cauchy's theorem and integral formula, and residues.

Differential Geometry. Chapter 13 begins differential geometry: tangent and cotangent bundles, vector fields, Riemannian metrics, geodesics, connections, covariant differentiation, curvature, and integration on manifolds elaborate the local linear structures and differential forms already present here.

De Rham Theory and Algebraic Topology. One especially direct continuation starts with the identity

$$d^2 = 0.$$

A form ω is called **closed** when

$$d\omega = 0,$$

and **exact** when

$$\omega = d\eta.$$

Every exact form is closed. De Rham cohomology records the extent to which the converse can fail globally:

$$H_{\text{dR}}^k(M) = \frac{\ker(d : \Omega^k(M) \longrightarrow \Omega^{k+1}(M))}{\text{im}(d : \Omega^{k-1}(M) \longrightarrow \Omega^k(M))}.$$

Thus it measures the obstruction to solving $\omega = d\eta$ on all of M . De Rham's theorem, not proved here, identifies these analytic groups with topological invariants and forms a central bridge to algebraic topology.

Example 13.75 (Closed does not mean globally exact). On $\mathbb{R}^2 \setminus \{0\}$ put

$$\eta = \frac{-y \, dx + x \, dy}{x^2 + y^2}.$$

Writing $P = -y/(x^2 + y^2)$ and $Q = x/(x^2 + y^2)$ gives $Q_x = (y^2 - x^2)/(x^2 + y^2)^2 = P_y$, so $d\eta = 0$. On the unit circle its pullback is dt , hence its integral is 2π (use the two injective semicircle patches). If $\eta = df$ globally, the line-integral FTC would make this integral zero. Thus it is closed but not exact. The missing origin prevents filling this loop by a disk in the domain. This example is a preview of a global obstruction, not a new cohomology theory.

Partial Differential Equations. Multivariable derivatives, divergence, integration by parts, and Stokes-type identities lead naturally to partial differential equations: the Laplace and Poisson equations, the heat and wave equations, weak derivatives, Sobolev spaces, and variational methods. Modern PDE joins the ideas of multivariable analysis with measure theory and functional analysis.

Probability. Modern probability provides a related application of measure theory: probability spaces, random variables, expectation as an integral, and convergence theorems give a rigorous language for random phenomena.

Appendix A

Optional: Permutations and Determinants over a Field

This appendix supplies the systematic permutation construction behind Chapter 10. A field F has addition, subtraction, multiplication, and division by nonzero elements obeying the familiar arithmetic laws, with $0 \neq 1$. No background in abstract algebra is assumed. Unless otherwise stated, vector spaces here are finite dimensional over F . Replace real scalars by elements of F in the vector-space axioms. Taking $F = \mathbb{R}$ recovers the main text; neither this appendix nor the next is a prerequisite for Chapters 11–13. The permutation exposition is influenced by the author’s algebra notes [4], specialized here to the elementary facts needed for determinants.

A.1 What Survives over an Arbitrary Field?

Span, independence, exchange, bases, dimension, linear maps, matrices, dual spaces, and multilinear maps use only field arithmetic. The Chapter 10 proofs therefore work over F . Determinants, tensor products, and exterior powers also belong to this algebraic theory, as the constructions below show.

A compatible total order, positivity, absolute value or norm, metric, topology, convergence, and orientation via determinant sign are additional structures, not consequences canonically supplied by field axioms. This does not mean fields cannot carry topologies or norms; it means these must be specified. Over $\mathbb{R}$ positive-definite symmetric bilinear inner products are natural. Over $\mathbb{C}$ the corresponding structure is Hermitian: conjugating and exchanging its arguments preserves its value, so $\langle v, v \rangle = \overline{\langle v, v \rangle}$ is real and positivity can be required there. Some other characteristic-zero fields support analogues when supplied with an ordering or a suitable involution (a field automorphism whose square is the identity). Characteristic zero alone guarantees neither such structure nor positivity. The Euclidean/Hermitian theory of inner-product spaces does not generalize canonically to arbitrary fields, although algebraic theories of bilinear, quadratic, Hermitian, and alternating forms remain meaningful. Positivity, distances, and completeness place inner-product theory in an analytic or geometric setting with additional scalar structure.

Example A.1 (Finite fields and characteristic two). For prime p , the residues $\mathbb{F}_p$ with arithmetic modulo p form a field, and $\mathbb{F}_p^n$ has the usual standard basis. The **characteristic** of a field is the least positive integer p such that $p \cdot 1 = 0$, or zero if there is no such integer. A positive characteristic is prime: if $p = ab$ with $1 < a, b < p$, then $(a \cdot 1)(b \cdot 1) = 0$ contradicts invertibility of nonzero factors. An ordered field has $1 > 0$; repeated addition would give $p \cdot 1 > 0$, contradicting $p \cdot 1 = 0$. Thus positive-characteristic fields cannot be ordered fields, and ordinary Euclidean

positivity is unavailable. Over $\mathbb{F}_2$, the symmetric bilinear expression $B(x, y) = x_1y_1 + x_2y_2$ has $B((1, 1), (1, 1)) = 1 + 1 = 0$ although $(1, 1) \neq 0$. A symmetric bilinear form is therefore not automatically an analytic inner product.

Alternating fundamentally means zero on repeated arguments. Expansion of two equal inputs $u + v$ still proves the interchange identity $A(u, v) = -A(v, u)$ (with any other slots fixed). In characteristic two, $-1 = 1$, so an alternating bilinear form is also symmetric. Symmetry alone does not force zero diagonal: $B(x, y) = xy$ on $\mathbb{F}_2$ has $B(1, 1) = 1$. The same warning applies to exterior products: $v \wedge v = 0$ is fundamental; the consequence $v \wedge w = -w \wedge v$ becomes $v \wedge w = w \wedge v$.

A.2 Cycles, Transpositions, and Parity

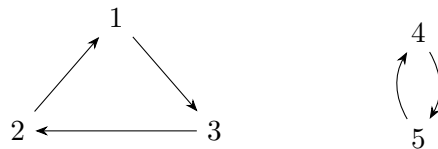
A permutation of $\{1, \dots, n\}$ is a bijection of this set to itself. Their set is denoted S_n . Composition is composition of bijections: $(\sigma\tau)(j) = \sigma(\tau(j))$. We read the right-hand permutation first. The identity fixes every entry, and σ^{-1} reverses all arrows of σ . No formal group theory is needed. A **cycle** $(a_1 a_2 \cdots a_k)$ sends each displayed entry to the next and the last back to the first, fixing all others.

A.2.1 Reading and composing bijections

A two-row display records inputs over their outputs:

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 1 & 2 & 5 & 4 \end{pmatrix}.$$

Its output list $(3, 1, 2, 5, 4)$ is shorter, but a cycle uses a different convention: $(1 \ 3 \ 2)$ means $1 \mapsto 3 \mapsto 2 \mapsto 1$. Thus $\sigma = (1 \ 3 \ 2)(4 \ 5)$. Starting the same cycle elsewhere does not change it: $(1 \ 3 \ 2) = (3 \ 2 \ 1) = (2 \ 1 \ 3)$. Reversing it does change the map and gives its inverse $(1 \ 2 \ 3)$. An omitted entry is fixed; it is not absent from the domain.



Example A.2 (Why order matters). Let $\alpha = (1 \ 2 \ 3)$ and $\beta = (1 \ 2)$ in S_3 . Follow each input through two arrows:

j	$\beta(j)$	$\alpha(\beta(j))$	$\alpha(j)$	$\beta(\alpha(j))$
1	2	3	2	1
2	1	2	3	3
3	3	1	1	2

Thus $\alpha\beta = (1 \ 3)$, but $\beta\alpha = (2 \ 3)$. By contrast $(1 \ 3 \ 2)$ and $(4 \ 5)$ act on disjoint sets of entries. For an entry in the first set, the second cycle does nothing, and conversely; every other entry is fixed by both. Hence disjoint cycles commute.

A.2.2 Following a permutation until it closes

Why should cycles describe every bijection? Start at one entry and keep following its image. There are finitely many entries, so eventually one repeats. Injectivity prevents the path from joining itself

anywhere except its starting point: if $\sigma^r(a) = \sigma^s(a)$ with $0 < r < s$, applying the inverse r times gives $a = \sigma^{s-r}(a)$, an earlier return. After closing one cycle, begin at an unused entry.

Proposition A.3 (Cycle decomposition). Every permutation decomposes into disjoint cycles, uniquely up to the order of the cycles and cyclic rotation of their entries. Every permutation is a product of transpositions, where a transposition is a two-entry cycle.

Proof. Idea. Follow each entry until it returns, then repeat with an unused entry. The list $a, \sigma(a), \sigma^2(a), \dots$ must repeat in the finite set. If the first repeated entry were $\sigma^i(a) = \sigma^j(a)$ with $0 < i < j$, applying σ^{-1} i times would give the earlier return $a = \sigma^{j-i}(a)$, a contradiction. Thus the first return is to a and determines its entire cycle, with its order forced by σ . If a later cycle started at an unused b and met an earlier cycle at $\sigma^r(b) = c$, applying σ^{-r} would put b in that earlier cycle, since following its arrows backward stays in it. This contradicts the choice of b . The cycles are disjoint and the finite procedure exhausts all entries. Any cycle decomposition must place a together with all its successive iterates in this forced order. Hence only the order of the disjoint cycles and the starting point used to write each cycle can vary. For a cycle use

$$(a_1 \ a_2 \ \cdots \ a_k) = (a_1 \ a_k)(a_1 \ a_{k-1}) \cdots (a_1 \ a_2),$$

as can be checked on every displayed entry. Combine these expressions for the disjoint cycles. $\square$

For example the permutation with list $(2, 3, 1, 5, 4)$ is $(1 \ 2 \ 3)(4 \ 5)$, and equals $(1 \ 3)(1 \ 2)(4 \ 5)$.

Example A.4 (A full decomposition and its inverse). The list $(4, 3, 2, 5, 1, 6)$ sends $1 \rightarrow 4 \rightarrow 5 \rightarrow 1$, $2 \rightarrow 3 \rightarrow 2$, and fixes 6. Hence

$$\sigma = (1 \ 4 \ 5)(2 \ 3) = (1 \ 5)(1 \ 4)(2 \ 3), \quad \sigma^{-1} = (1 \ 5 \ 4)(2 \ 3).$$

For a cycle $(a_1 \cdots a_k)$, the transposition formula can be verified without multiplying symbols formally. The rightmost swap takes a_1 to a_2 , which later swaps leave alone. For $2 \leq j < k$, the swap $(a_1 \ a_j)$ takes a_j to a_1 , and the next takes it to a_{j+1} . The final entry a_k goes to a_1 . Every unlisted entry stays fixed.

A.2.3 Adjacent swaps and the parity question

An **adjacent transposition** is $(i \ i+1)$. Swapping positions $i < j$ can be accomplished by moving the entry at i rightward to j and moving the displaced entry leftward back to i . Intermediate entries return to their original places. This uses $2(j-i) - 1$ adjacent swaps:

$$(i \ j) = (i \ i+1) \cdots (j-2 \ j-1)(j-1 \ j)(j-2 \ j-1) \cdots (i \ i+1).$$

For example $(1 \ 3) = (1 \ 2)(2 \ 3)(1 \ 2)$. The same permutation therefore has a one-swap and a three-swap expression. Inserting $(1 \ 2)(1 \ 2)$ produces still more expressions. Length is not intrinsic. The important issue is whether its parity is intrinsic: could the same permutation have both a two-swap and a three-swap expression?

We need parity to be independent of the chosen transpositions. The next proof uses formal polynomials with integer coefficients only as finite expressions in indeterminates; it does not assume any polynomial theory.

Theorem A.5 (Parity and sign). Any two transposition expressions for a permutation have the same parity of length. Define $\text{sgn}(\sigma) = (-1)^r$ using that parity. Then sign is multiplicative under composition.

Proof. Idea. A nonzero formal expression detects each swap by changing sign. Consider the Vandermonde expression $\Delta(t_1, \dots, t_n) = \prod_{i < j} (t_j - t_i)$ over the integers. Swapping adjacent variables t_i, t_{i+1} negates their mutual factor. Their factors involving any other variable exchange in pairs, so the whole expression is negated. Swapping positions $i < j$ can be done in $2(j-i)-1$ adjacent swaps, hence also negates Δ . Thus a decomposition of length r changes Δ to $(-1)^r \Delta$. Two decompositions of the same permutation perform the same substitution, so yield equal expressions. They cannot yield opposite signs: evaluating Δ at distinct integers shows it is nonzero and not its negative. Parity is therefore well defined. Concatenating decompositions proves $\text{sgn}(\sigma\tau) = \text{sgn}(\sigma)\text{sgn}(\tau)$. $\square$

Call permutations with sign 1 **even** and sign -1 **odd**. A k -cycle has sign $(-1)^{k-1}$. The parity is defined over the integers even when coefficients later lie in characteristic two; interpreting its signs in that field sends both $+1$ and -1 to 1 without changing the combinatorial parity theorem.

For clarity, in the adjacent-swap step of the parity proof the factors involving a third index $k < i$ are $(t_i - t_k)(t_{i+1} - t_k)$; after the swap these two factors exchange places. For $k > i+1$ the pair is $(t_k - t_i)(t_k - t_{i+1})$, with the same effect. Only $(t_{i+1} - t_i)$ changes sign. The test expression belongs to integer arithmetic, even if the determinant will later have coefficients in a different field. This separation avoids incorrectly trying to distinguish 1 and -1 inside a field of characteristic two.

The identity uses zero swaps and has sign 1. Reversing a swap expression gives an expression of the same length for the inverse, since each swap undoes itself. Therefore $\text{sgn}(\sigma^{-1}) = \text{sgn}(\sigma)$. A k -cycle uses $k-1$ swaps; a permutation with disjoint cycles of lengths $k_1, \dots, k_r$ has sign $(-1)^{\sum_j (k_j-1)}$. Fixed points contribute zero to this exponent.

A.2.4 Counting inversions instead of choosing swaps

Proposition A.6 (Inversions compute the same sign). An inversion of σ is a pair $i < j$ with $\sigma(i) > \sigma(j)$. Then $\text{sgn}(\sigma) = (-1)^{\text{inv}(\sigma)}$.

Proof. If the list is not increasing, it has an adjacent inverted pair. Exchanging that pair decreases the inversion count by one: their mutual inversion disappears and comparisons with all other entries have the same total count. After exactly $\text{inv}(\sigma)$ exchanges the list is increasing, hence is the identity. Parity now gives the formula. $\square$

Example A.7 (Inversions in a complete list). For $(3, 1, 2, 5, 4)$ the inverted position pairs are $(1, 2), (1, 3), (4, 5)$. There are three, so the permutation is odd. Sorting by adjacent exchanges makes the count visible:

$$31254 \longrightarrow 13254 \longrightarrow 12354 \longrightarrow 12345.$$

Its cycle expression $(1\ 3\ 2)(4\ 5)$ also gives sign $(-1)^2(-1) = -1$. The two calculations agree for a proved reason, not because a preferred decomposition was selected.

In the inversion proof, a third entry to the left compares with both exchanged entries before and after; the number of these two inversions is unchanged. A third entry to the right has the same property. Only the comparison between the adjacent entries reverses. Moreover an unsorted list must have adjacent entries out of order: if every neighbor increased, transitivity would make the entire list increasing.

A.2.5 Permutation matrices connect the notation to linear maps

The **permutation matrix** P_σ has column j equal to $e_{\sigma(j)}$. Applying it to basis vectors shows $P_\sigma P_\tau = P_{\sigma\tau}$ and $P_\sigma^{-1} = P_{\sigma^{-1}}$.

Example A.8 (Which way does a permutation matrix move coordinates?). For $\sigma = (1\ 2\ 3)$ the columns are e_2, e_3, e_1 , hence

$$P_\sigma = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad P_\sigma \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} x_3 \\ x_1 \\ x_2 \end{pmatrix}.$$

The old coordinate x_j moves to position $\sigma(j)$; the new coordinate in position i is $x_{\sigma^{-1}(i)}$. In the Leibniz sum only the selection $(2, 3, 1)$ has a nonzero product. Its sign is $+1$, so $\det P_\sigma = 1$. This example also explains why a permutation and its inverse can appear in different coordinate conventions without changing the determinant sign.

A.3 Constructing the Determinant Form

Let $(v_1, \dots, v_n)$ be a basis of V and write $x_j = \sum_i a_{ij} v_i$. The slot-by-slot uniqueness argument of [Theorem 10.112](#) uses only multilinearity and alternation, so works over F : each alternating form is determined by its value on this basis. We now give the field-valid existence argument in full.

Theorem A.9 (Existence over an arbitrary field). The formula

$$\omega(x_1, \dots, x_n) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{j=1}^n a_{\sigma(j), j}$$

defines the unique alternating n -linear form with $\omega(v_1, \dots, v_n) = 1$.

Proof. Idea. Every term is linear in each column; pair terms when two columns agree. Each product uses one coordinate from each input, so distributing sums or scalar multiples in one column proves multilinearity. If $x_r = x_s$ for $r \neq s$, pair σ with $\sigma \circ (r\ s)$. This is a fixed-point-free pairing. The products agree since the two coordinate columns agree, and the signs are opposite by parity. Their sum is zero in every field, including characteristic two where it is twice the same element. On the given basis, all terms except the identity term vanish, and that term is 1. For uniqueness, expanding an arbitrary form gives the same sum: repeated tuples vanish, and the surviving tuples are permutations with their forced signs. $\square$

In the standard basis this is $\det A$. It agrees with the main-text determinant when $F = \mathbb{R}$ by uniqueness, and gives $\det P_\sigma = \operatorname{sgn}(\sigma)$ interpreted in F . The uniqueness proof of multiplicativity, the invertibility criterion, the transpose formula, and basis independence in Chapter 10 remain valid over F . No order or positivity was used in them.

Proposition A.10 (Laplace expansion from the permutation sum). The minor and cofactor definitions and both expansions in [Theorem 10.119](#) are valid over F .

Proof. Group the terms of the Leibniz formula according to $\sigma(j) = i$ for a fixed column j . Remove that column choice and its selected row. The remaining bijection is a permutation of the remaining ordered rows and columns. Moving column j and row i to their first positions uses $j - 1 + i - 1$ swaps, so its full sign is $(-1)^{i+j}$ times the remaining permutation sign. The group of terms is exactly $a_{ij}(-1)^{i+j} M_{ij}$. Summing over i gives the column expansion. Transpose invariance gives the row expansion. $\square$

For example the sparse matrix of Chapter 10 has determinant $-6(8 - 3) = -30$, interpreted in F . The same calculation remains valid if this field element becomes zero; one must not infer real sign information or invertibility from an integer expression before interpreting it in F .

Exercise A.1. Find disjoint cycles, a transposition decomposition, sign, inversion count, and determinant of the permutation matrix for the list $(3, 1, 2, 5, 4)$.

Exercise A.2. Define the adjugate by $(\text{adj } A)_{ji} = C_{ij}(A)$. Using cofactor expansion of a matrix with one replaced row, prove $A \text{adj } A = (\det A)I$. Derive the inverse formula when $\det A \neq 0$. For $Ax = b$, use column multilinearity to derive $x_j = \det(A_1, \dots, b, \dots, A_n) / \det A$ (Cramer's rule).

Exercise A.3. In S_5 , compute both $(1\ 4\ 2)(2\ 3\ 5)$ and $(2\ 3\ 5)(1\ 4\ 2)$ by making a table of all five images. Give inverse lists and cycle expressions. Which entries witness noncommutation?

Exercise A.4. Give three distinct transposition expressions for $(1\ 4)$ and count their lengths. Explain what parity says, and what it does not say, about the smallest possible length of an expression.

Exercise A.5. For the reversing list $(n, n-1, \dots, 1)$ find the inversion count and sign. Compute its permutation matrix determinant for $n = 4, 5, 6$.

Exercise A.6. Over $\mathbb{F}_2$, compute the determinant of $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$. Explain why the answer gives an invertibility test but no positive or negative orientation. Distinguish permutation parity from its field-valued sign.

Appendix B

Optional: Tensor Products and Exterior Powers

This appendix develops constructions over a field F . Replace $\mathbb{R}$ by F in the algebraic definitions of Chapter 10; no topology or ordering is required. The spaces whose bases we count are finite dimensional. Quotient and tensor constructions also apply to arbitrary vector spaces, and the formal-symbol spaces used in the proofs are usually infinite dimensional. Only finitely many tensor factors or direct summands are considered. Taking $F = \mathbb{R}$ connects the constructions to forms, but the main text does not require this appendix. The proof architecture is adapted from the author's algebra notes, Chapter 8, specialized throughout to vector spaces.

The new skill is constructing maps on objects whose elements have many representations. We will distinguish a proposed formula from a well-defined map, and construction from the later proof that two existing maps are equal.

B.1 Quotient Vector Spaces and Universal Factorization

In $E = F^2$, suppose horizontal displacements are to be ignored. Then (x, y) and $(x + t, y)$ should describe the same object, a horizontal line. The second coordinate survives, whereas the first does not. This resembles the equivalence-class constructions of integers and rationals in Chapter 2 (Propositions 2.19 and 2.21): different representatives describe a single element. Here the differences we ignore form a vector subspace.

Definition B.1 (Cosets). For a subspace $R \subseteq E$, declare

$$x \sim y \iff x - y \in R.$$

The class of x is the coset $x + R = \{x + r : r \in R\}$. Write E/R for the set of these classes.

Proposition B.2 (Equivalence and equality of cosets). This is an equivalence relation, and $x + R = y + R$ exactly when $x - y \in R$.

Proof. $x - x = 0 \in R$ proves reflexivity. If $x - y \in R$, its negative $y - x$ is in R , proving symmetry. If $x - y, y - z \in R$, their sum $x - z$ is in R , proving transitivity. If $x - y \in R$, then $x + r = y + (x - y + r) \in y + R$ for every $r \in R$, giving one inclusion of cosets; interchange x, y for the other. If the cosets are equal, $x \in y + R$, so $x - y \in R$. $\square$

Theorem B.3 (The quotient vector space). The operations

$$(x + R) + (y + R) = (x + y) + R, \quad a(x + R) = ax + R$$

make E/R a vector space. The quotient map $q : E \rightarrow E/R$, $q(x) = x + R$, is linear and surjective, with kernel R .

Proof. First these are only proposed operations: the input classes do not specify unique representatives. Suppose $x' = x + r$ and $y' = y + s$ with $r, s \in R$. Then

$$(x' + y') - (x + y) = r + s \in R, \quad ax' - ax = ar \in R.$$

The coset criterion proves that changing either representative leaves each output class unchanged. Thus both operations are well defined.

The zero class is $0 + R = R$, and the additive inverse of $x + R$ is $-x + R$. For example the two association orders both give $(x + y + z) + R$, and $a((x + R) + (y + R)) = (ax + ay) + R = a(x + R) + a(y + R)$. Commutativity, the other distributive law, scalar associativity, and the scalar identity follow by the corresponding identities in E in exactly these representative calculations; their validity no longer depends on which representatives are used. The defining formulas give $q(x + y) = q(x) + q(y)$ and $q(ax) = aq(x)$. Every class is $q(x)$ for some x , so q is surjective. Finally $q(x) = R$ if and only if $x - 0 \in R$, giving $\ker q = R$. $\square$

Example B.4 (A legal rule and an illegal rule). For $R = \{(t, 0) : t \in F\} \subseteq F^2$, the rule $\mu((x, y) + R) = y$ is independent of representative. Its reverse is $\eta(y) = (0, y) + R$. Both are linear by the quotient operations; $\mu\eta(y) = y$, and $\eta\mu((x, y) + R) = (0, y) + R = (x, y) + R$ because their difference is horizontal. Thus $F^2/R \cong F$ by these specific maps. The rule $(x, y) + R \mapsto x$ is not a function on classes: $(0, 0)$ and $(1, 0)$ represent the same class but would give different outputs.

Theorem B.5 (Quotient universal factorization). For a linear $L : E \rightarrow U$, a linear map $\bar{L} : E/R \rightarrow U$ satisfying $L = \bar{L} \circ q$ exists if and only if $R \subseteq \ker L$. When it exists it is unique and is given by

$$\bar{L}(x + R) = L(x).$$

$$\begin{array}{ccc} E & \xrightarrow{q} & E/R \\ & \searrow L & \downarrow \bar{L} \\ & & U \end{array}$$

The diagram says both routes starting at $x \in E$ give $L(x) \in U$. All three arrows here are linear; the dashed one is the map to be constructed.

Proof. If $L = \bar{L}q$ and $r \in R$, then $q(r) = 0$ and $L(r) = \bar{L}(0) = 0$, proving necessity. For sufficiency suppose $R \subseteq \ker L$. If $x + R = y + R$, then $x - y \in R$, so $L(x) - L(y) = L(x - y) = 0$. The displayed rule is therefore independent of representative and defines $\bar{L}$. For classes $x + R, y + R$ and scalars a, b ,

$$\bar{L}(a(x + R) + b(y + R)) = L(ax + by) = a\bar{L}(x + R) + b\bar{L}(y + R).$$

Thus it has the required type, a linear map on the quotient, and $\bar{L}(q(x)) = L(x)$. For uniqueness, any map H with $Hq = L$ has $H(x + R) = H(q(x)) = L(x)$ on every class, since q is surjective. Hence $H = \bar{L}$. $\square$

The phrase “ L descends to the quotient” abbreviates this construction only after the relation-killing hypothesis has been checked.

Theorem B.6 (First isomorphism theorem). For a linear $T : V \rightarrow W$, the map

$$\Phi : V/\ker T \rightarrow \operatorname{im} T, \quad \Phi(v + \ker T) = T(v)$$

is a linear isomorphism.

Proof. Regard T as a map into its image. If $v + \ker T = v' + \ker T$, then $T(v - v') = 0$, so the values $T(v), T(v')$ agree. Thus Φ is well defined, and linearity follows by applying T to combinations of representatives. If $\Phi(v + \ker T) = 0$, then $v \in \ker T$, so the input class is zero; hence Φ is injective. Every y in the stated target is $T(v)$ for some v , so Φ is surjective.

The inverse can also be constructed explicitly. Given $y \in \operatorname{im} T$, choose any v with $T(v) = y$ and put $\Psi(y) = v + \ker T$. Two choices differ by an element of $\ker T$, so this class is independent of the choice. Choosing preimages v, v' shows that $av + bv'$ is a preimage of $ay + by'$, proving linearity. Finally $\Phi\Psi(y) = T(v) = y$ and $\Psi\Phi(v + \ker T) = v + \ker T$. $\square$

For $T(x, y, z) = (x - y, y - z)$, the quotient ignores precisely additions of (t, t, t) . The class of $(a, 0, -b)$ is the inverse image of (a, b) under this isomorphism; a different preimage gives the same class.

B.1.1 What a universal property does

A universal property is both a construction procedure and a uniqueness principle. For quotients, a map from E/R corresponds to a map from E that kills R . To construct the former we wrote the latter first, verified that it killed R , and then defined its value on a class. Uniqueness lets us compare maps without comparing all possible representatives individually. For tensors the input maps will be bilinear instead of relation-killing linear maps. Universality requires every eligible input map, into every target space, to factor in one and only one way.

Universal properties should explain constructions, not replace the constructions. Write the concrete input map and verify its hypotheses before invoking a factorization.

B.2 Bilinear Maps and the Tensor-Product Problem

Multiplication $b : F \times F \rightarrow F$, $b(x, y) = xy$, is linear in one variable when the other is held fixed. It is not linear on the product vector space: $b((1, 0) + (0, 1)) = 1$ whereas $b(1, 0) + b(0, 1) = 0$. This example works over every field. We seek a new source space that turns separate linearity into ordinary linearity.

Definition B.7 (Bilinearity). A map $b : V \times W \rightarrow U$ is bilinear if, for all vectors and scalars,

$$\begin{aligned} b(v + v', w) &= b(v, w) + b(v', w), \\ b(av, w) &= ab(v, w), \\ b(v, w + w') &= b(v, w) + b(v, w'), \\ b(v, aw) &= ab(v, w). \end{aligned}$$

These are two independent linearity requirements, one per slot.

Evaluation $(\lambda, v) \mapsto \lambda(v)$ is bilinear on $V^* \times V$. So is the rectangular table $(x, y) \mapsto (x_i y_j)_{ij}$. A tensor should retain all bilinear information, without imposing the extra coincidences of any one example such as a dot product.

Definition B.8 (Tensor product). A tensor product of V, W consists of a vector space T and a bilinear map $\tau : V \times W \rightarrow T$ with the following property:

- (a) For every vector space U over F ,
- (b) and every bilinear map $b : V \times W \rightarrow U$,
- (c) there exists a unique linear map $\tilde{b} : T \rightarrow U$ such that $\tilde{b} \circ \tau = b$.

Write $T = V \otimes_F W$ and $v \otimes w = \tau(v, w)$; the latter is an elementary tensor. We omit the subscript F when the field is understood.

$$\begin{array}{ccc} V \times W & \xrightarrow{\tau} & V \otimes W \\ & \searrow b & \downarrow \tilde{b} \\ & & U \end{array}$$

“Every” lets us choose U and b for the problem at hand. “Exists” gives an actual linear map on tensors, not just a formula on a pair. “Unique” says two linear maps with this same composite with τ must agree. The horizontal and diagonal arrows are bilinear; the vertical one is linear. The equation means both routes send (v, w) to $b(v, w)$.

Theorem B.9 (Uniqueness up to canonical isomorphism). If (T, τ) and (T', τ') are tensor products of V, W , there is a unique linear isomorphism $A : T \rightarrow T'$ satisfying $A\tau = \tau'$.

Proof. Construct A . Use the universal property of T with target $U = T'$ and input bilinear map $b = \tau'$. It gives a unique linear $A : T \rightarrow T'$ such that $A\tau = \tau'$. Temporarily distinguishing the spaces, this reads $A(v \otimes_T w) = v \otimes_{T'} w$.

Construct the reverse. Use the universal property of T' with target T and bilinear map τ . This gives a linear $C : T' \rightarrow T$ with $C\tau' = \tau$, or $C(v \otimes_{T'} w) = v \otimes_T w$.

First composite. Both CA and id_T are existing linear maps $T \rightarrow T$. Compute

$$(CA)\tau = C(A\tau) = C\tau' = \tau = \text{id}_T \tau.$$

The uniqueness clause for T , with target T and input map τ , says only one linear map has this composite. Therefore $CA = \text{id}_T$.

Second composite. Both AC and $\text{id}_{T'}$ are linear maps $T' \rightarrow T'$, and

$$(AC)\tau' = A(C\tau') = A\tau = \tau' = \text{id}_{T'} \tau'.$$

Uniqueness for T' , now with target T' and input map τ' , gives $AC = \text{id}_{T'}$. Hence A, C are mutual inverses. The same uniqueness used to construct A proves that no other isomorphism satisfying the stated input condition exists. $\square$

Only now call A canonical: its behavior on every pair uniquely determines it, without any basis choice. Two constructions need not be literally equal as sets to be identified by this particular isomorphism.

$$\begin{array}{ccc} T & \xrightleftharpoons[A]{C} & T' \\ & \nwarrow \tau \quad \nearrow \tau' & \\ & V \times W & \end{array}$$

B.3 Constructing the Tensor Product

B.3.1 Formal symbols before relations

Let E consist of finitely supported coefficient families indexed by pairs $(v, w) \in V \times W$. Write such a family as $\sum c_{v,w}[v, w]$. The symbol $[v, w]$ has coefficient 1 at that pair and zero elsewhere. These symbols form a basis by the definition of coefficient families; addition and scaling act on their coefficients. In particular $[v + v', w]$ is initially a single symbol, not the sum $[v, w] + [v', w]$. Even $[0, w]$ is a nonzero basis symbol at this stage.

Let R be the subspace spanned by all vectors of the following four types:

$$\begin{aligned} &[v + v', w] - [v, w] - [v', w], \\ &[av, w] - a[v, w], \\ &[v, w + w'] - [v, w] - [v, w'], \\ &[v, aw] - a[v, w]. \end{aligned}$$

Define $T = E/R$ and $\tau(v, w) = [v, w] + R$. Each displayed difference becomes zero in the quotient, so τ satisfies the four bilinear identities. No other types of relations have been imposed.

Theorem B.10 (Existence of the tensor product). The pair $(E/R, \tau)$ satisfies the tensor universal property.

Proof. Fix an arbitrary target U and bilinear $b : V \times W \rightarrow U$. First extend from the formal basis. Define

$$\hat{b} : E \rightarrow U, \quad \hat{b}\left(\sum c_{v,w}[v, w]\right) = \sum c_{v,w}b(v, w).$$

Each sum is finite, and its coefficients are uniquely specified in E . The formula is linear by distributing coefficients. At this stage bilinearity of b has not been used: any assignment of values to formal basis symbols extends linearly in this way.

Then check the relations. Bilinearity now gives all four checks:

$$\begin{aligned} \hat{b}([v + v', w] - [v, w] - [v', w]) &= b(v + v', w) - b(v, w) - b(v', w) = 0, \\ \hat{b}([av, w] - a[v, w]) &= b(av, w) - ab(v, w) = 0, \\ \hat{b}([v, w + w'] - [v, w] - [v, w']) &= b(v, w + w') - b(v, w) - b(v, w') = 0, \\ \hat{b}([v, aw] - a[v, w]) &= b(v, aw) - ab(v, w) = 0. \end{aligned}$$

Linearity makes $\hat{b}$ vanish on their whole span: $R \subseteq \ker \hat{b}$. Only now apply [Theorem B.5](#) to obtain the linear map $\tilde{b} : E/R \rightarrow U$, with

$$\tilde{b}(e + R) = \hat{b}(e), \quad \tilde{b}(v \otimes w) = b(v, w).$$

Thus $\tilde{b}\tau = b$. Any other linear H with this property has $Hq([v, w]) = b(v, w) = \hat{b}([v, w])$ on the formal basis, hence $Hq = \hat{b}$. Quotient uniqueness gives $H = \tilde{b}$. $\square$

The two stages solve different problems: basis extension gives a map before identifications; killing the relations allows it to survive identifications.

$$\begin{array}{ccc} E & \xrightarrow{q} & E/R \\ & \searrow \hat{b} & \downarrow \tilde{b} \\ & & U \end{array}$$

B.3.2 Spanning is not surjectivity of the input map

Every quotient class of a finite sum of symbols is a finite sum of elementary tensors. Therefore elementary tensors span. The universal map $\tau : V \times W \rightarrow V \otimes W$ need not be surjective: being a finite sum of elementary tensors is weaker than being one elementary tensor.

Proposition B.11 (Universality itself forces spanning). For any tensor product (T, τ) , $\text{span } \tau(V \times W) = T$.

Proof. Let S be that span. The quotient map $q : T \rightarrow T/S$ has $q\tau = 0$. The zero linear map $Z : T \rightarrow T/S$ also has $Z\tau = 0$. Tensor uniqueness with target T/S and zero bilinear input forces $q = Z$. But q is surjective, so T/S is the zero space. Hence every $t \in T$ belongs to S , as claimed. $\square$

Representations as sums need not be unique. For instance

$$(v + v') \otimes w = v \otimes w + v' \otimes w, \quad (av) \otimes w = v \otimes (aw).$$

The elementary factors and even the number of summands can change. A rule on a chosen representation therefore requires justification.

Working rule. To construct a map from a tensor product: define a concrete map on the independent variables, verify bilinearity, invoke the universal property, and obtain the induced linear map.

To prove two existing linear maps equal: first establish both maps, compare their values on elementary tensors, then use spanning. A formula on elementary tensors alone is not a definition of a linear map.

Example B.12 (Construct before evaluating). On $F^2 \times F^2$ let $b(v, w) = v_1w_1 + 2v_1w_2 - v_2w_1$. For fixed w it is $(w_1 + 2w_2)v_1 - w_1v_2$, linear in v ; for fixed v it is $(v_1 - v_2)w_1 + 2v_1w_2$, linear in w . Thus a unique linear $L : F^2 \otimes F^2 \rightarrow F$ exists with these input values. It sends $3e_1 \otimes e_2 + e_2 \otimes e_1$ to $6 - 1 = 5$, interpreted in F . By contrast the proposed rule $v \otimes w \mapsto v$ fails: $v \otimes 0 = 0$ for every v , whereas the proposed outputs vary. Its independent-variable map is not linear in w , so the working rule detects the problem in advance.

B.4 Tensor Bases and a First Canonical Isomorphism

Theorem B.13 (Tensor basis). If $(e_i)_{i=1}^m$ and $(f_j)_{j=1}^n$ are bases of V, W , the tensors $e_i \otimes f_j$ form a basis of $V \otimes W$. Its dimension is mn .

Proof. Spanning. Write $v = \sum_i x_i e_i$ and $w = \sum_j y_j f_j$. Bilinearity expands $v \otimes w = \sum_{i,j} x_i y_j (e_i \otimes f_j)$. Since elementary tensors span, so does the displayed finite list.

Independence. The coordinate maps $e^r : V \rightarrow F$ and $f^s : W \rightarrow F$ are already linear by uniqueness of coordinates in the two given bases. Define the scalar-valued input map $b_{rs}(v, w) = e^r(v)f^s(w)$. For example

$$b_{rs}(av + bw', w) = (ae^r(v) + be^r(w'))f^s(w) = ab_{rs}(v, w) + bb_{rs}(w', w),$$

and $b_{rs}(v, aw + bw') = ab_{rs}(v, w) + bb_{rs}(v, w')$ follows by expanding the second coordinate functional. Thus it is bilinear. Universality gives a linear $\ell_{rs} : V \otimes W \rightarrow F$. Only now compute

$$\ell_{rs}(e_i \otimes f_j) = \delta_{ri}\delta_{sj}.$$

Applying this existing map to $\sum c_{ij}e_i \otimes f_j = 0$ gives $c_{rs} = 0$. All coefficients vanish, proving independence. $\square$

We did not define ℓ_{rs} as “the coefficient extractor”: uniqueness of tensor coefficients was precisely what the proof had to establish. The bilinear detector constructed it without assuming the conclusion.

Example B.14 (A table, and a tensor that is not elementary). Choosing bases identifies tensors with rectangular coefficient tables. The maps sending $\sum c_{ij}e_i \otimes f_j$ to (c_{ij}) and sending (c_{ij}) back to that sum are linear and inverse: each composite restores every coefficient. An elementary tensor has table $(x_i y_j)$. For example $(e_1 + 2e_2) \otimes (f_1 - f_2)$ has table $\begin{pmatrix} 1 & -1 \\ 2 & -2 \end{pmatrix}$. The tensor $e_1 \otimes f_1 + e_2 \otimes f_2$ is not elementary: an elementary representation would require $x_1 y_1 = x_2 y_2 = 1$ and $x_1 y_2 = x_2 y_1 = 0$. The first two equations make all four factors nonzero, contradicting the last two. This proves that τ need not be surjective.

Proposition B.15 (Scalar multiplication linearized). There is a canonical isomorphism $\mu : F \otimes V \rightarrow V$ with $\mu(a \otimes v) = av$.

Proof. Start with $b : F \times V \rightarrow V$, $b(a, v) = av$. Field distributivity gives $b(a + a', v) = b(a, v) + b(a', v)$ and $b(ca, v) = cb(a, v)$; vector distributivity gives $b(a, v + v') = b(a, v) + b(a, v')$ and $b(a, cv) = cb(a, v)$. This bilinear map induces the linear μ . Define the reverse directly: $\eta : V \rightarrow F \otimes V$, $\eta(v) = 1 \otimes v$. The second tensor slot is linear, so η is linear. For every v , $\mu\eta(v) = 1v = v$. For an elementary tensor,

$$\eta\mu(a \otimes v) = 1 \otimes (av) = a(1 \otimes v) = a \otimes v.$$

The two existing linear maps $\eta\mu$ and the identity agree on a spanning set, hence agree everywhere. Thus both composites are identities. $\square$

A dimension count would say only that some isomorphism exists. It would not identify this particular one, whose action is specified by scalar multiplication.

B.5 Symmetry, Induced Maps, and Direct Sums

B.5.1 Interchanging factors

Proposition B.16 (Symmetry). The rule $v \otimes w \mapsto w \otimes v$ determines a canonical isomorphism $S : V \otimes W \rightarrow W \otimes V$.

Proof. Define $b(v, w) = w \otimes v$ on $V \times W$. For all $a, c \in F$,

$$\begin{aligned} b(av + cv', w) &= w \otimes (av + cv') = a(w \otimes v) + c(w \otimes v'), \\ b(v, aw + cw') &= (aw + cw') \otimes v = a(w \otimes v) + c(w' \otimes v). \end{aligned}$$

Thus the tensor universal property gives S . For the reverse, define $c(w, v) = v \otimes w$. Linearity of the first tensor slot gives linearity in v , and of the second slot gives linearity in w , so it induces $R : W \otimes V \rightarrow V \otimes W$. Then $RS(v \otimes w) = R(w \otimes v) = v \otimes w$, while $SR(w \otimes v) = S(v \otimes w) = w \otimes v$. Both composites are linear and elementary tensors span their respective domains; hence RS and SR are the two identity maps. $\square$

B.5.2 Applying maps to the two factors

For linear $f : V \rightarrow V'$ and $g : W \rightarrow W'$, define $b(v, w) = f(v) \otimes g(w)$. For the first slot,

$$b(av + cv', w) = (af(v) + cf(v')) \otimes g(w) = a b(v, w) + c b(v', w).$$

Expanding $g(aw + cw')$ and the second tensor slot proves the second linearity identity. The universal property therefore constructs $f \otimes g : V \otimes W \rightarrow V' \otimes W'$. For further linear maps f', g' , both $(f' \otimes g')(f \otimes g)$ and $(f'f) \otimes (g'g)$ now exist; their common value on $v \otimes w$ is $f'f(v) \otimes g'g(w)$. Spanning gives equality. The identity maps induce the identity. If f, g are isomorphisms, construct $f^{-1} \otimes g^{-1}$ by the same bilinear input procedure; the composition identities give both inverse composites, not merely equality of dimensions.

B.5.3 Distributing over a finite direct sum

Theorem B.17 (Binary distributivity). There are canonical isomorphisms

$$(U \oplus V) \otimes W \cong (U \otimes W) \oplus (V \otimes W), \quad W \otimes (U \oplus V) \cong (W \otimes U) \oplus (W \otimes V).$$

Proof. For the first, define $b((u, v), w) = (u \otimes w, v \otimes w)$. Componentwise addition and tensor linearity give

$$\begin{aligned} b(a(u, v) + c(u', v'), w) &= a b((u, v), w) + c b((u', v'), w), \\ b((u, v), aw + cw') &= a b((u, v), w) + c b((u, v), w'). \end{aligned}$$

Thus it induces a linear forward map Φ into the external direct sum.

For the reverse, define $b_U(u, w) = (u, 0) \otimes w$ and $b_V(v, w) = (0, v) \otimes w$. The component injections preserve linear combinations, and the tensor map is bilinear, so these are bilinear. They induce $j_U : U \otimes W \rightarrow (U \oplus V) \otimes W$ and $j_V : V \otimes W \rightarrow (U \oplus V) \otimes W$. By [Proposition 10.62](#), the formula $\Psi(t, s) = j_U(t) + j_V(s)$ defines a linear map on the direct sum.

For each elementary tensor in the first space,

$$\Psi\Phi((u, v) \otimes w) = (u, 0) \otimes w + (0, v) \otimes w = (u, v) \otimes w.$$

For the other composite,

$$\Phi\Psi(u \otimes w, 0) = (u \otimes w, 0), \quad \Phi\Psi(0, v \otimes w) = (0, v \otimes w).$$

The displayed elements span their respective source spaces, by tensor spanning and the decomposition of a tuple into its two injected components. Hence both composites are identities.

For the right-variable version, the concrete input is $c(w, (u, v)) = (w \otimes u, w \otimes v)$. Each component is linear in w and in the tuple, so c induces Φ_R . The bilinear maps $(w, u) \mapsto w \otimes (u, 0)$ and $(w, v) \mapsto w \otimes (0, v)$ induce two maps assembled as Ψ_R . Direct calculation gives

$$\Psi_R \Phi_R(w \otimes (u, v)) = w \otimes (u, 0) + w \otimes (0, v) = w \otimes (u, v).$$

The other composite fixes $(w \otimes u, 0)$ and $(0, w \otimes v)$. Their spanning property proves both identities here as well. $\square$

Corollary B.18 (Finite distributivity). For finite families there are canonical isomorphisms

$$\begin{aligned} \left(\bigoplus_i V_i \right) \otimes W &\cong \bigoplus_i (V_i \otimes W), \\ V \otimes \left(\bigoplus_j W_j \right) &\cong \bigoplus_j (V \otimes W_j), \\ \left(\bigoplus_i V_i \right) \otimes \left(\bigoplus_j W_j \right) &\cong \bigoplus_{i,j} (V_i \otimes W_j). \end{aligned}$$

Proof. For the first map use the bilinear input $((v_i), w) \mapsto (v_i \otimes w)_i$. The reverse is assembled from $(v_i, w) \mapsto \iota_i(v_i) \otimes w$. On an elementary tensor the reverse composite returns $\sum_i \iota_i(v_i) \otimes w = (v_i)_i \otimes w$; the forward composite fixes each injected elementary tensor. This proves both identities by spanning. In the second formula use $(v, (w_j)) \mapsto (v \otimes w_j)_j$ and reverse inputs $(v, w_j) \mapsto v \otimes \iota_j(w_j)$; the two composites give $v \otimes \sum_j \iota_j(w_j)$ and the original injected elementary tensors. For the third formula use $((v_i), (w_j)) \mapsto (v_i \otimes w_j)_{i,j}$, bilinear componentwise. Reverse maps are induced by $(v_i, w_j) \mapsto \iota_i(v_i) \otimes \iota_j(w_j)$ and assembled over the finite direct sum. The first composite expands to

$$\sum_{i,j} \iota_i(v_i) \otimes \iota_j(w_j) = \left(\sum_i \iota_i(v_i) \right) \otimes \left(\sum_j \iota_j(w_j) \right),$$

the original elementary tensor. The other fixes every (i, j) injected elementary tensor, hence every element. $\square$

The dimensions $(\sum_i \dim V_i)(\sum_j \dim W_j) = \sum_{i,j} \dim V_i \dim W_j$ are a consistency check after the maps have been constructed.

Example B.19 (Four components of a tensor). Take $A = F^2 \oplus F$ and $B = F \oplus F^2$, with injections $i : F^2 \rightarrow A$, $j : F \rightarrow A$, $r : F \rightarrow B$, and $s : F^2 \rightarrow B$. The isomorphism D of Corollary B.18 has target

$$(F^2 \otimes F) \oplus (F^2 \otimes F^2) \oplus (F \otimes F) \oplus (F \otimes F^2),$$

in that order. Let e_1, e_2 be the basis of the first F^2 and f_1, f_2 that of the second. On one elementary tensor it gives

$$D((e_1 + e_2, 1) \otimes (1, f_1)) = ((e_1 + e_2) \otimes 1, (e_1 + e_2) \otimes f_1, 1 \otimes 1, 1 \otimes f_1).$$

The inverse E is the already constructed sum of maps induced by the four pairs of injections. For the component tuple

$$z = (e_1 \otimes 1, e_2 \otimes f_1 + e_1 \otimes f_2, 1 \otimes 1, 1 \otimes f_2)$$

its value is the nontrivial sum

$$\begin{aligned} E(z) &= i(e_1) \otimes r(1) + i(e_2) \otimes s(f_1) + i(e_1) \otimes s(f_2) \\ &\quad + j(1) \otimes r(1) + j(1) \otimes s(f_2). \end{aligned}$$

Applying D to the five terms gives, respectively,

$$(e_1 \otimes 1, 0, 0, 0), (0, e_2 \otimes f_1, 0, 0), (0, e_1 \otimes f_2, 0, 0), \\ (0, 0, 1 \otimes 1, 0), (0, 0, 0, 1 \otimes f_2).$$

Their sum is z , checking DE on this input. For the first displayed elementary tensor, ED expands its four terms and recombines them as $(i(e_1 + e_2) + j(1)) \otimes (r(1) + s(f_1))$, the original tensor. Only after these maps and calculations do we count dimensions: $3 \cdot 3 = 2 + 4 + 1 + 2$. The count checks the construction but does not specify it.

B.6 Finite Multivariable Tensor Products

Three different input spaces occur, for example, when one evaluates a trilinear rule on three different kinds of data. Repeated copies of one space are only a special case. We first construct the space directly, without deciding how to parenthesize binary tensor products.

Definition B.20 (Multilinearity). A map $b : V_1 \times \cdots \times V_k \rightarrow U$ is multilinear if for each slot r , with all other inputs fixed,

$$b(v_1, \dots, v_r + v'_r, \dots, v_k) = b(v_1, \dots, v_r, \dots, v_k) + b(v_1, \dots, v'_r, \dots, v_k), \\ b(v_1, \dots, av_r, \dots, v_k) = ab(v_1, \dots, v_r, \dots, v_k).$$

Each input is varied separately; k is a positive finite integer.

Definition B.21 (Multivariable tensor product). A tensor product $V_1 \otimes \cdots \otimes V_k$ has a multilinear map $\tau : V_1 \times \cdots \times V_k \rightarrow V_1 \otimes \cdots \otimes V_k$ such that for every vector space U and every multilinear b into U , there is a unique linear $\tilde{b}$ with $\tilde{b}\tau = b$. Write $\tau(v_1, \dots, v_k) = v_1 \otimes \cdots \otimes v_k$.

$$\begin{array}{ccc} V_1 \times \cdots \times V_k & \xrightarrow{\tau} & V_1 \otimes \cdots \otimes V_k \\ & \searrow b & \downarrow \tilde{b} \\ & & U \end{array}$$

The horizontal and slant arrows are multilinear; the vertical arrow is linear. This distinguishes the Cartesian input space from the new vector space that represents multilinear rules.

Theorem B.22 (Construction and universality). The multivariable tensor product exists. Its elementary tensors span it.

Proof. Form the vector space E_k with basis symbols $[v_1, \dots, v_k]$. For every slot r , include in a subspace R the span of the relations

$$[v_1, \dots, v_r + v'_r, \dots, v_k] - [v_1, \dots, v_r, \dots, v_k] - [v_1, \dots, v'_r, \dots, v_k], \\ [v_1, \dots, av_r, \dots, v_k] - a[v_1, \dots, v_r, \dots, v_k].$$

The quotient E_k/R with the tuple-to-class map is multilinear because these relations become zero. Given multilinear b , define its formal linear extension by

$$\hat{b} \left(\sum c_{v_1, \dots, v_k} [v_1, \dots, v_k] \right) = \sum c_{v_1, \dots, v_k} b(v_1, \dots, v_k).$$

The two relation types in slot r map respectively to $b(\dots, v_r + v'_r, \dots) - b(\dots, v_r, \dots) - b(\dots, v'_r, \dots) = 0$ and $b(\dots, av_r, \dots) - ab(\dots, v_r, \dots) = 0$. This check holds for each r , so $R \subseteq \ker \hat{b}$. Quotient factorization gives $\tilde{b}$. Every quotient class is a finite sum of elementary tensors, proving spanning and uniqueness of a linear map with prescribed elementary values. This proves universality. The two-map uniqueness proof in [Theorem B.9](#) applies with multilinear tuple maps in place of bilinear pair maps. $\square$

Theorem B.23 (Multivariable tensor basis). If $(e_i^{(r)})_{i=1}^{n_r}$ is a basis of V_r , the tensors

$$e_{i_1}^{(1)} \otimes \cdots \otimes e_{i_k}^{(k)}, \quad 1 \leq i_r \leq n_r,$$

form a basis. Thus $\dim(V_1 \otimes \cdots \otimes V_k) = \prod_r n_r$.

Proof. Expanding every input in its basis and distributing one slot at a time expresses each elementary tensor in the displayed list, proving spanning. For a fixed index tuple $j = (j_1, \dots, j_k)$ let $e_r^{j_r}$ be the coordinate functional on V_r and define

$$b_j(v_1, \dots, v_k) = \prod_{r=1}^k e_r^{j_r}(v_r).$$

In slot r , the other factors are fixed scalars; linearity of $e_r^{j_r}$ proves linearity in that slot. Hence b_j is multilinear and induces a linear ℓ_j on the tensor product. Its value on the basis candidate indexed by i is $\prod_r \delta_{j_r i_r}$. Applied to a relation, it gives exactly the coefficient indexed by j . Every coefficient is zero, proving independence. If some $n_r = 0$, multilinearity makes every elementary tensor zero, and the empty basis and product dimension zero give the same conclusion. $\square$

Example B.24 (A three-factor rule becomes a functional). Consider three factor spaces with dimensions 2, 2, 3:

$$b : F^2 \times F^2 \times F^3 \rightarrow F, \quad b(x, y, z) = x_1 y_1 z_1 + x_2 y_1 z_2 + x_1 y_2 z_3.$$

Hold two inputs fixed. As a function of each remaining input the rule is

$$\begin{aligned} x &\longmapsto (y_1 z_1 + y_2 z_3)x_1 + (y_1 z_2)x_2, \\ y &\longmapsto (x_1 z_1 + x_2 z_2)y_1 + (x_1 z_3)y_2, \\ z &\longmapsto (x_1 y_1)z_1 + (x_2 y_1)z_2 + (x_1 y_2)z_3. \end{aligned}$$

Each is a fixed linear combination of coordinate functions; substituting $au + cv$ in its variable distributes into a times the value at u and c times the value at v . Thus all three linearity requirements hold. With target F and this input b , [Theorem B.22](#) constructs the unique linear functional

$$\tilde{b} : F^2 \otimes F^2 \otimes F^3 \rightarrow F, \quad \tilde{b}(x \otimes y \otimes z) = b(x, y, z).$$

Write e_i, f_j, g_k for the three standard bases. Its twelve basis values are

$$\tilde{b}(e_i \otimes f_j \otimes g_k) = \delta_{i1}\delta_{j1}\delta_{k1} + \delta_{i2}\delta_{j1}\delta_{k2} + \delta_{i1}\delta_{j2}\delta_{k3}.$$

Exactly the tensors indexed by $(1, 1, 1), (2, 1, 2), (1, 2, 3)$ have value 1; the other nine have value zero. For the finite sum

$$t = e_1 \otimes f_1 \otimes g_1 + e_2 \otimes f_2 \otimes g_2, \quad \tilde{b}(t) = 1 + 0 = 1.$$

This t is not elementary. In an elementary tensor the coefficients $c_{ijk} = x_i y_j z_k$ satisfy $c_{111}c_{222} = c_{122}c_{211}$. For t the left side is 1 and the right side is zero. Thus the new functional evaluates an input that cannot be supplied as one triple to b ; its value on a sum is independent of the chosen representation.

Definition B.25 (Tensor powers). Only now set $V^{\otimes k} = V \otimes \cdots \otimes V$, with k copies of V . Set $V^{\otimes 0} = F$ and identify the one-factor product with V . For the latter identification the input map $v \mapsto v$ induces the forward linear map, the reverse is $v \mapsto \tau(v)$, and both composites are identities on v and on the spanning $\tau(v)$ respectively.

B.6.1 Induced maps in all the factors

For linear $T_r : V_r \rightarrow W_r$ define

$$b(v_1, \dots, v_k) = T_1 v_1 \otimes \cdots \otimes T_k v_k.$$

In slot r , substitute $T_r(av_r + bv'_r) = aT_r v_r + bT_r v'_r$ and expand that slot of the target tensor. This verifies both linearity identities in every slot. The universal property constructs the linear map $\bigotimes_r T_r : \bigotimes_r V_r \rightarrow \bigotimes_r W_r$. Two existing composites, $\bigotimes_r (S_r T_r)$ and $(\bigotimes_r S_r)(\bigotimes_r T_r)$, take each elementary tensor to $S_1 T_1 v_1 \otimes \cdots \otimes S_k T_k v_k$; hence they agree by spanning. Identities induce identities. If each T_r is invertible, the induced map from the T_r^{-1} has both composites equal to the identity by these formulas. For repeated T write $T^{\otimes k}$; in degree zero use id_F .

B.7 Associativity: Constructing Both Nested Maps

The desired value $(u \otimes v) \otimes w \mapsto u \otimes (v \otimes w)$ looks like a change of parentheses. But a general input is a sum with nonunique decompositions, so the displayed value is a prescription, not yet a map. There are two tensor layers, and each needs a construction.

Theorem B.26 (Associativity). There is a canonical isomorphism $A : (U \otimes V) \otimes W \rightarrow U \otimes (V \otimes W)$ with the value above.

Proof. Put $X = U \otimes (V \otimes W)$. Fix $w \in W$ and define $b_w(u, v) = u \otimes (v \otimes w) \in X$. For example

$$\begin{aligned} b_w(au + bu', v) &= a(u \otimes (v \otimes w)) + b(u' \otimes (v \otimes w)), \\ b_w(u, av + bv') &= u \otimes (a(v \otimes w) + b(v' \otimes w)) = ab_w(u, v) + bb_w(u, v'). \end{aligned}$$

Thus b_w is bilinear and induces a linear $L_w : U \otimes V \rightarrow X$. Now compare two already existing linear maps on $U \otimes V$:

$$L_{aw+bw'}(u \otimes v) = u \otimes (v \otimes (aw + bw')) = (aL_w + bL_{w'})(u \otimes v).$$

Spanning gives $L_{aw+bw'} = aL_w + bL_{w'}$ on every tensor, not just elementary ones. Define $\beta(t, w) = L_w(t)$ on $(U \otimes V) \times W$. For fixed w it is linear because L_w is; for fixed t it is linear because of the map equality just proved. Only now apply the outer tensor universal property to obtain $A : (U \otimes V) \otimes W \rightarrow X$.

For the reverse put $Y = (U \otimes V) \otimes W$. Fix u and define $c_u(v, w) = (u \otimes v) \otimes w$. Linearity in v follows by expanding $u \otimes (av + bv')$, then the first outer slot; linearity in w follows by expanding the second outer slot. This bilinear map induces $M_u : V \otimes W \rightarrow Y$. For each elementary $v \otimes w$,

$$M_{au+bu'}(v \otimes w) = ((au + bu') \otimes v) \otimes w = (aM_u + bM_{u'})(v \otimes w).$$

Both sides are existing linear maps, so spanning gives equality everywhere. Consequently $\gamma(u, s) = M_u(s)$ is linear in u by that equality and linear in s by construction of M_u . It induces the reverse linear map $C : U \otimes (V \otimes W) \rightarrow (U \otimes V) \otimes W$.

The composites are now legitimate maps and satisfy

$$\begin{aligned} CA((u \otimes v) \otimes w) &= C(u \otimes (v \otimes w)) = (u \otimes v) \otimes w, \\ AC(u \otimes (v \otimes w)) &= A((u \otimes v) \otimes w) = u \otimes (v \otimes w). \end{aligned}$$

Nested elementary tensors span: first write an outer tensor as a sum of $t \otimes w$ and then write t as a sum of elementary $u \otimes v$; outer linearity distributes that sum. For the other nesting expand the second factor instead. Spanning proves both composites are identities. $\square$

Proposition B.27 (Comparison with the direct three-factor product). The directly constructed $D = U \otimes V \otimes W$ is canonically isomorphic to both parenthesized products.

Proof. The map $(u, v, w) \mapsto (u \otimes v) \otimes w$ is trilinear: the inner tensor identities handle slots one and two, and the outer identity handles slot three. Hence it induces $F : D \rightarrow (U \otimes V) \otimes W$. To construct G in reverse, fix w and induce $N_w : U \otimes V \rightarrow D$ from the bilinear map $(u, v) \mapsto u \otimes v \otimes w$. The existing maps $N_{aw+bw'}$ and $aN_w + bN_{w'}$ agree on elementary tensors by third-slot linearity in D , hence everywhere. Thus $(t, w) \mapsto N_w(t)$ is bilinear and induces G . The composites satisfy

$$\begin{aligned} GF(u \otimes v \otimes w) &= u \otimes v \otimes w, \\ FG((u \otimes v) \otimes w) &= (u \otimes v) \otimes w. \end{aligned}$$

Spanning makes these identities global. For the other parenthesization use AF with reverse GC , where A, C are the preceding inverse maps: $(GC)(AF) = G(CA)F = GF = \text{id}_D$ and $(AF)(GC) = A(FG)C = AC = \text{id}_X$. $\square$

This construction proves, rather than assumes, that both parenthesized products represent trilinear maps. For a finite number of factors repeat the fixed-input factorization at successive slots. Each step has the two inverse maps just constructed. Any two resulting maps with the same value on every elementary tuple agree by spanning; thus no ambiguity is created by suppressing parentheses. No infinite tensor product is being introduced.

B.8 Tensor–Hom Correspondences

B.8.1 A map whose values are maps

Let $\mathcal{L}(V, W)$ denote the vector space of linear maps, with pointwise addition and scalar multiplication as in Chapter 10. A map into this space returns a whole function, which can then be evaluated at a vector.

Proposition B.28 (Currying). There are inverse linear maps

$$C : \mathcal{L}(V \otimes W, U) \rightarrow \mathcal{L}(V, \mathcal{L}(W, U)), \quad D : \mathcal{L}(V, \mathcal{L}(W, U)) \rightarrow \mathcal{L}(V \otimes W, U).$$

The forward rule is $C(L)(v)(w) = L(v \otimes w)$.

Proof. First check the types in the forward rule:

$$\begin{aligned} C(L) &\in \mathcal{L}(V, \mathcal{L}(W, U)), \\ C(L)(v) &\in \mathcal{L}(W, U), \\ C(L)(v)(w) &\in U. \end{aligned}$$

For fixed v , $w \mapsto L(v \otimes w)$ is linear by second-slot linearity and linearity of L . Thus $C(L)(v)$ has the stated inner type. For all w , first-slot linearity gives $C(L)(av + bv')(w) = aC(L)(v)(w) + bC(L)(v')(w)$. Equality at every w proves equality of maps, so $v \mapsto C(L)(v)$ is linear. Finally $C(aL + bL')(v)(w) = aC(L)(v)(w) + bC(L')(v)(w)$; hence C itself is linear.

For the inverse, take a linear $A : V \rightarrow \mathcal{L}(W, U)$ and define $b_A(v, w) = A(v)(w)$. Since A is linear, $b_A(av + bv', w) = aA(v)(w) + bA(v')(w)$. Since $A(v)$ is a linear map, $b_A(v, aw + bw') = aA(v)(w) + bA(v)(w')$. Thus b_A is bilinear and induces the linear $D(A) : V \otimes W \rightarrow U$. Comparing existing maps on elementary tensors shows $D(aA + bA') = aD(A) + bD(A')$, so D is linear as well.

For every v, w , $C(D(A))(v)(w) = D(A)(v \otimes w) = A(v)(w)$, proving $CD(A) = A$. For every elementary tensor, $D(C(L))(v \otimes w) = C(L)(v)(w) = L(v \otimes w)$. Spanning proves $DC(L) = L$. These are both inverse identities. $\square$

B.8.2 A covector and a vector make a linear map

Proposition B.29 (Finite-dimensional tensor-map identification). If V is finite dimensional, there is a canonical isomorphism $J : V^* \otimes W \rightarrow \mathcal{L}(V, W)$ characterized by $J(\lambda \otimes w)(v) = \lambda(v)w$.

Proof. Begin with $b(\lambda, w)$ equal to the map $v \mapsto \lambda(v)w$. This is a linear map $V \rightarrow W$ because λ is linear. Evaluation at each v shows $b(a\lambda + b\mu, w) = ab(\lambda, w) + bb(\mu, w)$ and $b(\lambda, aw + bw') = ab(\lambda, w) + bb(\lambda, w')$ as equalities of maps. So b is bilinear with target $\mathcal{L}(V, W)$ and induces J .

Choose a basis (e_i) of V and dual basis (e^i) . Define the reverse by the finite sum

$$K(T) = \sum_i e^i \otimes T(e_i).$$

Pointwise linearity in T and tensor linearity show that K is linear. For $v = \sum_i e^i(v)e_i$,

$$J(K(T))(v) = \sum_i e^i(v)T(e_i) = T(v),$$

so $JK = \text{id}$. On an elementary tensor the other composite is

$$K(J(\lambda \otimes w)) = \sum_i e^i \otimes \lambda(e_i)w = \left(\sum_i \lambda(e_i)e^i \right) \otimes w = \lambda \otimes w.$$

The last covector equality follows by evaluation on every e_j . Spanning now proves $KJ = \text{id}$. Although a basis was used to write K , it is the unique inverse of the basis-free map J . Any other basis therefore writes the same inverse, not another isomorphism. $\square$

Finite dimension is used in the finite expansion of the identity on V . For example $e^1 \otimes w_1 + e^2 \otimes w_2$ represents the map $x_1e_1 + x_2e_2 \mapsto x_1w_1 + x_2w_2$, the basis-image rule of Chapter 10.

B.9 Exterior Powers: Imposing Alternation

B.9.1 Begin with two inputs

Signed area is bilinear in its two edge vectors and vanishes when they agree. Ordinary tensors retain $v \otimes v$; to model area-like rules we must impose a further relation. Define the subspace first:

$$R_2 = \text{span}\{v \otimes v : v \in V\} \subseteq V \otimes V.$$

Then set $\Lambda^2 V = (V \otimes V)/R_2$ and let $u \wedge v$ denote the class of $u \otimes v$. The two-input wedge map is bilinear, as a composite of a bilinear tensor map and a linear quotient map, and $v \wedge v = 0$. Expanding the repeated input $u + v$ gives

$$0 = (u + v) \wedge (u + v) = u \wedge v + v \wedge u,$$

so $u \wedge v = -v \wedge u$. In characteristic two the minus sign is a plus sign, but the repeated-input-zero relation still must be imposed.

For instance $(ae_1 + be_2) \wedge (ce_1 + de_2) = (ad - bc)e_1 \wedge e_2$. The determinant is already alternating bilinear, so its linear tensor factor kills R_2 and factors once more to a functional on $\Lambda^2 F^2$. That functional sends $e_1 \wedge e_2$ to 1, proving this wedge is nonzero. The general version of this two-stage argument follows next.

Definition B.30 (Alternating multilinear map). A multilinear $A : V^k \rightarrow U$ is alternating if

$$v_i = v_j \text{ for some } i \neq j \implies A(v_1, \dots, v_k) = 0.$$

For $k = 1$ the extra condition is empty. For $k \geq 2$, expansion with $u + v$ in two equal slots proves the interchange sign rule, just as above.

Definition B.31 (Exterior power). For $k \geq 1$, let $R_k \subseteq V^{\otimes k}$ be the span of elementary tensors having equal vectors in two distinct positions (so $R_1 = \{0\}$). Define

$$\Lambda^k V = V^{\otimes k} / R_k, \quad \iota_k(v_1, \dots, v_k) = q(v_1 \otimes \dots \otimes v_k).$$

Write this value $v_1 \wedge \dots \wedge v_k$, an elementary wedge. We identify $\Lambda^1 V$ with V , and set $\Lambda^0 V = F$.

Notice that R_k is a span. A relation is not required to be one repeated elementary tensor; it may be any finite linear combination of such tensors.

Theorem B.32 (Exterior universal property). The map ι_k is alternating multilinear. For every vector space U and every alternating multilinear $A : V^k \rightarrow U$, there exists a unique linear $\tilde{A} : \Lambda^k V \rightarrow U$ with $\tilde{A}\iota_k = A$.

$$\begin{array}{ccccc} V^k & \xrightarrow{\tau_k} & V^{\otimes k} & \xrightarrow{q} & \Lambda^k V \\ & \searrow A & \downarrow L & \swarrow \tilde{A} & \\ & & U & & \end{array}$$

Proof. The composite $q\tau_k$ is multilinear because τ_k is and q is linear. A repeated input puts $\tau_k(v_1, \dots, v_k)$ in R_k , so its quotient class is zero. Thus ι_k has the claimed type.

Tensor stage. Given A , use its multilinearity and [Theorem B.22](#) to construct a unique linear $L : V^{\otimes k} \rightarrow U$ with $L\tau_k = A$.

Relation check. On a generator $v_1 \otimes \cdots \otimes v_k$ of R_k , two inputs agree. Therefore $L(v_1 \otimes \cdots \otimes v_k) = A(v_1, \dots, v_k) = 0$ by alternation. Linearity makes L vanish on every finite combination of these generators, so $R_k \subseteq \ker L$.

Quotient stage. Apply [Theorem B.5](#) to obtain $\tilde{A} : \Lambda^k V \rightarrow U$, given by $\tilde{A}(t + R_k) = L(t)$. The relation check is precisely its representative-independence proof. Now $\tilde{A}\iota_k = \tilde{A}q\tau_k = L\tau_k = A$.

Two uniqueness clauses. If $H : \Lambda^k V \rightarrow U$ is linear with $H\iota_k = A$, then Hq and L are linear maps on $V^{\otimes k}$ with the same composite A with τ_k . Tensor uniqueness gives $Hq = L$. Quotient uniqueness then gives $H = \tilde{A}$. Existence used first multilinearity and then alternation; uniqueness uses the two stages in the same order. $\square$

Elementary wedges span because tensor elements are finite sums of elementary tensors and q is surjective. If another space with an alternating input map has the same universal property, construct A from its input map and C from the original one, as in [Theorem B.9](#). Both $CA\iota_k = \iota_k$ and $AC\iota'_k = \iota'_k$ hold, and the corresponding uniqueness clauses force both composites to be identities. Thus exterior powers too are determined up to the unique input-preserving isomorphism.

B.9.2 Expand, remove repeats, sort, and collect

For $n = 3, k = 2$, the full expansion is

$$\begin{aligned} & (a_1 e_1 + a_2 e_2 + a_3 e_3) \wedge (b_1 e_1 + b_2 e_2 + b_3 e_3) \\ &= a_1 b_1 e_1 \wedge e_1 + a_1 b_2 e_1 \wedge e_2 + a_1 b_3 e_1 \wedge e_3 \\ &\quad + a_2 b_1 e_2 \wedge e_1 + a_2 b_2 e_2 \wedge e_2 + a_2 b_3 e_2 \wedge e_3 \\ &\quad + a_3 b_1 e_3 \wedge e_1 + a_3 b_2 e_3 \wedge e_2 + a_3 b_3 e_3 \wedge e_3 \\ &= (a_1 b_2 - a_2 b_1) e_1 \wedge e_2 + (a_1 b_3 - a_3 b_1) e_1 \wedge e_3 + (a_2 b_3 - a_3 b_2) e_2 \wedge e_3. \end{aligned}$$

The three diagonal terms vanish. Sorting $e_2 \wedge e_1$, $e_3 \wedge e_1$, and $e_3 \wedge e_2$ changes each sign. Collecting the remaining pairs gives the three displayed coefficients. This proves spanning by the three increasing wedges; independence is a separate task.

Theorem B.33 (Exterior basis). For a basis $(e_1, \dots, e_n)$, the wedges $e_I = e_{i_1} \wedge \cdots \wedge e_{i_k}$ with $i_1 < \cdots < i_k$ form a basis of $\Lambda^k V$. Thus $\dim \Lambda^k V = \binom{n}{k}$, and the space is zero for $k > n$.

Proof. Spanning. Expand each input in its basis. Multilinearity expresses every elementary wedge as a sum of wedges of basis vectors. A repeated index gives zero. When indices are distinct, sort them into increasing order by interchanges, each contributing -1 . Elementary wedges already span, so the increasing wedges span the space.

Independence. For an increasing tuple $I = (i_1, \dots, i_k)$ define the concrete map

$$A_I(v_1, \dots, v_k) = \det(e^{i_r}(v_s))_{r,s=1}^k.$$

Changing v_s linearly changes column s linearly, so determinant multilinearity gives linearity in that slot. Equal inputs give equal columns and determinant zero. The determinant theory of Chapter 10 and Appendix A therefore proves alternating multilinearity over F . The exterior universal property gives a linear functional $\ell_I : \Lambda^k V \rightarrow F$. On an increasing e_J , if $I = J$ the matrix is the identity, so $\ell_I(e_I) = 1$. If $I \neq J$, one member of J is absent from I , giving a zero column and value zero. Applying ℓ_I to $\sum_J c_J e_J = 0$ gives $c_I = 0$. All coefficients vanish. There are $\binom{n}{k}$ increasing index sets; for $k > n$ none remain and the spanning space is zero. In degree zero the empty wedge is $1 \in F$, giving dimension one. $\square$

For instance $A_{13}(u, v) = u_1 v_3 - u_3 v_1$ induces values $0, 1, 0$ on $e_1 \wedge e_2, e_1 \wedge e_3, e_2 \wedge e_3$. The detector extracts a coefficient only after its existence as a functional has been proved.

B.9.3 Representations need not be unique

Even an elementary wedge has many presentations: $u \wedge (v + cu) = u \wedge v$, and for $c \neq 0$, $(cu) \wedge (c^{-1}v) = u \wedge v$. A general sum of elementary wedges need not reduce to one. In F^4 put $\omega = e_1 \wedge e_2 + e_3 \wedge e_4$. For an elementary $u \wedge v$ its coefficients $p_{ij} = u_i v_j - u_j v_i$ satisfy $p_{12}p_{34} - p_{13}p_{24} + p_{14}p_{23} = 0$. To verify this without assuming any extra algebra, expand:

$$\begin{aligned} p_{12}p_{34} &= u_1 u_3 v_2 v_4 - u_1 u_4 v_2 v_3 - u_2 u_3 v_1 v_4 + u_2 u_4 v_1 v_3, \\ -p_{13}p_{24} &= -u_1 u_2 v_3 v_4 + u_1 u_4 v_3 v_2 + u_3 u_2 v_1 v_4 - u_3 u_4 v_1 v_2, \\ p_{14}p_{23} &= u_1 u_2 v_4 v_3 - u_1 u_3 v_4 v_2 - u_4 u_2 v_1 v_3 + u_4 u_3 v_1 v_2. \end{aligned}$$

Commutativity pairs each term with its negative. For ω , the only nonzero increasing coefficients are $p_{12} = p_{34} = 1$, making that expression 1. Thus ω is not elementary, in any characteristic. Coordinates in a fixed wedge basis are unique; arbitrary elementary decompositions are not. This is why the product construction below must handle relations rather than assume a preferred decomposition.

B.9.4 Induced exterior maps and quotient compatibility

For linear $T : V \rightarrow W$ start from $b(v_1, \dots, v_k) = Tv_1 \wedge \dots \wedge Tv_k$. In each slot linearity of T followed by wedge multilinearity proves linearity; repeated $v_i = v_j$ give repeated target vectors and zero. The exterior universal property therefore constructs $\Lambda^k T : \Lambda^k V \rightarrow \Lambda^k W$. There is also a quotient description. The already constructed $T^{\otimes k}$ sends each repeated-input generator of $R_k(V)$ to a repeated-input tensor in $R_k(W)$. Thus $T^{\otimes k}(R_k(V)) \subseteq R_k(W)$, and the quotient rule is legal. Both linear maps from $V^{\otimes k}$ in the following square agree on every elementary tensor, with common value $Tv_1 \wedge \dots \wedge Tv_k$:

$$\begin{array}{ccc} V^{\otimes k} & \xrightarrow{T^{\otimes k}} & W^{\otimes k} \\ q_V \downarrow & & \downarrow q_W \\ \Lambda^k V & \xrightarrow{\Lambda^k T} & \Lambda^k W. \end{array}$$

Hence $q_W T^{\otimes k} = (\Lambda^k T) q_V$ everywhere by spanning. For further S , the existing maps $\Lambda^k(ST)$ and $(\Lambda^k S)(\Lambda^k T)$ agree on each elementary wedge, so agree globally. The identity induces the identity. When T is an isomorphism, construct $\Lambda^k(T^{-1})$ from its alternating input map; composition gives both inverse identities. In degree zero all induced maps here are id_F .

B.10 Constructing Exterior Multiplication

We want to concatenate two wedge lists, even when neither element is given by a unique list. For $p, q \geq 1$ the intended output lies in $X = \Lambda^{p+q}V$. We will first construct a map from one exterior power, then a map with values in a space of linear maps, and only then evaluate to obtain a bilinear product.

Lemma B.34 (Equality on spanning tuples). If two existing multilinear maps $A, B : V_1 \times \dots \times V_r \rightarrow U$ agree on tuples from spanning sets $S_1, \dots, S_r$, they agree everywhere.

Proof. Write $v_i = \sum_j a_{ij} s_{ij}$ with $s_{ij} \in S_i$. Expanding each slot gives for either map the finite sum over index tuples with coefficient $\prod_i a_{ij_i}$ and value at $(s_{1j_1}, \dots, s_{rj_r})$. Those values agree by hypothesis, so the sums agree. The lemma compares maps that already exist; it cannot prove that a proposed rule is a map. $\square$

Theorem B.35 (Well-defined exterior product). There is a unique bilinear map $m_{p,q} : \Lambda^p V \times \Lambda^q V \rightarrow \Lambda^{p+q} V$ whose value on two elementary wedges is their concatenated wedge.

Proof. First factor the u -variables. Fix the tuple $\mathbf{v} = (v_1, \dots, v_q)$ and define

$$b_{\mathbf{v}}(u_1, \dots, u_p) = u_1 \wedge \dots \wedge u_p \wedge v_1 \wedge \dots \wedge v_q \in X.$$

The universal $(p+q)$ -wedge map is linear in each u slot, and equal u entries make its value zero. Thus $b_{\mathbf{v}}$ is alternating p -linear. The exterior universal property constructs a linear $L_{\mathbf{v}} : \Lambda^p V \rightarrow X$ with this value on elementary wedges.

Verify that the resulting map-valued map is alternating. Consider

$$B : V^q \rightarrow \mathcal{L}(\Lambda^p V, X), \quad B(\mathbf{v}) = L_{\mathbf{v}}.$$

Each value has the stated type by the preceding construction. Replace one slot v_j by $av_j + bv'_j$. There are now two existing linear maps $L_{(\dots, av_j + bv'_j, \dots)}$ and $aL_{(\dots, v_j, \dots)} + bL_{(\dots, v'_j, \dots)}$. On an elementary $u_1 \wedge \dots \wedge u_p$, they agree by the linearity of slot $p+j$ in the long wedge. Since elementary p -wedges span, the two maps agree on every $\alpha \in \Lambda^p V$. This proves linearity in slot j of B , not just linearity of an isolated scalar value. If two v entries agree, $L_{\mathbf{v}}$ sends every elementary p -wedge to zero by alternation of the long wedge. It is then the zero linear map by spanning. Thus B is alternating multilinear with values in $\mathcal{L}(\Lambda^p V, X)$.

Factor the v -variables. Apply the exterior universal property again to obtain a linear map

$$H : \Lambda^q V \rightarrow \mathcal{L}(\Lambda^p V, X), \quad H(v_1 \wedge \dots \wedge v_q) = L_{\mathbf{v}}.$$

Define $m_{p,q}(\alpha, \beta) = H(\beta)(\alpha)$. For fixed β , $H(\beta)$ is linear, giving linearity in α . For fixed α , linearity of H gives

$$H(a\beta + b\beta')(\alpha) = (aH(\beta) + bH(\beta'))(\alpha) = aH(\beta)(\alpha) + bH(\beta')(\alpha).$$

Thus $m_{p,q}$ is bilinear and has the prescribed concatenation values. At no step did a general wedge require a unique elementary decomposition. For uniqueness, any other bilinear product with those values agrees on pairs from the spanning elementary wedges; [Lemma B.34](#) gives equality everywhere. $\square$

Now that this construction is complete, write $\alpha \wedge \beta = m_{p,q}(\alpha, \beta)$. When a degree is zero, define the product by scalar multiplication in $F = \Lambda^0 V$. The construction can be tracked by the types

$$V^q \longrightarrow \mathcal{L}(\Lambda^p V, \Lambda^{p+q} V) \longleftarrow \Lambda^q V,$$

where the left arrow is alternating multilinear and the right arrow is the linear factor H , evaluated at α only at the final stage.

Proposition B.36 (Associativity and graded commutativity). For homogeneous elements of degrees p, q, r ,

$$(\alpha \wedge \beta) \wedge \gamma = \alpha \wedge (\beta \wedge \gamma), \quad \alpha \wedge \beta = (-1)^{pq} \beta \wedge \alpha.$$

Proof. Both expressions in associativity are existing trilinear maps of α, β, γ , since the products just constructed are bilinear. On elementary wedges both give the same concatenation of the three lists. The spanning-tuple lemma proves equality everywhere. For commutativity, moving a block of p vectors past a block of q requires pq interchanges, giving sign $(-1)^{pq}$ on elementary pairs. Both sides are bilinear, so the same lemma extends this identity to arbitrary homogeneous elements. A zero degree is governed by scalar multiplication, for which associativity and the sign 1 hold directly. $\square$

B.11 The Exterior Algebra and Its Grading

Recall finite direct sums from [Section 10.9](#). For $\dim V = n$, define the exterior algebra as the vector space

$$\Lambda V = \bigoplus_{k=0}^n \Lambda^k V.$$

An element $\omega = \omega_0 + \cdots + \omega_n$ has a unique component in each degree, by the direct-sum construction. It is homogeneous of degree k if only that component can be nonzero. Extend the products above by

$$\left(\sum_p \alpha_p \right) \wedge \left(\sum_q \beta_q \right) = \sum_{p,q} \alpha_p \wedge \beta_q;$$

degrees above n are zero. The degree- k component of the product is $\sum_{p+q=k} \alpha_p \wedge \beta_q$. This finite sum is well defined because the homogeneous components are unique. Associativity on homogeneous elements and finite distribution prove associativity for arbitrary elements. The scalar 1 is an identity. “Algebra” here means a vector space with this bilinear associative multiplication; no additional algebra course is needed to use the construction.

Example B.37 (Compute with basis wedges). Write e_I with its indices increasing. If I, J overlap, $e_I \wedge e_J = 0$; otherwise sort their concatenation and attach its sign. Thus

$$\begin{aligned} & (2e_1 + e_3) \wedge (e_1 \wedge e_2 + 3e_2 \wedge e_4) \\ &= 6e_1 \wedge e_2 \wedge e_4 + e_3 \wedge e_1 \wedge e_2 + 3e_3 \wedge e_2 \wedge e_4 \\ &= e_1 \wedge e_2 \wedge e_3 + 6e_1 \wedge e_2 \wedge e_4 - 3e_2 \wedge e_3 \wedge e_4. \end{aligned}$$

The middle term needs two swaps, the last one swap. Over $\mathbb{R}$, the two-form $\omega = e_1 \wedge e_2 + e_3 \wedge e_4$ has $\omega \wedge \omega = 2e_1 \wedge e_2 \wedge e_3 \wedge e_4 \neq 0$: the two cross terms add because $(-1)^{2 \cdot 2} = 1$. In characteristic two this square is zero; the earlier coefficient obstruction, not this square test, proves non-elementarity over every field.

Example B.38 (Basis images still produce the matrix). For $T(e_1) = e_1 + e_2$, $T(e_2) = e_2$, $T(e_3) = 2e_3$, the images of $e_1 \wedge e_2$, $e_1 \wedge e_3$, $e_2 \wedge e_3$ are respectively $e_1 \wedge e_2$, $2e_1 \wedge e_3 + 2e_2 \wedge e_3$, and $2e_2 \wedge e_3$. The matrix is

$$[\Lambda^2 T] = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 2 & 2 \end{pmatrix}.$$

This is the Chapter 10 basis-image procedure applied to an exterior basis.

B.12 Duality: Alternating Rules and Linear Functionals

B.12.1 First identification: factor an alternating map

Write $\text{Alt}^k(V; F)$ for the vector space of alternating k -linear scalar-valued maps, with pointwise operations.

Proposition B.39. There is a canonical isomorphism $\text{Alt}^k(V; F) \cong (\Lambda^k V)^*$.

Proof. For $A \in \text{Alt}^k(V; F)$ use [Theorem B.32](#) to construct its unique linear factor $F(A) = \tilde{A} : \Lambda^k V \rightarrow F$. For the reverse define $G(\ell)(v_1, \dots, v_k) = \ell(v_1 \wedge \dots \wedge v_k)$. Linearity of ℓ preserves each slot's linearity and the zero value on repeated inputs, so $G(\ell)$ is indeed in $\text{Alt}^k(V; F)$. Pointwise evaluation shows G is linear; uniqueness of the factor, or evaluation on spanning elementary wedges, shows F is linear. For each tuple, $G(F(A))(v_1, \dots, v_k) = A(v_1, \dots, v_k)$, so GF is the identity. On each elementary wedge, $F(G(\ell))(v_1 \wedge \dots \wedge v_k) = \ell(v_1 \wedge \dots \wedge v_k)$. Both are linear functionals; spanning proves FG is the identity. In degree zero both spaces are F , with the identity identification. $\square$

B.12.2 Second identification: pair covector wedges with vector wedges

Theorem B.40 (Exterior duality). For finite-dimensional V , there is a canonical isomorphism $P : \Lambda^k(V^*) \rightarrow (\Lambda^k V)^*$ characterized by

$$P(\lambda_1 \wedge \dots \wedge \lambda_k)(v_1 \wedge \dots \wedge v_k) = \det(\lambda_i(v_j)).$$

Proof. First factor in the vector variables. Fix the covector list $\lambda = (\lambda_1, \dots, \lambda_k)$. The map $(v_1, \dots, v_k) \mapsto \det(\lambda_i(v_j))$ is multilinear by the column laws of the determinant and vanishes when two vector entries agree. It therefore induces $L_\lambda \in (\Lambda^k V)^*$.

Then factor in the covector variables. Regard $\lambda \mapsto L_\lambda$ as a map $(V^*)^k \rightarrow (\Lambda^k V)^*$. The determinant row laws show slot linearity and repeated-input zero after evaluation on every elementary vector wedge. The resulting functionals already exist. Their equality on those spanning wedges proves these identities as equalities of functionals. Thus this map is alternating multilinear, and the exterior universal property gives the linear P with the asserted value.

Construct the inverse. Choose a basis (e_i) of V , with dual (e^i) . Let $e_I = e_{i_1} \wedge \dots \wedge e_{i_k}$ and $e^I = e^{i_1} \wedge \dots \wedge e^{i_k}$ for increasing I . The determinant pairing gives $P(e^I)(e_J) = \delta_{IJ}$: matching lists give the identity matrix and distinct lists give a zero column. Define

$$Q : (\Lambda^k V)^* \rightarrow \Lambda^k(V^*), \quad Q(\ell) = \sum_I \ell(e_I) e^I.$$

This finite sum is linear in ℓ . For every basis wedge e_J ,

$$P(Q(\ell))(e_J) = \sum_I \ell(e_I) \delta_{IJ} = \ell(e_J).$$

Thus $PQ(\ell) = \ell$ by equality of functionals on a basis. In the reverse direction

$$Q(P(e^I)) = \sum_J P(e^I)(e_J) e^J = e^I.$$

Both sides are linear in the covector wedge, and the e^I form a basis, so $QP = \text{id}$. The basis writes the inverse explicitly, but the inverse is independent of the basis because it is the unique inverse of the canonical P . In degree zero the pairing is multiplication of scalars; if $k > n$ both spaces are zero and the empty sums give their unique maps. $\square$

Combining the two identifications recovers the alternating covectors called $\Lambda^k(V^*)$ in Chapter 13. A vector wedge is made from vectors; a covector wedge consumes them and returns a scalar. For example $(dx \wedge dz)((1, 2, 3), (0, 1, 4)) = 4$. The determinant pairing relates these roles without confusing them or dividing by a factorial.

Proposition B.41 (Compatibility with the shuffle product). Let V be finite dimensional over $\mathbb{R}$. Denote by

$$J_k : \Lambda^k(V^*)_{\text{abstract}} \rightarrow \text{Alt}^k(V; \mathbb{R})$$

the determinant-pairing identification obtained above. For degrees p, q ,

$$J_{p+q}(\alpha \wedge \beta) = J_p(\alpha) \wedge_{\text{shuffle}} J_q(\beta).$$

Proof. Both sides are existing bilinear maps in α, β : the abstract product was constructed in Section B.10, the maps J_k by duality, and the shuffle product in Section 13.6. It suffices by Lemma B.34 to take $\alpha = \lambda_1 \wedge \cdots \wedge \lambda_p$ and $\beta = \mu_1 \wedge \cdots \wedge \mu_q$. Evaluate on $v_1, \dots, v_{p+q}$. The left side is the determinant whose rows are the values of $\lambda_1, \dots, \lambda_p, \mu_1, \dots, \mu_q$ on these vectors. Expand by the columns assigned to the first p rows. For each (p, q) -shuffle σ the contribution is

$$\text{sgn}(\sigma) \det(\lambda_i(v_{\sigma(j)}))_{i,j=1}^p \det(\mu_i(v_{\sigma(p+j)}))_{i,j=1}^q.$$

Summing gives precisely the shuffle definition on the right. This is the determinant/concatenation calculation already explained in Chapter 13. Equality on every vector tuple proves equality of these alternating forms; spanning now extends it to all α, β . Degree zero uses scalar multiplication and the empty determinant 1; degrees above $\dim V$ give zero. $\square$

Appendix B constructs an exterior algebra through quotients, whereas Chapter 13 constructs it through alternating covectors and shuffles. The canonical maps J_k identify their products as well as their vector spaces, so they give the same exterior algebra after this identification.

B.13 The Top Exterior Power and the Determinant

For $\dim V = n$, the basis theorem gives $\dim \Lambda^n V = 1$, with basis $e_1 \wedge \cdots \wedge e_n$. Any endomorphism of this one-dimensional space multiplies by a scalar. Which scalar does $\Lambda^n T$ use? Write $T(e_j) = \sum_i a_{ij} e_i$ and expand every slot:

$$T e_1 \wedge \cdots \wedge T e_n = \sum_{i_1, \dots, i_n} a_{i_1 1} \cdots a_{i_n n} e_{i_1} \wedge \cdots \wedge e_{i_n}.$$

Repeated indices vanish; the remaining choices are permutations. Sorting them gives their permutation signs, so the coefficient of the one basis wedge is exactly the Leibniz sum already proved for the determinant in Chapter 10. Therefore

$$\Lambda^n T = (\det T) \text{id}_{\Lambda^n V}.$$

This is a retrospective characterization of that same determinant, not a new definition. Rescaling the one basis wedge does not change the scalar by which a linear map acts. Compatibility of induced maps with composition then also explains $\det(ST) = \det S \det T$. In dimension zero the induced map on F is the identity, agreeing with the empty determinant 1.

Over $\mathbb{R}$ with $n > 0$, a nonzero top covector ω is a volume form. It selects the orientation of bases on which its value is positive. Replacing it by a positive multiple preserves that orientation, whereas changing its magnitude changes the normalization of volume. Evaluation on a basis and the preceding determinant computation give $T^* \omega = (\det T) \omega$. This joins the constructions: a linear map acts on vectors, its top exterior map records the signed scaling, and a top covector evaluates that scaling. The scalar identity works over every field; the language of positive orientation uses real order.

Exercises

Exercise B.1. Let $R = \text{span}(1, 1)$ in F^2 . Determine which of $x + y$ and $x - y$ factor through F^2/R , including characteristic two. Construct an isomorphism of the quotient with F , its reverse, and both composites.

Exercise B.2. Let $R \subseteq E$, $S \subseteq H$ be subspaces and $L : E \rightarrow H$ linear with $L(R) \subseteq S$. Construct $\bar{L} : E/R \rightarrow H/S$ by using the quotient universal property on $q_H L$. Verify the commutative square and prove uniqueness. Explain why the hypothesis is necessary.

Exercise B.3. Construct the map $F^2 \otimes F^2 \rightarrow F^2$ from $b(x, y) = (x_1 y_2, x_2 y_1)$, checking both slots first. Evaluate it on $(e_1 - e_2) \otimes (e_1 + 2e_2)$ using two different expansions.

Exercise B.4. A proposed real-valued rule on $\mathbb{R}^2 \otimes \mathbb{R}^2$ sends $v \otimes w$ to $\|v\|_2 \|w\|_2$. Compare the equal tensors $(-v) \otimes w = v \otimes (-w)$ and the additivity requirement on $(v + (-v)) \otimes w$. Decide whether the proposed values can be those of a linear functional, and identify the failed slot-linearity condition.

Exercise B.5. Suppose an extra relation $v \otimes w = w \otimes v$ is imposed on $F^2 \otimes F^2$. Use $b(v, w) = v_1 w_2$ to show the resulting quotient does not satisfy the ordinary tensor universal property.

Exercise B.6. Construct symmetry maps and induced tensor maps independently, then prove $S_{V', W'}(f \otimes g) = (g \otimes f)S_{V, W}$. Name the common value on an elementary tensor and explain why comparison is legitimate only after construction.

Exercise B.7. In [Example B.19](#), consider the component tuple $(0, e_1 \otimes f_1 + e_2 \otimes f_2, 0, 0)$. Send it backward explicitly and show that the result is not an elementary tensor of $A \otimes B$. Use coordinate projections to isolate a two-by-two coefficient table and its nonzero determinant. Explain why a dimension count alone cannot detect this.

Exercise B.8. For $A : V \rightarrow \mathcal{L}(W, U)$, distinguish the types of A , $A(v)$, and $A(v)(w)$. Reconstruct its tensor factor and prove the two currying inverse identities.

Exercise B.9. Expand $(e_1 + e_2) \wedge (e_2 + e_3)$ in F^3 , and recover the three coefficients with determinant detectors constructed from independent inputs. Compute its wedge with $e_1 + e_3$.

Exercise B.10. For $T(e_1) = e_1 - e_3$, $T(e_2) = 2e_2$, $T(e_3) = e_1 + e_3$ over $\mathbb{R}$, compute $\Lambda^2 T$ by its basis images and compare $\Lambda^3 T$ with $\det T$. Verify the quotient compatibility square on an elementary tensor.

Exercise B.11. In the construction of exterior multiplication, prove explicitly that the map $\mathbf{v} \mapsto L_{\mathbf{v}}$ takes equal v entries to the zero linear map. State where spanning, existence, and alternation are used.

Exercise B.12. For $\alpha = e^1 \wedge e^3$ and $\xi = (e_1 + e_2) \wedge (e_2 + 2e_3)$, compute the determinant pairing and identify the spaces containing α , ξ , and the answer. Construct the inverse duality map for this basis.

Exercise B.13. Over $\mathbb{R}$, set $E = \mathbb{R}^3$, $R = \text{span}(1, 1, 0)$, and $L(x, y, z) = (x - y, z)$. Construct the factor $E/R \rightarrow \mathbb{R}^2$ and an explicit inverse, checking both composites. If instead $R = \text{span}(1, 0, 0)$, show exactly where the same proposed quotient rule fails.

Exercise B.14. For the functional in [Example B.24](#), characterize its kernel by the twelve tensor-basis coefficients and give a basis of the kernel. Explain why starting from a formula on elementary tensors without first checking the three slots would leave a gap in this argument.

References and Further Reading

- [1] Robert G. Bartle. *The Elements of Real Analysis*. Second edition, John Wiley & Sons, 1976. A broad classical companion for Cartesian-space topology, sequences and function convergence, asymptotic and order notation, Riemann integration and convergence, finite-dimensional differentiation, and multivariable integration and change of variables. The manifold and differential-form development instead uses Munkres, Spivak, Tu, and Lee as its geometric references.
- [2] Stephen H. Friedberg, Arnold J. Insel, and Lawrence E. Spence. *Linear Algebra*. Fifth edition, Pearson. Chapters 1–2 provide the organizational benchmark for vector spaces, replacement, bases, linear maps, and their matrix representations. The analysis main text selects only the finite-dimensional real foundations.
- [3] Tom M. Apostol. *Calculus, Volume 2*. Second edition, Wiley. Further reading for the integration of linear algebra with multivariable calculus and a structural approach to determinants.
- [4] Tommy Chen. *Algebra lecture notes*. Author’s local revision draft, consulted September 12, 2026. Theorem 5.12.3, “Existence and uniqueness of the determinant form,” supplies the slot-by-slot expansion pattern adapted in Chapter 10. Section 1.5 guides the concrete permutation discussion in Appendix A. Chapter 8 guides the universal-property workflows and explanatory structure of Appendix B, specialized here to vector spaces over fields.
- [5] John M. Lee. *Introduction to Smooth Manifolds*. Graduate Texts in Mathematics 218, second edition, Springer, 2013. Cited for the external manifold-refinement theorem used in Chapter 13; also a technical reference for tangent spaces, boundary charts, orientation, partitions of unity, and integration.
- [6] Terence Tao. *Analysis I*. Springer. Foundations and rational Cauchy constructions; useful for checking the order of logical dependencies in Chapters 2–4.
- [7] Stephen Abbott. *Understanding Analysis*. Second edition, Springer, 2015. A companion for motivation, counterexamples, and learning to write the elementary proofs in Part I.
- [8] Kenneth A. Ross. *Elementary Analysis: The Theory of Calculus*. Second edition, Springer, 2013. Elementary rigor and graduated practice with sequences, continuity, and one-variable calculus.
- [9] Charles C. Pugh. *Real Mathematical Analysis*. Second edition, Springer, 2015. Geometric viewpoints on analysis, finite-dimensional topology, and multivariable differentiation.
- [10] Vladimir A. Zorich. *Mathematical Analysis I–II*. Springer. A broader analysis perspective, with geometric interpretations of multivariable derivatives, integration, and classical vector analysis.

- [11] James R. Munkres. *Analysis on Manifolds*. Addison–Wesley, 1991. A systematic companion for multivariable differentiation, integration, and the route from forms to Stokes.
- [12] Michael Spivak. *Calculus on Manifolds*. W. A. Benjamin, 1965. A compact account of the unity of the classical integral theorems; best read after the more detailed developments here.
- [13] Loring W. Tu. *An Introduction to Manifolds*. Second edition, Springer, 2011. Concrete differential forms, charts, tangent spaces, orientation, and integration; a natural next companion for Chapter 13.
- [14] Walter Rudin. *Principles of Mathematical Analysis*. International Series in Pure and Applied Mathematics, third edition, McGraw–Hill, 1976. Further reading for real analysis.
- [15] James R. Munkres. *Topology*. Second edition, Prentice Hall, 2000. Further reading for the general-topology background used sparingly in Part II.
- [16] Michael Spivak. *Calculus*. Fourth edition, Publish or Perish, 2008. Further reading for rigorous single-variable calculus.

Attribution note. Except for the external refinement theorem explicitly cited in Chapter 13, the principal results in these notes are proved internally. The references listed here provide verification and further reading; they are not claims that all of the manuscript’s proofs or exposition were adapted from those sources. The author’s algebra notes [4] are acknowledged locally for the determinant expansion pattern, their influence on the permutation exposition, and Appendix B’s proof architecture adapted from their Chapter 8. These specific adaptations do not describe the origin of the entire manuscript.

A reading route. For a second pass through foundations, pair Tao with Abbott or Ross. For Part II, use Pugh or Zorich for geometry and Munkres for a systematic calculus-on-manifolds treatment. Read Spivak for the compressed unifying picture after the details are familiar. Tu offers a gradual continuation into manifolds; Lee supplies a more extensive technical reference. None of their larger scopes is a prerequisite for these notes. For the algebraic foundations, compare Chapters 1–2 of Friedberg–Insel–Spence [2]; for determinants and their calculus role, compare Apostol [3] and Munkres. The optional appendices isolate the deeper algebraic constructions.